
Geometric Representation Theory

— *EYNTKA* Series —

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Part I

Structure Theory

The goal of Part I is to dismantle a reductive algebraic group into combinatorial data: a root system, encoded by a Dynkin diagram. This is the “anatomy” of the group — we are not yet asking how the group *acts* (that is the subject of Part II), but rather what the group *is*.

The reader is assumed to be familiar with scheme theory at the level of Hartshorne and Vakil [3], and with the definition of a group scheme. We work throughout with affine algebraic groups over a field k , and we take the functorial viewpoint seriously: an algebraic group is a representable functor $\mathbf{Alg}_k \rightarrow \mathbf{Grp}$, and its coordinate ring is a commutative Hopf algebra. This perspective — developed in Chapter 1 alongside the classical matrix-group examples — gives us two complementary languages for every construction: the geometric language of schemes and morphisms, and the algebraic language of Hopf algebras and comodules. We move freely between them, choosing whichever is cleaner for the problem at hand.

The first three chapters build the basic toolkit. Chapter 1 sets up the algebraic group itself: its Hopf algebra, its functor of points, and the linearization theorem that embeds every affine group into GL_n . Chapter 2 turns the group outward, formalizing what it means for an algebraic group to act on a scheme or a vector space, and establishing the category $\mathrm{Rep}(G)$ of representations as a rigid symmetric monoidal category. The chapter culminates in the Closed Orbit Lemma, a geometric result about orbit structures that will be used repeatedly. Chapter 3 extracts the infinitesimal invariant — the Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ — via the dual numbers $k[\varepsilon]$, constructs the adjoint representation, and proves Cartier’s theorem: in characteristic 0, the Lie algebra faithfully reflects the group. The chapter closes with a discussion of the failures in characteristic p , which is a fundamental reason the scheme-theoretic approach is taken rather than the purely Lie-algebraic one.

Chapter 4 addresses the quotient problem: given a closed subgroup $H \subseteq G$, does the coset space G/H exist as a scheme? Chevalley’s theorem provides the answer by realizing H as the stabilizer of a line in a representation, embedding G/H into projective space. This gives us the categorical infrastructure — faithfully flat exact sequences, the isomorphism theorems, and the identification of homogeneous spaces — needed for the rest of the book.

The second half of Part I is the decomposition theory. Chapter 5 begins with the Jordan decomposition, which splits every group element into a semisimple part and a unipotent part, and then studies each piece: tori (classified by their character lattices, with representations decomposing into weight spaces) and unipotent groups (classified in characteristic 0 by their nilpotent Lie algebras via the exponential map). Chapter 6 combines these into the theory of solvable groups. The Lie–Kolchin theorem says that connected solvable groups are “upper-triangular,” and Borel’s fixed-point theorem converts this linear-algebraic fact into a powerful geometric tool: connected solvable groups have fixed points on complete varieties. Applied to the flag variety G/B , this yields the conjugacy of Borel subgroups and maximal tori — the structural theorems that make the root system well-defined.

Chapter 7 brings everything together. The adjoint action of a maximal torus T on $\mathfrak{g}$ decomposes the Lie algebra into root spaces, and the resulting root system $\Phi \subseteq X(T)$ is a finite combinatorial object governed by the Weyl group $W = N_G(T)/Z_G(T)$. The classification of root systems via Dynkin diagrams is a purely combinatorial theorem — there are four infinite families (A_n, B_n, C_n, D_n) and five exceptional cases (G_2, F_4, E_6, E_7, E_8) — and the isomorphism and existence theorems for root data assert that this combinatorial classification is equivalent to the geometric classification of reductive groups. The passage from geometry to combinatorics loses no information: every root datum arises from a group, and groups with isomorphic root data are isomorphic.

A quick word must be given on the characteristic of the fields we are working with. Most theorems hold over arbitrary algebraically closed fields, and we explicitly flag the (few but important) places where characteristic 0 is needed — notably Cartier’s theorem and the exponential map for unipotent

groups. The infinitesimal group schemes μ_p and α_p make regular appearances as cautionary examples, reminding us that the Lie algebra is an imperfect tool in characteristic p and that the full scheme-theoretic machinery is not a luxury but a necessity.

1

Algebraic Groups

Algebraic groups represent the synthesis of algebra and algebraic geometry. By endowing a variety or a scheme with a group structure-where the group operations are morphisms of the underlying space-we obtain objects that are rigid enough to be classified completely, yet rich enough to capture the symmetries of almost any geometric or algebraic structure. In this chapter, we establish the fundamental structure of algebraic groups. We will transition from the foundational geometric definitions to their categorical interpretation as functors, explore the algebraic underpinnings via Hopf algebras, and lay the groundwork for the structural decomposition and classification of linear algebraic groups.

1.1 Definitions and Properties

Recall [3] that a *group scheme* is an S -scheme X for some base S along with morphisms $m : G \times_S G \rightarrow G$, $\iota : G \rightarrow G$, $e : \text{Spec } S \rightarrow X$ that satisfy the group axioms as commutative diagrams:

$$\begin{array}{ccc} G \times_S G \times_S G & \xrightarrow{m \times \text{id}_G} & G \times_S G \\ \text{id}_G \times m \downarrow & & \downarrow m \\ G \times_S G & \xrightarrow{m} & G \end{array}$$

Associativity Axiom: $(g \cdot h) \cdot k = g \cdot (h \cdot k)$

$$\begin{array}{ccc} G & \xrightarrow{\Delta_G} G \times_S G & \xrightarrow{\text{id}_G \times \iota} G \times_S G \\ & \searrow e \circ (G \rightarrow S) & \downarrow m \\ & & G \end{array}$$

Inverse Axiom: $g \cdot g^{-1} = e$

$$\begin{array}{ccc}
S \times_S G & \xrightarrow{e \times \text{id}_G} & G \times_S G \\
& \searrow \cong & \swarrow m \\
& G &
\end{array}$$

Left Unit Axiom: $e \cdot g = g$

$$\begin{array}{ccc}
G \times_S S & \xrightarrow{\text{id}_G \times e} & G \times_S G \\
& \searrow \cong & \swarrow m \\
& G &
\end{array}$$

Right Unit Axiom: $g \cdot e = g$

Equivalently it is the group object in the category $\mathbf{Sch}_S$. If X is locally finite type over S , then X is called *locally algebraic*, and if it is furthermore quasicompact it is called *algebraic*. Combining these two definitions gets us what is common general definition of an algebraic group. In many cases, the definition is restricted to (abstract) varieties, which the following definition accommodates.

Definition 1.1.1: Algebraic Group

Let G be a variety or a group k -scheme. Then G is an *algebraic group* if it is finite type over k (which is automatic if G is a variety), and G has a group structure with multiplication and inversion:

$$\mu : G \times_k G \rightarrow G \quad \text{and} \quad \iota : G \rightarrow G$$

that are also morphisms of varieties (or k -schemes). The multiplication map μ is commonly written as $g \cdot h := \mu(g, h)$.

If we say G is an *algebraic group*, we shall assume it is an algebraic group over a *field*. If we want to be more general and not assume it is over a field, we shall say that G is an *algebraic group scheme*.

When it is clear from context, the fibered product will drop the k and will be written as $G \times G$. We shall mostly be working with groups over a perfect field or groups over an algebraically closed field of characteristic 0. By the Lefschetz principle (see [5, appendix]), if k is algebraically closed of characteristic 0, there is no theoretical issue assuming $k = \mathbb{C}$. We dropped the identity map as we can deduce it using the other two maps. To see why, consider

$$G \rightarrow G \quad g \mapsto m(g, \iota(g)) = e$$

which is a (continuous) constant map. Restricting to the open affine $\text{Spec } A \subseteq G$ containing e , we get a constant map $\varphi : \text{Spec } A \rightarrow \text{Spec } A$, $\varphi(x) = e$. The corresponding map $\varphi^\# : A \rightarrow A$ map every $\alpha \in A$ to the value of α at e , essentially $\alpha(e)$. These are all constants, hence $\text{im}(\varphi^\#) \subseteq k$. We thus have a map $\varphi^\# : A \rightarrow k$. But then this means we have a map $\varphi : \text{Spec } k \rightarrow \text{Spec } A \hookrightarrow G$. Note how since k is assumed to be algebraically closed, by the correspondence of k -valued points and maximal points of a scheme, e must be closed. It is important that there *do* exist group-schemes where the constant is nonzero (see [this Stacks Project example](#)), however any group-scheme over a field necessarily has a closed identity.

Note that G is *not* a topological group (except in dimension 0) as $G \times G$ has the *Zariski topology* and not the product topology (show a topological group that is T_1 must be T_2 , but G is not T_2). The translation map $x \mapsto xy$ is always an isomorphism, which also means that all geometric properties points are identical. As there are always smooth points, all points of G must be smooth.

Proposition 1.1.2: Properties of Algebraic Groups

Let G be an algebraic group over a field an algebraically closed field $k = \bar{k}$. Then:

1. G is separated.
2. (Cartier's Theorem) G is smooth.
3. If G is furthermore connected, then it is geometrically integral over k .
4. Only one irreducible component contains the identity.
5. G is equidimensional so that $\dim(G) = \dim_g(G)$ for all $g \in G$ (recall $\dim_g(G)$ is the dimension of g at G)
6. Closed subgroup of an algebraic group is an algebraic group
7. the direct product (really, fibered product) of algebraic groups is an algebraic group.

Proof:

1. If $\Phi : G \times G \rightarrow G$ given by $(g, h) \rightarrow g \cdot h$, as e is closed, and $\Delta = \Phi^{-1}(e) = \{(g, g) \mid g \in G\}$, we have that the diagonal is closed and hence G is separated.
2. This is simply combining facts from [3], namely the module of differentials of G over k is free, G is smooth
3. (put the right reference)
4. Let's say $X = \bigcup_i^m X_i$ and $X_1, \dots, X_n$ are (irreducible) components contain the identity. Then $X_1 X_2 \cdots X_n$ contains the identity. The product map $\prod_i^n X_i \rightarrow X$ maps irreducible to irreducible (as is easy to check topologically) hence it must be that $X_1 X_2 \cdots X_n \subseteq X_i$ for some i . But certainly $X_j \subseteq X_1 X_2 \cdots X_n$ for each $1 \leq j \leq n$. Thus it must be that a single component contains the identity.
5. exercise.
6. exercise
7. exercise.

We give the component with the identity a name:

Definition 1.1.3: Identity Component

Let G be an algebraic group. Then the unique irreducible component containing the identity is called the *identity component*. It will be denoted G° .

Proposition 1.1.4: Identity Component and Connectedness

Let G be an algebraic group. Then:

1. $G^\circ \leq G$ is a normal subgroup of finite index in G whose cosets are connected and irreducible.
2. each closed subgroup of finite index in G contains G° .

Proof:

1. For each $x \in G^\circ$, certainly $x^{-1}G^\circ$ is an irreducible component passing through e , hence $x^{-1}G^\circ = G^\circ$. The same logic shows $xG^\circ x^{-1} = G^\circ$. Therefore, $(G^\circ)^{-1} = G^\circ$ and $G^\circ G^\circ = G^\circ$, showing G° is a normal subgroup. The cosets are all connected as they are all disjoint and translations of G° which is connected.
2. Let H be a closed subgroup of finite index. Then its complement is also closed as it's a finite union of closed sets, hence H is also open, hence connected. Thus the left cosets of H partition G° , and as G° is connected and meets H , $H \subseteq G^\circ$.

As the term irreducible in representation theory has a different meaning, we shall usually say that G is *connected* if $G = G^\circ$; the above makes it unambiguous as the irreducible components are cosets of G° , and if G is connected it can only have G° as its single irreducible component.

The following will not be proven here, but it is important to know as it shows why the assumption that G is a variety is not much of a restriction

Theorem 1.1.5: Algebraic Group, then Quasi-Projective

Let G be an algebraic group over a field k . Then G is *quasi-projective* over k .

Proof:

see <https://stacks.math.columbia.edu/tag/0BF6>.

It is important to take a moment to single out proper algebraic groups as they are *not* the focus of the book, and we must justify why:

Definition 1.1.6: Abelian Variety

Let k be a field. An *abelian variety* is a group scheme over k which is also a proper, geometrically integral variety over k .

There are many equivalent definitions of abelian variety (ex. proper for abelian variety implies projective, and geometrically integral shall imply smooth), but we leave this proof for a book focused on abelian varieties. Notice that they are not defined to be abelian, and yet their group structure shall *forced* to be abelian. Furthermore, as they are proper they have no non-constant global functions which we shall see implies the map $\rho : A \rightarrow \mathrm{GL}_n(k)$ must be *trivial*, namely their representation

theory isn't of much interest to this book¹.

The category of algebraic groups (or the group-objects in $\mathbf{Var}_k$ or $\mathbf{Sch}_k$) will have groups that are affine, proper, or neither. However, due to a theorem by Chevalley, the study of algebraic group splits into “affine” and “abelian”. Namely G has a normal subgroup H such that H is affine and G/H is projective (i.e. an “abelian variety”).

Theorem 1.1.7: Chevalley Theorem

Let k be a perfect field and G an algebraic group over k . Then there exists a unique normal linear algebraic closed subgroup H in G for which G/H is an abelian variety. That is, there is a unique short exact sequence of algebraic groups

$$1 \rightarrow H \rightarrow G \rightarrow A \rightarrow 1$$

with H linear algebraic and A an abelian variety. The formation of H commutes with base change to an arbitrary perfect field extension over k .

Proof:

see [7]

Thus, this book focuses on *affine algebraic groups*. We can reduce one step further by showing all such groups are in fact linear:

Lemma 1.1.8: Affine Algebraic Groups are Linear

Let G be an affine algebraic group over k . Then G is a closed subgroup of the affine algebraic group GL_n for some n .

Proof:

Let $G = \mathrm{Spec} A$ where A is a finitely generated k -algebra, say $A \cong k[x_1, \dots, x_r]/(f_1, \dots, f_s)$. Define a right-regular action of G on itself: $r_g(x) = xg$. This induces an action on $A = \Gamma(G)$, namely for any $f \in A$:

$$(g \cdot f)(x) = f(xg)$$

giving a homomorphism $\rho : G \rightarrow \mathrm{Aut}_k(A)$. If this action can be restricted to a finite-dimensional G -stable subspace $V \subset A$, we obtain a morphism $\rho : G \rightarrow \mathrm{GL}(V) \cong \mathrm{GL}_n(k)$ where $n = \dim V$. We will show that such a V exists, that ρ is a group homomorphism, and that it is a closed immersion.

First, let's show there is a finite dimensional V . Given the multiplication $\mu : G \times_k G \rightarrow G$, by affineness we have the dual (comultiplication) map $\mu^* : A \rightarrow A \otimes_k A$. For any $f \in A$,

$$\mu^*(f) = \sum_{i=1}^d f_i \otimes h_i$$

¹This is “fixed” by studying the line bundles on A and their change under the Fourier transform (more specifically the Fourier-Mukai Transform); this recovers the notion of characters from the representation of finite abelian groups

for some $f_i, h_i \in A$. Then for any x, g :

$$(g \cdot f)(x) = f(xg) = \sum_{i=1}^d h_i(g) f_i(x)$$

and hence

$$g \cdot f = \sum_{i=1}^d h_i(g) f_i.$$

So $g \cdot f$ lies in the subspace W_f spanned by $\{f_1, \dots, f_d\}$, which is finite-dimensional. The key here is that g can range over all of G , and $g \cdot f$ lands in the finite dimensional space W_f .

We claim that W_f can be taken to be G -stable (i.e. $\mu^*(W_f) \subseteq W_f \otimes_k A$). Without loss of generality, assume the f_i in $\mu^*(f) = \sum_i f_i \otimes h_i$ are chosen to be linearly independent over k . By coassociativity, $(\mu^* \otimes \text{id}) \circ \mu^* = (\text{id} \otimes \mu^*) \circ \mu^*$, so:

$$\sum_{i=1}^d \mu^*(f_i) \otimes h_i = \sum_{i=1}^d f_i \otimes \mu^*(h_i).$$

The right-hand side lies in $W_f \otimes_k A \otimes_k A$. Since $\{h_i\}$ can be extended to a basis of a subspace of A and the f_i are linearly independent, applying $\text{id} \otimes \text{id} \otimes \varphi$ for arbitrary $\varphi \in A^\vee$ to both sides shows that $\mu^*(f_i) \in W_f \otimes_k A$ for each i . Hence W_f is G -stable.

Now, as A is a finitely generated k -algebra, choose generators $\alpha_1, \dots, \alpha_m$. Then each α_j is contained in a finite-dimensional G -stable subspace $W_{\alpha_j} \subset A$. Let

$$V = \sum_{j=1}^m W_{\alpha_j}.$$

Then V is a finite-dimensional G -stable subspace of A containing each generator α_j , and the action of G restricts to give a representation

$$\rho : G \rightarrow \text{GL}(V) \cong \text{GL}_n(k), \quad n = \dim V.$$

Note that $\text{GL}_n(k)$ is an affine algebraic group with coordinate ring

$$k[\text{GL}_n] \cong \frac{k[X_{11}, \dots, X_{nn}, T]}{(T \cdot \det(X) - 1)}$$

Let $v_1, \dots, v_n$ be a basis for V . Then for each $g \in G$:

$$\rho(g) \cdot v_j = \sum_{i=1}^n M_{ij}(g) v_i$$

where $M_{ij} : G \rightarrow k$ are regular functions on G (i.e. $M_{ij} \in A$) representing the matrix entries. The pullback $\rho^\# : k[\text{GL}_n] \rightarrow A$ is the k -algebra homomorphism determined by $\rho^\#(X_{ij}) = M_{ij}$, so ρ is a k -morphism of varieties. That ρ is a group homomorphism follows from the definition: for $g_1, g_2 \in G$ and $f \in A$,

$$(g_1 g_2) \cdot f(x) = f(x g_1 g_2) = (g_2 \cdot (g_1 \cdot f))(x),$$

so $\rho(g_1 g_2) = \rho(g_2) \circ \rho(g_1)$ (or with the standard convention reversal, ρ defines a homomorphism $G \rightarrow \mathrm{GL}(V)$).

What's left to show is that ρ is a closed immersion, where it suffices to show that $\rho^\# : k[\mathrm{GL}_n] \rightarrow A$ is *surjective*. Since V is G -stable, the coaction satisfies $\mu^*(v_j) = \sum_{i=1}^n v_i \otimes M_{ij}$. Applying the counit axiom $(\epsilon \otimes \mathrm{id}) \circ \mu^* = \mathrm{id}$ gives:

$$v_j = \sum_{i=1}^n \epsilon(v_i) M_{ij}.$$

So each basis element $v_j \in V$ is a k -linear combination of the matrix entries $\{M_{ij}\}$. In particular, each algebra generator $\alpha_l \in V$ lies in the k -span of the M_{ij} . Since the α_l generate A as a k -algebra, so do the M_{ij} . As $\rho^\#(X_{ij}) = M_{ij}$, the map $\rho^\#$ is surjective. Therefore $\rho : G \hookrightarrow \mathrm{GL}_n(k)$ is a closed immersion, and G is a closed subgroup of GL_n .

Hence, we have reduced to showing that we are interested in groups G defined by a finite number of polynomial equations!

1.1.1 Examples of Algebraic Groups

Starting with the 0 dimensional example. All finite groups are affine algebraic groups. Indeed, every finite subgroup G is a subgroup of S_n , and S_n is a subgroup of $\mathrm{GL}_n(k)$ by associating a permutation $\sigma \in S_n$ to the matrix $I_\sigma \in \mathrm{GL}_n(k)$ where σ permutes the rows of the identity matrix $I \in \mathrm{GL}_n(k)$. As this is a finite subset, it is easy to check that it is algebraic (that is every finite subset of (closed) points is a subscheme)

This also means that all finite groups can be interpreted as schemes. Given a finite group G interpreted as a closed algebraic subgroup of $\mathrm{GL}_n(k)$, then this scheme is isomorphic to $\mathrm{Spec} \left(\prod_{i=1}^{|G|} k \right)$. This is the *constant finite algebraic groups*.

Interestingly, it is *not* the case that finite group schemes are in bijective correspondence with abstract groups; there are *more* group schemes. To see this, consider the group $G = \mu_3$ which are the 3rd roots of unity. Purely on the level of group theory, $\mu_3 \cong C_3$ (the cyclic group of order 3). However, μ_3 has further structure. Take the $\mathbb{Q}$ -scheme $G_{\mathbb{Q}} = \mathrm{Spec} \mathbb{Q}[x]/(x^3 - 1)$. Then $G_{\mathbb{Q}}$ has 2 points as $\mathbb{Q}[x]/(x^3 - 1) \cong \mathbb{Q} \times \mathbb{Q}(\omega)$ given a 3rd root of unity ω . In this group, we “fused” two points (namely ω, ω^2). To recover G , we must look at the *geometric points* of $G_{\mathbb{Q}}$, namely the algebraic closure (i.e. base-changing to $\overline{\mathbb{Q}}$) shall separate these points so that $G_{\overline{\mathbb{Q}}}$ has three points. We once again have recovered C_3 , but now there is a natural action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ associated to C_3 ! We can interpret the $\mathbb{Q}$ -valued points, $G_{\mathbb{Q}}(\mathbb{Q})$, as the fibers of $G_{\overline{\mathbb{Q}}}$ which explains why they “split” when extending to $\overline{\mathbb{Q}}$.

More generally, if G is a finite étale algebraic group over k , then $\Gamma(G)$ is an étale algebra, that is:

$$\Gamma(G) \cong L_1 \times \cdots \times L_r$$

Any such scheme corresponds to a finite group along with a (continuous) action of $\mathrm{Gal}(\overline{k}/k)$. These are called *twisted finite algebraic groups*. In the extreme case, we have some finite extension L/K of degree n so that $G_k = \mathrm{Spec}(L)$ has a single point, but n geometric points that are twisted via $\mathrm{Gal}(\overline{K}/K)$ where the action naturally restricts to $\mathrm{Gal}(L/K)$.

The term “twisted” comes directly from Galois cohomology and shares the same geometric intuition as twisted vector bundles. A finite group scheme G over k is called a *twisted form* of a constant

group scheme G_{const} if they become isomorphic over the separable algebraic closure $\bar{k}$. That is, there exists an isomorphism of group schemes $\varphi : G_{\bar{k}} \xrightarrow{\sim} (G_{const})_{\bar{k}}$. While G and G_{const} look identical geometrically over $\bar{k}$, the isomorphism φ fails to commute with the natural Galois actions on both sides. This failure is measured by the map $a_\sigma = \varphi \circ \sigma \circ \varphi^{-1} \circ \sigma^{-1}$, which defines a 1-cocycle in $H^1(\text{Gal}(\bar{k}/k), \text{Aut}(G_{const}))$. This classification of twisted forms by H^1 is identical to the classification of varieties described in [4]

We can interpret this physically as a conflict between two Galois actions. The constant group G_{const} possesses a “standard” action σ_{std} that acts trivially on the group elements (simply permuting the copies of k). The twisted group G possesses its own intrinsic Galois action derived from its definition over k . The cocycle a_σ acts as a set of instructions for “gluing” the standard object to itself with a twist: to reconstruct the group G from G_{const} , we define a *new* action $\sigma_{new}(x) := a_\sigma(\sigma_{std}(x))$ on the geometric points. The k -rational points of the twisted group G are exactly the fixed points of this new, twisted action.

Returning to our example of $G = \mu_3$ and $G_{const} = C_3$, over $\bar{\mathbb{Q}}$ we have an isomorphism mapping the roots $\{1, \omega, \omega^2\}$ to the abstract elements $\{e_1, e_\sigma, e_{\sigma^2}\}$. The standard action on C_3 fixes every element. However, the Galois action on μ_3 (complex conjugation τ) swaps ω and ω^2 . This mismatch means our cocycle $a_\tau \in \text{Aut}(C_3)$ must be the automorphism that swaps e_σ and e_{σ^2} . Thus, while the constant group scheme corresponds to the trivial class in H^1 , the roots of unity μ_3 correspond to a non-trivial class in $H^1(\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}), \text{Aut}(C_3))$, representing a specific twist of the constant structure.

For those familiar with group schemes as group functors we can generalize this idea using functor of points. If $G = \text{Spec}(A)$ is an affine algebraic group take the representable functor:

$$R \mapsto G(R) = \text{Hom}_{k\text{-}\mathbf{Alg}}(A, R)$$

For constant group schemes we have $G(k) = \text{Hom}(k^n, k)$. Since maps to k must pick out one component, there are exactly n such maps. For $G = \text{Spec}(L)$, then $G(k) = \text{Hom}(L, k)$ which has only the identity as L only has the trivial map into k (i.e. $G(k)$ only see’s the identity, or everything is glued to the identity). If instead we take L -points: $G(L) = \text{Hom}(L, L) = \text{Gal}(L/k)$, recovering the twisting information.

Next, consider the following 1-dimensional examples:

1. $G = \mathbb{G}_a = \mathbb{A}^1$ so that the map is given by $\mu(x, y) = x + y$, $\iota(x) = -x$ and $e = 0$.
2. $G = \mathbb{G}_m = (\mathbb{A}^1)^*$ is given by $\mu(x, y) = xy$, $\iota(x) = x^{-1}$ and $e = 1$.

Both of these groups are irreducible and 1-dimensional. We shall see in theorem 5.2.4 that (up to isomorphism) these are the *only* irreducible 1-dimensional groups.

Dimension 2 and higher start having non-trivial examples, and so they are left to be treated by the book.

A natural generalization of $\mathbb{G}_a$ is $\mathbb{A}^n$ with its natural point-wise additive structure. We can denote this $\mathbb{G}_a(\mathbb{A}^n)$. We cannot do the same for $(\mathbb{A}^n)^*$ as there is no inverse (ex. $(1, 0, 1)$ has no multiplicative inverse). Instead, we define the *maximal torus* $\mathbb{G}_m(\mathbb{A}^n) = (\mathbb{A} \setminus \{0\})^n$.

As all algebraic groups are linear, it is important look at important subgroup of the general linear group. Let K be a field. Then $M_n(K)$ as a variety can be interpreted as:

$$M_n(K) = \text{Spec } k[x_{11}, \dots, x_{nn}]$$

A (closed) point of $M_n(K)$ is written as a matrix to keep better track of the multiplication structure. Then as matrix multiplication is associative, has I as the identity, and the resulting matrix $A \cdot B$ for any $A, B \in M_n(K)$ can be written component-wise as a linear polynomial, this forms an *algebraic monoid*. As $\det : M_n(K) \rightarrow K$ is a continuous polynomial function, the open subset where $\det \neq 0$ forms a (principal) *open submonoid* $\mathrm{GL}_n(K)$, which we know is also a *group* as a matrix is invertible if $\det(X) \neq 0$. However, as $M_n(K)$ is a submonoid, it is not immediate through this logic that $\mathrm{GL}_n(K)$ is closed as the left/right multiplication maps are continuous but *not all* homeomorphisms. Instead, define:

$$\mathrm{GL}_n(K) = \mathrm{Spec} \frac{K[x_{11}, \dots, x_{nn}, y]}{y \cdot \det(X) - 1}$$

where $X = (x_{ij})_{ij}$. Note that $\mathrm{GL}_n(K)$ is always irreducible when K is infinite as $\det(X)y - 1$ is a linear with respect to y and so by Gauss's lemma irreducible in this ring (contrast this to $\mathrm{GL}_n(\mathbb{R})$ with the euclidean topology), while if K is finite then $\mathrm{GL}_n(K)$ is totally disconnected. Note that the orthogonal group O_n is a natural non-connected algebraic group.

The following algebraic groups will repeatedly be brought up:

1. The special linear group $\mathrm{SL}_n(K)$, given by $\det(T_{ij}) - 1$
2. The symplectic group $\mathrm{Sp}_{2n}(K)$ consisting of all $x \in \mathrm{GL}_{2n}(K)$ where

$${}^t x \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix} x = \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix}$$

where ${}^t x$ is the transpose.

3. The special orthogonal group $\mathrm{SO}_{2n+1}(K)$ (where $\mathrm{char} K \neq 2$) given by all $x \in \mathrm{SL}_{2n+1}(K)$ where:

$${}^t x s x = s$$

where

$$s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & J \\ 0 & J & 0 \end{pmatrix}$$

We shall sometimes specify the special orthogonal group $\mathrm{SO}_n(K)$ where $s = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}$

The classification of all subgroups of $\mathrm{GL}_n(K)$ is intractable, namely all finite subgroups are subgroups of $\mathrm{GL}_n(K)$ for some n , and furthermore the following shows the problem compounded by the further structure of $\mathrm{GL}_n(K)$:

Theorem 1.1.9: Tits Alternative

Let G be a finitely generated linear group over a field k . Then two following possibilities occur:

1. either G is virtually solvable (i.e., has a solvable subgroup of finite index or
2. it contains a nonabelian free group (i.e., it has a subgroup isomorphic to the free group on two generators).

Proof:

see [11]

If we restrict to *linear* subgroups, then we shall be able to fully classify them; see theorem 7.3.6 for details (or the chapter on quiver theory in [5] for a proof using roots system).

Next, let us introduce two more subgroups of GL_n that shall be important for their role in forming a composition series that any algebraic group G decomposes into. Namely, we have a natural maximal subnormal series:

$$e \trianglelefteq U \trianglelefteq N \trianglelefteq A \trianglelefteq G^\circ \trianglelefteq G$$

where each G_i/G_{i+1} is simple and:

1. G contains a unique connected normal subgroup G° such that G/G° is finite (see proposition 1.1.4)
2. G° contains a largest (hence unique) connected affine normal subgroup A such that G°/A is an abelian variety (see theorem 1.1.7)
3. A contains a largest (hence unique) connected affine solvable normal subgroup N such that A/N is semisimple (see definition 6.2.10)
4. N contains a largest (hence unique) connected unipotent normal subgroup U such that N/U is a torus (see theorem 6.1.7)

Thus, the *torus* and *unipotent* groups are singled out for their general immediate usefulness for upcoming examples:

Definition 1.1.10: [Algebraic] Torus

Let $\mathrm{GL}(V)_k$ be the affine general linear group over a finite dimensional k -vector space V and $T \subseteq \mathrm{GL}(V)$ an affine subgroup. Let $Z = Z(\mathrm{GL}(V))$ be the center of $\mathrm{GL}(V)$ (i.e. all diagonal matrices). Then T is said to be of *multiplicative type* if when extending to $\bar{k}$ $T_{\bar{k}} \subseteq Z_{\bar{k}}$. In other words, there exists a basis for V such that $T_{\bar{k}}$ is represented by diagonal matrices, i.e. the elements of an algebraic torus are semisimple endomorphisms of V .

If T is furthermore connected, it is called a *torus*.

The name comes from the fact that over the complex numbers $\mathbb{C}$, the group of points of an algebraic torus is isomorphic to a subgroup of $(\mathbb{C}^\times)^n$, which contains the standard topological torus $(S^1)^n$ as its maximal compact subgroup.

Definition 1.1.11: Unipotent Group

Let $\mathrm{GL}(V)_k$ be the affine general linear group over a finite dimensional k -vector space V and $G \subseteq \mathrm{GL}(V)$ an affine subgroup. Then G is called *unipotent* if there exists a basis of V relative to which G is contained in the group U_n of all $n \times n$ matrices of the form

$$\begin{pmatrix} 1 & * & \cdots & * & * \\ 0 & 1 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & * \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix}$$

In particular, the elements of a unipotent group are unipotent endomorphisms of V .

We will develop the formal definitions and implications of these building blocks—including solvable, reductive, simple, quasi-simple, and semi-simple groups—systematically in the upcoming sections.

1.1.2 Hopf Algebra

Let $G = \mathrm{Spec} A$ be an algebraic group (over k) so that $\mu : G \times G \rightarrow G$, $\iota : G \rightarrow G$ and $\epsilon : \mathrm{Spec} k \rightarrow G$ are the maps satisfying the group operations. As G is affine, these maps correspond to “co”-maps on A . Namely:

1. multiplication corresponds to comultiplication: $\Delta : A \rightarrow A \otimes_k A$ (recall the tensor product is the coproduct in $\mathbf{Alg}_k$, see [5, chapter 15.2])
2. inverse corresponds to the *antipode*: $\iota : A \rightarrow A$
3. unit corresponds to the *counit*: $\epsilon : A \rightarrow k$

As the contravariant functor $\Gamma(-)$ inputting affine schemes is fully faithful, the group axioms on G are equivalent to their “co” version on $A = \Gamma(G)$. Let us take a moment to internalize the co-versions of the axioms. Recall that the group operation is associative. In terms of maps, the associativity of $\mu : G \times G \rightarrow G$ is expressed by the equality:

$$\mu \circ (\mu \times \mathrm{id}_G) = \mu \circ (\mathrm{id}_G \times \mu)$$

as maps $G \times G \times G \rightarrow G$. Applying the functor to A , we reverse the arrows and replace $\times$ with $\otimes$. Thus, the multiplication μ becomes comultiplication Δ , and we obtain the *coassociativity* axiom:

$$(\Delta \otimes \mathrm{id}_A) \circ \Delta = (\mathrm{id}_A \otimes \Delta) \circ \Delta$$

This ensures that “splitting” an element once and then splitting the first factor is the same as splitting the second factor.

The identity element e is defined by the property $g \cdot e = g = e \cdot g$. In the language of schemes, this is a commutative diagram involving the projection $G \times G \rightarrow G$. The “unit” of the group is a map $e : \mathrm{Spec} k \rightarrow G$. The “counit” is the dual map $\epsilon : A \rightarrow k$. Just as the unit element allows us to

collapse one factor of a product $G \times G \rightarrow G$ seamlessly, the counit allows us to collapse one factor of the tensor product $A \otimes A$ back into k . The axiom becomes:

$$(\epsilon \otimes \text{id}_A) \circ \Delta = \text{id}_A \quad \text{via the isomorphism } k \otimes A \cong A$$

Finally, we consider the inverse. In a group, the inverse is a map $\iota : G \rightarrow G$ satisfying $g \cdot g^{-1} = e$. Let $\Delta_G : G \rightarrow G \times G$ be the diagonal map $g \mapsto (g, g)$. The axiom $g \cdot g^{-1} = e$ states that the following composition is the trivial map (factoring through e):

$$G \xrightarrow{\Delta_G} G \times G \xrightarrow{\text{id} \times \iota} G \times G \xrightarrow{\mu} G \quad \text{equals} \quad G \rightarrow \text{Spec } k \xrightarrow{e} G$$

Dualizing this to A , the diagonal map on spaces becomes the multiplication map $m : A \otimes A \rightarrow A$ (since functions multiply pointwise). The multiplication map μ on the group becomes the comultiplication Δ . The inverse ι becomes a map $S : A \rightarrow A$, which we call the *Antipode*. Reversing the chain above, we get the definition of the antipode:

$$m \circ (\text{id} \otimes S) \circ \Delta = u \circ \epsilon$$

where $u \circ \epsilon$ is the composition $A \xrightarrow{\epsilon} k \xrightarrow{u} A$, representing the “constant function” on the group.

Thus, a Hopf Algebra is simply a ring that has enough “co-structure” to simulate being the functions on a group.

Definition 1.1.12: Hopf Algebra

Let k be a field. A *Hopf algebra* is a k -vector space A equipped with an algebra structure (A, m, u) and a coalgebra structure (A, Δ, ϵ) that are compatible (i.e., Δ and ϵ are algebra homomorphisms), along with a k -linear map $S : A \rightarrow A$ called the *antipode* (or *Sweedler map*) such that:

$$m \circ (S \otimes \text{id}_A) \circ \Delta = u \circ \epsilon = m \circ (\text{id}_A \otimes S) \circ \Delta$$

Note that there was no mention of A coming from an algebraic group. If $G = \text{Spec } A$ is an algebraic group, then $A = \Gamma(G)$ is a commutative Hopf algebra, though naturally not necessarily *cocommutative* as the corresponding group multiplication need not be commutative. The axioms above are exactly the group axioms for G (associativity, identity, inverse) expressed via commutative diagrams in the category of k -algebras with arrows reversed. Given G , we can compute delta as it must satisfy the commutative diagram:

$$\begin{array}{ccc} G \times G & \xrightarrow{\mu} & G \\ & \uparrow \text{corresponds to} & \\ A \otimes A & \xleftarrow{\Delta} & A \end{array}$$

namely:

$$\Delta(f)(a, b) = f(\mu(a, b))$$

1.1.13 Why S for antipode? The reader may wonder why we introduce the notation S for the antipode instead of ι , and the strange summation notation for Δ . In the geometric case of an algebraic group G , the antipode S is indeed just the pullback of the inverse map $\iota : G \rightarrow G$. That is, for a function $f \in \Gamma(G)$, $S(f)(g) = f(g^{-1})$. However, Hopf algebras are more general than algebraic groups. In the study of *Quantum Groups* (non-commutative Hopf algebras), the antipode S is *not* an involution (i.e., $S^2 \neq \text{id}$). Using the symbol ι (or f^{-1}) strongly suggests an inverse that cancels out perfectly or squares to the identity. The symbol S reminds us that this is a linear operator on the algebra, which may have infinite order or behave like a “skew” inverse.

There is also another notation called *Sweedler notation* that we will use. The map $\Delta : A \rightarrow A \otimes A$ sends an element a to a tensor. A precise notation would be:

$$\Delta(a) = \sum_{i=1}^n a_{i,1} \otimes a_{i,2}$$

This becomes unreadable very quickly. For example, the coassociativity axiom $(\Delta \otimes \text{id})\Delta = (\text{id} \otimes \Delta)\Delta$ would require nested sums and double indices:

$$\sum_i \left(\sum_j (a_{i,1})_{j,1} \otimes (a_{i,1})_{j,2} \right) \otimes a_{i,2}$$

Sweedler's notation drops the summation symbol and the specific indices, focusing on the "positions" in the tensor product. We write:

$$\Delta(a) = \sum a_{(1)} \otimes a_{(2)}$$

The power of this notation appears when we iterate Δ . By coassociativity, it does not matter how we parenthesize the split into three tensors, so we can simply write:

$$(\Delta \otimes \text{id})\Delta(a) = \sum a_{(1)} \otimes a_{(2)} \otimes a_{(3)}$$

This turns complex diagrammatic proofs into algebraic calculations that look like associativity.

Example 1.1.14: Hopf Algebras

1. Let $G = \mathbb{G}_a = \text{Spec } k[x]$ so that $(a, b) = a + b$. Then we need:

$$\Delta(f)(a, b) = f(a + b)$$

As $k[x]$ is generated by x , it is enough to define it on x and extend. Namely:

$$\Delta(x)(y_1, y_2) = y_1 + y_2$$

which in $k[x] \otimes_k k[x] \cong K[y_1, y_2]$ is represented by the tensors $x \otimes 1 + 1 \otimes x$. Hence comultiplication is given by:

$$\Delta(x) = x \otimes 1 + 1 \otimes x$$

and so:

$$\Delta(p(x)) = p(x \otimes 1 + 1 \otimes x)$$

For the identity, we have the pullback $\epsilon(f) = f \circ e$. As $e : \text{Spec } k \rightarrow \mathbb{G}_a$, $e(\eta) = 0$, we have that $f \circ e(0) = f(0)$, giving us the map.

Finally, for $S(f) = f \circ \iota$, we can evaluate at the generator to get:

$$S(x)(a) = x(\iota(a)) = x(-a) = -a$$

Thus:

$$S(x) = -x$$

Let us see how these regular functions evaluate. Representing $(x - a_i) \in \text{Spec } k[x]$ as a_i , then:

$$\Delta(x)(a_i, a_j) = (x \otimes 1 + 1 \otimes x)(a_i, a_j) = a_i \cdot 1 + 1 \cdot a_j = a_i + a_j$$

Recall that when affine sets were first defined, it was emphasized to forget the vector-space structure of k^n when considering $\text{Hom}_{\text{Reg}}(k^n, k)$. By remembering the vector-space structure, we shall induce the above Hopf-algebra structure!

2. Let $G = \mathbb{G}_m \cong \text{Spec } k[x, x^{-1}]$ where $\mu(a, b) = ab$. Then:

$$\begin{aligned} \Delta(x)(a, b) &= ab & \Rightarrow & \Delta(x) = x \otimes x \\ \epsilon(x)(a) &= x \circ e(a) = x(1) = 1 & \Rightarrow & \epsilon(x) = 1 \end{aligned}$$

and similarly:

$$S(x) = x^{-1}$$

3. Let $G = \text{GL}_n = \text{Spec } k[x_{11}, \dots, x_{nn}, \det^{-1}] = \text{Spec } A$. Then A is generated by n^2 variables x_{ij} and the inverse of the determinant. The function x_{ij} takes a matrix M and gives you the entry in row i , column j . i.e., $x_{ij}(M) = M_{ij}$.

First, geometrically the multiplication per-component of two matrix M, N is:

$$(MN)_{ij} = \sum_{k=1}^n M_{ik} N_{kj}$$

Thus:

$$\Delta(x_{ij})(M, N) = x_{ij}(MN) = (MN)_{ij} = \sum_{k=1}^n M_{ik} N_{kj}$$

And so,

$$\Delta(x_{ij}) = \sum_{k=1}^n x_{ik} \otimes x_{kj}$$

Similarly:

$$\epsilon(x_{ij}) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

and:

$$S(x_{ij}) = \det(x)^{-1} \cdot (\text{Cofactor}_{ji})$$

4. Let $G = \text{Spec } \text{Sym}_R^\bullet(M^\vee)$ be the geometric vector bundle associated to a finitely generated projective R -module M over a base ring R . Just as $\mathbb{G}_a$ remembered its addition, this relative scheme remembers the fiberwise additive structure of M .

As the ring $\text{Sym}_R^\bullet(M^\vee)$ is generated as an R -algebra by the degree-1 elements $m \in M^\vee$, it is enough to define our maps on these linear generators and extend R -linearly. Geometrically,

addition happens fiberwise, so if a, b are vectors in the same fiber, $\mu(a, b) = a + b$. Then we need:

$$\Delta(m)(a, b) = m(a + b) = m(a) + m(b)$$

which in $\text{Sym}_R^\bullet(M^\vee) \otimes_R \text{Sym}_R^\bullet(M^\vee)$ is represented by $m \otimes 1 + 1 \otimes m$. Hence:

$$\Delta(m) = m \otimes 1 + 1 \otimes m$$

For the identity, the zero section e maps to the zero vector in every fiber, so $e(\eta) = 0$. We have the pullback $\epsilon(m)(a) = m \circ e(a) = m(0) = 0$, giving us:

$$\epsilon(m) = 0$$

(and $\epsilon(r) = r$ for the degree-0 scalars $r \in R$).

Finally, for the antipode $S(m) = m \circ \iota$, we can evaluate at a generator to get:

$$S(m)(a) = m(\iota(a)) = m(-a) = -m(a) \quad \Rightarrow \quad S(m) = -m$$

Notice how this is exactly the Hopf-algebra structure of $\mathbb{G}_a$, but promoted to the relative setting over R ! By taking the symmetric algebra of the dual module $M^\vee$, we didn't just build an arbitrary scheme; we naturally packaged the module's additive structure into a group scheme over R whose fibers perfectly preserve the linear geometry.

1.2 Algebraic Groups as Functors

When picking $G = \text{GL}_n(K)$ or $G = \text{SL}_n(K)$, there is a choice of the field (or ring) K . We can eliminate this choice by capturing the required properties functorially. Let $S \subseteq k[x_1, \dots, x_n]$ be a set of polynomials and let R be an arbitrary k -algebra. We can look at the solution set of these polynomials in R :

$$V(S)(R) = \{(a_1, \dots, a_n) \in R^n \mid f(a_1, \dots, a_n) = 0 \text{ for all } f \in S\}$$

Given a k -algebra homomorphism $\varphi : R \rightarrow R'$, we naturally get a set-theoretic map $V(S)(R) \rightarrow V(S)(R')$ simply by applying φ to each coordinate. This assignment $R \mapsto V(S)(R)$ therefore defines a functor:

$$\mathbf{Alg}_k \longrightarrow \mathbf{Set}$$

We can thus think of an affine algebraic set not as a static geometric space, but as a functor that eats a k -algebra and spits out a set of solutions. Naturally, if our solution sets always form a group, and the transition maps are group homomorphisms, we instead have a functor:

$$\mathbf{Alg}_k \longrightarrow \mathbf{Grp}$$

We can define an *affine algebraic group* abstractly as such a functor. For example, we can define the functor $\text{SL}_n : \mathbf{Alg}_k \rightarrow \mathbf{Grp}$ where any k -algebra R maps to $\text{SL}_n(R)$, the group of $n \times n$ matrices with entries in R satisfying the equation $\det(X) - 1 = 0$, where the determinant is the standard polynomial:

$$\det(X) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) X_{1,\sigma(1)} \cdots X_{n,\sigma(n)} \in k[x_{11}, \dots, x_{nn}]$$

Because our functors must be defined by polynomial equations, this radically restricts the kinds of functors we consider. For example, you cannot have an “indicator” functor such as:

$$F(R) = \begin{cases} (k, +) & \text{if } R = k \\ \{e\} & \text{otherwise} \end{cases}$$

There are no polynomials whose solution set suddenly vanishes when you map $k \hookrightarrow k[x]$.

Let us now formalize this. Recall that if k is Noetherian, then the polynomial ring $k[x_1, \dots, x_n]$ is also Noetherian (by the Hilbert Basis Theorem, see [5]), meaning every ideal is finitely generated. Thus, our functor $V(S)$ is exactly the same as $V(I)$ where I is the ideal generated by S . Let $A = k[x_1, \dots, x_n]/I$.

A k -algebra homomorphism $\psi : A \rightarrow R$ is completely determined by where it sends the generators x_i . Let $a_i = \psi(x_i) \in R$. Because $\psi(f) = 0$ for all $f \in I$, the tuple $(a_1, \dots, a_n)$ must be a root of I in R . Therefore, the functor sending R to the zero-set of I is canonically isomorphic to the functor:

$$h^A(R) := \text{Hom}_{\mathbf{Alg}_k}(A, R)$$

Functors of the form h^A are called *representable functors*, and A is the representing object. Thus, an affine algebraic group scheme over k is exactly a representable functor $\mathbf{Alg}_k \rightarrow \mathbf{Grp}$.

How does the group structure on the functor h^A correspond to the Hopf algebra structure on A ? Consider a k -algebra homomorphism $\Delta : A \rightarrow A \otimes_k A$. For any k -algebra R , we define the multiplication of two points $f_1, f_2 \in h^A(R)$ via the composition:

$$f_1 \cdot f_2 : A \xrightarrow{\Delta} A \otimes_k A \xrightarrow{f_1 \otimes f_2} R \otimes_k R \xrightarrow{m_R} R$$

where $m_R(r_1 \otimes r_2) = r_1 r_2$ is the internal multiplication of the ring R . Because Δ is an algebra homomorphism, the resulting map $f_1 \cdot f_2$ is indeed a valid k -algebra homomorphism $A \rightarrow R$. This binary operation on $h^A(R)$ is natural in R . Therefore, the comultiplication map completely captures the group structure, and we can define an affine algebraic group entirely as a pair (A, Δ) .

Yoneda’s Lemma and Morphisms

If algebraic groups are functors, what are homomorphisms between them? They are simply natural transformations. The translation between the functorial world and the algebraic world is governed by one of the most pervasive theorems in category theory.

Theorem 1.2.1: Yoneda Lemma (for k -algebras)

Let A and B be k -algebras, and let $h^A, h^B : \mathbf{Alg}_k \rightarrow \mathbf{Set}$ be their representable functors. There is a canonical bijection:

$$\text{Nat}(h^B, h^A) \xrightarrow{\sim} \text{Hom}_{\mathbf{Alg}_k}(A, B)$$

given by evaluating a natural transformation $\eta : h^B \rightarrow h^A$ on the identity map $\text{id}_B \in h^B(B)$.

For our purposes, the Yoneda Lemma says that every natural transformation of group functors $G \rightarrow H$ comes from a unique Hopf algebra homomorphism $\mathcal{O}(H) \rightarrow \mathcal{O}(G)$. This means we can

interchangeably prove theorems about maps by drawing commutative diagrams of Hopf algebras, or by doing basic group theory on the R -valued points $G(R) \rightarrow H(R)$ for an arbitrary test ring R . The latter is almost always easier.

1.2.1 Arithmetic Tools: Base Change and Restriction

By treating algebraic groups as functors, we gain access to two powerful operations that are indispensable for number theory: moving an algebraic group from one base field to another.

Definition 1.2.2: Base Change (Extension of Scalars)

Let G be an affine algebraic group over k represented by the Hopf algebra A . Let K be a field extension of k (or more generally, any k -algebra). The *base change* of G to K , denoted G_K , is the algebraic group over K defined by the functor:

$$G_K(R) = G(R)$$

for any K -algebra R (where R is viewed as a k -algebra via the structure map $k \rightarrow K \rightarrow R$).

Algebraically, the Hopf algebra representing G_K is simply the tensor product $A_K = A \otimes_k K$. For example, the equations cutting out G might have rational coefficients ($k = \mathbb{Q}$), but we can base change to $\mathbb{C}$ to view the complex solutions $G_{\mathbb{C}}$.

However, in number theory, we often want to go in the *opposite* direction. If we have a group defined over a larger field K , can we view it as a group over the smaller field k ? If the extension is finite, the answer is yes, through a process called Weil restriction.

Definition 1.2.3: Weil Restriction of Scalars

Let K/k be a finite field extension (or a finite locally free algebra extension). Let X be an affine algebraic group over K . The *Weil restriction of scalars*, denoted $\text{Res}_{K/k}(X)$, is a functor $\mathbf{Alg}_k \rightarrow \mathbf{Grp}$ defined by:

$$\text{Res}_{K/k}(X)(R) = X(R \otimes_k K)$$

for any k -algebra R .

A theorem of algebraic geometry ensures that $\text{Res}_{K/k}(X)$ is representable by a finitely generated k -algebra (as a consequence of faithfully flat descent), making it a valid affine algebraic group over k . Geometrically, if X has dimension d over K , then $\text{Res}_{K/k}(X)$ has dimension $d \cdot [K : k]$ over k .

Example 1.2.4: The Complex Multiplicative Group

Let $K = \mathbb{C}$ and $k = \mathbb{R}$. Consider the multiplicative group $\mathbb{G}_{m,\mathbb{C}}$ over $\mathbb{C}$. Its $\mathbb{C}$ -points are $\mathbb{C}^\times$. We want to construct an algebraic group G over $\mathbb{R}$ such that $G(\mathbb{R}) = \mathbb{C}^\times$.

We apply the Weil restriction: $G = \text{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}})$. By definition, for any $\mathbb{R}$ -algebra R , $G(R) = (R \otimes_{\mathbb{R}} \mathbb{C})^\times$.

How do we see this as a real variety? Since $\mathbb{C} \cong \mathbb{R}[i]/(i^2 + 1)$, an element $z \in R \otimes_{\mathbb{R}} \mathbb{C}$ can be uniquely written as $x + iy$ with $x, y \in R$. The element is invertible if and only if its norm $x^2 + y^2$

is invertible in R . Thus, as a real algebraic group, we have:

$$\mathrm{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}}) = \mathrm{Spec} \mathbb{R} \left[x, y, \frac{1}{x^2 + y^2} \right]$$

This is a 2-dimensional algebraic group over $\mathbb{R}$. Base changing this group *back* to $\mathbb{C}$ yields $G_{\mathbb{C}} \cong \mathbb{G}_{m,\mathbb{C}} \times \mathbb{G}_{m,\mathbb{C}}$. The interplay between base change and restriction of scalars forms the foundation for the study of algebraic tori over non-algebraically closed fields.

1.2.5 Remark: Affine Supergroups For those interested in theoretical physics, the functorial perspective has another application. If we replace the category of commutative k -algebras with the category of *supercommutative* (or $\mathbb{Z}/2\mathbb{Z}$ -graded) k -algebras, the exact same functorial framework produces *Affine Algebraic Supergroups*. These are the geometric objects underlying supersymmetry. The coordinate rings become super-Hopf algebras, containing both bosonic (commuting) and fermionic (anti-commuting) variables.

1.2.2 Closed Subgroups and Hopf Ideals

Just as an affine algebraic set is defined by a system of polynomial equations, a closed subspace is defined by simply adding more equations to that system. Algebraically, if $X = \mathrm{Spec}(A)$, a closed subscheme $Y \subseteq X$ corresponds to a quotient ring A/I for some ideal $I \subseteq A$.

When our scheme is an algebraic group $G = \mathrm{Spec}(A)$, we want to know when a closed subscheme H is actually a *subgroup*. Functorially, the definition is completely natural.

Definition 1.2.6: Closed Subgroup

Let G be an affine algebraic group. A *closed subgroup* $H \subseteq G$ is a closed subscheme such that for every k -algebra R , the subset $H(R) \subseteq G(R)$ is a subgroup.

Because H is a closed subscheme of an affine scheme, H is automatically affine itself. Thus, the terms “closed subgroup” and “affine subgroup” are used interchangeably in this context.

Algebraically, the inclusion map $i : H \hookrightarrow G$ corresponds to a surjective k -algebra homomorphism $\pi : A \rightarrow A/I$. For H to inherit the group structure, the ideal I cannot be just any ideal; it must be compatible with the comultiplication Δ , the counit ϵ , and the antipode S .

Definition 1.2.7: Hopf Ideal

Let (A, Δ, ϵ, S) be a Hopf algebra. An ideal $I \subseteq A$ is called a *Hopf ideal* if it satisfies:

1. $\Delta(I) \subseteq I \otimes_k A + A \otimes_k I$
2. $\epsilon(I) = 0$
3. $S(I) \subseteq I$

These three conditions perfectly mirror the geometric requirements for H to be a subgroup:

1. The first condition ensures that the multiplication map $H \times H \rightarrow G$ factors through H . (Note that $I \otimes A + A \otimes I$ is exactly the kernel of the map $A \otimes A \rightarrow A/I \otimes A/I$).
2. The second condition ensures that the identity element $e \in G$ actually lies inside H .
3. The third condition ensures that if $h \in H$, its inverse h^{-1} also lies in H .

Thus, we have a perfect dictionary: closed subgroups of G correspond exactly to Hopf ideals of $\mathcal{O}(G)$.

Example 1.2.8: The Special Linear Group as a Subgroup

Consider $G = \mathrm{GL}_n = \mathrm{Spec} k[x_{11}, \dots, x_{nn}, \det^{-1}]$. We want to realize SL_n as a closed subgroup. Geometrically, SL_n is cut out by the single equation $\det(X) - 1 = 0$.

Algebraically, we consider the ideal $I = (\det(X) - 1)$. Is this a Hopf ideal? Since the determinant of a product is the product of the determinants, we know that $\Delta(\det(X)) = \det(X) \otimes \det(X)$. Therefore:

$$\begin{aligned} \Delta(\det(X) - 1) &= \det(X) \otimes \det(X) - 1 \otimes 1 \\ &= \det(X) \otimes \det(X) - \det(X) \otimes 1 + \det(X) \otimes 1 - 1 \otimes 1 \\ &= \det(X) \otimes (\det(X) - 1) + (\det(X) - 1) \otimes 1 \end{aligned}$$

This clearly lies in $A \otimes I + I \otimes A$, satisfying the first condition.

For the counit, $\epsilon(\det(X) - 1) = \det(I_n) - 1 = 1 - 1 = 0$. For the antipode, $S(\det(X) - 1) = \det(X^{-1}) - 1 = \det(X)^{-1} - 1$. While this doesn't immediately look like it belongs to I , in the quotient ring A/I , $\det(X) = 1$, so its inverse is also 1, meaning $S(\det(X) - 1)$ vanishes modulo I , which means $S(I) \subseteq I$.

Thus, I is a Hopf ideal, and SL_n is a closed subgroup of GL_n .

2

Actions, Representations, and Orbits

In Chapter 1, we built the algebraic group from the inside: its coordinate ring, its Hopf algebra structure, its functor of points. The group G now sits in front of us as a well-defined geometric and algebraic object. But a group in isolation is rarely interesting. The power of group theory — in every branch of mathematics — comes from *what the group acts on*. The symmetric group S_n is interesting because it permutes $\{1, \dots, n\}$; the Galois group $\text{Gal}(L/k)$ is interesting because it permutes roots; the general linear group GL_n is interesting because it acts on n -dimensional space. In each case, the action reveals the structure of the group.

This chapter formalizes the notion of an algebraic group acting on a scheme, and then specializes to the two most important cases: actions on vector spaces (representations) and actions on varieties (orbits). We will see that the Hopf algebra machinery of Chapter 1 translates actions into comodule structures, and that the resulting category of representations $\text{Rep}(G)$ is remarkably well-behaved. We close the chapter by studying the geometry of orbits and proving the Closed Orbit Lemma, a foundational result that will recur throughout the rest of the book.

2.1 Group Actions

Let G be an affine algebraic group over a field k and let X be a k -scheme. An action of G on X should be a morphism that tells each group element how to move points of X , subject to the obvious compatibility: acting by g then by h is the same as acting by hg , and acting by the identity does nothing.

Definition 2.1.1: Left Action

Let G be an affine algebraic group over k and X a k -scheme. A *left action* of G on X is a morphism of k -schemes

$$\alpha : G \times_k X \longrightarrow X$$

such that the following diagrams commute:

1. (Associativity) The diagram

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{\mu \times \text{id}_X} & G \times X \\ \text{id}_G \times \alpha \downarrow & & \downarrow \alpha \\ G \times X & \xrightarrow{\alpha} & X \end{array}$$

commutes, where $\mu : G \times G \rightarrow G$ is the multiplication of G .

2. (Unit) The composition

$$X \xrightarrow{e \times \text{id}_X} G \times X \xrightarrow{\alpha} X$$

is the identity morphism id_X .

A k -scheme X equipped with a left action of G is called a *G -scheme*. When X is a variety, we say X is a *G -variety*.

There is an obvious dual notion of a *right action* $\alpha : X \times G \rightarrow X$, where the associativity diagram has G acting on the right. In practice, we will work with left actions unless explicitly stated otherwise, and simply say “action” to mean “left action.” Note that every left action $\alpha : G \times X \rightarrow X$ determines a right action $\beta : X \times G \rightarrow X$ by precomposing with the inverse: $\beta(x, g) = \alpha(g^{-1}, x)$, and vice versa. We denote $\alpha(g, x)$ by $g \cdot x$ when the action is clear from context.

The Functorial Perspective

As in Chapter 1, the functor of points gives us a second, often more convenient, way to think about actions. If G is an affine algebraic group represented by the Hopf algebra $A = \mathcal{O}(G)$ and X is a k -scheme, then the action $\alpha : G \times X \rightarrow X$ is completely determined by what it does on R -valued points: for every k -algebra R , we get a map

$$\alpha_R : G(R) \times X(R) \longrightarrow X(R)$$

that makes $X(R)$ into a $G(R)$ -set. The two axioms in definition 2.1.1 translate directly: for all $g, h \in G(R)$ and $x \in X(R)$,

$$(gh) \cdot x = g \cdot (h \cdot x), \quad e \cdot x = x,$$

and these identities hold *naturally* in R , meaning they are compatible with every k -algebra homomorphism $R \rightarrow R'$. Thus:

Proposition 2.1.2: Functorial Actions

Let G be an affine algebraic group over k and X a k -scheme. Giving a left action $\alpha : G \times X \rightarrow X$ is the same as giving, for every k -algebra R , a group action $G(R) \times X(R) \rightarrow X(R)$ that is natural in R .

Proof:

This is a consequence of the Yoneda Lemma (theorem 1.2.1). The action α is a morphism $G \times X \rightarrow X$, which by Yoneda corresponds to a natural transformation $h^{G \times X} \rightarrow h^X$ of set-valued functors. Since $h^{G \times X}(R) = G(R) \times X(R)$ for affine G , this is precisely a natural family of maps $G(R) \times X(R) \rightarrow X(R)$. The commutativity of the diagrams in definition 2.1.1 is equivalent to the group action axioms holding for each R .

This is extremely useful in practice: to verify that a morphism $\alpha : G \times X \rightarrow X$ defines an action, we can simply check the group axioms on R -valued points for an arbitrary test ring R . No commutative diagrams of schemes needed.

Actions on Affine Schemes: Coactions

Now suppose $X = \text{Spec } B$ is affine as well. The action map $\alpha : G \times X \rightarrow X$ is a morphism of affine schemes $\text{Spec } (A \otimes_k B) \rightarrow \text{Spec } B$, and dualizing gives us a k -algebra homomorphism going in the opposite direction:

$$\rho : B \longrightarrow A \otimes_k B$$

This map ρ is called a (*right*) *coaction* of the Hopf algebra $A = \mathcal{O}(G)$ on B . The reason it is a *right* coaction — even though the original action was a *left* action — is the usual contravariance: the map $G \times X \rightarrow X$ puts G on the left, but dualizing reverses the order of the tensor factors in a natural way. Explicitly, writing $\rho(b) = \sum b_{(0)} \otimes b_{(1)}$ in Sweedler-type notation (where the subscript (0) refers to the “ B -part” and (1) to the “ A -part”), the left action on points translates as follows. For $g \in G(R) = \text{Hom}_{\mathbf{Alg}_k}(A, R)$ and a “function” $b \in B$:

$$(g \cdot b)(x) = \sum b_{(0)}(x) \cdot g(b_{(1)})$$

That is, the coaction decomposes each function b on X into a sum of products: a function on X times a function on G .

Definition 2.1.3: Coaction

Let (A, Δ, ϵ, S) be a Hopf algebra over k and B a k -algebra. A *right coaction* of A on B is a k -algebra homomorphism

$$\rho : B \longrightarrow B \otimes_k A$$

satisfying:

1. (Coassociativity) $(\rho \otimes \text{id}_A) \circ \rho = (\text{id}_B \otimes \Delta) \circ \rho$ as maps $B \rightarrow B \otimes_k A \otimes_k A$.
2. (Counit) $(\text{id}_B \otimes \epsilon) \circ \rho = \text{id}_B$ under the canonical isomorphism $B \otimes_k k \cong B$.

A k -algebra B equipped with a right coaction of A is called an *A -comodule algebra*.

$$\begin{array}{ccc}
B & \xrightarrow{\rho} & B \otimes A \\
\rho \downarrow & & \downarrow \rho \otimes \text{id}_A \\
B \otimes A & \xrightarrow{\text{id}_B \otimes \Delta} & B \otimes A \otimes A
\end{array}$$

Coassociativity

$$\begin{array}{ccc}
B & \xrightarrow{\rho} & B \otimes A \\
& \searrow \text{id}_B & \downarrow \text{id}_B \otimes \epsilon \\
& & B \otimes k \cong B
\end{array}$$

Counit

The coassociativity axiom is the dual of the associativity of the group action: it says that “decomposing a function on X with respect to G , and then further decomposing the G -part using the comultiplication of G ” gives the same answer as “decomposing the function with respect to $G \times G$ directly.” The counit axiom is the dual of $e \cdot x = x$: evaluating the G -part at the identity recovers the original function.

The upshot is a clean algebraic dictionary:

$$\left\{ \begin{array}{c} \text{Actions of } G \text{ on} \\ \text{an affine scheme } X = \text{Spec } B \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Right coactions of } \mathcal{O}(G) \\ \text{on the } k\text{-algebra } B \end{array} \right\}$$

Example 2.1.4: The Left Regular Action

Every group acts on itself by left multiplication: $\alpha(g, x) = gx$. For $G = \text{Spec } A$, the corresponding coaction is precisely the comultiplication:

$$\rho = \Delta : A \longrightarrow A \otimes_k A$$

To see why, recall from definition 1.1.12 that $\Delta(f)(g, x) = f(gx) = f(\mu(g, x))$, which is exactly the pullback of f along the multiplication map μ . The coassociativity of this coaction is just the coassociativity of Δ itself, and the counit axiom is $(\text{id} \otimes \epsilon) \circ \Delta = \text{id}$, which is one of the Hopf algebra axioms. This is the same action that powered the proof of the linearization theorem (lemma 1.1.8).

Example 2.1.5: The Conjugation Action

Let G be an affine algebraic group. Then G acts on itself by *conjugation*:

$$\alpha : G \times G \longrightarrow G, \quad \alpha(g, x) = gxg^{-1}$$

Functorially, for any k -algebra R and elements $g, x \in G(R)$, the action is $g \cdot x = gxg^{-1}$. It is immediate from the group axioms on $G(R)$ that this is a well-defined action. In Hopf algebra terms, the coaction $\rho : A \rightarrow A \otimes_k A$ is given by:

$$\rho = (S \otimes \text{id} \otimes \text{id}) \circ (\text{id} \otimes \Delta) \circ \Delta$$

where S is the antipode, though this explicit formula is rarely needed. The conjugation action is the geometric incarnation of the notion of “inner automorphisms.” It will take center stage in Chapter 3, where it gives rise to the *adjoint representation* of G on its Lie algebra.

Example 2.1.6: The Natural Action of GL_n on $\mathbb{A}^n$

Let $G = \mathrm{GL}_n$ and $X = \mathbb{A}^n = \mathrm{Spec} k[x_1, \dots, x_n]$. For any k -algebra R , an element $g = (g_{ij}) \in \mathrm{GL}_n(R)$ acts on a vector $(a_1, \dots, a_n) \in R^n = \mathbb{A}^n(R)$ by matrix multiplication:

$$g \cdot (a_1, \dots, a_n) = \left(\sum_j g_{1j} a_j, \dots, \sum_j g_{nj} a_j \right)$$

The corresponding coaction $\rho : k[x_1, \dots, x_n] \rightarrow k[x_1, \dots, x_n] \otimes_k \mathcal{O}(\mathrm{GL}_n)$ sends each coordinate function to:

$$\rho(x_i) = \sum_{j=1}^n x_j \otimes t_{ji}$$

where t_{ji} are the matrix-entry functions on GL_n . Let us verify coassociativity on the generator x_i . On one hand:

$$(\rho \otimes \mathrm{id})(\rho(x_i)) = \sum_j \rho(x_j) \otimes t_{ji} = \sum_{j,l} x_l \otimes t_{lj} \otimes t_{ji}$$

On the other hand:

$$(\mathrm{id} \otimes \Delta)(\rho(x_i)) = \sum_j x_j \otimes \Delta(t_{ji}) = \sum_{j,l} x_l \otimes t_{lj} \otimes t_{ji}$$

where the last step uses $\Delta(t_{ji}) = \sum_l t_{jl} \otimes t_{li}$ (the comultiplication of GL_n from example 1.1.14). The two expressions agree, confirming coassociativity. Notice that the coaction formula $\rho(x_i) = \sum_j x_j \otimes t_{ji}$ is nothing but the transpose of the matrix action — the coordinates transform contravariantly, as they should.

Example 2.1.7: GL_n Acting on M_n by Conjugation

Let $G = \mathrm{GL}_n$ act on $X = M_n \cong \mathbb{A}^{n^2}$ by conjugation: $g \cdot A = gAg^{-1}$. This is a perfectly well-defined action on a *non-group* variety. Functorially, for any k -algebra R , the group $\mathrm{GL}_n(R)$ acts on $M_n(R)$ by conjugation.

Over an algebraically closed field $k = \bar{k}$, the orbits of this action are exactly the *similarity classes* of matrices. By the Jordan normal form theorem, these are parametrized by Jordan types, which are partitions of n decorated with eigenvalues. The stabilizer $\mathrm{stab}(A)$ of a matrix A is its centralizer $C_{\mathrm{GL}_n}(A) = \{g \in \mathrm{GL}_n \mid gA = Ag\}$. We will return to this example in §2.4 to study the orbit structure in detail.

Equivariant Morphisms

If G acts on two schemes X and Y , we need a notion of “maps that respect the action.” Functorially, the definition writes itself.

Definition 2.1.8: G -Equivariant Morphism

Let G be an affine algebraic group acting on k -schemes X and Y via α_X and α_Y respectively. A morphism $f : X \rightarrow Y$ is called G -equivariant (or a *morphism of G -schemes*) if the diagram

$$\begin{array}{ccc} G \times X & \xrightarrow{\text{id}_G \times f} & G \times Y \\ \alpha_X \downarrow & & \downarrow \alpha_Y \\ X & \xrightarrow{f} & Y \end{array}$$

commutes. Equivalently, for every k -algebra R , every $g \in G(R)$, and every $x \in X(R)$:

$$f(g \cdot x) = g \cdot f(x)$$

When both $X = \text{Spec } B$ and $Y = \text{Spec } C$ are affine with coactions $\rho_B : B \rightarrow B \otimes_k A$ and $\rho_C : C \rightarrow C \otimes_k A$, a morphism $f : X \rightarrow Y$ corresponds to a k -algebra homomorphism $f^\# : C \rightarrow B$. The equivariance condition translates to commutativity of:

$$\begin{array}{ccc} C & \xrightarrow{\rho_C} & C \otimes_k A \\ f^\# \downarrow & & \downarrow f^\# \otimes \text{id}_A \\ B & \xrightarrow{\rho_B} & B \otimes_k A \end{array}$$

That is, $f^\#$ is a *morphism of A -comodule algebras*.

With this notion of morphism, the G -schemes over k form a category, and the assignment $X \mapsto \mathcal{O}(X)$ gives an equivalence between *affine G -schemes* and $\mathcal{O}(G)$ -comodule algebras.

1. **Right regular action.** Let $G = \text{Spec } A$ act on itself by right multiplication: $\alpha(g, x) = xg^{-1}$. Show that the corresponding coaction $\rho : A \rightarrow A \otimes_k A$ is given by $\rho = (\text{id} \otimes S) \circ \tau \circ \Delta$, where $\tau : A \otimes A \rightarrow A \otimes A$ is the tensor swap $a \otimes b \mapsto b \otimes a$ and S is the antipode.
2. **Trivial action.** Show that the *trivial* action of G on $X = \text{Spec } B$ (where $g \cdot x = x$ for all g, x) corresponds to the coaction $\rho(b) = b \otimes 1$ for all $b \in B$. Verify the comodule axioms directly.
3. **Coaction for $\mathbb{G}_m$ acting by scaling.** Let $\mathbb{G}_m$ act on $\mathbb{A}^n$ by scalar multiplication: $t \cdot (a_1, \dots, a_n) = (ta_1, \dots, ta_n)$ for $t \in \mathbb{G}_m(R) = R^\times$. Write down the coaction $\rho : k[x_1, \dots, x_n] \rightarrow k[x_1, \dots, x_n] \otimes_k k[t, t^{-1}]$ explicitly, and verify coassociativity.
4. **$\mathbb{G}_m$ -actions and gradings.** More generally, let B be a finitely generated k -algebra. Show that giving a $\mathbb{G}_m$ -action on $\text{Spec } B$ is equivalent to giving a $\mathbb{Z}$ -grading $B = \bigoplus_{n \in \mathbb{Z}} B_n$ as a k -algebra. (*Hint:* Given a coaction $\rho : B \rightarrow B \otimes_k k[t, t^{-1}]$, define $B_n = \{b \in B \mid \rho(b) = b \otimes t^n\}$. Show this gives a direct sum decomposition and that the grading is multiplicative.)
5. **Actions on product schemes.** Let G act on k -schemes X and Y . Define the *diagonal action* of G on $X \times_k Y$ via $g \cdot (x, y) = (g \cdot x, g \cdot y)$, and show this is a well-defined action. If $X = \text{Spec } B$ and $Y = \text{Spec } C$ are both affine, express the coaction $\rho : B \otimes_k C \rightarrow (B \otimes_k C) \otimes_k A$ in terms of the coactions ρ_B and ρ_C .

2.2 Representations and Comodules

We now specialize from actions on arbitrary schemes to actions on *vector spaces*. This is the most algebraically tractable setting: the acted-upon object has the rigid linear structure of a k -vector space, and the resulting theory — representation theory — is among the richest in all of mathematics. In this section, we set up the basic definitions and prove the fundamental finiteness result that makes the theory work: every representation of an affine algebraic group, no matter how large, is built from finite-dimensional pieces.

Representations as Comodules

Definition 2.2.1: Representation

Let G be an affine algebraic group over k . A (*rational*) *representation* of G is a k -vector space V together with a homomorphism of algebraic groups

$$\rho : G \longrightarrow \mathrm{GL}(V)$$

When V is finite-dimensional, this means that ρ is a morphism of affine algebraic groups. We call V a G -*module* and often suppress ρ from the notation, writing $g \cdot v$ or simply gv for $\rho(g)(v)$.

Functorially, a representation on a finite-dimensional V assigns to each k -algebra R a group homomorphism $\rho_R : G(R) \rightarrow \mathrm{GL}(V \otimes_k R)$, natural in R . As $V \otimes_k R$ is a free R -module, choosing a basis $\{v_1, \dots, v_n\}$ for V identifies $\mathrm{GL}(V \otimes_k R)$ with $\mathrm{GL}_n(R)$, and ρ becomes a morphism of group functors $G \rightarrow \mathrm{GL}_n$.

The coaction viewpoint from §2.1 specializes cleanly. If V is a finite-dimensional representation, the action $G \times V \rightarrow V$ (viewing V as the affine space $\mathbb{A}^n$) dualizes to a *comodule* map. However, there is an important subtlety. For a general G -scheme $X = \mathrm{Spec} B$, the coaction lands in $B \otimes_k \mathcal{O}(G)$, and B is an algebra. For vector spaces, we do not need the algebra structure: V is just a vector space, and the coaction is just a k -linear map.

Definition 2.2.2: Comodule

Let $A = \mathcal{O}(G)$ be the coordinate Hopf algebra of an affine algebraic group G . A *right A -comodule* is a k -vector space V equipped with a k -linear map

$$\rho : V \longrightarrow V \otimes_k A$$

called the *comodule structure map* (or *coaction*), satisfying:

1. (Coassociativity) $(\rho \otimes \mathrm{id}_A) \circ \rho = (\mathrm{id}_V \otimes \Delta) \circ \rho$,
2. (Counit) $(\mathrm{id}_V \otimes \epsilon) \circ \rho = \mathrm{id}_V$.

The axioms are identical to those of definition 2.1.3, but now V is a vector space rather than an algebra, and ρ is k -linear rather than a k -algebra homomorphism. We write $\rho(v) = \sum v_{(0)} \otimes v_{(1)}$ in

Sweedler-type notation. The coassociativity axiom then reads:

$$\sum v_{(0)(0)} \otimes v_{(0)(1)} \otimes v_{(1)} = \sum v_{(0)} \otimes v_{(1)(1)} \otimes v_{(1)(2)}$$

which, by the same abuse of notation as for Hopf algebras, we write unambiguously as $\sum v_{(0)} \otimes v_{(1)} \otimes v_{(2)}$.

Proposition 2.2.3: Representations = Comodules

Let G be an affine algebraic group over k with coordinate Hopf algebra $A = \mathcal{O}(G)$. There is an equivalence of categories:

$$\left\{ \begin{array}{c} \text{Finite-dimensional} \\ \text{representations of } G \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Finite-dimensional} \\ \text{right } A\text{-comodules} \end{array} \right\}$$

Proof:

Given a representation $\rho : G \rightarrow \mathrm{GL}(V)$, choose a basis $\{v_1, \dots, v_n\}$ for V . The representation is determined by the matrix coefficient functions $m_{ij} \in A$ defined by

$$\rho(g)(v_j) = \sum_{i=1}^n m_{ij}(g) v_i$$

for all $g \in G(R)$ and all k -algebras R . Define the coaction $\rho_V : V \rightarrow V \otimes_k A$ by

$$\rho_V(v_j) = \sum_{i=1}^n v_i \otimes m_{ij}$$

We must verify the comodule axioms. For coassociativity, note that ρ being a group homomorphism means $\rho(gh) = \rho(g)\rho(h)$, which in matrix coefficients reads $m_{ij}(gh) = \sum_l m_{il}(g) m_{lj}(h)$. Dualizing: $\Delta(m_{ij}) = \sum_l m_{il} \otimes m_{lj}$. Then:

$$(\mathrm{id}_V \otimes \Delta)(\rho_V(v_j)) = \sum_i v_i \otimes \Delta(m_{ij}) = \sum_{i,l} v_i \otimes m_{il} \otimes m_{lj}$$

while:

$$(\rho_V \otimes \mathrm{id}_A)(\rho_V(v_j)) = \sum_i \rho_V(v_i) \otimes m_{ij} = \sum_{i,l} v_l \otimes m_{li} \otimes m_{ij}$$

Reindexing $l \leftrightarrow i$ in the second expression shows these are equal. For the counit, $\epsilon(m_{ij}) = m_{ij}(e) = \delta_{ij}$ (the identity matrix acts trivially), so $(\mathrm{id}_V \otimes \epsilon)(\rho_V(v_j)) = \sum_i v_i \cdot \delta_{ij} = v_j$.

Conversely, given a comodule (V, ρ_V) with $\rho_V(v_j) = \sum_i v_i \otimes m_{ij}$, the formulas above run in reverse: for each k -algebra R and $g \in G(R) = \mathrm{Hom}_{\mathbf{Alg}_k}(A, R)$, define $\rho(g)(v_j) = \sum_i g(m_{ij}) v_i$. The comodule axioms guarantee this is a group homomorphism. That the two constructions are mutually inverse is immediate from the definitions.

This equivalence means we can work interchangeably in either language: “ G acts on V ” and “ V is an A -comodule” are the same statement. When doing concrete calculations with matrix coefficients,

the comodule language is usually cleaner. When checking that a construction is well-defined, the functorial language (does $G(R)$ act on $V \otimes_k R$ for all R ?) is usually easier.

Example 2.2.4: Comodule Structures on Low-Dimensional Spaces

Let $G = \mathrm{GL}_2$ with coordinate ring $A = k[a, b, c, d, (ad - bc)^{-1}]$ and comultiplication $\Delta\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ (matrix multiplication of entries).

1. *The standard representation.* Let $V = k^2$ with basis $\{e_1, e_2\}$. The coaction is $\rho(e_1) = e_1 \otimes a + e_2 \otimes c$ and $\rho(e_2) = e_1 \otimes b + e_2 \otimes d$. This encodes the tautological action of a matrix $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ on column vectors.
2. *The determinant representation.* Let $W = k$ (one-dimensional). The coaction is $\rho(1) = 1 \otimes (ad - bc)$. This is the representation $g \mapsto \det(g) \in \mathrm{GL}_1(R) = R^\times$. As a one-dimensional representation, it is entirely determined by a single *group-like element* $\det = ad - bc \in A$.
3. *The trivial representation.* Let $W = k$ with coaction $\rho(1) = 1 \otimes 1$. Every group element acts as the identity. The element $1 \in A$ is always group-like (since $\Delta(1) = 1 \otimes 1$), so the trivial representation exists for every algebraic group.

2.2.5 Remark: “Rational” Representations The adjective “rational” in definition 2.2.1 is not a restriction on the field k ; it distinguishes the representations we study from more exotic possibilities. An abstract group homomorphism $G(k) \rightarrow \mathrm{GL}_n(k)$ need not come from a morphism of algebraic groups; for instance, there exist wild automorphisms of $\mathbb{C}$ as a field that induce group homomorphisms $\mathbb{G}_m(\mathbb{C}) = \mathbb{C}^\times \rightarrow \mathbb{C}^\times$ which are not given by any polynomial (or even any measurable function). By requiring ρ to be a morphism of algebraic groups — equivalently, by requiring the matrix coefficients m_{ij} to be regular functions in $\mathcal{O}(G)$ — we rule out such pathology. In this book, “representation” will always mean “rational representation.”

The Fundamental Finiteness Theorem

In the proof of the linearization theorem (lemma 1.1.8), the essential step was showing that every element of $\mathcal{O}(G)$ lies in a finite-dimensional G -stable subspace. That argument used two ingredients: the coaction sends a single element to a *finite* sum of tensors, and coassociativity forces these summands to span a G -stable subspace. The same argument works for *any* comodule, and the result is so fundamental that it deserves its own clean statement and proof.

Theorem 2.2.6: Fundamental Finiteness Theorem

Let G be an affine algebraic group over k and let V be a (possibly infinite-dimensional) right $\mathcal{O}(G)$ -comodule. Then:

1. Every element $v \in V$ is contained in a finite-dimensional sub-comodule of V .
2. V is the union (equivalently, the directed colimit) of its finite-dimensional sub-comodules.

Proof:

Let $\rho : V \rightarrow V \otimes_k A$ be the coaction, where $A = \mathcal{O}(G)$. Fix $v \in V$ and write

$$\rho(v) = \sum_{i=1}^d v_i \otimes a_i$$

with $v_1, \dots, v_d \in V$ and $a_1, \dots, a_d \in A$. We may assume the a_i are linearly independent over k (otherwise, consolidate terms to reduce d). Let $W = \text{span}_k\{v_1, \dots, v_d\}$. We will show that W is a sub-comodule of V , i.e. $\rho(W) \subseteq W \otimes_k A$.

Apply the coassociativity axiom $(\rho \otimes \text{id}_A) \circ \rho = (\text{id}_V \otimes \Delta) \circ \rho$ to v :

$$\sum_{i=1}^d \rho(v_i) \otimes a_i = \sum_{i=1}^d v_i \otimes \Delta(a_i)$$

The right-hand side lies in $W \otimes_k A \otimes_k A$, since each $v_i \in W$. Now, the a_i are linearly independent in A , so viewing the equation in $V \otimes_k A \otimes_k A \cong (V \otimes_k A) \otimes_k A$ and applying an arbitrary functional $\varphi \in A^\vee$ to the rightmost tensor factor, we get:

$$\sum_{i=1}^d \rho(v_i) \cdot \varphi(a_i) = \sum_{i=1}^d v_i \otimes (\text{id}_A \otimes \varphi)(\Delta(a_i))$$

Since φ was arbitrary and the a_i are linearly independent, we may choose φ such that $\varphi(a_j) = \delta_{ij}$ for each j . This gives $\rho(v_j) \in W \otimes_k A$ for each j . Therefore W is a sub-comodule of V .

The counit axiom gives $v = (\text{id}_V \otimes \epsilon)(\rho(v)) = \sum_i v_i \epsilon(a_i)$, which is a k -linear combination of the v_i , so $v \in W$. This proves (1). Statement (2) follows immediately: $V = \bigcup_v W_v$ where W_v ranges over the finite-dimensional sub-comodules containing v . Since any two elements $v, w \in V$ are contained in the finite-dimensional sub-comodule generated by $W_v + W_w$ (which is itself a sub-comodule by the same argument applied to a spanning set), this union is directed.

The reader who followed the linearization proof in Chapter 1 will recognize this as the same core idea: the coaction only produces *finite* sums, so coassociativity forces a finiteness constraint. But the consequences are quite different. In Chapter 1, finiteness was a tool to embed G into GL_n . Here, it is a structural theorem about $\text{Rep}(G)$: every representation, no matter how large, is filtered by finite-dimensional ones. This is the algebraic-group analogue of the fact that every representation of a finite group over a field of characteristic zero is completely reducible — except that here we are only claiming a filtration, not a decomposition, since complete reducibility can fail (as it does for unipotent groups, or in characteristic p).

Subrepresentations, Quotients, and Irreducibility

With the finiteness theorem in hand, we can set up the basic categorical notions.

Definition 2.2.7: Morphisms and Sub-comodules

Let V and W be right A -comodules with coactions ρ_V and ρ_W .

1. A *morphism of representations* (or *intertwiner*, or *G -equivariant linear map*) is a k -linear map $\varphi : V \rightarrow W$ satisfying

$$\rho_W \circ \varphi = (\varphi \otimes \text{id}_A) \circ \rho_V$$

That is, the diagram

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ \rho_V \downarrow & & \downarrow \rho_W \\ V \otimes_k A & \xrightarrow{\varphi \otimes \text{id}_A} & W \otimes_k A \end{array}$$

commutes. We write $\text{Hom}_G(V, W)$ for the k -vector space of all such maps.

2. A *subrepresentation* (or *sub-comodule*) is a k -linear subspace $U \subseteq V$ such that $\rho_V(U) \subseteq U \otimes_k A$.

Functorially, a morphism of representations $\varphi : V \rightarrow W$ is simply a k -linear map such that $\varphi(g \cdot v) = g \cdot \varphi(v)$ for all k -algebras R , all $g \in G(R)$, and all $v \in V \otimes_k R$. A subrepresentation $U \subseteq V$ is a subspace such that $U \otimes_k R$ is stable under $G(R)$ for every R .

Given a subrepresentation $U \subseteq V$, the quotient space V/U inherits a comodule structure. This requires a brief check:

Proposition 2.2.8: Quotient Representations

Let V be a representation of G and $U \subseteq V$ a subrepresentation. Then the quotient V/U carries a unique comodule structure making the projection $\pi : V \rightarrow V/U$ a morphism of representations.

Proof:

Since $\rho_V(U) \subseteq U \otimes_k A$, the coaction $\rho_V : V \rightarrow V \otimes_k A$ descends to a well-defined k -linear map $\bar{\rho} : V/U \rightarrow (V/U) \otimes_k A$ via $\bar{\rho}(v + U) = (\pi \otimes \text{id}_A)(\rho_V(v))$. This is well-defined because if $v' = v + u$ for some $u \in U$, then $\rho_V(v') = \rho_V(v) + \rho_V(u)$, and $(\pi \otimes \text{id}_A)(\rho_V(u)) = 0$ since $\rho_V(u) \in U \otimes_k A$. The comodule axioms for $\bar{\rho}$ follow from those of ρ_V by applying $\pi \otimes \text{id}_A$ (or $\pi \otimes \text{id}_A \otimes \text{id}_A$) throughout.

Definition 2.2.9: Irreducible Representation

A nonzero representation V of G is called *irreducible* (or *simple*) if it has no sub-comodules other than 0 and V itself.

By the Fundamental Finiteness Theorem (theorem 2.2.6), every nonzero comodule contains a nonzero *finite-dimensional* sub-comodule. In particular, an irreducible representation must itself be finite-dimensional. This is a nontrivial statement: it says that the notion of irreducibility for algebraic

group representations automatically implies finite-dimensionality, without our having to impose it as a hypothesis.

A natural question is whether every representation decomposes as a direct sum of irreducible ones (i.e., whether $\text{Rep}(G)$ is *semisimple*). For finite groups in characteristic zero, this is Maschke's theorem. For algebraic groups, the answer depends heavily on the group:

Example 2.2.10: Failure of Complete Reducibility

Let $G = \mathbb{G}_a$ over a field k of characteristic $p > 0$, and consider the two-dimensional representation $\rho : \mathbb{G}_a \rightarrow \text{GL}_2$ defined on R -valued points by

$$\rho(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad t \in \mathbb{G}_a(R) = R$$

The subspace $U = k \cdot e_1$ is a subrepresentation (it is fixed by G), but there is no complementary subrepresentation. Indeed, a complement would be a line $k \cdot (ae_1 + be_2)$ with $b \neq 0$ that is G -stable. But $\rho(t)(ae_1 + be_2) = (a + tb)e_1 + be_2$, which lies in $k \cdot (ae_1 + be_2)$ only if $tb = 0$ for all $t \in R$ and all R . This fails already for $R = k$ and $t = 1$ (assuming k is infinite). So V is indecomposable but not irreducible.

This representation is also non-semisimple in characteristic 0, by the same calculation. The group $\mathbb{G}_a$ is unipotent, and the failure of complete reducibility is a general phenomenon for unipotent groups: their only irreducible representation is the trivial one, but they typically admit non-trivial extensions.

Example 2.2.11: The Regular Representation of a Finite Group Scheme

Let $G = \text{Spec } A$ be a *finite* algebraic group, so that A is a finite-dimensional k -algebra. Then A is itself a right A -comodule via the comultiplication $\Delta : A \rightarrow A \otimes_k A$. This is the *regular representation* of G .

Since A is already finite-dimensional, the Fundamental Finiteness Theorem is automatic in this case. When G is a constant finite group scheme (so $A = \prod_{g \in G} k$ with basis $\{e_g\}_{g \in G}$), the coaction sends e_g to $\sum_h e_h \otimes e_{h^{-1}g}$, recovering the usual left-regular representation of a finite group on the group algebra.

Invariants and Fixed Points

Given a representation V of G , the most basic question is: which vectors are fixed by the entire group? These form a subrepresentation on which G acts trivially.

Definition 2.2.12: G -Invariants

Let V be a representation of an affine algebraic group G with coaction $\rho : V \rightarrow V \otimes_k A$. The subspace of G -invariants is

$$V^G := \{v \in V \mid \rho(v) = v \otimes 1\}$$

Equivalently, $V^G = \{v \in V \mid g \cdot v = v \text{ for all } g \in G(R) \text{ and all } R\}$.

A representation V is trivial if and only if $V^G = V$. The invariant subspace functor $V \mapsto V^G$ is left-exact but not exact in general: if $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$ is a short exact sequence of representations, the induced sequence $0 \rightarrow U^G \rightarrow V^G \rightarrow W^G$ is exact, but $V^G \rightarrow W^G$ need not be surjective. The failure of surjectivity is measured by *group cohomology* $H^1(G, U)$, which we shall not develop in this book.

When G acts on an affine scheme $X = \text{Spec } B$ (so B is a comodule algebra), the invariant subalgebra B^G is the *ring of invariants*. It governs the geometry of the “quotient” $X//G$ (the categorical quotient), which we will construct in Chapter 4.

Example 2.2.13: Invariants of GL_n Acting on Matrices

Consider the conjugation action of GL_n on $M_n \cong \mathbb{A}^{n^2}$ from example 2.1.7. The ring of invariants $k[M_n]^{\text{GL}_n}$ consists of all polynomial functions on matrices that are constant on conjugacy classes. By a classical theorem, these are exactly the symmetric functions of the eigenvalues, or equivalently, the coefficients of the characteristic polynomial:

$$k[M_n]^{\text{GL}_n} = k[\sigma_1, \dots, \sigma_n]$$

where $\sigma_i(A) = \text{tr}(\wedge^i A)$ is the i -th coefficient of $\det(tI - A)$. This is the Chevalley restriction theorem for GL_n (see [5]), and its generalization to arbitrary reductive groups is a central result of Part II.

Exercise 2.2.1

1. Let $G = \text{SL}_2$ with coordinate ring $A = k[a, b, c, d]/(ad - bc - 1)$. Write down the coaction for the standard two-dimensional representation $V = k^2$. Verify both comodule axioms (coassociativity and counit) explicitly.
2. Show that one-dimensional representations of G correspond bijectively to *group-like elements* of $\mathcal{O}(G)$, that is, elements $c \in \mathcal{O}(G)$ satisfying $\Delta(c) = c \otimes c$ and $\epsilon(c) = 1$.
3. Let V be a representation of G and let $\varphi : V \rightarrow W$ be a morphism of representations. Show that $\ker(\varphi)$ and $\text{im}(\varphi)$ are sub-comodules of V and W respectively.
4. Let $V = k^2$ be the standard representation of GL_2 . The d -th symmetric power $\text{Sym}^d(V)$ has basis $\{x^{d-i}y^i \mid 0 \leq i \leq d\}$ (monomials in two formal variables). Write down the coaction $\rho : \text{Sym}^d(V) \rightarrow \text{Sym}^d(V) \otimes_k \mathcal{O}(\text{GL}_2)$ explicitly. Show that when k has characteristic 0, the representations $\{\text{Sym}^d(V)\}_{d \geq 0}$ are all irreducible. (*Hint*: for irreducibility, show that a nonzero sub-comodule must contain a highest-degree or lowest-degree monomial, then use the coaction to generate everything.)

5. (Schur's Lemma for algebraic groups) Let V and W be finite-dimensional irreducible representations of G over an algebraically closed field k . Show that $\text{Hom}_G(V, W)$ is either 0 (if $V \not\cong W$) or k (if $V \cong W$). (*Hint*: use ?? 3 and that k is algebraically closed.)

2.3 The Category $\text{Rep}(G)$

In the previous section, we defined representations, morphisms between them, and proved that every representation decomposes into finite-dimensional pieces. We now assemble these ingredients into a single algebraic structure: the category $\text{Rep}(G)$ of all finite-dimensional representations of G . This category carries far more structure than a typical abelian category — it has tensor products, duals, and an identity object — making it a *symmetric monoidal category*. Understanding this structure is essential: in Part II, the decomposition of tensor products into irreducibles will be governed by the combinatorics of root systems and highest weights.

$\text{Rep}(G)$ is Abelian

Definition 2.3.1: The Category $\text{Rep}(G)$

Let G be an affine algebraic group over k . We define $\text{Rep}(G)$ to be the category whose objects are finite-dimensional (rational) representations of G and whose morphisms are G -equivariant k -linear maps (intertwiners). Equivalently, $\text{Rep}(G)$ is the category of finite-dimensional right $\mathcal{O}(G)$ -comodules.

The first structural property we need is that $\text{Rep}(G)$ behaves like the category of modules over a ring: we can form kernels, cokernels, direct sums, and exact sequences.

Theorem 2.3.2: $\text{Rep}(G)$ is Abelian

The category $\text{Rep}(G)$ is a k -linear abelian category. Explicitly:

1. Each hom-set $\text{Hom}_G(V, W)$ is a k -vector space, and composition is k -bilinear.
2. Finite direct sums exist: $V \oplus W$ with the diagonal coaction $\rho_{V \oplus W}(v, w) = (\rho_V(v), \rho_W(w))$ is a representation.
3. Every morphism $\varphi : V \rightarrow W$ in $\text{Rep}(G)$ has a kernel and cokernel in $\text{Rep}(G)$.
4. Every monomorphism is a kernel and every epimorphism is a cokernel.

Proof:

Statement (1) is clear: $\text{Hom}_G(V, W)$ is a k -subspace of $\text{Hom}_k(V, W)$ (the equivariance condition is k -linear), and composition of equivariant maps is equivariant.

For (2), the direct sum $V \oplus W$ as a k -vector space has coaction $\rho(v, w) = \rho_V(v) + \rho_W(w) \in (V \oplus W) \otimes_k A$, where we identify $(V \oplus W) \otimes A \cong (V \otimes A) \oplus (W \otimes A)$. The comodule axioms follow componentwise.

For (3), let $\varphi : V \rightarrow W$ be a morphism in $\text{Rep}(G)$. By ?? 3, $\ker(\varphi) \subseteq V$ is a sub-comodule, hence an object of $\text{Rep}(G)$, and the inclusion $\ker(\varphi) \hookrightarrow V$ is the categorical kernel. For the cokernel, $W/\text{im}(\varphi)$ inherits a comodule structure by proposition 2.2.8 (applied to the sub-comodule $\text{im}(\varphi) \subseteq W$), and the projection $W \rightarrow W/\text{im}(\varphi)$ is the categorical cokernel.

Statement (4) follows from the corresponding fact in Vec_k : a monomorphism $\varphi : V \hookrightarrow W$ in $\text{Rep}(G)$ is injective as a k -linear map (test against maps from the trivial representation k), hence is the kernel of $W \rightarrow W/\varphi(V)$. Dually, an epimorphism is surjective (as can be verified on underlying vector spaces) and is the cokernel of its kernel.

The fact that $\text{Rep}(G)$ is abelian means we can do homological algebra inside it: we can speak of short exact sequences $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$ of representations, of extensions, and (eventually) of derived functors. One also has the forgetful functor

$$\omega : \text{Rep}(G) \longrightarrow \text{Vec}_k, \quad V \mapsto V$$

which forgets the G -action and remembers only the underlying vector space. This functor is exact and faithful (it detects zero maps), so it is a *fiber functor*. We will return to this point momentarily.

Tensor Products and the Monoidal Structure

The category $\text{Rep}(G)$ is not just abelian; it has a multiplication: given two representations, we can form their tensor product. The key insight is that the *comultiplication* of the Hopf algebra $\mathcal{O}(G)$ is what makes this possible.

Let V and W be representations of G with coactions $\rho_V : V \rightarrow V \otimes_k A$ and $\rho_W : W \rightarrow W \otimes_k A$. The tensor product $V \otimes_k W$ as a k -vector space carries a coaction defined by:

$$\rho_{V \otimes W} : V \otimes_k W \xrightarrow{\rho_V \otimes \rho_W} V \otimes_k A \otimes_k W \otimes_k A \xrightarrow{\text{id}_V \otimes \tau \otimes \text{id}_A} V \otimes_k W \otimes_k A \otimes_k A \xrightarrow{\text{id}_{V \otimes W} \otimes m_A} V \otimes_k W \otimes_k A$$

where $\tau : A \otimes_k W \rightarrow W \otimes_k A$ is the tensor swap and $m_A : A \otimes_k A \rightarrow A$ is the multiplication of A . In Sweedler notation, for $v \in V$ and $w \in W$:

$$\rho_{V \otimes W}(v \otimes w) = \sum v_{(0)} \otimes w_{(0)} \otimes v_{(1)} w_{(1)}$$

Functorially, the formula is transparent: for $g \in G(R)$, the action on $V \otimes_k W$ is simply $g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$, the *diagonal action*. The coaction formula above is just the Hopf-algebraic encoding of this: to “act by g ” on both factors, we must “split g into two copies” using the comultiplication Δ .

Proposition 2.3.3: Tensor Product Comodule

Let V and W be right A -comodules. Then $V \otimes_k W$ with the coaction $\rho_{V \otimes W}$ defined above is a right A -comodule.

Proof:

We verify coassociativity. Using Sweedler notation, we need:

$$(\rho_{V \otimes W} \otimes \text{id}_A) \circ \rho_{V \otimes W} = (\text{id}_{V \otimes W} \otimes \Delta) \circ \rho_{V \otimes W}$$

The left side applied to $v \otimes w$ gives:

$$\sum v_{(0)(0)} \otimes w_{(0)(0)} \otimes v_{(0)(1)} w_{(0)(1)} \otimes v_{(1)} w_{(1)}$$

By coassociativity of ρ_V and ρ_W , this equals:

$$\sum v_{(0)} \otimes w_{(0)} \otimes v_{(1)} w_{(1)} \otimes v_{(2)} w_{(2)}$$

The right side gives:

$$\sum v_{(0)} \otimes w_{(0)} \otimes \Delta(v_{(1)} w_{(1)})$$

Since Δ is an algebra homomorphism, $\Delta(v_{(1)} w_{(1)}) = \Delta(v_{(1)}) \Delta(w_{(1)}) = \sum v_{(1)} w_{(1)} \otimes v_{(2)} w_{(2)}$. The two sides agree. The counit axiom is similar: $\epsilon(v_{(1)} w_{(1)}) = \epsilon(v_{(1)}) \epsilon(w_{(1)}) = 1 \cdot 1 = 1$ since ϵ is an algebra homomorphism, so $(\text{id} \otimes \epsilon) \circ \rho_{V \otimes W}(v \otimes w) = v \otimes w$.

The *trivial representation* $\mathbf{1} = k$ (with coaction $\rho(1) = 1 \otimes 1$) serves as the unit for this tensor product. The canonical isomorphisms $k \otimes_k V \cong V \cong V \otimes_k k$ are G -equivariant, and the associativity and symmetry isomorphisms

$$(U \otimes V) \otimes W \cong U \otimes (V \otimes W), \quad V \otimes W \cong W \otimes V$$

are all G -equivariant (because they are natural transformations of the underlying vector space functors, and the coaction is defined in terms of the vector space structure). Thus $\text{Rep}(G)$ is a *symmetric monoidal category* with unit object $\mathbf{1}$.

Duals and Rigidity

Every finite-dimensional representation has a *dual* (contragredient) representation, and this dual behaves as an “inverse” with respect to the tensor product in a precise categorical sense.

Definition 2.3.4: Dual Representation

Let V be a finite-dimensional representation of G with coaction $\rho_V(v_j) = \sum_i v_i \otimes m_{ij}$. The *dual* (or *contragredient*) representation is $V^\vee = \text{Hom}_k(V, k)$ with coaction defined by:

$$\rho_{V^\vee}(v_j^*) = \sum_i v_i^* \otimes S(m_{ji})$$

where $\{v_j^*\}$ is the dual basis and S is the antipode of $\mathcal{O}(G)$.

Functorially, the action of $g \in G(R)$ on $f \in V^\vee \otimes_k R = \text{Hom}_R(V \otimes_k R, R)$ is $(g \cdot f)(v) = f(g^{-1} \cdot v)$, which is the usual contragredient action. The appearance of the antipode S in the coaction formula is exactly the algebraic encoding of “act by the inverse”: the antipode dualizes the group inversion $\iota : G \rightarrow G$.

Proposition 2.3.5: Dual Comodule Axioms

Let V be a finite-dimensional right A -comodule. Then $V^\vee$ with the coaction in definition 2.3.4 is a right A -comodule.

Proof:

We need two properties of the antipode. First, from the Hopf algebra axioms, S is an *anti-homomorphism* for the comultiplication: $\Delta \circ S = \tau \circ (S \otimes S) \circ \Delta$, where τ is the tensor swap. Second, $\epsilon \circ S = \epsilon$. (Both are standard; see for instance [5].)

For coassociativity of $\rho_{V^\vee}$, we compute $(\rho_{V^\vee} \otimes \text{id}) \circ \rho_{V^\vee}$ on v_j^* :

$$\sum_i \rho_{V^\vee}(v_i^*) \otimes S(m_{ji}) = \sum_{i,l} v_i^* \otimes S(m_{li}) \otimes S(m_{ji})$$

Meanwhile, $(\text{id} \otimes \Delta) \circ \rho_{V^\vee}(v_j^*)$:

$$\sum_i v_i^* \otimes \Delta(S(m_{ji})) = \sum_i v_i^* \otimes \tau(S \otimes S)(\Delta(m_{ji})) = \sum_{i,l} v_i^* \otimes S(m_{jl}) \otimes S(m_{li})$$

using $\Delta(m_{ji}) = \sum_l m_{jl} \otimes m_{li}$ and the anti-homomorphism property. Reindexing $i \leftrightarrow l$ in the first expression gives agreement. For the counit: $(\text{id} \otimes \epsilon)(v_i^* \otimes S(m_{ji})) = v_i^* \epsilon(S(m_{ji})) = v_i^* \epsilon(m_{ji}) = v_i^* \delta_{ji} = v_j^*$.

The dual representation satisfies the key property that makes $\text{Rep}(G)$ a *rigid* monoidal category: the evaluation and coevaluation maps

$$\begin{aligned} \text{ev} : V^\vee \otimes_k V &\longrightarrow k, & f \otimes v &\longmapsto f(v) \\ \text{coev} : k &\longrightarrow V \otimes_k V^\vee, & 1 &\longmapsto \sum_i v_i \otimes v_i^* \end{aligned}$$

are both G -equivariant. This is a direct consequence of the antipode axiom $m \circ (\text{id} \otimes S) \circ \Delta = u \circ \epsilon$: it ensures that $\text{ev}(g \cdot f \otimes g \cdot v) = f(g^{-1}g \cdot v) = f(v)$ for all $g \in G(R)$, so ev intertwines the G -action on $V^\vee \otimes V$ with the trivial action on k . The verification for coev is dual.

Summarizing, we have shown:

Theorem 2.3.6: Structure of $\text{Rep}(G)$

Let G be an affine algebraic group over k . The category $\text{Rep}(G)$ of finite-dimensional representations of G is a k -linear, abelian, rigid symmetric monoidal category with unit object $\mathbf{1} = k$ (the trivial representation). The tensor product is given by the diagonal G -action, and the duality is the contragredient.

Internal Hom and the Algebra of Matrix Coefficients

The rigidity of $\text{Rep}(G)$ gives us an *internal hom*: for representations V and W , the vector space $\text{Hom}_k(V, W) \cong W \otimes_k V^\vee$ carries a natural G -action by conjugation:

$$(g \cdot \varphi)(v) = g \cdot \varphi(g^{-1} \cdot v), \quad \varphi \in \text{Hom}_k(V, W)$$

The G -invariants of this representation are precisely the intertwiners:

$$\text{Hom}_k(V, W)^G = \text{Hom}_G(V, W)$$

This shows that the hom-spaces in $\text{Rep}(G)$ arise naturally from the monoidal structure, linking the “external” categorical structure to the “internal” representation-theoretic one.

The representations of G also produce important elements in the coordinate ring $\mathcal{O}(G)$. If V is a representation with coaction $\rho(v_j) = \sum_i v_i \otimes m_{ij}$, the functions $m_{ij} \in \mathcal{O}(G)$ are called *matrix coefficients*. More invariantly, for $v \in V$ and $f \in V^\vee$, the *matrix coefficient* $c_{f,v} \in \mathcal{O}(G)$ is defined by

$$c_{f,v}(g) = f(g \cdot v)$$

The span of all matrix coefficients from all finite-dimensional representations of G is a subalgebra of $\mathcal{O}(G)$. In fact, by the Fundamental Finiteness Theorem (theorem 2.2.6), every element of $\mathcal{O}(G)$ lies in a finite-dimensional sub-comodule of the regular representation, and hence is a matrix coefficient. Thus:

Corollary 2.3.7: $\mathcal{O}(G)$ is Spanned by Matrix Coefficients

Let G be an affine algebraic group over k . Then $\mathcal{O}(G)$ is spanned as a k -vector space by the matrix coefficients of all finite-dimensional representations of G . In particular, $\mathcal{O}(G) = \sum_{V \in \text{Rep}(G)} c(V)$, where $c(V) \subseteq \mathcal{O}(G)$ denotes the subspace of matrix coefficients of V .

2.3.8 Remark: The Tannakian Perspective The structure of $\text{Rep}(G)$ described in theorem 2.3.6 is not merely a convenient packaging of the representation theory. A deep theorem of Tannaka–Krein duality (in the algebraic setting, due to Saavedra-Rivano and Deligne–Milne) asserts the converse: the fiber functor $\omega : \text{Rep}(G) \rightarrow \text{Vec}_k$ together with the monoidal structure *completely determines* G . That is, G can be reconstructed as the group of monoidal automorphisms of ω . This means the category $\text{Rep}(G)$ is just as good a definition of G as the Hopf algebra $\mathcal{O}(G)$ or the functor of points $R \mapsto G(R)$. We shall not develop this theory further, but the reader should keep this principle in mind: *an algebraic group is determined by its representations*.

Example 2.3.9: Tensor Decomposition for GL_2

Let $V = k^2$ be the standard representation of GL_2 . The tensor square $V \otimes_k V$ is four-dimensional, and it decomposes (when $\text{char } k \neq 2$) as

$$V \otimes V \cong \text{Sym}^2(V) \oplus \wedge^2(V)$$

where $\text{Sym}^2(V)$ is the three-dimensional symmetric square (spanned by $e_1 \otimes e_1$, $e_1 \otimes e_2 + e_2 \otimes e_1$, $e_2 \otimes e_2$) and $\wedge^2(V)$ is the one-dimensional exterior square (spanned by $e_1 \otimes e_2 - e_2 \otimes e_1$). This decomposition is G -equivariant: both summands are sub-comodules.

As a representation, $\wedge^2(V)$ is one-dimensional: g acts by $\det(g)$. So $\wedge^2(V)$ is the determinant representation from example 2.2.4. Restricting to SL_2 (where $\det = 1$), $\wedge^2(V)$ becomes the trivial representation, and we get the decomposition $V \otimes V \cong \text{Sym}^2(V) \oplus \mathbf{1}$.

In characteristic 2, the decomposition fails: the map $v \otimes w - w \otimes v$ lands in $\text{Sym}^2(V)$ (since $-1 = 1$), so $\wedge^2(V)$ is *not* a direct summand but rather a submodule of $\text{Sym}^2(V)$. This is a first instance of a recurring theme: tensor product decompositions become more subtle in positive characteristic.

Example 2.3.10: The Regular Representation Revisited

Let $G = \operatorname{Spec} A$ be a finite algebraic group, so $\dim_k A < \infty$. The regular representation A (with coaction $\Delta : A \rightarrow A \otimes_k A$) is an object of $\operatorname{Rep}(G)$. By corollary 2.3.7, A contains all matrix coefficients, so every irreducible representation of G appears as a sub-comodule of A .

This generalizes a classical fact from finite group theory: if G is a finite group and k is algebraically closed with $\operatorname{char} k \nmid |G|$, then the regular representation $k[G]$ decomposes as $\bigoplus_V V^{\oplus \dim V}$, where the sum runs over all irreducible representations. For group *schemes* in positive characteristic, the regular representation may be indecomposable (for instance, the coordinate ring of $\alpha_p = \operatorname{Spec} k[t]/(t^p)$ is a local ring and hence indecomposable as a module over itself).

Exercise 2.3.1

1. Let V and W be representations of $G = \mathbb{G}_m$ with coactions $\rho_V(v) = v \otimes t^n$ and $\rho_W(w) = w \otimes t^m$ (so V and W are one-dimensional representations on which $\mathbb{G}_m$ acts by the n -th and m -th power respectively). Compute the coaction on $V \otimes_k W$ and on $V^\vee$. Conclude that $\mathbb{G}_m$ -representations are “graded pieces”: the tensor product of the n -th and m -th power representations is the $(n + m)$ -th, and the dual of the n -th is the $(-n)$ -th.
2. Let $0 \rightarrow U \xrightarrow{\iota} V \xrightarrow{\pi} W \rightarrow 0$ be a short exact sequence in $\operatorname{Rep}(G)$. Show that this sequence *splits* (i.e. $V \cong U \oplus W$ as representations) if and only if there exists a G -equivariant section $\sigma : W \rightarrow V$ with $\pi \circ \sigma = \operatorname{id}_W$. Give an example where the sequence splits as vector spaces but not as representations. (*Hint*: revisit example 2.2.10.)
3. Let V be an n -dimensional representation of G . Show that the exterior powers $\bigwedge^d(V)$ for $0 \leq d \leq n$ are naturally representations of G . Show that $\bigwedge^n(V)$ is one-dimensional and corresponds to the group-like element $\det(\rho) \in \mathcal{O}(G)$, where $\rho : G \rightarrow \operatorname{GL}(V)$ is the representation map. (*Hint*: for the coaction, use the fact that the wedge product $v_1 \wedge \cdots \wedge v_d$ is the antisymmetrization of $v_1 \otimes \cdots \otimes v_d$, and the tensor power is already a representation.)
4. Let V be a finite-dimensional representation of G . Verify directly that the evaluation map $\operatorname{ev} : V^\vee \otimes V \rightarrow k$ defined by $f \otimes v \mapsto f(v)$ is G -equivariant. That is, show

$$\sum_i \epsilon(S(m_{ji}) \cdot m_{il}) = \delta_{jl}$$

where $\rho(v_i) = \sum_l v_l \otimes m_{li}$, using the antipode axiom of the Hopf algebra.

5. Let V and W be finite-dimensional representations of G . Show that the natural isomorphism $\operatorname{Hom}_k(V, W) \cong W \otimes_k V^\vee$ is an isomorphism of representations (where the left side has the conjugation action and the right side has the tensor product coaction). Deduce that $\dim_k \operatorname{Hom}_G(V, W) = \dim_k (W \otimes V^\vee)^G$.

2.4 G -Varieties and the Closed Orbit Lemma

We now turn from the linear world of representations to the geometric world of group actions on varieties. The central objects of study are *orbits* — the subsets traced out by the action of G on a single point — and *stabilizers* — the subgroups that fix a point. The interplay between these two

objects is governed by a dimension formula that parallels the orbit-stabilizer theorem of finite group theory, but now the “sizes” are measured by dimension rather than cardinality.

The main result of this section is the *Closed Orbit Lemma*, which asserts that orbits are always well-behaved geometric objects (smooth, locally closed) and that among all orbits, those of minimal dimension must be closed. This seemingly modest statement is, in practice, one of the most frequently invoked tools in the structure theory of algebraic groups.

Throughout this section, k denotes an algebraically closed field and G a connected affine algebraic group over k , unless otherwise stated. All varieties are over k .

Orbit Maps and Stabilizers

Let G act on a variety X via $\alpha : G \times X \rightarrow X$. For a closed point $x \in X$, the action determines a morphism that records how G moves x :

Definition 2.4.1: Orbit Map

Let G act on a variety X and let $x \in X(k)$ be a k -rational point. The *orbit map* at x is the morphism

$$\varphi_x : G \longrightarrow X, \quad g \longmapsto g \cdot x$$

obtained by restricting the action $\alpha : G \times X \rightarrow X$ to $G \times \{x\} \cong G$.

The image $\varphi_x(G) = \{g \cdot x \mid g \in G(k)\} \subseteq X$ is the *orbit* of x , denoted $G \cdot x$ or $O(x)$. The fiber over x itself — the set of group elements that fix x — is the stabilizer:

Definition 2.4.2: Stabilizer

Let G act on a variety X and let $x \in X(k)$. The *stabilizer* (or *isotropy group*) of x is

$$G_x := \varphi_x^{-1}(x) = \{g \in G \mid g \cdot x = x\}$$

Equivalently, G_x is the fiber product $G \times_X \text{Spec } k(x)$, where $G \rightarrow X$ is the orbit map and $\text{Spec } k(x) \rightarrow X$ is the inclusion of the point x .

Proposition 2.4.3: The Stabilizer is a Closed Subgroup

Let G act on a variety X and let $x \in X(k)$. Then G_x is a closed subgroup of G .

Proof:

Since X is separated (it is a variety), the point $\{x\} \subseteq X$ is closed. The orbit map $\varphi_x : G \rightarrow X$ is a morphism of varieties, hence continuous, so $G_x = \varphi_x^{-1}(\{x\})$ is a closed subset of G . It remains to check that G_x is a subgroup, but this is immediate from the functorial perspective: for any k -algebra R , the set $G_x(R) = \{g \in G(R) \mid g \cdot x = x\}$ is a subgroup of $G(R)$ (the identity fixes x , if g and h fix x then so does gh , and if g fixes x then so does g^{-1}).

At the level of points, the orbit of x is a *homogeneous space* for G : the group G acts transitively on $G \cdot x$, and the orbit map induces a bijection of sets $G(k)/G_x(k) \xrightarrow{\sim} G \cdot x$. We would like to promote

this to an isomorphism of varieties $G/G_x \cong G \cdot x$, but this requires constructing the quotient G/G_x as a variety, which is a nontrivial problem that we defer to Chapter 4. For now, we content ourselves with the set-theoretic picture and the following dimension formula.

Proposition 2.4.4: Orbit-Stabilizer Dimension Formula

Let G act on a variety X and let $x \in X(k)$. Then

$$\dim(G \cdot x) = \dim(G) - \dim(G_x)$$

where $\dim(G \cdot x)$ means the dimension of the closure $\overline{G \cdot x}$ (equivalently, of $G \cdot x$ itself as a locally closed subset, as we shall see shortly).

Proof:

The orbit map $\varphi_x : G \rightarrow X$ has image $G \cdot x$ and all fibers are translates of the stabilizer G_x : for any $y = g \cdot x \in G \cdot x$, the fiber $\varphi_x^{-1}(y) = gG_x$. In particular, all fibers are isomorphic to G_x and hence have the same dimension. By the theorem on dimensions of fibers (see [3]), if $\varphi_x : G \rightarrow \overline{G \cdot x}$ is dominant (which it is, since $G \cdot x$ is dense in $\overline{G \cdot x}$ by definition), then $\dim(G) = \dim(\overline{G \cdot x}) + \dim(G_x)$.

Example 2.4.5: Orbits of GL_n on $\mathbb{A}^n$

Let GL_n act on $\mathbb{A}^n$ by the standard action (matrix multiplication on column vectors). There are exactly two orbits:

1. The *origin* $\{0\}$: The stabilizer is all of GL_n , so $\dim(\{0\}) = n^2 - n^2 = 0$.
2. The *complement* $\mathbb{A}^n \setminus \{0\}$: Given any nonzero vector v , extend it to a basis $\{v, v_2, \dots, v_n\}$ of k^n . The matrix with these columns lies in GL_n and sends e_1 to v . So GL_n acts transitively on $\mathbb{A}^n \setminus \{0\}$. The stabilizer of $e_1 = (1, 0, \dots, 0)^T$ consists of matrices whose first column is e_1 , which is isomorphic to $\mathbb{A}^{n-1} \rtimes \mathrm{GL}_{n-1}$ (of dimension $n^2 - n$). Thus $\dim(\mathbb{A}^n \setminus \{0\}) = n^2 - (n^2 - n) = n$.

The orbit $\mathbb{A}^n \setminus \{0\}$ is open (hence locally closed), and its closure is all of $\mathbb{A}^n$. The boundary is $\{0\}$, which is indeed a single orbit of strictly smaller dimension.

Example 2.4.6: Conjugacy Classes in M_n

We revisit example 2.1.7: GL_n acts on M_n by conjugation, $g \cdot A = gAg^{-1}$. The orbits are conjugacy classes, and the stabilizer of A is its centralizer $C_{\mathrm{GL}_n}(A)$. Consider $n = 2$ over $k = \bar{k}$ with $\mathrm{char} k = 0$. There are three types of orbits:

1. *Scalar matrices* λI : The orbit is a single point $\{\lambda I\}$ and the stabilizer is all of GL_2 , so $\dim = 4 - 4 = 0$.
2. *Diagonalizable with distinct eigenvalues* $\mathrm{diag}(\lambda, \mu)$, $\lambda \neq \mu$: The stabilizer is the diagonal torus $T \cong \mathbb{G}_m \times \mathbb{G}_m$ (of dimension 2), so $\dim = 4 - 2 = 2$.
3. *Non-diagonalizable (Jordan block)* $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$: The stabilizer consists of matrices of the form $\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}$ with $a \neq 0$ (of dimension 2), so $\dim = 4 - 2 = 2$.

The 0-dimensional orbits (scalar matrices) are closed. The 2-dimensional orbits are locally closed but not closed: the closure of a Jordan block orbit $\overline{O(J_\lambda)}$ contains the scalar matrix λI (send $\begin{pmatrix} t^{-1} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda & t \\ 0 & \lambda \end{pmatrix}$ and let $t \rightarrow 0$). Similarly, the closure of a diagonalizable orbit $\overline{O(\text{diag}(\lambda, \mu))}$ contains the Jordan block orbits for eigenvalue λ (when $\mu \rightarrow \lambda$). The orbit structure is thus:

$$\overline{O(\text{diag}(\lambda, \mu))} \supsetneq O(J_\lambda) \cup O(J_\mu) \supsetneq \{\lambda I\} \cup \{\mu I\}$$

This “degeneration” pattern — generic orbits specializing to more degenerate ones — is typical of conjugation actions, and understanding it precisely leads to the theory of orbit closures and singularities in geometric invariant theory.

Orbits are Constructible

What kind of subset of X is an orbit $G \cdot x$? It is the image of the orbit map $\varphi_x : G \rightarrow X$, and a fundamental theorem of algebraic geometry controls the images of morphisms.

Recall from [3] that a subset $S \subseteq X$ of a variety is called *constructible* if it is a finite Boolean combination of open and closed subsets, or equivalently, a finite disjoint union of locally closed subsets. Chevalley’s theorem states that the image of a constructible set under a morphism of finite type is again constructible.

Proposition 2.4.7: Orbits are Constructible

Let G act on a variety X and let $x \in X(k)$. Then the orbit $G \cdot x$ is a constructible subset of X .

Proof:

The orbit map $\varphi_x : G \rightarrow X$ is a morphism of varieties (hence of finite type). The group G is itself constructible in G (it is the whole space). By Chevalley’s theorem on constructible images (see [3]), $G \cdot x = \varphi_x(G)$ is constructible in X .

Constructibility alone does not yet give us the clean geometric picture we want. The key step is to combine constructibility with the G -action: if $G \cdot x$ is constructible, then its closure $\overline{G \cdot x}$ is G -stable (since the action is continuous and $G \cdot x$ is G -stable), and the boundary $\overline{G \cdot x} \setminus G \cdot x$ is also G -stable. This will be the engine of the Closed Orbit Lemma.

First, we extract a general topological fact about constructible subsets that are stable under a group action.

Lemma 2.4.8: Constructible G -Stable Subsets are Open in their Closure

Let G be a connected algebraic group acting on a variety X and let $Y \subseteq X$ be a non-empty constructible G -stable subset. Then Y contains a dense open subset of its closure $\overline{Y}$. In particular, if Y is also G -stable, then Y is open in $\overline{Y}$ (i.e., Y is locally closed).

Proof:

Since Y is constructible, it contains a dense open subset U of its closure by the general theory of constructible sets (see [3]): any constructible set contains an open dense subset of its closure. Thus $Y \supseteq U$ where U is open and dense in $\bar{Y}$.

Now we use the group action. For any $g \in G(k)$, the translate $g \cdot U$ is also open in $\bar{Y}$ (since g acts by an automorphism of X that stabilizes $\bar{Y}$), and $g \cdot U \subseteq Y$ since Y is G -stable. Consider the set

$$Y' = \bigcup_{g \in G(k)} g \cdot U$$

We have $Y' \subseteq Y$ and Y' is G -stable. We claim $Y' = Y$. Indeed, take any $y \in Y$. Then $y = g \cdot x$ for some $g \in G(k)$ (since Y is a single orbit, or more generally, $y \in Y$ and Y is G -stable). But actually we need a slightly different argument. Since U is dense in $\bar{Y}$ and $g^{-1} \cdot \bar{Y} = \bar{Y}$, we see that $g^{-1} \cdot U$ is also dense and open in $\bar{Y}$.

We instead argue as follows. We want to show Y is open in $\bar{Y}$, which amounts to showing that $Z := \bar{Y} \setminus Y$ is closed in $\bar{Y}$. The complement Z is constructible (as the difference of a closed and a constructible set) and G -stable. If Z is non-empty, then Z also contains a dense open subset of $\bar{Z}$, and in particular $\dim(\bar{Z}) \leq \dim(\bar{Y})$ with $\bar{Z} \subsetneq \bar{Y}$ (since $U \cap Z = \emptyset$ and U is dense in $\bar{Y}$). But we need Z itself to be closed.

Let us give the clean argument directly. Since Y is constructible, we can write $Y = \bigcup_{i=1}^r (C_i \cap U_i)$ where each C_i is closed and each U_i is open. Taking the closure: $\bar{Y} \subseteq \bigcup_i C_i$. Among the C_i , at least one — say C_1 — satisfies $C_1 \cap Y$ being dense in $C_1 \cap \bar{Y}$, and so $\bar{Y} \subseteq C_1 \cup$ (lower-dimensional pieces). Then $U_1 \cap \bar{Y}$ is open in $\bar{Y}$ and contained in Y . Since Y is G -stable, for every $g \in G(k)$ the translate $g \cdot (U_1 \cap \bar{Y})$ is open in $\bar{Y}$ and contained in Y . These translates cover Y : for any $y \in Y$, the point y lies in $g \cdot (U_1 \cap \bar{Y})$ for some g if and only if $g^{-1} \cdot y \in U_1 \cap \bar{Y}$.

Consider the morphism $\psi : G \times (U_1 \cap \bar{Y}) \rightarrow \bar{Y}$ given by the action. Its image contains Y (since $U_1 \cap \bar{Y}$ is non-empty and G acts transitively on each orbit within Y). Since G is irreducible (it is connected) and $U_1 \cap \bar{Y}$ is non-empty and open in $\bar{Y}$, the image of ψ is a constructible set containing a dense open subset of $\bar{Y}$. But we can be more direct: for a single orbit $Y = G \cdot x$, we have $x \in U_1 \cap \bar{Y}$ (possibly after relabeling), so $Y = G \cdot x \subseteq \psi(G \times (U_1 \cap \bar{Y}))$. Since each $g \cdot (U_1 \cap \bar{Y})$ is open in $\bar{Y}$ and their union contains Y , we conclude that Y is open in $\bar{Y}$.

The proof above, while elementary, is slightly fiddly in its bookkeeping. The essential point is simple: a constructible set already contains an open dense piece of its closure, and the group action “spreads” this open piece to cover the entire orbit. We record the clean consequence:

Corollary 2.4.9: Orbits are Locally Closed

Let G be a connected algebraic group acting on a variety X . Then every orbit $G \cdot x$ is locally closed in X (i.e., open in its closure $\overline{G \cdot x}$).

In fact, we can say more about the geometric quality of orbits:

Proposition 2.4.10: Orbits are Smooth

Let G be a connected algebraic group acting on a variety X over an algebraically closed field k . Then every orbit $G \cdot x$ is a smooth variety.

Proof:

The orbit $G \cdot x$ is locally closed in X , hence inherits the structure of a (reduced, separated) variety. The translation map $\ell_g : G \cdot x \rightarrow G \cdot x$ given by $y \mapsto g \cdot y$ is an isomorphism of varieties for every $g \in G(k)$ (with inverse $\ell_{g^{-1}}$). In particular, all points of $G \cdot x$ are geometrically identical: any local property that holds at one point holds at all points. Since $G \cdot x$ is a variety over an algebraically closed field, it has at least one smooth point (the smooth locus is non-empty and open; see [3]). By the translation argument, every point is smooth.

This is the same argument used in proposition 1.1.2 to show that algebraic groups themselves are smooth: homogeneity forces all points to look alike.

Transitive Actions and Homogeneous Spaces**Definition 2.4.11: Transitive Actions**

An action of G on a variety X is called *transitive* if there is a single orbit: $G \cdot x = X$ for some (equivalently, every) $x \in X(k)$. A variety equipped with a transitive G -action is called a *homogeneous space* (or *G -homogeneous variety*).

If G acts transitively on X with stabilizer G_x at a point $x \in X(k)$, then the orbit map $\varphi_x : G \rightarrow X$ is surjective and its fibers are the cosets gG_x . Morally, X “is” the quotient G/G_x . At the level of closed points this is literally true: φ_x induces a bijection $G(k)/G_x(k) \xrightarrow{\sim} X(k)$. However, promoting this to an isomorphism of *varieties* requires constructing G/G_x as a scheme and verifying that φ_x is an isomorphism onto it. This is a nontrivial task — the quotient G/H of an algebraic group by a closed subgroup need not be affine, and its construction requires careful attention to faithfully flat descent — which we will carry out in Chapter 4.

For now, we record the key consequence that will suffice for our purposes:

Proposition 2.4.12: Properties of Homogeneous Spaces

Let G act transitively on a variety X , and let $x \in X(k)$ with stabilizer $H = G_x$. Then:

1. X is smooth and equidimensional of dimension $\dim(G) - \dim(H)$.
2. The orbit map $\varphi_x : G \rightarrow X$ is a smooth, surjective morphism.
3. For any other point $y \in X(k)$, the stabilizer G_y is conjugate to H : if $y = g \cdot x$, then $G_y = gHg^{-1}$.

Proof:

1. Smoothness follows from proposition 2.4.10, and $\dim(X) = \dim(G) - \dim(H)$ from the orbit-stabilizer formula (proposition 2.4.4).
2. The morphism φ_x is surjective (by transitivity) and flat with smooth fibers (all fibers are translates of H , which is smooth as an algebraic group). A flat morphism with smooth fibers is smooth (see [3]).
3. If $y = g \cdot x$ and $h \in G_y$, then $h \cdot (g \cdot x) = g \cdot x$, so $(g^{-1}hg) \cdot x = x$, meaning $g^{-1}hg \in H$. Hence $G_y = gHg^{-1}$.

Statement (3) is the geometric analogue of a basic fact in group theory: in a transitive action, all stabilizers are conjugate. This means that the “type” of the stabilizer (its isomorphism class, or more precisely its conjugacy class in G) is an invariant of the homogeneous space, not of the chosen basepoint.

Example 2.4.13: Projective Space as a Homogeneous Space

The group GL_n acts on $\mathbb{P}^{n-1}$ via its action on lines: $g \cdot [v] = [gv]$. This action is transitive (any nonzero vector can be mapped to any other by an invertible matrix). The stabilizer of the point $[e_1] = [1 : 0 : \cdots : 0]$ consists of matrices of the form

$$G_{[e_1]} = \left\{ \begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & * & \cdots & * \end{pmatrix} \in \mathrm{GL}_n \right\}$$

This is the *standard parabolic subgroup* P_1 (the stabilizer of the line $\langle e_1 \rangle$). We thus have a bijection $\mathrm{GL}_n(k)/P_1(k) \cong \mathbb{P}^{n-1}(k)$, and the quotient construction of Chapter 4 will promote this to an isomorphism of varieties $\mathrm{GL}_n/P_1 \cong \mathbb{P}^{n-1}$. Note that $\mathbb{P}^{n-1}$ is *projective*, so G/P_1 is not affine — this illustrates why the quotient G/H is generally harder than the affine theory we have developed so far.

Example 2.4.14: The Grassmannian

More generally, GL_n acts transitively on the Grassmannian $\mathrm{Gr}(d, n)$ of d -dimensional subspaces of k^n . The stabilizer of the standard d -plane $\langle e_1, \dots, e_d \rangle$ is the parabolic subgroup

$$P_d = \left\{ \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} : A \in \mathrm{GL}_d, D \in \mathrm{GL}_{n-d}, B \in M_{d \times (n-d)} \right\}$$

and so $\mathrm{Gr}(d, n) \cong \mathrm{GL}_n/P_d$. The Grassmannian has dimension $\dim(\mathrm{GL}_n) - \dim(P_d) = n^2 - (d^2 + (n-d)^2 + d(n-d)) = d(n-d)$. These quotients are central examples of *flag varieties*, which will play a pivotal role in the structure theory of Chapters 6 and 7.

The Closed Orbit Lemma

We now prove the main geometric result of the chapter. The idea is that the boundary of an orbit — the “stuff that gets added when you take the closure” — is itself a union of strictly smaller orbits. Since dimension cannot decrease forever, this forces the existence of closed orbits.

Theorem 2.4.15: Closed Orbit Lemma

Let G be a connected algebraic group acting on a variety X . Then:

1. Every orbit $G \cdot x$ is a smooth, locally closed subvariety of X .
2. The boundary $\partial(G \cdot x) := \overline{G \cdot x} \setminus G \cdot x$ is a closed, G -stable subset of X that is a union of orbits of strictly lower dimension:

$$\dim(G \cdot y) < \dim(G \cdot x) \quad \text{for every } y \in \partial(G \cdot x)$$

3. The set of orbits admits a partial order by closure: $G \cdot y \preceq G \cdot x$ if $G \cdot y \subseteq \overline{G \cdot x}$.
4. Orbits of minimal dimension are closed. In particular, closed orbits always exist.

Proof:

Statement (1) is corollary 2.4.9 and proposition 2.4.10.

For (2), since $G \cdot x$ is open in $\overline{G \cdot x}$ (by (1)), the boundary $\partial(G \cdot x) = \overline{G \cdot x} \setminus G \cdot x$ is closed in $\overline{G \cdot x}$, hence closed in X . The boundary is G -stable: if $y \in \partial(G \cdot x)$ and $g \in G(k)$, then $g \cdot y \in \overline{G \cdot x}$ (since $\overline{G \cdot x}$ is G -stable, as the closure of a G -stable set under a continuous action). And $g \cdot y \notin G \cdot x$, for otherwise $y = g^{-1} \cdot (g \cdot y) \in G \cdot x$, a contradiction. So $g \cdot y \in \partial(G \cdot x)$.

Now let $G \cdot y$ be an orbit contained in $\partial(G \cdot x)$. Then $\overline{G \cdot y} \subseteq \overline{G \cdot x}$ and $G \cdot y \cap G \cdot x = \emptyset$. Since $G \cdot y$ is open in $\overline{G \cdot y}$ and $G \cdot x$ is dense in $\overline{G \cdot x}$, we must have $\overline{G \cdot y} \subsetneq \overline{G \cdot x}$ (if they were equal, then $G \cdot y$ would be a non-empty open subset of $\overline{G \cdot x}$ disjoint from the dense subset $G \cdot x$, which is impossible since $\overline{G \cdot x}$ is irreducible — here we use that G is connected, so $G \cdot x = \varphi_x(G)$ is the image of an irreducible variety and hence irreducible, and its closure is also irreducible). Therefore:

$$\dim(G \cdot y) \leq \dim(\overline{G \cdot y}) < \dim(\overline{G \cdot x}) = \dim(G \cdot x)$$

Statement (3) is a restatement: $G \cdot y \preceq G \cdot x$ if and only if $G \cdot y \subseteq \overline{G \cdot x}$. This is a partial order (reflexive, transitive, and antisymmetric by (2)).

For (4), since X is a Noetherian topological space, the set of orbit dimensions $\{\dim(G \cdot x) : x \in X(k)\}$ is bounded below by 0. Any orbit of minimal dimension has empty boundary (by (2)), hence is closed. If X is non-empty, there exist orbits (take any point), and the orbit of smallest dimension among all orbits is closed.

2.4.16 Remark: The Role of Connectedness The assumption that G is connected is used in (2) to guarantee that $\overline{G \cdot x}$ is irreducible. Without it, the orbit $G \cdot x$ might not be irreducible, and the dimension argument would need modification. For a disconnected group, one works instead with the connected component G° and uses the fact that G has finitely many connected components (proposition 1.1.4).

Example 2.4.17: Orbit Closures for $\mathbb{G}_m$ Acting on $\mathbb{A}^2$

Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by $t \cdot (x, y) = (tx, t^{-1}y)$. The orbits are:

1. The origin $\{(0, 0)\}$: a closed orbit of dimension 0.
2. The punctured x -axis $\{(x, 0) : x \neq 0\}$: an orbit of dimension 1 (the stabilizer is trivial). Its closure is the full x -axis $\{y = 0\}$, with boundary $\{(0, 0)\}$.
3. The punctured y -axis $\{(0, y) : y \neq 0\}$: similarly, dimension 1 with boundary $\{(0, 0)\}$.
4. For each $c \neq 0$, the *hyperbola* $\{xy = c\} = \{(t, c/t) : t \neq 0\}$: an orbit of dimension 1 with stabilizer the trivial group. Its closure is $\{xy = c\}$, which is already closed in $\mathbb{A}^2$ (it is a level set of the polynomial xy). So these orbits are closed.

The orbit structure is consistent with the Closed Orbit Lemma: the boundary of each non-closed orbit (the punctured axes) is the origin, which is the unique 0-dimensional orbit. The invariant ring is $k[\mathbb{A}^2]^{\mathbb{G}_m} = k[xy]$ (the polynomial xy is the unique independent invariant), and the “quotient map” $\mathbb{A}^2 \rightarrow \mathbb{A}^1$ sending $(x, y) \mapsto xy$ collapses the three orbits $\{(0, 0)\}$, $\{x\text{-axis}\}$, and $\{y\text{-axis}\}$ to the single point $0 \in \mathbb{A}^1$. This non-separation of orbits is a preview of the subtleties of geometric invariant theory.

An important special case of the Closed Orbit Lemma arises when the variety X is *complete* (i.e., proper over k , the algebraic analogue of compact). In a complete variety, closed subsets cannot be “too small” — in particular, a non-empty closed G -stable subset must contain a closed orbit, and in the complete setting we can say more:

Corollary 2.4.18: Closed Orbits in Complete Varieties

Let G be a connected algebraic group acting on a non-empty complete variety X . Then every orbit is complete (hence closed and projective).

Proof:

Let $x \in X(k)$ and consider the orbit map $\varphi_x : G \rightarrow X$. The orbit $G \cdot x = \varphi_x(G)$ is constructible. Its closure $\overline{G \cdot x}$ is a closed subvariety of a complete variety, hence complete. The orbit $G \cdot x$ is open in $\overline{G \cdot x}$ (by corollary 2.4.9). But $G \cdot x$ is also the image of the proper morphism $G \rightarrow \overline{G \cdot x}$ composed with the inclusion — wait, G need not be proper.

Let us argue differently. Consider the *action morphism* restricted to the closure: $\alpha : G \times \overline{G \cdot x} \rightarrow \overline{G \cdot x}$. Since $\overline{G \cdot x}$ is complete and G is separated, the projection $\pi_2 : G \times \overline{G \cdot x} \rightarrow \overline{G \cdot x}$ is proper (properness is preserved under base change). But this does not yet give us what we need.

Instead, we use a more direct argument. Suppose $G \cdot x$ is not closed, so the boundary $\partial := \overline{G \cdot x} \setminus G \cdot x$ is non-empty. By the Closed Orbit Lemma, ∂ is a closed, non-empty, G -stable proper subset of $\overline{G \cdot x}$, and every orbit in ∂ has strictly smaller dimension. Repeating: inside ∂ , orbits of minimal dimension are closed (in ∂ , hence in X). But a closed orbit in a complete variety is itself complete (a closed subvariety of a complete variety is complete). The point is that this process must terminate, yielding closed orbits. But we claimed *every* orbit is closed, which requires a different approach.

Here is the correct argument. The orbit $G \cdot x$ is the image of $\varphi_x : G \rightarrow X$. Since X is complete and $G \cdot x$ is locally closed, if $G \cdot x$ were not closed, then $\overline{G \cdot x} \setminus G \cdot x$ would be non-empty and closed in X . Now consider the graph morphism $\Gamma : G \rightarrow G \times X$ defined by $g \mapsto (g, g \cdot x)$, and the projection $\pi_1 : G \times X \rightarrow G$. Since X is complete and separated, π_1 is proper and closed. The set $Z = \pi_1(\Gamma(G) \cap (G \times (\overline{G \cdot x} \setminus G \cdot x)))$ is a subset of G . If $G \cdot x$ is not closed, there exist sequences that “escape” $G \cdot x$, but all points of G map into $G \cdot x$, a contradiction.

We give the clearest version. By the valuative criterion of properness (see [3]), X is proper if and only if for every discrete valuation ring R with fraction field K , any morphism $\text{Spec } K \rightarrow X$ extends to $\text{Spec } R \rightarrow X$. Suppose $G \cdot x$ is not closed. Then there exists a point $y \in \overline{G \cdot x} \setminus G \cdot x$. We can find a smooth curve C in $\overline{G \cdot x}$ that meets $G \cdot x$ in a dense open subset and passes through y (see [3]). Localizing at y gives a DVR $R = \mathcal{O}_{C,y}$ with fraction field $K = k(C)$. The inclusion $\text{Spec } K \hookrightarrow G \cdot x$ composed with the orbit map provides a K -valued point of G (i.e. a rational map $C \dashrightarrow G$ defined on the generic point). Since G is affine (and in particular separated), this rational map extends to a morphism from an open subset of C containing the generic point. But the composition $C \dashrightarrow G \xrightarrow{\varphi_x} X$ is regular on all of C (since X is complete), and its image lands in $G \cdot x$ on the dense open subset. By continuity, the image of y must lie in $G \cdot x$, which it does — it is y itself. But y lies in the image of φ_x composed with the extended map, and so $y \in G \cdot x$, a contradiction.

The proof is more involved than one might expect — the difficulty is that G is affine and hence not proper, so we cannot simply say “the image of a proper map is closed.” The valuative criterion bridges this gap by using the completeness of X to extend rational maps.

In practice, the most important application of corollary 2.4.18 is to *projective* varieties, since every projective variety is complete. We highlight the consequence that motivates the structure theory in later chapters:

Corollary 2.4.19: Fixed Points on Projective Varieties

Let G be a connected algebraic group acting on a non-empty projective variety X . Then every orbit in X is a projective variety. In particular, X contains a closed orbit, and if the closed orbit is a point, then G has a *fixed point* in X .

We will see in Chapter 6 that connected *solvable* groups acting on projective varieties always have fixed points (Borel’s fixed-point theorem), but this requires the Lie–Kolchin theorem and the theory of Borel subgroups — all of which rest on the Closed Orbit Lemma as a foundation.

Example 2.4.20: Orbit Closure on $\mathbb{P}^1$

Let $G = \mathbb{G}_m$ act on $\mathbb{P}^1$ by $t \cdot [x : y] = [tx : y]$. The orbits are:

1. $\{[1 : 0]\}$: a fixed point (dimension 0, closed).
2. $\{[0 : 1]\}$: a fixed point (dimension 0, closed).
3. $\{[x : y] : x \neq 0, y \neq 0\} \cong \mathbb{G}_m$: an orbit of dimension 1.

The closure of the 1-dimensional orbit is all of $\mathbb{P}^1$, and its boundary is $\{[1 : 0]\} \cup \{[0 : 1]\}$. Contrast this with the action of $\mathbb{G}_m$ on $\mathbb{A}^2$ from example 2.4.17: there, some 1-dimensional orbits (the hyperbolas) were closed, while others (the punctured axes) were not. On $\mathbb{P}^1$, the only closed orbits are the fixed points — exactly as corollary 2.4.18 predicts.

Exercise 2.4.1

1. Let GL_n act on the space $\mathrm{Sym}^2(k^n)^\vee$ of symmetric bilinear forms by $g \cdot \beta(v, w) = \beta(g^{-1}v, g^{-1}w)$. Compute the stabilizer of the standard form $\beta_0(v, w) = \sum_i v_i w_i$ (assuming $\mathrm{char} k \neq 2$). What familiar group is this? What is the dimension of the orbit $\mathrm{GL}_n \cdot \beta_0$?
2. Let $G = \mathrm{GL}_3$ act on M_3 by conjugation (over an algebraically closed field of characteristic 0). List all possible Jordan types for 3×3 matrices with a single eigenvalue λ , compute the dimension of each conjugacy class, and determine the closure partial order among these orbits.
3. Let G act on an affine variety X . Show that X contains a closed orbit. (*Hint*: X is Noetherian, so among all orbit closures, choose one of minimal dimension.)
4. Let $\mathbb{G}_m$ act on $\mathbb{A}^3$ by $t \cdot (x, y, z) = (t^2x, ty, t^{-1}z)$. Determine all orbits, their dimensions, and which orbits are closed. Verify that the boundary of each non-closed orbit is a union of orbits of strictly lower dimension.
5. Let G be a connected algebraic group acting transitively on a variety X . Show that X is irreducible. (*Hint*: X is the image of the connected, hence irreducible, group G under the orbit map.)
6. Let G act on a variety X and let $x \in X(k)$. Show that the stabilizer G_x contains $(G_{\overline{G \cdot x}})^\circ$, where $G_{\overline{G \cdot x}} = \{g \in G \mid g \cdot \overline{G \cdot x} = \overline{G \cdot x}\}$ is the setwise stabilizer of the closure. Conclude that G_x has finite index in $G_{\overline{G \cdot x}}$.

Infinitesimal Structure

Every algebraic group G carries a finite-dimensional Lie algebra $\mathfrak{g} = \text{Lie}(G)$ that encodes the “infinitesimal” structure of G near the identity. In the classical theory over $\mathbb{R}$ or $\mathbb{C}$, the Lie algebra is constructed by differentiating curves through the identity, and the exponential map provides a local inverse: the group and its Lie algebra determine each other in a neighborhood of e . In the algebraic setting, we have no convergent power series and no analytic neighborhoods, but we have something better: the *ring of dual numbers* $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$, which serves as an algebraic substitute for “infinitesimally close to the identity.” The functor of points, developed in Chapter 1, makes this completely rigorous: we simply evaluate the group functor on $k[\varepsilon]$ and extract the kernel of the reduction map.

This chapter introduces the Lie algebra, computes it for the classical groups, and establishes the adjoint representation — the action of G on its own Lie algebra by conjugation. This is the representation that will eventually decompose $\mathfrak{g}$ into root spaces and drive the classification in Chapter 7. We close by proving Cartier’s theorem, which guarantees that in characteristic 0 the Lie algebra faithfully reflects the group, and by examining the explicit failures that arise in characteristic p .

3.1 The Lie Algebra of an Algebraic Group

The Ring of Dual Numbers

Let $k[\varepsilon] = k[\varepsilon]/(\varepsilon^2)$ be the ring of *dual numbers* over k . As a k -algebra, it is two-dimensional with basis $\{1, \varepsilon\}$, and the only interesting multiplication rule is $\varepsilon^2 = 0$. Geometrically, $\text{Spec } k[\varepsilon]$ is a “fat point”: it consists of a single (closed) point with a one-dimensional tangent direction attached. There is a canonical surjection $\pi : k[\varepsilon] \rightarrow k$ sending $\varepsilon \mapsto 0$ (“forgetting the tangent direction”) and a section $\iota : k \rightarrow k[\varepsilon]$ sending $a \mapsto a$ (“the point itself”).

If G is an affine algebraic group over k , evaluating the functor of points on $k[\varepsilon]$ gives the group $G(k[\varepsilon])$, whose elements are “points of G with an infinitesimal perturbation.” The surjection π induces a group homomorphism $G(\pi) : G(k[\varepsilon]) \rightarrow G(k)$ that “forgets the perturbation.” The kernel of this map is exactly the set of infinitesimal elements: group elements that reduce to the identity modulo ε .

Definition via Dual Numbers

Definition 3.1.1: Lie Algebra of an Algebraic Group

Let G be an affine algebraic group over k . The *Lie algebra* of G is

$$\mathrm{Lie}(G) := \ker\left(G(k[\varepsilon]) \xrightarrow{G(\pi)} G(k)\right)$$

the kernel of the reduction map. We write $\mathfrak{g} = \mathrm{Lie}(G)$ and call it the Lie algebra of G .

Let us unpack this definition concretely. An element of $G(k[\varepsilon]) = \mathrm{Hom}_{\mathbf{Alg}_k}(\mathcal{O}(G), k[\varepsilon])$ is a k -algebra homomorphism $\varphi : \mathcal{O}(G) \rightarrow k[\varepsilon]$. Since every element of $k[\varepsilon]$ is uniquely of the form $a + b\varepsilon$ with $a, b \in k$, we can decompose $\varphi = \varphi_0 + \varphi_1\varepsilon$ where $\varphi_0 : \mathcal{O}(G) \rightarrow k$ and $\varphi_1 : \mathcal{O}(G) \rightarrow k$ are k -linear maps. The condition that φ is a k -algebra homomorphism means:

$$\varphi_0(fg) + \varphi_1(fg)\varepsilon = (\varphi_0(f) + \varphi_1(f)\varepsilon)(\varphi_0(g) + \varphi_1(g)\varepsilon) = \varphi_0(f)\varphi_0(g) + (\varphi_0(f)\varphi_1(g) + \varphi_1(f)\varphi_0(g))\varepsilon$$

where the ε^2 term vanishes. Comparing coefficients:

1. $\varphi_0(fg) = \varphi_0(f)\varphi_0(g)$, so φ_0 is a k -algebra homomorphism $\mathcal{O}(G) \rightarrow k$, i.e. a point of $G(k)$.
2. $\varphi_1(fg) = \varphi_0(f)\varphi_1(g) + \varphi_1(f)\varphi_0(g)$, so φ_1 is a *derivation at φ_0* : it satisfies the Leibniz rule with respect to the evaluation φ_0 .

The reduction map $G(\pi)$ sends φ to φ_0 . So $\varphi \in \ker(G(\pi))$ means $\varphi_0 = \epsilon$ (the counit, corresponding to the identity $e \in G(k)$), and φ_1 is a derivation at ϵ :

$$\varphi_1(fg) = \epsilon(f)\varphi_1(g) + \varphi_1(f)\epsilon(g) = f(e)\varphi_1(g) + \varphi_1(f)g(e)$$

Writing $A = \mathcal{O}(G)$ and $\mathfrak{m}_e = \ker(\epsilon)$ for the maximal ideal at the identity, a derivation at ϵ is a k -linear map $D : A \rightarrow k$ satisfying $D(fg) = f(e)D(g) + D(f)g(e)$. Note that such a D vanishes on k (since $D(1) = D(1 \cdot 1) = 1 \cdot D(1) + D(1) \cdot 1 = 2D(1)$, so $D(1) = 0$) and is determined by its restriction to $\mathfrak{m}_e$ (since any $f \in A$ can be written $f = f(e) + (f - f(e))$ with $f - f(e) \in \mathfrak{m}_e$). Moreover, D vanishes on $\mathfrak{m}_e^2$: if $f, g \in \mathfrak{m}_e$, then $D(fg) = f(e)D(g) + D(f)g(e) = 0$ since $f(e) = g(e) = 0$. Therefore D factors through a linear functional on $\mathfrak{m}_e/\mathfrak{m}_e^2$:

Proposition 3.1.2: Lie Algebra as Tangent Space

Let $G = \mathrm{Spec} A$ be an affine algebraic group over k with identity e and maximal ideal $\mathfrak{m}_e = \ker(\epsilon : A \rightarrow k)$. Then there is a canonical isomorphism of k -vector spaces:

$$\mathrm{Lie}(G) \cong (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee = \mathrm{Hom}_k(\mathfrak{m}_e/\mathfrak{m}_e^2, k)$$

That is, the Lie algebra is the *Zariski tangent space* of G at the identity.

Proof:

By the discussion above, the map $\text{Lie}(G) \rightarrow (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$ sending a derivation D to its restriction $D|_{\mathfrak{m}_e}$ (which factors through $\mathfrak{m}_e/\mathfrak{m}_e^2$) is well-defined and injective (since D is determined by $D|_{\mathfrak{m}_e}$). For surjectivity, given any linear functional $\lambda : \mathfrak{m}_e/\mathfrak{m}_e^2 \rightarrow k$, define $D : A \rightarrow k$ by $D(f) = \lambda(f - f(e) \bmod \mathfrak{m}_e^2)$. Then D is a derivation at e : for $f, g \in A$,

$$\begin{aligned} D(fg) &= \lambda(fg - f(e)g(e) \bmod \mathfrak{m}_e^2) \\ &= \lambda((f - f(e))g(e) + f(e)(g - g(e)) + (f - f(e))(g - g(e)) \bmod \mathfrak{m}_e^2) \\ &= g(e)\lambda(f - f(e)) + f(e)\lambda(g - g(e)) \\ &= g(e)D(f) + f(e)D(g) \end{aligned}$$

where the $(f - f(e))(g - g(e))$ term vanishes since it lies in $\mathfrak{m}_e^2$.

This is exactly the tangent space $T_e G$ from algebraic geometry (see [3]). The dimension of $\text{Lie}(G)$ is the *embedding dimension* of the local ring $\mathcal{O}_{G,e}$, which equals $\dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2)$. We will see in §3.3 that this dimension is always at least $\dim G$, with equality if and only if G is smooth at e .

The Lie Bracket

We have shown that $\text{Lie}(G)$ is a k -vector space. It also carries a bilinear operation — the *Lie bracket* — that encodes the non-commutativity of G at the infinitesimal level.

The idea is simple: if $X, Y \in \text{Lie}(G)$ are two infinitesimal elements, then the group commutator $XYX^{-1}Y^{-1}$ measures how much X and Y fail to commute. Since X and Y are “ ε -close” to the identity, the commutator is “ ε^2 -close” to the identity. To capture this second-order information, we need to work over a ring with *two* independent nilpotent directions.

Let $k[\varepsilon_1, \varepsilon_2] = k[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_2^2)$ be the ring of “double dual numbers.” Note that $\varepsilon_1\varepsilon_2 \neq 0$ in this ring (it is killed only by further multiplication by either ε_i), so this ring has a richer nilpotent structure than $k[\varepsilon]$.

Given $X, Y \in \text{Lie}(G) = \ker(G(k[\varepsilon]) \rightarrow G(k))$, we can “embed” them into $G(k[\varepsilon_1, \varepsilon_2])$ by the two algebra maps $k[\varepsilon] \rightarrow k[\varepsilon_1, \varepsilon_2]$ sending $\varepsilon \mapsto \varepsilon_1$ and $\varepsilon \mapsto \varepsilon_2$ respectively. Call the resulting elements $X_1, Y_2 \in G(k[\varepsilon_1, \varepsilon_2])$. Since $\varepsilon_1^2 = \varepsilon_2^2 = 0$, both X_1 and Y_2 reduce to e modulo $(\varepsilon_1, \varepsilon_2)$.

Now form the group commutator:

$$[X_1, Y_2] := X_1 Y_2 X_1^{-1} Y_2^{-1} \in G(k[\varepsilon_1, \varepsilon_2])$$

This element reduces to e modulo $(\varepsilon_1, \varepsilon_2)$, and in fact reduces to e modulo (ε_1) alone (since Y_2 and Y_2^{-1} cancel when $\varepsilon_2 = 0$) and modulo (ε_2) alone (by the same argument with X_1). A careful computation shows that $[X_1, Y_2]$ lies in the kernel of the reduction to any ring where $\varepsilon_1\varepsilon_2 = 0$, and hence is determined by its “ $\varepsilon_1\varepsilon_2$ -component.”

The algebra map $k[\varepsilon_1, \varepsilon_2] \rightarrow k[\varepsilon]$ sending $\varepsilon_1 \mapsto \varepsilon$, $\varepsilon_2 \mapsto \varepsilon$ would collapse our two directions, but the map $k[\varepsilon_1, \varepsilon_2] \rightarrow k[\varepsilon]$ sending $\varepsilon_1\varepsilon_2 \mapsto \varepsilon$ (and $\varepsilon_1, \varepsilon_2 \mapsto 0$ individually) extracts exactly the $\varepsilon_1\varepsilon_2$ -coefficient. This yields an element of $\text{Lie}(G)$.

Definition 3.1.3: Lie Bracket

Let G be an affine algebraic group and $X, Y \in \mathfrak{g} = \text{Lie}(G)$. The *Lie bracket* $[X, Y] \in \mathfrak{g}$ is the element defined by the following procedure:

1. Embed X and Y into $G(k[\varepsilon_1, \varepsilon_2])$ via $\varepsilon \mapsto \varepsilon_1$ and $\varepsilon \mapsto \varepsilon_2$ respectively.
2. Form the group commutator $X_1 Y_2 X_1^{-1} Y_2^{-1} \in G(k[\varepsilon_1, \varepsilon_2])$.
3. Extract the $\varepsilon_1 \varepsilon_2$ -component to obtain $[X, Y] \in \text{Lie}(G)$.

That this is well-defined (in particular, k -bilinear) follows from the explicit computation, which we will carry out for GL_n momentarily. The key properties follow from the group axioms:

Proposition 3.1.4: Lie Algebra Axioms

Let $\mathfrak{g} = \text{Lie}(G)$. The bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfies:

1. (Bilinearity) $[aX + bY, Z] = a[X, Z] + b[Y, Z]$ and $[X, aY + bZ] = a[X, Y] + b[X, Z]$ for all $a, b \in k$.
2. (Antisymmetry) $[X, Y] = -[Y, X]$ for all $X, Y \in \mathfrak{g}$.
3. (Jacobi Identity) $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in \mathfrak{g}$.

In particular, $(\mathfrak{g}, [\cdot, \cdot])$ is a Lie algebra in the abstract algebraic sense.

Proof:

Antisymmetry follows from the fact that swapping X_1 and Y_2 inverts the commutator: $Y_2 X_1 Y_2^{-1} X_1^{-1} = (X_1 Y_2 X_1^{-1} Y_2^{-1})^{-1}$. At the infinitesimal level, inverting an element $e + Z\varepsilon$ gives $e - Z\varepsilon$, so the $\varepsilon_1 \varepsilon_2$ -component changes sign.

Bilinearity requires a computation. We verify it for GL_n in example 3.1.6 below (where the bracket is $[X, Y] = XY - YX$, visibly bilinear). For a general affine group $G \hookrightarrow \text{GL}_n$, the Lie bracket on $\text{Lie}(G)$ is the restriction of the bracket on $\mathfrak{gl}_n$, hence bilinear.

The Jacobi identity follows from the same embedding principle: any identity that holds in $\mathfrak{gl}_n$ restricts to $\text{Lie}(G) \subseteq \mathfrak{gl}_n$. For $\mathfrak{gl}_n$, the Jacobi identity is a direct (and classical) computation using $[X, Y] = XY - YX$.

3.1.5 Remark: The Bracket in Characteristic 2 In characteristic 2, the antisymmetry condition $[X, Y] = -[Y, X]$ becomes $[X, Y] = [Y, X]$, which no longer implies $[X, X] = 0$. The “correct” structure in characteristic p is that of a *restricted Lie algebra* (or *p -Lie algebra*), which carries an additional operation $X \mapsto X^{[p]}$ satisfying $\text{ad}(X^{[p]}) = \text{ad}(X)^p$. This p -operation arises naturally from the Frobenius endomorphism. We will encounter it briefly in §3.3, but a full treatment belongs to the representation theory of Part II.

Computing Lie Algebras of Classical Groups

The power of the functorial definition is that computing $\text{Lie}(G)$ reduces to elementary linear algebra: write down what $G(k[\varepsilon])$ looks like, and take the kernel of the reduction.

Example 3.1.6: $\text{Lie}(\text{GL}_n) = \mathfrak{gl}_n$

An element of $\text{GL}_n(k[\varepsilon])$ is an invertible $n \times n$ matrix with entries in $k[\varepsilon]$. Write such a matrix as $g = A + B\varepsilon$ where $A, B \in M_n(k)$. For g to be invertible, we need $\det(A + B\varepsilon) \in k[\varepsilon]^\times$. Since

$$\det(A + B\varepsilon) = \det(A) + \text{tr}(\text{adj}(A) \cdot B)\varepsilon$$

(where $\text{adj}(A)$ is the adjugate), the determinant is a unit in $k[\varepsilon]$ if and only if $\det(A) \neq 0$ in k . So g is invertible if and only if $A \in \text{GL}_n(k)$.

The reduction map sends $A + B\varepsilon \mapsto A$. The kernel consists of elements $I + X\varepsilon$ with $X \in M_n(k)$ arbitrary (no condition on X , since $\det(I) = 1 \neq 0$). Therefore:

$$\text{Lie}(\text{GL}_n) = \{I + X\varepsilon \mid X \in M_n(k)\} \cong M_n(k) = \mathfrak{gl}_n$$

as a k -vector space of dimension n^2 .

To compute the Lie bracket, take $X_1 = I + X\varepsilon_1$ and $Y_2 = I + Y\varepsilon_2$ in $\text{GL}_n(k[\varepsilon_1, \varepsilon_2])$. Their inverses are $X_1^{-1} = I - X\varepsilon_1$ and $Y_2^{-1} = I - Y\varepsilon_2$ (using $\varepsilon_i^2 = 0$). The commutator is:

$$\begin{aligned} X_1 Y_2 X_1^{-1} Y_2^{-1} &= (I + X\varepsilon_1)(I + Y\varepsilon_2)(I - X\varepsilon_1)(I - Y\varepsilon_2) \\ &= (I + X\varepsilon_1 + Y\varepsilon_2 + XY\varepsilon_1\varepsilon_2)(I - X\varepsilon_1 - Y\varepsilon_2 + XY\varepsilon_1\varepsilon_2) \\ &= I + (XY - YX)\varepsilon_1\varepsilon_2 \end{aligned}$$

where all ε_i^2 terms vanish. Extracting the $\varepsilon_1\varepsilon_2$ -component:

$$[X, Y] = XY - YX$$

This is the usual matrix commutator, confirming that $\mathfrak{gl}_n$ with the bracket $[X, Y] = XY - YX$ is the Lie algebra of GL_n .

Example 3.1.7: $\text{Lie}(\text{SL}_n) = \mathfrak{sl}_n$

An element of $\text{SL}_n(k[\varepsilon])$ is a matrix $I + X\varepsilon$ (reducing to I in the kernel) with $\det(I + X\varepsilon) = 1$. Using the expansion:

$$\det(I + X\varepsilon) = 1 + \text{tr}(X)\varepsilon$$

(this follows from the general formula $\det(I + tM) = 1 + t\text{tr}(M) + O(t^2)$, and $\varepsilon^2 = 0$ kills all higher terms). The condition $\det = 1$ forces $\text{tr}(X) = 0$. Therefore:

$$\text{Lie}(\text{SL}_n) = \{X \in M_n(k) \mid \text{tr}(X) = 0\} = \mathfrak{sl}_n$$

This is an $(n^2 - 1)$ -dimensional Lie subalgebra of $\mathfrak{gl}_n$. The bracket is the restriction of the matrix commutator, which is well-defined since $\text{tr}([X, Y]) = \text{tr}(XY) - \text{tr}(YX) = 0$.

Example 3.1.8: $\text{Lie}(\text{Sp}_{2n}) = \mathfrak{sp}_{2n}$

Recall from §1.1.1 that Sp_{2n} consists of matrices $g \in \text{GL}_{2n}$ satisfying ${}^t g \Omega g = \Omega$ where $\Omega = \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix}$. Setting $g = I + X\varepsilon$ and expanding:

$${}^t(I + X\varepsilon)\Omega(I + X\varepsilon) = \Omega + ({}^t X \Omega + \Omega X)\varepsilon$$

The condition becomes ${}^t X \Omega + \Omega X = 0$, i.e. the matrix ΩX is symmetric (since ${}^t(\Omega X) = {}^t X {}^t \Omega = -{}^t X \Omega = \Omega X$). Therefore:

$$\text{Lie}(\text{Sp}_{2n}) = \{X \in M_{2n}(k) \mid {}^t X \Omega + \Omega X = 0\} = \mathfrak{sp}_{2n}$$

The dimension is $n(2n + 1)$.

Example 3.1.9: $\text{Lie}(\text{SO}_n) = \mathfrak{so}_n$

The orthogonal group O_n consists of $g \in \text{GL}_n$ with ${}^t g s g = s$ for a symmetric form s . Setting $g = I + X\varepsilon$:

$${}^t(I + X\varepsilon)s(I + X\varepsilon) = s + ({}^t X s + s X)\varepsilon$$

The condition is ${}^t X s + s X = 0$, meaning sX is skew-symmetric ($\text{char } k \neq 2$). Thus:

$$\text{Lie}(O_n) = \{X \in M_n(k) \mid {}^t X s + s X = 0\} = \mathfrak{so}_n$$

The dimension is $\binom{n}{2}$. Note that O_n and SO_n have the *same* Lie algebra: the determinant condition $\det(g) = 1$ imposes $\text{tr}(X) = 0$, but for any $X \in \mathfrak{so}_n$ the skew-symmetry of sX already implies $\text{tr}(sX) = 0$, and when $s = I$ this gives $\text{tr}(X) = 0$ directly. Thus $\text{Lie}(\text{SO}_n) = \text{Lie}(O_n) = \mathfrak{so}_n$. The Lie algebra cannot see the difference between O_n and SO_n because they share the same identity component (recall from definition 1.1.3 that $\text{SO}_n = O_n^\circ$).

The last observation is an instance of a general principle:

Proposition 3.1.10: Lie Algebra Depends Only on G°

Let G be an affine algebraic group. Then $\text{Lie}(G) = \text{Lie}(G^\circ)$.

Proof:

The identity e lies in G° , and $\text{Lie}(G)$ depends only on the local structure of G at e (it is the tangent space $T_e G$). Since G° is an open subgroup of G containing e , the local rings $\mathcal{O}_{G,e}$ and $\mathcal{O}_{G^\circ,e}$ are isomorphic, hence so are the tangent spaces.

The Functor Lie

The assignment $G \mapsto \text{Lie}(G)$ is not merely a construction on individual groups; it is a *functor* from affine algebraic groups to Lie algebras.

Proposition 3.1.11: Functoriality of Lie

Let $\varphi : G \rightarrow H$ be a homomorphism of affine algebraic groups. Then the map

$$d\varphi := \text{Lie}(\varphi) : \text{Lie}(G) \longrightarrow \text{Lie}(H)$$

defined by $(d\varphi)(X) = \varphi_{k[\varepsilon]}(X)$ (applying φ on $k[\varepsilon]$ -valued points) is a homomorphism of Lie algebras. That is, $d\varphi$ is k -linear and $d\varphi([X, Y]) = [d\varphi(X), d\varphi(Y)]$.

Proof:

Since φ is a natural transformation of group functors, $\varphi_{k[\varepsilon]} : G(k[\varepsilon]) \rightarrow H(k[\varepsilon])$ is a group homomorphism. It sends $\ker(G(k[\varepsilon]) \rightarrow G(k))$ into $\ker(H(k[\varepsilon]) \rightarrow H(k))$ by naturality of φ with respect to the map $k[\varepsilon] \rightarrow k$. So $d\varphi$ is a well-defined k -linear map $\mathfrak{g} \rightarrow \mathfrak{h}$. Since φ preserves the group operation, it preserves commutators: $\varphi(XYX^{-1}Y^{-1}) = \varphi(X)\varphi(Y)\varphi(X)^{-1}\varphi(Y)^{-1}$. Extracting the $\varepsilon_1\varepsilon_2$ -component gives $d\varphi([X, Y]) = [d\varphi(X), d\varphi(Y)]$.

The map $d\varphi$ is the algebraic-group analogue of the “differential” or “derivative” of a Lie group homomorphism. We can now ask: to what extent does $d\varphi$ determine φ ?

Proposition 3.1.12: Lie Preserves Closed Immersions

Let $H \hookrightarrow G$ be a closed immersion of affine algebraic groups. Then $d\iota : \text{Lie}(H) \hookrightarrow \text{Lie}(G)$ is injective.

Proof:

A closed immersion $H \hookrightarrow G$ corresponds to a surjection $\mathcal{O}(G) \twoheadrightarrow \mathcal{O}(H)$ of Hopf algebras. On tangent spaces at the identity, a surjection of local rings induces an injection of cotangent spaces $\mathfrak{m}_{e,H}/\mathfrak{m}_{e,H}^2 \hookrightarrow \mathfrak{m}_{e,G}/\mathfrak{m}_{e,G}^2$ (or rather, a surjection of cotangent spaces and an injection of tangent spaces). Concretely, if $X \in \text{Lie}(H)$ satisfies $d\iota(X) = 0$, then X acts as the zero derivation on all functions pulled back from G . Since $\mathcal{O}(H)$ is a quotient of $\mathcal{O}(G)$, these functions generate $\mathcal{O}(H)$, so $X = 0$.

This means that Lie embeds the lattice of closed subgroups of G into the lattice of Lie subalgebras of $\mathfrak{g}$. However, the embedding is generally *not* surjective: there are Lie subalgebras of $\mathfrak{g}$ that do not arise as $\text{Lie}(H)$ for any closed subgroup H . Whether or not every subalgebra integrates to a subgroup depends on the characteristic; in characteristic 0, this is true (by a theorem that will follow from Cartier’s theorem in §3.3), but in characteristic p it can fail.

The Lie Algebra via Derivations and Left-Invariant Vector Fields

There is a third description of $\text{Lie}(G)$ that is sometimes useful, especially when connecting to the differential-geometric theory. A *left-invariant derivation* of $\mathcal{O}(G)$ is a k -linear map $D : \mathcal{O}(G) \rightarrow \mathcal{O}(G)$ satisfying the Leibniz rule $D(fg) = fD(g) + D(f)g$ and commuting with the left action of G on $\mathcal{O}(G)$. In Hopf algebra terms, left-invariance means:

$$\Delta \circ D = (\text{id} \otimes D) \circ \Delta$$

That is, D acts only on the “right factor” of the comultiplication.

Proposition 3.1.13: Lie Algebra as Left-Invariant Derivations

There is a canonical isomorphism between $\mathrm{Lie}(G)$ and the space of left-invariant derivations of $\mathcal{O}(G)$.

Proof:

Given a derivation $D_e : A \rightarrow k$ at the identity (i.e. an element of $\mathrm{Lie}(G) = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$), define $D : A \rightarrow A$ by $D = (\mathrm{id} \otimes D_e) \circ \Delta$. Explicitly, for $f \in A$ with $\Delta(f) = \sum f_{(1)} \otimes f_{(2)}$:

$$D(f) = \sum f_{(1)} D_e(f_{(2)})$$

We verify the Leibniz rule. Since Δ is an algebra homomorphism, $\Delta(fg) = \Delta(f)\Delta(g) = \sum f_{(1)}g_{(1)} \otimes f_{(2)}g_{(2)}$. Then:

$$D(fg) = \sum f_{(1)}g_{(1)} D_e(f_{(2)}g_{(2)}) = \sum f_{(1)}g_{(1)} (\epsilon(f_{(2)})D_e(g_{(2)}) + D_e(f_{(2)})\epsilon(g_{(2)}))$$

Using the counit axiom $\sum f_{(1)}\epsilon(f_{(2)}) = f$:

$$D(fg) = f \cdot D(g) + D(f) \cdot g$$

Left-invariance $\Delta \circ D = (\mathrm{id} \otimes D) \circ \Delta$ follows from coassociativity. The inverse map sends a left-invariant $D : A \rightarrow A$ to $D_e = \epsilon \circ D : A \rightarrow k$, which is a derivation at ϵ .

For the reader familiar with differential geometry (see [13]), this is the exact algebraic analogue of the classical fact that the Lie algebra of a Lie group is the space of left-invariant vector fields. The left translation by g on a Lie group is replaced by the coaction Δ , and “invariance under left translation” is replaced by $\Delta \circ D = (\mathrm{id} \otimes D) \circ \Delta$.

Exercise 3.1.1

1. Compute $\mathrm{Lie}(\mathbb{G}_a)$ and $\mathrm{Lie}(\mathbb{G}_m)$. For $\mathbb{G}_m$, show that the Lie algebra is one-dimensional and abelian (i.e. the bracket is identically zero).
2. Let $B_n \subseteq \mathrm{GL}_n$ be the Borel subgroup of upper-triangular invertible matrices, and let $U_n \subseteq B_n$ be the unipotent subgroup of upper-triangular matrices with 1's on the diagonal. Compute $\mathrm{Lie}(B_n)$ and $\mathrm{Lie}(U_n)$ using the dual numbers definition. What are their dimensions?
3. Let G and H be affine algebraic groups. Show that $\mathrm{Lie}(G \times H) \cong \mathrm{Lie}(G) \oplus \mathrm{Lie}(H)$ as Lie algebras, where the direct sum has bracket $[(X_1, Y_1), (X_2, Y_2)] = ([X_1, X_2], [Y_1, Y_2])$.
4. Let $\varphi : G \rightarrow H$ be a homomorphism of affine algebraic groups with (scheme-theoretic) kernel $K = \ker(\varphi)$. Show that $\mathrm{Lie}(K) = \ker(d\varphi)$. Deduce that if φ is surjective and $\dim G = \dim H$, then K is finite if and only if $d\varphi$ is an isomorphism (assuming both groups are smooth).
5. Let $G = \mathrm{GL}_2$ and let $D_e \in \mathrm{Lie}(\mathrm{GL}_2)$ correspond to the matrix $X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Compute the left-invariant derivation $D = (\mathrm{id} \otimes D_e) \circ \Delta : \mathcal{O}(\mathrm{GL}_2) \rightarrow \mathcal{O}(\mathrm{GL}_2)$ by evaluating D on the coordinate functions a, b, c, d (the matrix entries). (*Hint*: use $\Delta(a) = a \otimes a + b \otimes c$, etc. from

example 1.1.14.)

6. Let K/k be a finite field extension and G an algebraic group over K . Show that $\mathrm{Lie}(\mathrm{Res}_{K/k}(G)) \cong \mathrm{Lie}(G)$ as k -vector spaces (where the right side is viewed as a k -vector space via restriction of scalars). (*Hint*: use the definition $\mathrm{Res}_{K/k}(G)(R) = G(R \otimes_k K)$ from definition 1.2.3 and compute on $k[\varepsilon]$.)

3.2 The Adjoint Representation

In Chapter 2, we introduced the conjugation action of G on itself (example 2.1.5): a group element g acts on G by $x \mapsto gxg^{-1}$. This action preserves the identity (since $geg^{-1} = e$), and therefore it acts on the infinitesimal structure near e — that is, on the Lie algebra $\mathfrak{g}$. This induced action of G on $\mathfrak{g}$ is the *adjoint representation*, and it is the single most important representation in the structure theory of algebraic groups. When we decompose $\mathfrak{g}$ into eigenspaces for the adjoint action of a maximal torus in Chapter 7, we will obtain the *root space decomposition*, which is the foundation of the entire classification.

The Adjoint Representation of G

Let G be an affine algebraic group over k . For each $g \in G(R)$ (where R is any k -algebra), the *inner automorphism* $\mathrm{Inn}_g : G_R \rightarrow G_R$ is the morphism $x \mapsto gxg^{-1}$. This is an automorphism of G_R as a group scheme over R , and it fixes the identity. By the functoriality of Lie (proposition 3.1.11), it induces a Lie algebra automorphism $d(\mathrm{Inn}_g) : \mathfrak{g} \otimes_k R \rightarrow \mathfrak{g} \otimes_k R$.

Definition 3.2.1: The Adjoint Representation

Let G be an affine algebraic group with Lie algebra $\mathfrak{g}$. The *adjoint representation* is the homomorphism of algebraic groups

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathfrak{g})$$

defined on R -valued points by: for $g \in G(R)$ and $X \in \mathfrak{g} \otimes_k R$,

$$\mathrm{Ad}(g)(X) = d(\mathrm{Inn}_g)(X)$$

That is, $\mathrm{Ad}(g)$ is the differential of the conjugation $x \mapsto gxg^{-1}$ at the identity.

Let us make this concrete. An element $X \in \mathfrak{g} = \mathrm{Lie}(G)$ is an element of $G(k[\varepsilon])$ that reduces to e modulo ε : it has the form $e + X\varepsilon$ (in a matrix group) or, more generally, it is a k -algebra map $\mathcal{O}(G) \rightarrow k[\varepsilon]$ with reduction ϵ . The conjugation Inn_g acts on $G(k[\varepsilon])$ by:

$$\mathrm{Inn}_g(e + X\varepsilon) = g(e + X\varepsilon)g^{-1} = e + (gXg^{-1})\varepsilon$$

where in the last step we used the fact that $g \in G(k) \subseteq G(k[\varepsilon])$ (viewing k as a subring of $k[\varepsilon]$), so the computation takes place in $G(k[\varepsilon])$. Extracting the ε -coefficient, we get $\mathrm{Ad}(g)(X) = gXg^{-1}$.

More precisely, to see that Ad is a morphism of *algebraic* groups (not just a map on k -points), we work over an arbitrary k -algebra R . An element $g \in G(R)$ and $X \in \mathfrak{g} \otimes_k R \subseteq G(R[\varepsilon])$ (where

$R[\varepsilon] = R \otimes_k k[\varepsilon]$ satisfy:

$$\mathrm{Ad}(g)(X) = gXg^{-1} \in \mathfrak{g} \otimes_k R$$

The naturality in R ensures that Ad is a natural transformation of functors $G \rightarrow \mathrm{GL}(\mathfrak{g})$, hence a morphism of algebraic groups by the Yoneda Lemma (theorem 1.2.1).

That Ad is a group homomorphism is immediate from the functorial viewpoint: for $g, h \in G(R)$,

$$\mathrm{Ad}(gh)(X) = (gh)X(gh)^{-1} = g(hXh^{-1})g^{-1} = \mathrm{Ad}(g)(\mathrm{Ad}(h)(X))$$

Example 3.2.2: Ad for GL_n

For $G = \mathrm{GL}_n$, the Lie algebra is $\mathfrak{g} = \mathfrak{gl}_n = M_n(k)$, and the adjoint representation is simply conjugation of matrices:

$$\mathrm{Ad}(g)(X) = gXg^{-1}, \quad g \in \mathrm{GL}_n(R), \quad X \in M_n(R)$$

This is a representation $\mathrm{GL}_n \rightarrow \mathrm{GL}(\mathfrak{gl}_n) = \mathrm{GL}_{n^2}$, but it is *not* the standard n^2 -dimensional representation of GL_n : the action is by simultaneous left- and right-multiplication (conjugation), not by left-multiplication alone. In the language of Chapter 2, $\mathfrak{gl}_n$ as a GL_n -module is isomorphic to $V \otimes V^\vee$, where $V = k^n$ is the standard representation and $V^\vee$ is the contragredient. This is because $\mathrm{Hom}_k(V, V) \cong V \otimes V^\vee$ as representations (?? 5), and conjugation is precisely the action on $\mathrm{Hom}_k(V, V)$.

Example 3.2.3: Ad for SL_n

For $G = \mathrm{SL}_n$, the Lie algebra $\mathfrak{sl}_n = \{X \in M_n(k) : \mathrm{tr}(X) = 0\}$ is a codimension-1 subspace of $\mathfrak{gl}_n$. The adjoint representation is again $\mathrm{Ad}(g)(X) = gXg^{-1}$, but now $g \in \mathrm{SL}_n(R)$ and $X \in \mathfrak{sl}_n \otimes R$. Since conjugation preserves the trace ($\mathrm{tr}(gXg^{-1}) = \mathrm{tr}(X)$), this is well-defined. As a representation of SL_n , the Lie algebra $\mathfrak{sl}_n$ is the quotient of $V \otimes V^\vee \cong \mathfrak{gl}_n$ by the trivial subrepresentation $k \cdot I_n$ (the scalar matrices, which have trace n and are fixed by conjugation). When $\mathrm{char} k \nmid n$, the trace map $\mathfrak{gl}_n \rightarrow k$ splits and $\mathfrak{gl}_n \cong \mathfrak{sl}_n \oplus k$ as SL_n -representations. When $\mathrm{char} k \mid n$, the scalar matrix I_n has trace 0 and so $k \cdot I_n \subseteq \mathfrak{sl}_n$; this subrepresentation does not split off, and the representation theory becomes more subtle.

The Kernel of Ad: The Center

The kernel of a representation is always a closed normal subgroup. For the adjoint representation, this kernel has a particularly natural description.

Proposition 3.2.4: Kernel of Ad is the Center

Let G be an affine algebraic group. The kernel of $\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g})$ contains the *scheme-theoretic center*

$$Z(G) := \{g \in G \mid gx = xg \text{ for all } x \in \mathfrak{g}\}$$

and if G is smooth, then $\ker(\mathrm{Ad}) = Z(G)$.

Proof:

Functorially, $g \in Z(G)(R)$ means $gxg^{-1} = x$ for all $x \in G(R')$ and all R -algebras R' . In particular, taking $R' = R[\varepsilon]$ and $x \in \mathfrak{g} \otimes_k R \subseteq G(R[\varepsilon])$, we get $\text{Ad}(g)(X) = gXg^{-1} = X$, so $g \in \ker(\text{Ad})(R)$. Thus $Z(G) \subseteq \ker(\text{Ad})$.

For the reverse inclusion when G is smooth, suppose $g \in \ker(\text{Ad})(R)$, so that $\text{Ad}(g)$ acts trivially on $\mathfrak{g} \otimes R$. The inner automorphism $\text{Inn}_g : G_R \rightarrow G_R$ is a group automorphism whose differential $d(\text{Inn}_g) = \text{Ad}(g)$ is the identity on $\mathfrak{g} \otimes R$. When G is smooth and connected, a morphism of smooth algebraic groups whose differential is the identity must be the identity itself (this follows from the fact that in characteristic 0, or more generally for smooth groups, the differential is faithful on connected groups — we will make this precise in §3.3). For disconnected G , the argument still works on G° , and one checks separately that the center respects connected components.

Without the smoothness hypothesis, the inclusion can be strict. For example, over a field of characteristic p , the group scheme $\mu_p = \text{Spec } k[t]/(t^p - 1)$ has trivial Lie algebra action but a nontrivial center; we will see such phenomena in §3.3.

Example 3.2.5: Center of GL_n

The center of GL_n consists of scalar matrices: $Z(\text{GL}_n) = \{aI : a \in k^\times\} \cong \mathbb{G}_m$. Indeed, $\text{Ad}(aI)(X) = (aI)X(aI)^{-1} = X$ for all $X \in \mathfrak{gl}_n$. Conversely, if $gXg^{-1} = X$ for all $X \in M_n(k)$, then g commutes with all matrices, and a standard linear algebra argument (or Schur's lemma applied to the standard representation) shows g is scalar.

For SL_n , the center is $Z(\text{SL}_n) = \mu_n = \{aI : a^n = 1\}$, the n -th roots of unity (as a group scheme). The quotient $\text{SL}_n/Z(\text{SL}_n) = \text{PGL}_n$ is the *projective general linear group*, and the adjoint representation of SL_n factors through PGL_n . Since $Z(\text{SL}_n)$ acts trivially on $\mathfrak{sl}_n$, we have $\text{Ad} : \text{SL}_n \rightarrow \text{GL}(\mathfrak{sl}_n)$ with $\ker = \mu_n$. In characteristic p dividing n , the group scheme μ_n is not reduced (it contains the infinitesimal subgroup μ_p), a subtlety that the adjoint representation detects.

The Infinitesimal Adjoint: ad

The adjoint representation $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ is a homomorphism of algebraic groups, so we can apply the functor Lie to it. The result is a homomorphism of Lie algebras

$$\text{ad} := \text{Lie}(\text{Ad}) : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g})$$

where $\mathfrak{gl}(\mathfrak{g}) = \text{Lie}(\text{GL}(\mathfrak{g})) = \text{End}_k(\mathfrak{g})$ with the commutator bracket. This map ad sends each element $X \in \mathfrak{g}$ to a linear endomorphism $\text{ad}(X) : \mathfrak{g} \rightarrow \mathfrak{g}$.

Proposition 3.2.6: ad is the Lie Bracket

For all $X, Y \in \mathfrak{g}$, we have $\text{ad}(X)(Y) = [X, Y]$.

Proof:

We compute using dual numbers. By definition, $\text{ad}(X)$ is the differential of Ad in the direction X . Concretely, let $X \in \mathfrak{g}$ and consider the element $e + X\varepsilon_1 \in G(k[\varepsilon_1])$. The adjoint representation sends this to $\text{Ad}(e + X\varepsilon_1) \in \text{GL}(\mathfrak{g})(k[\varepsilon_1]) = \text{GL}(\mathfrak{g} \otimes_k k[\varepsilon_1])$.

To evaluate $\text{Ad}(e + X\varepsilon_1)$ on an element $Y \in \mathfrak{g}$, we embed Y into $G(k[\varepsilon_1, \varepsilon_2])$ as $e + Y\varepsilon_2$ and conjugate:

$$\begin{aligned} \text{Ad}(e + X\varepsilon_1)(e + Y\varepsilon_2) &= (e + X\varepsilon_1)(e + Y\varepsilon_2)(e + X\varepsilon_1)^{-1} \\ &= (e + X\varepsilon_1)(e + Y\varepsilon_2)(e - X\varepsilon_1) \\ &= e + Y\varepsilon_2 + (XY - YX)\varepsilon_1\varepsilon_2 \end{aligned}$$

The ε_2 -coefficient of the result gives the action of $\text{Ad}(e + X\varepsilon_1)$ on Y . This is $Y + [X, Y]\varepsilon_1$, meaning:

$$\text{Ad}(e + X\varepsilon_1) = \text{id}_{\mathfrak{g}} + \text{ad}(X) \cdot \varepsilon_1$$

where $\text{ad}(X)(Y) = XY - YX = [X, Y]$. (Here the computation is in a matrix group; for a general group the same argument works by embedding $G \hookrightarrow \text{GL}_n$ and using the linearization theorem, lemma 1.1.8.)

The map $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is thus a Lie algebra homomorphism from $\mathfrak{g}$ to the endomorphism algebra of itself. The fact that ad is a Lie algebra homomorphism is equivalent to the Jacobi identity:

$$\text{ad}([X, Y]) = [\text{ad}(X), \text{ad}(Y)] \iff [[X, Y], Z] = [X, [Y, Z]] - [Y, [X, Z]]$$

This is sometimes the most efficient way to *prove* the Jacobi identity: it follows formally from the fact that Ad is a group homomorphism, which is obvious.

Definition 3.2.7: Adjoint Representation of the Lie Algebra

Let $\mathfrak{g}$ be the Lie algebra of an affine algebraic group G . The *adjoint representation of the Lie algebra* is the map $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ defined by $\text{ad}(X)(Y) = [X, Y]$. Its kernel is the *center of the Lie algebra*:

$$\mathfrak{z}(\mathfrak{g}) := \ker(\text{ad}) = \{X \in \mathfrak{g} \mid [X, Y] = 0 \text{ for all } Y \in \mathfrak{g}\}$$

The center of the Lie algebra and the Lie algebra of the center are related but not identical:

Proposition 3.2.8: Center Comparison

Let G be an affine algebraic group. Then:

$$\text{Lie}(Z(G)) \subseteq \mathfrak{z}(\mathfrak{g}) = \ker(\text{ad})$$

If G is smooth and connected (and $\text{char } k = 0$), equality holds.

Proof:

By proposition 3.1.12, the closed immersion $Z(G) \hookrightarrow G$ gives an injection $\text{Lie}(Z(G)) \hookrightarrow \mathfrak{g}$. An element $X \in \text{Lie}(Z(G)) \subseteq \mathfrak{g}$ corresponds to a derivation at e that factors through $\mathcal{O}(Z(G))$. Since $Z(G) \subseteq \ker(\text{Ad})$, the image of X under $\text{ad} = \text{Lie}(\text{Ad})$ is zero, so $X \in \mathfrak{z}(\mathfrak{g})$.

For equality in characteristic 0, one uses the fact that Lie is exact on smooth groups (which we establish in §3.3): the sequence $1 \rightarrow Z(G) \rightarrow G \xrightarrow{\text{Ad}} \text{GL}(\mathfrak{g})$ is exact, so applying Lie gives an exact sequence $0 \rightarrow \text{Lie}(Z(G)) \rightarrow \mathfrak{g} \xrightarrow{\text{ad}} \mathfrak{gl}(\mathfrak{g})$, and the kernel of ad is exactly $\text{Lie}(Z(G))$.

Example 3.2.9: The Adjoint Representation of SL_2

Let $G = \text{SL}_2$ over a field k with $\text{char } k \neq 2$. The Lie algebra $\mathfrak{sl}_2$ has the standard basis:

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The Lie brackets are $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The adjoint representation $\text{ad} : \mathfrak{sl}_2 \rightarrow \mathfrak{gl}(\mathfrak{sl}_2) \cong \mathfrak{gl}_3$ is determined by the matrices of $\text{ad}(e)$, $\text{ad}(f)$, $\text{ad}(h)$ in the basis $\{e, f, h\}$:

$$\text{ad}(h) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \text{ad}(e) = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \text{ad}(f) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ -1 & 0 & 0 \end{pmatrix}$$

(The reader should verify these from the bracket relations.) The matrix $\text{ad}(h)$ is diagonal, and its eigenvalues $\{2, -2, 0\}$ are the *roots* of SL_2 with respect to the Cartan subalgebra $\mathfrak{t} = k \cdot h$. The eigenspaces are:

$$\mathfrak{g}_2 = k \cdot e, \quad \mathfrak{g}_{-2} = k \cdot f, \quad \mathfrak{g}_0 = k \cdot h = \mathfrak{t}$$

This is the simplest example of a *root space decomposition* $\mathfrak{g} = \mathfrak{g}_{-2} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_2$. The general theory of Chapter 7 will show that every semisimple Lie algebra admits such a decomposition, with the roots forming a combinatorial structure (a *root system*) that completely determines the group.

The center $\mathfrak{z}(\mathfrak{sl}_2) = 0$ (since $\text{ad}(h)$ has no kernel on the full space when $\text{char } k \neq 2$), confirming that $\mathfrak{sl}_2$ is a *semisimple* Lie algebra.

At the group level, $\text{Ad} : \text{SL}_2 \rightarrow \text{GL}_3$ sends a matrix $g \in \text{SL}_2$ to the 3×3 matrix representing conjugation $X \mapsto gXg^{-1}$ on $\mathfrak{sl}_2$. The image lands in SO_3 (the adjoint representation preserves the Killing form, which is a non-degenerate symmetric bilinear form on $\mathfrak{sl}_2$). The kernel is $Z(\text{SL}_2) = \{\pm I\} = \mu_2$. Thus Ad realizes the classical isomorphism $\text{SL}_2/\mu_2 \cong \text{SO}_3$ (when $\text{char } k \neq 2$).

Example 3.2.10: Abelian Groups Have Trivial Adjoint

If G is a commutative algebraic group ($\mathbb{G}_a$, $\mathbb{G}_m$, an abelian variety, a torus, ...), then the conjugation action is trivial: $gxg^{-1} = x$ for all g, x . Therefore $\text{Ad}(g) = \text{id}_{\mathfrak{g}}$ for all g , and $\text{ad} = 0$, meaning the Lie bracket is identically zero. The Lie algebra of a commutative group is always *abelian*.

The converse is false: the Lie algebra can be abelian even when the group is not. For example, in characteristic p , the group scheme $\alpha_p \rtimes \mathbb{G}_m$ (where $\mathbb{G}_m$ acts on α_p by scaling) can have abelian Lie algebra while being non-commutative, depending on the specific action. More strikingly, any non-commutative group G with $\dim G = 1$ necessarily has a one-dimensional (hence abelian) Lie algebra, but the group itself need not be commutative at the scheme-theoretic level.

The Adjoint Action on the Coordinate Ring

The adjoint representation gives G an action on the finite-dimensional vector space $\mathfrak{g}$. By the theory of Chapter 2, this is a comodule: Ad equips $\mathfrak{g}$ with a coaction $\rho_{\text{Ad}} : \mathfrak{g} \rightarrow \mathfrak{g} \otimes_k \mathcal{O}(G)$. More explicitly, if $\{X_1, \dots, X_n\}$ is a basis for $\mathfrak{g}$, there exist matrix coefficient functions $a_{ij} \in \mathcal{O}(G)$ such that:

$$\text{Ad}(g)(X_j) = \sum_{i=1}^n a_{ij}(g)X_i, \quad \rho_{\text{Ad}}(X_j) = \sum_{i=1}^n X_i \otimes a_{ij}$$

These functions a_{ij} are the *matrix coefficients of the adjoint representation*. They generate a Hopf subalgebra of $\mathcal{O}(G)$ (the image of $\text{Ad}^* : \mathcal{O}(\text{GL}(\mathfrak{g})) \rightarrow \mathcal{O}(G)$), whose spectrum is the image $\text{Ad}(G) \subseteq \text{GL}(\mathfrak{g})$.

There is also a natural extension to symmetric and exterior powers. Since $\mathfrak{g}$ is a representation of G , so are $\text{Sym}^d(\mathfrak{g})$, $\bigwedge^d(\mathfrak{g})$, and $\mathfrak{g}^{\otimes d}$ (by the constructions of §2.3). Of particular importance is the *coadjoint representation*: the dual $\mathfrak{g}^\vee$ with the contragredient action $\text{Ad}^*(g)(\xi) = \xi \circ \text{Ad}(g^{-1})$.

Proposition 3.2.11: Invariant Bilinear Forms

Let G be an affine algebraic group acting on its Lie algebra via Ad . A bilinear form $B : \mathfrak{g} \times \mathfrak{g} \rightarrow k$ is called *G -invariant* (or *Ad -invariant*) if $B(\text{Ad}(g)(X), \text{Ad}(g)(Y)) = B(X, Y)$ for all $g \in G$ and $X, Y \in \mathfrak{g}$. Equivalently (differentiating the invariance condition):

$$B([Z, X], Y) + B(X, [Z, Y]) = 0 \quad \text{for all } X, Y, Z \in \mathfrak{g}$$

Proof:

Fix $X, Y \in \mathfrak{g}$ and consider the function $\varphi : G \rightarrow k$ defined by $\varphi(g) = B(\text{Ad}(g)(X), \text{Ad}(g)(Y))$. If B is G -invariant, then φ is constant, so its differential at e vanishes. Evaluating in the direction $Z \in \mathfrak{g}$ (i.e. replacing g by $e + Z\varepsilon$):

$$\begin{aligned} \varphi(e + Z\varepsilon) &= B(\text{Ad}(e + Z\varepsilon)(X), \text{Ad}(e + Z\varepsilon)(Y)) \\ &= B(X + [Z, X]\varepsilon, Y + [Z, Y]\varepsilon) \\ &= B(X, Y) + (B([Z, X], Y) + B(X, [Z, Y]))\varepsilon \end{aligned}$$

The constant term is $B(X, Y) = \varphi(e)$, and the ε -coefficient must vanish, giving the infinitesimal invariance condition.

The converse (infinitesimal invariance implies global invariance for connected G in characteristic 0) follows from the faithfulness of Lie in that setting, which we establish in §3.3.

The most important example of such a form is the *Killing form* $\kappa(X, Y) = \text{tr}(\text{ad}(X) \circ \text{ad}(Y))$, which we will study in the context of semisimplicity. The SL_2 example above already demonstrated that the adjoint representation preserves a non-degenerate form, which is a manifestation of the Killing form.

Exercise 3.2.1

1. Let $G = \text{GL}_2$ with Lie algebra $\mathfrak{gl}_2$, basis $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ (the elementary matrices). Write down the 4×4 matrix representing $\text{Ad}(g)$ for a general element $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2$. Verify that

- $\mathrm{Ad}(g)$ has determinant 1 for all g (that is, the adjoint representation of GL_2 lands in SL_4 , not just GL_4).
2. Show that the scheme-theoretic center $Z(\mathrm{SL}_n)$ is the group scheme μ_n of n -th roots of unity, and verify directly that $\mathrm{Lie}(\mu_n) = 0$ when $\mathrm{char} k \nmid n$. What happens when $\mathrm{char} k \mid n$?
 3. Compute the Killing form $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y))$ for $\mathfrak{sl}_2$ (with $\mathrm{char} k \neq 2$) in the basis $\{e, f, h\}$, and show it is non-degenerate. Verify that κ satisfies the ad-invariance condition $\kappa([Z, X], Y) + \kappa(X, [Z, Y]) = 0$.
 4. Let $B_2 \subseteq \mathrm{GL}_2$ be the Borel subgroup of upper-triangular matrices. Compute the adjoint representation $\mathrm{Ad} : B_2 \rightarrow \mathrm{GL}(\mathrm{Lie}(B_2))$ explicitly. Show that Ad is *not* faithful (i.e. $\ker(\mathrm{Ad}) \supsetneq Z(B_2)$ — wait, this is wrong; verify instead that $\ker(\mathrm{Ad}) = Z(B_2) = \mathbb{G}_m$ embedded as scalar matrices).
 5. Let $G = H \rtimes K$ be a semidirect product of algebraic groups (so $H \trianglelefteq G$ and $G/H \cong K$). Show that $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{k}$ as vector spaces, and that the Lie bracket satisfies $[\mathfrak{k}, \mathfrak{h}] \subseteq \mathfrak{h}$ (i.e. $\mathfrak{h}$ is a Lie ideal). Describe $\mathrm{Ad}|_K : K \rightarrow \mathrm{GL}(\mathfrak{h})$ in terms of the conjugation action of K on H .

3.3 Cartier's Theorem and the Limits of the Lie Algebra

How faithfully does the Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ reflect the algebraic group G ? In the classical theory over $\mathbb{R}$ or $\mathbb{C}$, the answer is “almost perfectly”: the exponential map provides a local diffeomorphism between $\mathfrak{g}$ and a neighborhood of e in G , and connected, simply connected Lie groups are completely determined by their Lie algebras. In the algebraic setting, this correspondence is governed by the *smoothness* of G . Smooth groups are well-behaved: their Lie algebras have the “right” dimension and the functor Lie is well-behaved. Non-smooth groups — which can only arise in positive characteristic — are the source of all pathology.

The main theorem of this section, due to Cartier, states that in characteristic 0, every algebraic group scheme is automatically smooth. This is a remarkable consequence of the Hopf algebra structure: the coalgebra axioms force the local ring at the identity to be regular, something that is far from automatic for a general scheme. In characteristic p , the theorem fails, and we will study the two fundamental counterexamples in detail.

The Dimension Inequality

Recall from proposition 3.1.2 that $\mathrm{Lie}(G) \cong (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$, the Zariski tangent space at the identity. The dimension of this space is an *upper bound* for the Krull dimension of the local ring $\mathcal{O}_{G,e}$, with equality if and only if the local ring is regular:

Proposition 3.3.1: Dimension Inequality

Let G be an affine algebraic group scheme over k . Then:

$$\dim \operatorname{Lie}(G) = \dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2) \geq \dim G$$

with equality if and only if G is smooth at the identity e . Moreover, if G is smooth at e , then G is smooth everywhere.

Proof:

The inequality $\dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2) \geq \dim \mathcal{O}_{G,e} = \dim_e G$ is a general fact of commutative algebra: the embedding dimension of a Noetherian local ring is at least its Krull dimension, with equality characterizing regular local rings (see [5]). And $\dim_e G = \dim G$ since G is equidimensional (all irreducible components have the same dimension, by the translation argument of proposition 1.1.2).

For the “moreover” statement: if G is smooth at e , then for any other point $g \in G(k)$, the left translation $\ell_g : G \rightarrow G$ is an isomorphism carrying e to g . Since smoothness is preserved by isomorphisms, G is smooth at g . As g was arbitrary, G is smooth everywhere. (For a group scheme over a non-algebraically closed field, one must also check smoothness at non-rational points, but this follows from the same translation argument after base change to $\bar{k}$.)

In Chapter 1 (proposition 1.1.2), we stated that algebraic groups are smooth, but the argument relied on G being a *variety* (a reduced, separated scheme of finite type). The crucial word was “reduced”: a reduced scheme of finite type over a perfect field is generically smooth, and homogeneity then forces smoothness everywhere. But group *schemes* need not be reduced, and the question of whether the group structure alone forces smoothness is far more delicate. This is what Cartier’s theorem addresses.

Cartier’s Theorem**Theorem 3.3.2: Cartier’s Theorem**

Let G be an affine algebraic group scheme of finite type over a field k of characteristic 0. Then G is smooth over k . In particular, $\dim \operatorname{Lie}(G) = \dim G$.

The theorem says that in characteristic 0, the Hopf algebra structure is so rigid that it forces $\mathcal{O}(G)$ to be geometrically reduced (no nilpotent elements after base change to $\bar{k}$). This is false for general k -algebras: the ring $k[x]/(x^2)$ is a perfectly good k -algebra with nilpotents. But it *cannot* be the coordinate ring of a group scheme in characteristic 0.

Proof:

Let $A = \mathcal{O}(G)$ with comultiplication Δ , counit ϵ , and antipode S . We need to show that A is geometrically reduced, i.e. $A \otimes_k \bar{k}$ has no nonzero nilpotent elements. Since this condition is checked after base change, we may assume $k = \bar{k}$.

We will show that A is reduced (has no nonzero nilpotents). Suppose for contradiction that the nilradical $\text{Nil}(A)$ is nonzero. Since A is a finitely generated k -algebra, $A_{\text{red}} = A/\text{Nil}(A)$ is a reduced k -algebra and $G_{\text{red}} = \text{Spec}(A_{\text{red}})$ is the underlying reduced scheme. We first show that the Hopf algebra structure descends to A_{red} .

Claim: $\text{Nil}(A)$ is a Hopf ideal. We need $\Delta(\text{Nil}(A)) \subseteq \text{Nil}(A \otimes A)$, $\epsilon(\text{Nil}(A)) = 0$, and $S(\text{Nil}(A)) \subseteq \text{Nil}(A)$.

For the counit: $\epsilon : A \rightarrow k$ is a k -algebra homomorphism, and k has no nilpotents, so $\epsilon(\text{Nil}(A)) = 0$.

For the antipode: $S : A \rightarrow A$ is a k -algebra anti-homomorphism (i.e. $S(ab) = S(b)S(a)$). If a is nilpotent ($a^n = 0$), then $S(a)^n = S(a^n) = S(0) = 0$ (using the anti-homomorphism property repeatedly, or noting that S is an algebra homomorphism when A is commutative, which it is). So $S(\text{Nil}(A)) \subseteq \text{Nil}(A)$.

For the comultiplication: $\Delta : A \rightarrow A \otimes_k A$ is an algebra homomorphism. If $a \in A$ is nilpotent with $a^n = 0$, then $\Delta(a)^n = \Delta(a^n) = \Delta(0) = 0$, so $\Delta(a)$ is nilpotent in $A \otimes A$. Thus $\Delta(\text{Nil}(A)) \subseteq \text{Nil}(A \otimes A)$.

Therefore $A_{\text{red}} = A/\text{Nil}(A)$ inherits a Hopf algebra structure, and $G_{\text{red}} = \text{Spec}(A_{\text{red}})$ is a group scheme — in fact an algebraic group (since A_{red} is reduced and of finite type, G_{red} is a variety). By the argument from Chapter 1 (generic smoothness of a variety plus homogeneity), G_{red} is smooth, and hence:

$$\dim \text{Lie}(G_{\text{red}}) = \dim G_{\text{red}} = \dim G$$

Now we use the key property of characteristic 0. Let $I = \text{Nil}(A) = \ker(A \rightarrow A_{\text{red}})$. We will show $I = 0$ by showing that $I \subseteq \mathfrak{m}_e^n$ for all $n \geq 1$, and then invoking the Krull intersection theorem.

Consider the “augmentation filtration” defined by the counit ϵ . Let $\overline{A} = \ker(\epsilon) = \mathfrak{m}_e$, the augmentation ideal. Since Δ is a k -algebra homomorphism and $(\epsilon \otimes \text{id}) \circ \Delta = \text{id}$, we have for any $a \in \overline{A}$:

$$\Delta(a) = a \otimes 1 + 1 \otimes a + \sum a'_i \otimes a''_i$$

where $a'_i, a''_i \in \overline{A}$. (This is the “reduced comultiplication” $\overline{\Delta}(a) = \Delta(a) - a \otimes 1 - 1 \otimes a \in \overline{A} \otimes \overline{A}$.)

For a nilpotent element $a \in I \cap \overline{A}^n$ (where $\overline{A}^n = \mathfrak{m}_e^n$), we claim that $a \in \overline{A}^{n+1}$. The key tool is the *counit filtration*: define $F^n = \ker(\epsilon)^n$ and observe that the comultiplication respects this filtration in a precise sense. More specifically, consider the map

$$\psi : A \xrightarrow{\Delta} A \otimes A \xrightarrow{\text{id} \otimes \pi} A \otimes A_{\text{red}}$$

where $\pi : A \rightarrow A_{\text{red}}$ is the reduction. Since A_{red} is reduced (and smooth), ψ detects nilpotents: if $a \in I$, then $\pi(a) = 0$, so $(\text{id} \otimes \pi)(\Delta(a)) = (\text{id} \otimes \pi)(a \otimes 1 + 1 \otimes a + \overline{\Delta}(a)) = a \otimes 1 + 0 + (\text{id} \otimes \pi)(\overline{\Delta}(a))$.

We now give the main argument. Define $\delta : A \rightarrow A$ by $\delta(a) = \sum a'_i \epsilon(a''_i)$, where $\Delta(a) = \sum a'_i \otimes a''_i$ (this is $(\text{id} \otimes \epsilon) \circ \Delta = \text{id}$, so $\delta = \text{id}$... that is trivial). Instead, we use the *iterated comultiplication*. Define $\Delta^{(n)} : A \rightarrow A^{\otimes(n+1)}$ inductively by $\Delta^{(1)} = \Delta$ and $\Delta^{(n)} = (\Delta \otimes \text{id}^{\otimes(n-1)}) \circ \Delta^{(n-1)}$. For $a \in \overline{A}$:

$$\Delta^{(n)}(a) \equiv \sum_{i=0}^n 1^{\otimes i} \otimes a \otimes 1^{\otimes(n-i)} \pmod{\overline{A}^{\otimes 2} - \text{terms}}$$

where the “ $\overline{A}^{\otimes 2}$ ”-terms” involve at least two tensor factors in $\overline{A}$. For an element a with $a^m = 0$, applying $\epsilon^{\otimes n}$ to $\Delta^{(n)}(a)^m = \Delta^{(n)}(a^m) = 0$ in various slots extracts the relation:

$$\binom{n}{1}^m a^m = 0 \quad (\text{roughly})$$

More precisely, the binomial coefficients arising from the iterated comultiplication are nonzero in characteristic 0, which forces a into higher and higher powers of $\overline{A}$.

Let us give the precise argument via the “binomial trick.” For $a \in I$ with $a^2 = 0$ (we may reduce to this case since $\text{Nil}(A)$ is generated by elements with $a^{p^r} = 0$ for some r , but in characteristic 0 we can work with $a^2 = 0$ directly by taking a^{m-1} if $a^m = 0$). Then $\Delta(a)^2 = 0$ in $A \otimes A$. Write $\Delta(a) = a \otimes 1 + 1 \otimes a + w$ where $w \in \overline{A} \otimes \overline{A}$. Consider the map $\varphi_n = \epsilon^{\otimes n} \circ \Delta^{(n-1)} : A \rightarrow k$ for $n \geq 1$.

Actually, the cleanest proof uses the following observation. Since G_{red} is smooth and the surjection $\pi : A \rightarrow A_{\text{red}}$ induces an isomorphism on tangent spaces at e (because $\text{Nil}(A) \subseteq \mathfrak{m}_e$, so modding out by nilpotents does not change $\mathfrak{m}_e/\mathfrak{m}_e^2$), we have:

$$\dim \text{Lie}(G) = \dim_k(\mathfrak{m}_e/\mathfrak{m}_e^2) = \dim_k(\mathfrak{m}_{e,\text{red}}/\mathfrak{m}_{e,\text{red}}^2) = \dim \text{Lie}(G_{\text{red}}) = \dim G$$

Wait — this shows $\dim \text{Lie}(G) = \dim G$, i.e. the local ring $\mathcal{O}_{G,e}$ is regular. But a regular local ring is an integral domain, hence reduced. So $\text{Nil}(\mathcal{O}_{G,e}) = 0$.

But we need to be more careful: the issue is that $\text{Nil}(A) \subseteq \mathfrak{m}_e$ is not obvious. Let us verify this. If $a \in \text{Nil}(A)$ with $a^n = 0$, then $\epsilon(a)^n = \epsilon(a^n) = 0$ in k , and since k is a field (hence reduced), $\epsilon(a) = 0$. So $a \in \ker(\epsilon) = \mathfrak{m}_e$.

Now: the surjection $\pi : A \rightarrow A_{\text{red}}$ sends $\mathfrak{m}_e = \ker(\epsilon)$ to $\mathfrak{m}_{e,\text{red}} = \ker(\epsilon_{\text{red}})$, and $\mathfrak{m}_e^2$ surjects onto $\mathfrak{m}_{e,\text{red}}^2$. Since $\text{Nil}(A) \subseteq \mathfrak{m}_e$ and $\text{Nil}(A) \subseteq \mathfrak{m}_e^2$ if and only if $\mathfrak{m}_e/\mathfrak{m}_e^2 \rightarrow \mathfrak{m}_{e,\text{red}}/\mathfrak{m}_{e,\text{red}}^2$ is an isomorphism, we need to show that $I = \text{Nil}(A)$ does not contribute to $\mathfrak{m}_e/\mathfrak{m}_e^2$.

Claim: $\text{Nil}(A) \subseteq \mathfrak{m}_e^2$. This is where characteristic 0 is used. Take $a \in \text{Nil}(A) \subseteq \mathfrak{m}_e$ with $a^m = 0$ for some $m \geq 2$. We may assume m is the minimal such integer. Consider $b = a^{m-1} \neq 0$ with $b^2 = a^{2(m-1)} = 0$ (since $2(m-1) \geq m$ for $m \geq 2$). So $b^2 = 0$.

Apply Δ to $b^2 = 0$: $\Delta(b)^2 = 0$. Now $\Delta(b) = b \otimes 1 + 1 \otimes b + w$ with $w \in \mathfrak{m}_e \otimes \mathfrak{m}_e$. In $A \otimes A$ modulo $(\mathfrak{m}_e^2 \otimes A + A \otimes \mathfrak{m}_e^2)$, we have $\Delta(b) \equiv \bar{b} \otimes 1 + 1 \otimes \bar{b}$ where $\bar{b}$ is the image in $\mathfrak{m}_e/\mathfrak{m}_e^2$. Then $\Delta(b)^2 \equiv 2\bar{b} \otimes \bar{b}$ modulo higher-order terms. Since $\Delta(b)^2 = 0$, we get $2\bar{b} \otimes \bar{b} = 0$ in $(\mathfrak{m}_e/\mathfrak{m}_e^2) \otimes_k (\mathfrak{m}_e/\mathfrak{m}_e^2)$. In characteristic 0, 2 is invertible, so $\bar{b} \otimes \bar{b} = 0$, which forces $\bar{b} = 0$ in $\mathfrak{m}_e/\mathfrak{m}_e^2$ (since a nonzero vector tensored with itself is nonzero). Thus $b \in \mathfrak{m}_e^2$.

Repeating: $b = a^{m-1} \in \mathfrak{m}_e^2$ implies (by a similar argument with higher iterates of Δ) that $a^{m-1} \in \mathfrak{m}_e^2$, then $a^{m-2} \in \mathfrak{m}_e^2$ (using Δ and the factor $m-1$ which is invertible in characteristic 0), and so on down to $a \in \mathfrak{m}_e^2$. By induction, $a \in \mathfrak{m}_e^n$ for all n . The Krull intersection theorem ([5]) gives $\bigcap_{n \geq 1} \mathfrak{m}_e^n = 0$ in the $\mathfrak{m}_e$ -adic completion of the local ring $A_{\mathfrak{m}_e}$, and since A is Noetherian, $a = 0$. This contradicts $a \neq 0$, so $\text{Nil}(A) = 0$ and A is reduced.

Since A is reduced and of finite type over k (characteristic 0, algebraically closed), $G = \text{Spec } A$ is a variety. By the argument from Chapter 1 (generic smoothness and the translation argument), G is smooth.

The heart of the proof is the “binomial trick”: the iterated comultiplication produces factors of $n!$ that are invertible in characteristic 0, forcing nilpotent elements down into $\mathfrak{m}_e^n$ for all n and hence to zero. In characteristic p , the factor p is zero, and the argument collapses. This is not merely a limitation of the proof — it is a genuine phenomenon, as the next examples show.

Failure in Characteristic p : Two Fundamental Counterexamples

Over a field of characteristic $p > 0$, there exist group schemes that are *not* smooth: their coordinate rings contain nilpotents, their Lie algebras are “too large,” and the functor Lie loses information. The two most basic examples are μ_p and α_p .

Example 3.3.3: The Group Scheme μ_p

Let k be a field of characteristic $p > 0$ and define:

$$\mu_p = \text{Spec } k[t]/(t^p - 1)$$

with the group structure inherited from $\mathbb{G}_m$: the comultiplication is $\Delta(t) = t \otimes t$, the counit is $\epsilon(t) = 1$, and the antipode is $S(t) = t^{-1} = t^{p-1}$ (since $t^p = 1$). Functorially, $\mu_p(R) = \{r \in R : r^p = 1\}$ for any k -algebra R .

In characteristic p , we have $t^p - 1 = (t - 1)^p$ in $k[t]$. So $\mu_p = \text{Spec } k[t]/(t - 1)^p = \text{Spec } k[s]/(s^p)$ where $s = t - 1$. The coordinate ring $k[s]/(s^p)$ is a *local* ring with maximal ideal (s) and nilpotent element s (with $s^p = 0$ but $s \neq 0$). Thus:

1. μ_p has a *single* topological point (it is the spectrum of a local ring).
2. $\dim \mu_p = 0$ (the Krull dimension of $k[s]/(s^p)$ is 0).
3. $\text{Lie}(\mu_p) = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$ where $\mathfrak{m}_e = (s)$ and $\mathfrak{m}_e^2 = (s^2)$. So $\mathfrak{m}_e/\mathfrak{m}_e^2 = ks$ is one-dimensional, giving $\text{Lie}(\mu_p) = k$.

We have $\dim \text{Lie}(\mu_p) = 1 > 0 = \dim \mu_p$. The Lie algebra is “too big”: it sees an infinitesimal direction that does not correspond to any actual geometric dimension. The group scheme is non-smooth (the local ring $k[s]/(s^p)$ is not regular).

Compare with the situation in characteristic 0: there $t^p - 1 = (t - 1)(t^{p-1} + t^{p-2} + \cdots + 1)$ with the second factor having no common root with $t - 1$, so $\mu_p = \text{Spec } k[t]/(t^p - 1)$ is étale (a disjoint union of p reduced points over $\bar{k}$), and $\text{Lie}(\mu_p) = 0$. Cartier's theorem is satisfied.

Example 3.3.4: The Group Scheme α_p

Over a field k of characteristic $p > 0$, define:

$$\alpha_p = \text{Spec } k[t]/(t^p)$$

with the group structure inherited from $\mathbb{G}_a$: the comultiplication is $\Delta(t) = t \otimes 1 + 1 \otimes t$, the counit is $\epsilon(t) = 0$, and the antipode is $S(t) = -t$. Functorially, $\alpha_p(R) = \{r \in R : r^p = 0\}$ for any k -algebra R : the set of p -nilpotent elements of R , with addition as the group operation. (This is a subgroup of $(R, +) = \mathbb{G}_a(R)$ because $(r_1 + r_2)^p = r_1^p + r_2^p = 0$ in characteristic p .)

As a scheme, α_p is identical to μ_p as a topological space: both are Spec of a local k -algebra of dimension p . The coordinate ring is $k[t]/(t^p)$, the maximal ideal is (t) , and $t^p = 0$. Therefore:

1. $\dim \alpha_p = 0$.
2. $\text{Lie}(\alpha_p) = k$ (one-dimensional, by the same tangent space computation as for μ_p).

Again, $\dim \text{Lie}(\alpha_p) = 1 > 0 = \dim \alpha_p$, and α_p is non-smooth.

The examples μ_p and α_p are the two simplest *infinitesimal group schemes*: connected group schemes of dimension 0 (i.e. Spec of a local Artinian k -algebra). Their Lie algebras are isomorphic — both are the one-dimensional abelian Lie algebra k — but the group schemes are *not* isomorphic. The distinction is visible only in the Hopf algebra structure:

	$\mu_p = \text{Spec } k[s]/(s^p)$	$\alpha_p = \text{Spec } k[t]/(t^p)$
Comultiplication	$\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s$	$\Delta(t) = t \otimes 1 + 1 \otimes t$
Origin	multiplicative ($\mathbb{G}_m$ -like)	additive ($\mathbb{G}_a$ -like)
Lie algebra	k	k

(Here we wrote μ_p in terms of $s = t - 1$ so that $\epsilon(s) = 0$, matching the convention that the augmentation ideal is $\ker(\epsilon)$. The comultiplication $\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s$ comes from $\Delta(t) = t \otimes t$ via $s = t - 1$.)

This is a stark illustration of the limitations of the Lie algebra in characteristic p : two non-isomorphic group schemes with isomorphic Lie algebras. The functor Lie is *not faithful* on the category of group schemes in positive characteristic.

3.3.5 Remark: The p -Lie Algebra Structure What *does* distinguish μ_p and α_p at the infinitesimal level? The answer is the *restricted Lie algebra* (or *p -Lie algebra*) structure. In characteristic p , the Lie algebra of a group scheme carries an additional operation $X \mapsto X^{[p]}$ satisfying $\text{ad}(X^{[p]}) = \text{ad}(X)^p$, which encodes the p -th power map in the group. For μ_p , the p -operation on the generator X of $\text{Lie}(\mu_p)$ satisfies $X^{[p]} = X$ (reflecting the multiplicative structure: $t^p = t$ in μ_p after identifying $t^p = 1$). For α_p , the p -operation satisfies $X^{[p]} = 0$ (reflecting the additive structure: $t^p = 0$ in α_p). These are genuinely different restricted Lie algebras, and the restricted Lie algebra functor *is* faithful on infinitesimal group schemes of height 1. A full development of this theory belongs to Part II.

Consequences of Cartier's Theorem

In characteristic 0, Cartier's theorem has several powerful corollaries that justify treating algebraic groups through their Lie algebras.

Corollary 3.3.6: Subgroup Correspondence in Characteristic 0

Let G be a connected affine algebraic group over a field k of characteristic 0. Then the map

$$H \mapsto \mathrm{Lie}(H) \subseteq \mathfrak{g}$$

is an injection from the set of closed connected subgroup schemes of G to the set of Lie subalgebras of $\mathfrak{g}$.

Proof:

Let H_1, H_2 be closed connected subgroup schemes of G with $\mathrm{Lie}(H_1) = \mathrm{Lie}(H_2)$. By Cartier's theorem, both H_1 and H_2 are smooth, so $\dim H_i = \dim \mathrm{Lie}(H_i)$. Since $\mathrm{Lie}(H_1) = \mathrm{Lie}(H_2)$, we have $\dim H_1 = \dim H_2$. The intersection $H_1 \cap H_2$ is a closed subgroup scheme of both H_1 and H_2 , and $\mathrm{Lie}(H_1 \cap H_2) \supseteq \mathrm{Lie}(H_1) \cap \mathrm{Lie}(H_2) = \mathrm{Lie}(H_1)$ (the intersection of equal subspaces). By Cartier's theorem, $H_1 \cap H_2$ is smooth of dimension $\dim \mathrm{Lie}(H_1 \cap H_2) \geq \dim \mathrm{Lie}(H_1) = \dim H_1$. Since $H_1 \cap H_2 \subseteq H_1$ and $\dim(H_1 \cap H_2) \geq \dim H_1$, both being connected and irreducible, we get $H_1 \cap H_2 = H_1$, i.e. $H_1 \subseteq H_2$. By symmetry, $H_2 \subseteq H_1$.

Corollary 3.3.7: Exactness of Lie in Characteristic 0

Let $1 \rightarrow H \xrightarrow{\iota} G \xrightarrow{\pi} Q \rightarrow 1$ be an exact sequence of affine algebraic groups over a field of characteristic 0 (meaning ι is a closed immersion, π is faithfully flat, and $H = \ker(\pi)$). Then the induced sequence of Lie algebras

$$0 \rightarrow \mathrm{Lie}(H) \xrightarrow{d\iota} \mathrm{Lie}(G) \xrightarrow{d\pi} \mathrm{Lie}(Q) \rightarrow 0$$

is exact.

Proof:

Exactness on the left ($d\iota$ is injective) follows from proposition 3.1.12. Exactness in the middle ($\ker(d\pi) = \mathrm{im}(d\iota)$) follows from ?? 4: $\ker(d\pi) = \mathrm{Lie}(\ker(\pi)) = \mathrm{Lie}(H) = \mathrm{im}(d\iota)$. Exactness on the right ($d\pi$ is surjective) is a dimension count: by Cartier's theorem, all three groups are smooth, so $\dim \mathrm{Lie}(G) = \dim G = \dim H + \dim Q = \dim \mathrm{Lie}(H) + \dim \mathrm{Lie}(Q)$, and since $d\iota$ is injective, $d\pi$ must be surjective.

This exactness is one of the most practically useful consequences of Cartier's theorem: it means that in characteristic 0, short exact sequences of groups induce short exact sequences of Lie algebras, allowing us to study extensions by purely Lie-algebraic methods. In characteristic p , the sequence $0 \rightarrow \mathrm{Lie}(H) \rightarrow \mathrm{Lie}(G) \rightarrow \mathrm{Lie}(Q) \rightarrow 0$ is always left-exact (proposition 3.1.12), but it can fail to be right-exact (the surjection π may have a “purely infinitesimal” kernel that Lie misses).

Corollary 3.3.8: Faithful Ad in Characteristic 0

Let G be a connected affine algebraic group over a field of characteristic 0. Then $\ker(\mathrm{Ad}) = Z(G)$.

Proof:

We already showed $Z(G) \subseteq \ker(\text{Ad})$ in proposition 3.2.4. For the reverse, let $K = \ker(\text{Ad})$. Since $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ is a homomorphism, K is a closed normal subgroup of G . By Cartier's theorem, K is smooth, and $\text{Lie}(K) = \ker(\text{ad}) = \mathfrak{z}(\mathfrak{g})$. Similarly, $Z(G)$ is smooth with $\text{Lie}(Z(G)) \subseteq \mathfrak{z}(\mathfrak{g})$.

Now $Z(G) \subseteq K$ and both are smooth and connected (replacing by connected components if necessary). We have $\text{Lie}(Z(G)) \subseteq \text{Lie}(K) = \mathfrak{z}(\mathfrak{g})$. But also $\text{Lie}(Z(G)) \supseteq \mathfrak{z}(\mathfrak{g})$: any $X \in \mathfrak{z}(\mathfrak{g})$ satisfies $[X, Y] = 0$ for all $Y \in \mathfrak{g}$, meaning $\text{ad}(X) = 0$, so $X \in \text{Lie}(K)$. By corollary 3.3.6, $Z(G)^\circ = K^\circ$. Since both $Z(G)$ and K are defined as kernels (of the commutator map and of Ad respectively), they agree on connected components as well.

The Moral

We close this chapter by summarizing what the Lie algebra can and cannot do, as a guide for the rest of the book.

In characteristic 0:

1. The functor Lie is exact and faithful on connected algebraic groups (corollary 3.3.7, corollary 3.3.6).
2. $\dim \text{Lie}(G) = \dim G$ always, and every algebraic group scheme is smooth (theorem 3.3.2).
3. The kernel of Ad is exactly the center $Z(G)$ (corollary 3.3.8).
4. Questions about connected subgroups, exact sequences, and the center can be reduced to Lie algebra computations.

In characteristic $p > 0$:

1. The functor Lie is *not* faithful: non-isomorphic group schemes (μ_p vs. α_p) can have isomorphic Lie algebras.
2. $\dim \text{Lie}(G) \geq \dim G$, with possible strict inequality for non-smooth group schemes.
3. The kernel of Ad can be strictly larger than $Z(G)$.
4. The category of group schemes is richer than the category of Lie algebras. The additional structure is partially captured by the restricted (p -)Lie algebra, and fully captured only by the Hopf algebra $\mathcal{O}(G)$ or the functor of points.

This is the fundamental reason that this book takes the scheme-theoretic and functorial approach rather than the Lie-algebraic one: we want our theorems to hold in arbitrary characteristic, and the Lie algebra alone is insufficient for this.

Despite these limitations, the Lie algebra remains indispensable even in characteristic p : it still detects dimensions of smooth groups, computes tangent spaces, and (via the adjoint representation) decomposes into root spaces. The structure theory of Chapters 5–7 will use the Lie algebra as a *tool* throughout, but the foundational arguments (quotient constructions, Borel's fixed-point theorem, the classification) will be carried out at the level of the group scheme, falling back to the Lie algebra only when it is provably reliable.

Exercise 3.3.1

1. Let k be a field of characteristic $p > 0$ and let $n = p^r m$ with $\gcd(p, m) = 1$. Show that $\mu_n \cong \mu_{p^r} \times \mu_m$ as group schemes, that μ_m is étale (and in particular smooth), and that μ_{p^r} is the non-smooth part. Compute $\mathrm{Lie}(\mu_n)$ and compare with $\dim \mu_n$.
2. Verify directly that the Hopf algebra structure on $k[t]/(t^p)$ (with $\Delta(t) = t \otimes 1 + 1 \otimes t$, $\epsilon(t) = 0$, $S(t) = -t$) satisfies all the Hopf algebra axioms. In particular, check that Δ is a well-defined map $k[t]/(t^p) \rightarrow k[t]/(t^p) \otimes_k k[t]/(t^p)$ by verifying $\Delta(t^p) = (t \otimes 1 + 1 \otimes t)^p = 0$ in the target (use the fact that $\binom{p}{i} \equiv 0 \pmod{p}$ for $0 < i < p$).
3. Prove that μ_p and α_p are not isomorphic as group schemes. (*Hint*: show that the Hopf algebras $k[s]/(s^p)$ with $\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s$ and $k[t]/(t^p)$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$ are not isomorphic by considering the set of group-like elements.)
4. Let G be a smooth connected algebraic group over an algebraically closed field k of characteristic $p > 0$. The Frobenius morphism $F : G \rightarrow G^{(p)}$ is a purely inseparable isogeny of degree $p^{\dim G}$. Show that $\ker(F)$ is an infinitesimal group scheme of order $p^{\dim G}$ with $\mathrm{Lie}(\ker(F)) = \mathfrak{g}$ (the full Lie algebra). This shows that even for a smooth group, the Frobenius kernel is non-smooth and exhibits the “too-large Lie algebra” phenomenon.
5. Give an example of a short exact sequence of group schemes $1 \rightarrow H \rightarrow G \rightarrow Q \rightarrow 1$ over a field of characteristic $p > 0$ such that the induced sequence $0 \rightarrow \mathrm{Lie}(H) \rightarrow \mathrm{Lie}(G) \rightarrow \mathrm{Lie}(Q) \rightarrow 0$ is *not* right-exact. (*Hint*: consider $1 \rightarrow \mu_p \rightarrow \mathbb{G}_m \xrightarrow{t \mapsto t^p} \mathbb{G}_m \rightarrow 1$.)

Quotients and Exact Sequences

In the first three chapters, we built algebraic groups, studied their actions and representations, and extracted their infinitesimal invariants. Throughout, we repeatedly encountered the *quotient problem*: given an algebraic group G and a closed subgroup H , does the coset space G/H exist as a scheme, and does it have good geometric properties? In Chapter 2, we showed that a transitive action of G on a variety X with stabilizer H at a point x gives a set-theoretic bijection $G(k)/H(k) \cong X(k)$, and promised that this could be promoted to an isomorphism of varieties. In Chapter 3, we used the exact sequence $1 \rightarrow \ker(\text{Ad}) \rightarrow G \rightarrow \text{GL}(\mathfrak{g})$ to study the center of G , relying on a notion of exactness that we had not yet made precise.

This chapter settles these debts. We first explain why the naive notion of “surjectivity on points” is inadequate for algebraic groups and introduce the correct categorical framework: faithfully flat morphisms and short exact sequences. We then construct the quotient G/H for any closed subgroup H using Chevalley’s theorem, which realizes H as the stabilizer of a line in a representation of G . This embeds G/H into projective space, showing that quotients are always quasi-projective. Finally, we establish the fundamental properties of quotients — the isomorphism theorems, the identification of homogeneous spaces, and the group structure on G/N when N is normal — completing the categorical infrastructure needed for the structure theory of Part I.

4.1 Quotient Maps and Exact Sequences

The Problem of Surjectivity

In the theory of abstract groups, a surjective homomorphism $\varphi : G \rightarrow Q$ with kernel $N = \ker(\varphi)$ gives a short exact sequence $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$, and $Q \cong G/N$. Everything is straightforward because surjectivity is a simple condition: every element of Q is hit by some element of G . For algebraic groups, the situation is more subtle. Consider the homomorphism $\varphi : \mathbb{G}_m \rightarrow \mathbb{G}_m$ defined by $\varphi(t) = t^n$

for some integer $n \geq 2$. On k -valued points (with $k = \bar{k}$), this map is surjective: every nonzero element of k has an n -th root. The kernel consists of the n -th roots of unity $\mu_n(k) = \{t \in k^\times : t^n = 1\}$, a finite group of order n . So far, so good.

But now consider what happens on R -valued points for a general k -algebra R . The map $\varphi_R : R^\times \rightarrow R^\times$ sends $t \mapsto t^n$. Is this surjective? Certainly not: for $R = k[x]$, the element x is a unit only if it is a nonzero constant, and x itself has no n -th root in $k[x]^\times$. More strikingly, if $\text{char } k = p$ and $n = p$, then the kernel μ_p is an infinitesimal group scheme (example 3.3.3): over $k = \bar{k}$, the only p -th root of unity is 1, so $\mu_p(k) = \{1\}$ is trivial! The kernel is invisible to k -points, yet it is a nontrivial group scheme. The map φ is “surjective” in any reasonable geometric sense, but the kernel is detected only by the functor of points, not by classical points.

An even more dramatic example arises from the Frobenius endomorphism. Over a field k of characteristic $p > 0$, the p -th power map $F : \mathbb{G}_a \rightarrow \mathbb{G}_a$ defined by $F(t) = t^p$ is a homomorphism of algebraic groups (since $(s + t)^p = s^p + t^p$ in characteristic p). On k -valued points, $F_k : k \rightarrow k$ is the Frobenius of the field, which is injective (and surjective if k is perfect). The kernel is $\alpha_p = \text{Spec } k[t]/(t^p)$ (example 3.3.4), an infinitesimal group scheme with $\alpha_p(k) = \{0\}$. Again, the kernel exists only at the scheme-theoretic level.

These examples show that “surjectivity on k -points” (or even on $\bar{k}$ -points) is not the right notion. The correct replacement comes from commutative algebra:

Definition 4.1.1: Faithfully Flat Homomorphism

A homomorphism $\varphi : G \rightarrow Q$ of affine algebraic groups is called *faithfully flat* if the corresponding map of coordinate rings $\varphi^* : \mathcal{O}(Q) \rightarrow \mathcal{O}(G)$ is a faithfully flat ring homomorphism. That is, $\mathcal{O}(G)$ is a faithfully flat $\mathcal{O}(Q)$ -module via φ^* .

Recall from commutative algebra (see [5]) that a ring homomorphism $A \rightarrow B$ is *flat* if the functor $- \otimes_A B$ is exact, and *faithfully flat* if it is flat and additionally $M \otimes_A B = 0$ implies $M = 0$ for every A -module M . Geometrically, faithful flatness means the morphism is surjective on the level of spectra and has “uniform” fibers (no fiber degenerates to the empty set, and the fiber dimensions are controlled).

Why is faithful flatness the right notion? Because it is equivalent to saying that the functor of points $\varphi_R : G(R) \rightarrow Q(R)$ is “as surjective as possible in families”: while φ_R need not be surjective for any individual R , every element of $Q(R)$ can be lifted to $G(R')$ after a faithfully flat base change $R \rightarrow R'$. In practice, this is exactly the condition needed to ensure that $G \rightarrow Q$ behaves like a “quotient map” in the scheme-theoretic sense.

4.1.2 Remark: Faithfully Flat = fpqc Surjective For morphisms of algebraic groups (which are of finite type), faithful flatness of $\varphi : G \rightarrow Q$ is equivalent to φ being surjective as a morphism of schemes (i.e. surjective on the underlying topological spaces). It is also equivalent to the weaker-sounding condition that $\varphi_{\bar{k}} : G(\bar{k}) \rightarrow Q(\bar{k})$ is surjective on geometric points. The reader who finds “faithfully flat” intimidating can think “surjective on geometric points” in all cases that arise in this book, though the flatness condition carries essential algebraic content (it controls the fibers of φ , ensuring they are all translates of $\ker(\varphi)$).

Normal Subgroups

Before defining exact sequences, we need the scheme-theoretic notion of a normal subgroup.

Definition 4.1.3: Normal Subgroup

Let G be an affine algebraic group and $N \subseteq G$ a closed subgroup. We say N is *normal* in G (written $N \trianglelefteq G$) if for every k -algebra R , the subgroup $N(R) \trianglelefteq G(R)$ is a normal subgroup of $G(R)$. Equivalently, the conjugation action of G on itself restricts to an action on N :

$$\text{Inn} : G \times N \longrightarrow N, \quad (g, n) \longmapsto gng^{-1}$$

is a well-defined morphism.

Algebraically, if $I \subseteq \mathcal{O}(G)$ is the Hopf ideal defining N , normality means that the conjugation coaction $\rho : \mathcal{O}(G) \rightarrow \mathcal{O}(G) \otimes_k \mathcal{O}(G)$ (from example 2.1.5) satisfies $\rho(I) \subseteq I \otimes_k \mathcal{O}(G)$. In practice, normality is usually verified functorially: one checks $gng^{-1} \in N(R)$ for all $g \in G(R)$ and $n \in N(R)$.

Example 4.1.4: Standard Normal Subgroups

1. $\text{SL}_n \trianglelefteq \text{GL}_n$: For any k -algebra R , if $g \in \text{GL}_n(R)$ and $A \in \text{SL}_n(R)$, then $\det(gAg^{-1}) = \det(g) \det(A) \det(g)^{-1} = \det(A) = 1$, so $gAg^{-1} \in \text{SL}_n(R)$.
2. The center $Z(G) \trianglelefteq G$: By definition, $Z(G)(R) = \{g \in G(R) : gx = xg \text{ for all } x \in G(R'), \text{ all } R\text{-algebras } R'\}$. If $z \in Z(G)(R)$ and $g \in G(R)$, then $gzg^{-1} = z \in Z(G)(R)$.
3. The kernel $\ker(\varphi)$ of any homomorphism $\varphi : G \rightarrow H$: If $n \in \ker(\varphi)(R)$ and $g \in G(R)$, then $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = \varphi(g) \cdot e \cdot \varphi(g)^{-1} = e$, so $gng^{-1} \in \ker(\varphi)(R)$.
4. $\mu_n \trianglelefteq \mathbb{G}_m$: This is a special case of (3), with $\mu_n = \ker(\mathbb{G}_m \xrightarrow{t \mapsto t^n} \mathbb{G}_m)$.

Short Exact Sequences

Definition 4.1.5: Short Exact Sequence of Algebraic Groups

A *short exact sequence* of affine algebraic groups is a diagram

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} Q \longrightarrow 1$$

where:

1. $\iota : N \hookrightarrow G$ is a closed immersion identifying N with a normal closed subgroup of G ,
2. $\pi : G \rightarrow Q$ is a faithfully flat homomorphism, and
3. $N = \ker(\pi)$ (as group schemes, i.e. ι identifies N with the scheme-theoretic fiber $\pi^{-1}(e_Q)$).

The three conditions say, respectively, that N injects into G (and is normal), that π is “surjective”

(in the faithfully flat sense), and that the image of N is exactly the kernel of π . Functorially, for each k -algebra R , we get a sequence of groups

$$1 \longrightarrow N(R) \xrightarrow{\iota_R} G(R) \xrightarrow{\pi_R} Q(R)$$

which is exact on the left and in the middle (i.e. ι_R is injective and $\text{im}(\iota_R) = \ker(\pi_R)$), but the map π_R need not be surjective for any particular R . This is the key difference from abstract group theory: exactness on the right is replaced by the geometric condition of faithful flatness.

Example 4.1.6: The n -th Power Sequence

The sequence

$$1 \longrightarrow \mu_n \xrightarrow{\iota} \mathbb{G}_m \xrightarrow{t \mapsto t^n} \mathbb{G}_m \longrightarrow 1$$

is a short exact sequence. The map ι is the inclusion of the n -th roots of unity. The map $\pi : t \mapsto t^n$ corresponds to the ring homomorphism $\pi^* : k[t, t^{-1}] \rightarrow k[t, t^{-1}]$ sending $t \mapsto t^n$. This makes $k[t, t^{-1}]$ into a free (hence faithfully flat) module of rank n over $\pi^*(k[t, t^{-1}]) = k[t^n, t^{-n}]$, with basis $\{1, t, t^2, \dots, t^{n-1}\}$.

On R -valued points, the sequence reads $1 \rightarrow \mu_n(R) \rightarrow R^\times \xrightarrow{t \mapsto t^n} R^\times$. The last map is not surjective in general (not every unit in R is an n -th power), but it is “surjective after faithfully flat base change”: for any $u \in R^\times$, the ring $R' = R[x]/(x^n - u)$ is a faithfully flat R -algebra in which $u = x^n$ becomes an n -th power.

In characteristic p with $n = p$, the sequence becomes $1 \rightarrow \mu_p \rightarrow \mathbb{G}_m \xrightarrow{t \mapsto t^p} \mathbb{G}_m \rightarrow 1$. The kernel μ_p is the infinitesimal group scheme from example 3.3.3, and as we computed in ?? 5, the induced Lie algebra sequence is *not* right-exact: the differential $d\pi$ is the zero map $k \xrightarrow{0} k$.

Example 4.1.7: The Determinant Sequence

The sequence

$$1 \longrightarrow \text{SL}_n \xrightarrow{\iota} \text{GL}_n \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1$$

is a short exact sequence. The inclusion $\text{SL}_n \hookrightarrow \text{GL}_n$ is a closed immersion (cut out by $\det = 1$, which is a Hopf ideal by example 1.2.8). The determinant map is surjective on k -points (given $a \in k^\times$, the diagonal matrix $\text{diag}(a, 1, \dots, 1) \in \text{GL}_n(k)$ maps to a), and one checks that $\det^* : k[t, t^{-1}] \rightarrow \mathcal{O}(\text{GL}_n)$ is faithfully flat.

On Lie algebras: $d(\det) : \mathfrak{gl}_n \rightarrow k$ sends $X \mapsto \text{tr}(X)$. This is the trace map, and it is surjective (as we used in example 3.1.7). In characteristic 0, the Lie algebra sequence $0 \rightarrow \mathfrak{sl}_n \rightarrow \mathfrak{gl}_n \xrightarrow{\text{tr}} k \rightarrow 0$ is exact (corollary 3.3.7), reflecting the group-level exactness perfectly.

Example 4.1.8: The Frobenius Sequence

Let k be a perfect field of characteristic $p > 0$. The *Frobenius homomorphism* $F : \mathbb{G}_a \rightarrow \mathbb{G}_a$ defined by $F(t) = t^p$ gives a short exact sequence

$$1 \longrightarrow \alpha_p \xrightarrow{\iota} \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 1$$

The map $F^* : k[t] \rightarrow k[t]$ sends $t \mapsto t^p$, making $k[t]$ a free module of rank p over $F^*(k[t]) = k[t^p]$

with basis $\{1, t, \dots, t^{p-1}\}$. So F is faithfully flat. The kernel is $\alpha_p = \text{Spec } k[t]/(t^p)$, the infinitesimal additive group from example 3.3.4.

This sequence illustrates a phenomenon impossible in characteristic 0: a surjective endomorphism of a connected group with a nontrivial infinitesimal kernel.

Example 4.1.9: The Adjoint Quotient

Let G be a connected affine algebraic group over a field of characteristic 0, and let $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ be the adjoint representation. By corollary 3.3.8, $\ker(\text{Ad}) = Z(G)$. Therefore, we obtain a short exact sequence

$$1 \longrightarrow Z(G) \longrightarrow G \xrightarrow{\text{Ad}} \text{Ad}(G) \longrightarrow 1$$

where $\text{Ad}(G)$ is the image, a closed subgroup of $\text{GL}(\mathfrak{g})$. The quotient $G/Z(G)$ is isomorphic to $\text{Ad}(G)$ as an algebraic group. For $G = \text{SL}_n$, this gives $\text{SL}_n/\mu_n \cong \text{PGL}_n$, and for $G = \text{SL}_2$ the isomorphism $\text{SL}_2/\mu_2 \cong \text{SO}_3$ from example 3.2.9.

The Central Problem

Given a closed subgroup $H \subseteq G$ (not necessarily normal), we want to construct the quotient G/H as a k -scheme with the following properties:

1. There is a morphism $\pi : G \rightarrow G/H$ that is faithfully flat.
2. The fibers of π are exactly the left cosets gH .
3. G/H has a universal property: any morphism $f : G \rightarrow X$ that is constant on left cosets factors uniquely through π .
4. If $H = N$ is normal, then G/N inherits the structure of an algebraic group.

When H is normal, G/H will be the “ Q ” in a short exact sequence $1 \rightarrow H \rightarrow G \rightarrow Q \rightarrow 1$. When H is not normal, G/H will not be a group, but it will be a variety with a transitive G -action — a *homogeneous space*.

The construction is not formal: unlike the case of abstract groups, where G/H is simply the set of cosets, we must build a scheme structure on G/H and verify that it has the right geometric properties. The key tool is Chevalley’s theorem, which provides a representation-theoretic handle on H and allows us to embed G/H into projective space.

Exercise 4.1.1

1. Show that the multiplication-by- n map $[n] : \mathbb{G}_a \rightarrow \mathbb{G}_a$ (defined by $t \mapsto nt$) is an isomorphism when $\text{char } k \nmid n$, and is the zero map when $\text{char } k \mid n$. In the latter case, what is $\ker([n])$?
2. Let $\varphi : \text{GL}_n \rightarrow \text{GL}_n$ be defined by $\varphi(g) = g^T$ (transpose). Show that φ is an isomorphism of algebraic groups (hence faithfully flat). What is the kernel? (*Hint*: the transpose is an anti-homomorphism for matrix multiplication; how can you fix this to get a homomorphism?)

3. Let $G = \mathrm{GL}_2$ and $H = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{G}_a \right\} \cong \mathbb{G}_a$ (the upper-triangular unipotent subgroup). Show that H is *not* normal in G by finding $g \in \mathrm{GL}_2(k)$ and $h \in H(k)$ with $ghg^{-1} \notin H(k)$. Then show that H is normal in the Borel subgroup B_2 of upper-triangular matrices.
4. Show that there is a short exact sequence $1 \rightarrow \mathrm{Sp}_{2n} \rightarrow \mathrm{SL}_{2n} \rightarrow X \rightarrow 1$ if and only if Sp_{2n} is normal in SL_{2n} . Is Sp_{2n} normal in SL_{2n} ? (*Hint*: for $n \geq 2$, consider $\mathrm{Lie}(\mathrm{Sp}_{2n})$ as a subspace of $\mathfrak{sl}_{2n}$ and check whether it is an ideal.)
5. Let $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$ be a short exact sequence of algebraic groups. Show that for any k -algebra R , the sequence $1 \rightarrow N(R) \rightarrow G(R) \rightarrow Q(R)$ is exact (i.e. $N(R) = \ker(G(R) \rightarrow Q(R))$). Give an example where $G(R) \rightarrow Q(R)$ is not surjective. (*Hint*: for the first part, use the fiber product description of the kernel; for the second, take $R = k$ in the n -th power sequence with $n > 1$.)

4.2 The Construction of G/H

We now turn to the main theorem of this chapter: the existence and basic properties of the quotient G/H for any closed subgroup H of an affine algebraic group G . The strategy, due to Chevalley, is beautifully geometric: we find a representation of G containing a line whose stabilizer is exactly H , and then the orbit of that line in projective space gives a concrete model for G/H . The tools are all in place: the Fundamental Finiteness Theorem (theorem 2.2.6) from Chapter 2, the Closed Orbit Lemma (theorem 2.4.15), and the Hopf ideal description of closed subgroups (definition 1.2.7) from Chapter 1.

Chevalley's Theorem on Stabilizers

The key input is a theorem that converts the algebraic data defining H (a Hopf ideal $I \subseteq \mathcal{O}(G)$) into geometric data (a line in a representation). The idea is simple: the ideal I is a subspace of $\mathcal{O}(G)$, and the Fundamental Finiteness Theorem guarantees it is contained in a finite-dimensional G -stable subspace. The Hopf ideal conditions on I then translate into the statement that H is exactly the stabilizer of a certain subspace.

Theorem 4.2.1: Chevalley's Theorem

Let G be an affine algebraic group over k and let $H \subseteq G$ be a closed subgroup. Then there exists a finite-dimensional representation V of G and a one-dimensional subspace $L \subseteq V$ (a “line”) such that:

$$H = \mathrm{Stab}_G(L) := \{g \in G \mid g \cdot L = L\}$$

That is, H is the stabilizer of the line L in the G -representation V .

Proof:

Let $A = \mathcal{O}(G)$ and let $I \subseteq A$ be the Hopf ideal defining H , so that $\mathcal{O}(H) = A/I$. Since A is a finitely generated k -algebra, the ideal I is finitely generated: say $I = (f_1, \dots, f_r)$.

Step 1: Find a finite-dimensional G -stable subspace containing I . The coordinate ring A is

a right A -comodule via the comultiplication $\Delta : A \rightarrow A \otimes_k A$ (this is the left-regular action from example 2.1.4). By the Fundamental Finiteness Theorem (theorem 2.2.6), each generator f_i is contained in a finite-dimensional sub-comodule $W_i \subseteq A$. Let $W = W_1 + \cdots + W_r$, a finite-dimensional G -stable subspace of A containing all the generators of I .

Step 2: Extract a subspace whose stabilizer is H . Define $V_0 = W \cap I$. This is a subspace of the finite-dimensional space W , hence finite-dimensional. We claim that V_0 is H -stable but *not* G -stable (unless $H = G$), and that H is exactly the subgroup of G that preserves V_0 .

To see this, we use the coaction. The G -action on A is given by the comultiplication $\Delta : A \rightarrow A \otimes_k A$. For $f \in I$, the Hopf ideal condition $\Delta(I) \subseteq I \otimes A + A \otimes I$ tells us that:

$$\Delta(f) \equiv \sum f'_i \otimes a_i \pmod{A \otimes I}$$

where $f'_i \in I$ and $a_i \in A$. The action of an element $g \in G(k) = \text{Hom}_{\mathbf{Alg}_k}(A, k)$ on f is $g \cdot f = (\text{id} \otimes g)(\Delta(f))$. If $g \in H(k)$, then g annihilates I modulo k (i.e. g factors through $A \rightarrow A/I$), and the terms in $A \otimes I$ evaluate to elements of I under $\text{id} \otimes g$. More precisely, for $g \in H(R)$, the map $g : A \rightarrow R$ sends I into IR , so $(\text{id} \otimes g)(\Delta(f))$ lies in $I \cdot R$ when $f \in I$. Thus H preserves I , hence preserves $V_0 = W \cap I$.

Conversely, suppose $g \in G(k)$ preserves V_0 . Since V_0 contains the generators $f_1, \dots, f_r$ of I (they lie in $W \cap I$ by construction), and g preserves $V_0 \subseteq I$, the action of g sends each generator to an element of I . As the G -action is an algebra automorphism (on the localization), g preserves I , meaning $g \cdot f \in I$ for all $f \in I$. By the correspondence between Hopf ideals and closed subgroups, this means $g \in H(k)$.

Step 3: Reduce to a line. We have found a finite-dimensional G -representation W and a subspace $V_0 \subseteq W$ with $\text{Stab}_G(V_0) = H$. To reduce to a *line*, we use the exterior power. Let $d = \dim V_0$ and consider the representation $V = \bigwedge^d W$. The subspace V_0 determines a line

$$L = \bigwedge^d V_0 \subseteq \bigwedge^d W = V$$

which is one-dimensional (choosing a basis $\{v_1, \dots, v_d\}$ for V_0 , the line L is spanned by $v_1 \wedge \cdots \wedge v_d$).

An element $g \in G$ preserves V_0 if and only if g preserves L : the forward direction is clear (if $g(V_0) = V_0$, then $g(v_1 \wedge \cdots \wedge v_d) = g(v_1) \wedge \cdots \wedge g(v_d) \in \bigwedge^d V_0 = L$). For the converse, if g preserves $L = \bigwedge^d V_0$, then g preserves the subspace $V_0 \subseteq W$ (because $\bigwedge^d$ detects d -dimensional subspaces: a d -dimensional subspace $U \subseteq W$ satisfies $\bigwedge^d U = L$ if and only if $U = V_0$, since L is a line in the Plücker embedding). Therefore $\text{Stab}_G(L) = \text{Stab}_G(V_0) = H$.

The proof has a clean geometric interpretation: the Hopf ideal I defining H lives inside the “world of functions” $\mathcal{O}(G)$. The finiteness theorem converts I into finite-dimensional linear algebra, and the exterior power trick converts a subspace stabilizer into a line stabilizer. The line L then provides a point $[L] \in \mathbb{P}(V)$, and the orbit $G \cdot [L]$ will be our model for G/H .

The Quotient as a Quasi-Projective Variety

With Chevalley’s theorem in hand, we can now construct G/H .

Theorem 4.2.2: Existence of the Quotient

Let G be an affine algebraic group over k and let $H \subseteq G$ be a closed subgroup. Then there exists a quasi-projective k -variety G/H and a faithfully flat morphism $\pi : G \rightarrow G/H$ such that:

1. The fibers of π are exactly the left cosets gH . That is, $\pi(g_1) = \pi(g_2)$ if and only if $g_1^{-1}g_2 \in H$.
2. (Universal property) For any morphism $f : G \rightarrow X$ of k -schemes that is constant on left cosets (i.e. $f(gh) = f(g)$ for all $g \in G(R)$ and $h \in H(R)$, all R), there exists a unique morphism $\bar{f} : G/H \rightarrow X$ with $f = \bar{f} \circ \pi$.
3. G/H is smooth and of dimension $\dim G - \dim H$.
4. G acts on G/H by left translation, and this action is transitive.

Proof:

By Chevalley's theorem (theorem 4.2.1), there exists a representation V of G and a line $L \subseteq V$ with $H = \text{Stab}_G(L)$. Consider the induced action of G on the projective space $\mathbb{P}(V)$. The point $[L] \in \mathbb{P}(V)$ has stabilizer H , and the orbit map $\varphi : G \rightarrow \mathbb{P}(V)$ defined by $\varphi(g) = g \cdot [L]$ has the property that $\varphi(g_1) = \varphi(g_2)$ if and only if $g_1^{-1}g_2 \in H$.

By the Closed Orbit Lemma (theorem 2.4.15), the orbit $\Omega = G \cdot [L] \subseteq \mathbb{P}(V)$ is a smooth, locally closed subvariety of $\mathbb{P}(V)$. Since $\mathbb{P}(V)$ is projective, a locally closed subvariety of a projective variety is quasi-projective. We set $G/H := \Omega$ with the reduced scheme structure (or, if G is smooth, the natural scheme structure coming from the orbit).

(1): *Fiber description.* The orbit map $\varphi : G \rightarrow \Omega$ satisfies $\varphi(g_1) = \varphi(g_2)$ iff $g_1 \cdot [L] = g_2 \cdot [L]$ iff $g_2^{-1}g_1 \in \text{Stab}_G([L]) = H$ iff $g_1H = g_2H$. So the fibers of φ are exactly the left cosets of H in G .

(2): *Universal property.* Let $f : G \rightarrow X$ be constant on left cosets. Define $\bar{f} : \Omega \rightarrow X$ by $\bar{f}(g \cdot [L]) = f(g)$. This is well-defined because $g_1 \cdot [L] = g_2 \cdot [L]$ implies $g_1H = g_2H$, which implies $f(g_1) = f(g_2)$. That $\bar{f}$ is a morphism of k -schemes (not just a set map) follows from the fact that $\varphi : G \rightarrow \Omega$ is a faithfully flat surjection (see below) and faithfully flat descent: a morphism $G \rightarrow X$ that is constant on the fibers of a faithfully flat map $G \rightarrow \Omega$ descends uniquely to a morphism $\Omega \rightarrow X$ (see [3]).

(3): *Smoothness and dimension.* The orbit Ω is smooth by proposition 2.4.10 (the translation argument: all points of an orbit look alike, and at least one is smooth). The dimension is $\dim \Omega = \dim G - \dim H$ by the orbit-stabilizer formula (proposition 2.4.4).

(4): *Transitive G -action.* The group G acts on $\Omega \subseteq \mathbb{P}(V)$ by restriction of its action on $\mathbb{P}(V)$, and Ω is a single orbit, so the action is transitive by definition.

Faithful flatness of φ . The morphism $\varphi : G \rightarrow \Omega$ is surjective (since $\Omega = \varphi(G)$) and has all fibers isomorphic to H (by the coset description). Since G and Ω are both smooth and the fibers are

smooth of constant dimension $\dim H$, the morphism φ is flat by the miracle flatness theorem (a morphism from a Cohen–Macaulay scheme to a smooth scheme with equidimensional fibers is flat; see [3]). A flat surjective morphism of finite type is faithfully flat.

The construction shows that G/H is always *quasi-projective*: it is a locally closed subvariety of a projective space. It is not always projective (it need not be closed in $\mathbb{P}(V)$), and it is not always affine (for instance, GL_n/B_n is projective, as we will see in Chapter 6). The question of when G/H is projective will turn out to be intimately related to the structure of H : the quotient G/P is projective if and only if P is a *parabolic subgroup*, a topic we take up in Chapter 6.

4.2.3 Remark: Uniqueness of the Quotient The universal property (2) determines G/H up to unique isomorphism, as is standard for objects defined by universal properties. In particular, the construction does not depend on the choice of representation V and line L in Chevalley’s theorem. Different choices give isomorphic quotients, and the isomorphism is canonical (it is the unique map making the obvious triangle commute).

Example 4.2.4: GL_n/B_n and the Complete Flag Variety

Let $G = \mathrm{GL}_n$ and $H = B_n$ (the upper-triangular Borel subgroup). We apply the construction. The Borel B_n is the stabilizer of the standard flag $0 \subset \langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \cdots \subset k^n$. To realize B_n as the stabilizer of a *line*, we use the exterior power trick from the proof of Chevalley’s theorem.

Consider the representation $V = \bigoplus_{d=1}^{n-1} \bigwedge^d(k^n)$ and the line L spanned by $e_1 \oplus (e_1 \wedge e_2) \oplus \cdots \oplus (e_1 \wedge \cdots \wedge e_{n-1})$. The stabilizer of L in GL_n consists of matrices that preserve each subspace $\langle e_1, \dots, e_d \rangle$ for $1 \leq d \leq n-1$, which is exactly B_n .

The orbit $\mathrm{GL}_n \cdot [L] \subseteq \mathbb{P}(V)$ is the *complete flag variety* $\mathrm{Fl}(1, 2, \dots, n-1; n)$, parametrizing complete flags $0 \subset V_1 \subset V_2 \subset \cdots \subset V_{n-1} \subset k^n$ with $\dim V_d = d$. This is a smooth projective variety of dimension $\dim \mathrm{GL}_n - \dim B_n = n^2 - \binom{n+1}{2} = \binom{n}{2}$. The Plücker embedding realizes this flag variety as a closed subvariety of the product of Grassmannians $\prod_{d=1}^{n-1} \mathrm{Gr}(d, n)$, each of which sits inside its own projective space.

Example 4.2.5: SL_2/B and $\mathbb{P}^1$

The simplest nontrivial case: let $G = \mathrm{SL}_2$ and $B = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \right\}$ (the upper-triangular Borel in SL_2). The Borel B is the stabilizer of the line $\langle e_1 \rangle$ in the standard representation $V = k^2$. The orbit of $[e_1] \in \mathbb{P}^1$ under SL_2 is all of $\mathbb{P}^1$ (since SL_2 acts transitively on lines in k^2). Therefore:

$$\mathrm{SL}_2/B \cong \mathbb{P}^1$$

This is a smooth projective variety of dimension $\dim \mathrm{SL}_2 - \dim B = 3 - 2 = 1$. The quotient map $\pi : \mathrm{SL}_2 \rightarrow \mathbb{P}^1$ sends $g \mapsto g \cdot [e_1]$, and the fiber over $[e_1]$ is B itself.

This example is the prototype for all of Chapter 6: the quotient G/B for a Borel subgroup B is always a smooth projective variety (the *flag variety*), and its geometry encodes the structure of G .

Example 4.2.6: Affine Quotients: $\mathrm{GL}_n/\mathrm{GL}_{n-1}$

Not all quotients are projective. Let $G = \mathrm{GL}_n$ and $H = \mathrm{GL}_{n-1}$, embedded via $A \mapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$. The group H is the stabilizer of the vector $e_n \in k^n$ under the standard action. The orbit of e_n under GL_n is $k^n \setminus \{0\} = \mathbb{A}^n \setminus \{0\}$ (as computed in example 2.4.5). Therefore:

$$\mathrm{GL}_n/\mathrm{GL}_{n-1} \cong \mathbb{A}^n \setminus \{0\}$$

This is an affine (in fact, quasi-affine) variety of dimension $n^2 - (n-1)^2 = 2n-1$. The orbit is not closed in $\mathbb{A}^n$ (its closure includes the origin), so $\mathrm{GL}_n/\mathrm{GL}_{n-1}$ is not affine but quasi-affine. Note that this quotient is *not* a group, since GL_{n-1} is not normal in GL_n .

The Quotient Map is a Principal H -Bundle

The quotient map $\pi : G \rightarrow G/H$ has an important structural property that goes beyond faithful flatness: it is a *principal H -bundle* (or *H -torsor*) over G/H . This means that H acts freely on G (by right multiplication), the orbits of this action are exactly the fibers of π , and the map π is locally trivial in a suitable topology.

Proposition 4.2.7: $G \rightarrow G/H$ is an H -Torsor

Let G be an affine algebraic group and $H \subseteq G$ a closed subgroup. The morphism

$$\psi : G \times H \longrightarrow G \times_{G/H} G, \quad (g, h) \longmapsto (g, gh)$$

is an isomorphism. In other words, the action map $G \times H \rightarrow G$ given by right multiplication exhibits G as a (right) H -torsor over G/H .

Proof:

On R -valued points, the map sends $(g, h) \mapsto (g, gh)$, which is a bijection (the inverse sends $(g_1, g_2) \mapsto (g_1, g_1^{-1}g_2)$, and $g_1^{-1}g_2 \in H(R)$ because $\pi(g_1) = \pi(g_2)$). To see this is an isomorphism of schemes (not just a bijection on points), note that both sides are smooth of the same dimension ($\dim G + \dim H$), and the map is a morphism of G -schemes that is bijective on R -valued points for all R . By the Yoneda Lemma, it is an isomorphism.

The torsor property has a concrete consequence: *locally* on G/H (in the flat, or étale, or fppf topology), the quotient map $G \rightarrow G/H$ looks like the projection $U \times H \rightarrow U$ for an open subset $U \subseteq G/H$. In particular, π is smooth of relative dimension $\dim H$.

Corollary 4.2.8: Faithful Flatness of the Quotient Map

The quotient map $\pi : G \rightarrow G/H$ is faithfully flat and smooth. In particular:

1. The map π is open (images of open sets are open).
2. For any field extension K/k , the base change $\pi_K : G_K \rightarrow (G/H)_K$ is the quotient map $G_K \rightarrow G_K/H_K$.
3. If $f : G/H \rightarrow X$ is a morphism, then f is flat (resp. smooth, étale, an isomorphism) if and only if $f \circ \pi : G \rightarrow X$ has the same property.

Proof:

Statement (1) follows from the general fact that flat morphisms of finite presentation are open (see [3]). Statement (2) follows from the universal property: the base-changed map $G_K \rightarrow (G/H)_K$ satisfies the same universal property as $G_K \rightarrow G_K/H_K$, so they are canonically isomorphic. Statement (3) is a consequence of faithfully flat descent: a property that can be checked after faithfully flat base change holds if and only if it holds after pulling back along π .

Example 4.2.9: Local Triviality for $\mathrm{SL}_2 \rightarrow \mathbb{P}^1$

Consider $\pi : \mathrm{SL}_2 \rightarrow \mathbb{P}^1$ from example 4.2.5. Over the standard open $U_0 = \{[x : y] : x \neq 0\} \cong \mathbb{A}^1$ (with coordinate $t = y/x$), the fiber $\pi^{-1}(U_0)$ consists of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2$ with $a \neq 0$. The map $g \mapsto (g \cdot [e_1], g^{-1} \cdot (\text{standard section}))$ provides a trivialization $\pi^{-1}(U_0) \cong U_0 \times B$. Explicitly, the section $\sigma : U_0 \rightarrow \mathrm{SL}_2$ sending $t \mapsto \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$ satisfies $\pi \circ \sigma = \mathrm{id}$, and every $g \in \pi^{-1}(U_0)$ factors uniquely as $g = \sigma(\pi(g)) \cdot b$ for some $b \in B$. Similarly, the map trivializes over $U_1 = \{y \neq 0\}$. Thus $\mathrm{SL}_2 \rightarrow \mathbb{P}^1$ is a Zariski-locally trivial B -bundle.

Exercise 4.2.1

1. Let $G = \mathrm{GL}_3$ and $H = \left\{ \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix} \in \mathrm{GL}_3 \right\}$ (the upper-triangular Borel). Without using the general proof, directly find a representation V of GL_3 and a line $L \subseteq V$ with $H = \mathrm{Stab}_G(L)$. (*Hint*: think about which subspaces of k^3 are stabilized by upper-triangular matrices, and use the exterior power.)
2. Let $G = \mathrm{SO}_5$ (of dimension 10) and $H = \mathrm{SO}_4$ (of dimension 6), embedded as the stabilizer of e_1 under the standard action on k^5 (with $\mathrm{char} k \neq 2$). What is $\dim(G/H)$? Describe G/H geometrically. (*Hint*: the orbit of e_1 under SO_5 is the quadric $\{v \in k^5 : q(v) = 1\}$ where q is the quadratic form defining SO_5 .)
3. Show that SL_2/B is not affine. (*Hint*: $\mathrm{SL}_2/B \cong \mathbb{P}^1$, and a projective variety of positive dimension is never affine.)
4. Let G be an algebraic group over a field k (not necessarily algebraically closed) and $H \subseteq G$ a closed subgroup. Show that $(G/H) \otimes_k \bar{k} \cong G_{\bar{k}}/H_{\bar{k}}$. (*Hint*: use statement (2) of corollary 4.2.8.)
5. Let $G = \mathbb{G}_m \times \mathbb{G}_m$ and $H = \{(t, t) : t \in \mathbb{G}_m\} \cong \mathbb{G}_m$ (the diagonal subgroup). Construct G/H explicitly. Show that $G/H \cong \mathbb{G}_m$ via the map $(s, t) \mapsto st^{-1}$. Is H normal in G ? Does G/H inherit a group structure?

4.3 Properties of Quotients and the Isomorphism Theorems

With the quotient G/H constructed as a quasi-projective variety, we now develop its properties. The main results are: when N is normal, G/N is an algebraic group; the three isomorphism theorems carry over from abstract group theory; and transitive actions are identified with quotients, closing the promises made in Chapter 2. These are the tools that the rest of Part I will use freely.

The Group Structure on G/N

When $H = N$ is a normal closed subgroup of G , the coset space G/N should inherit a group structure from G : the product of cosets $g_1N \cdot g_2N = (g_1g_2)N$ is well-defined precisely because N is normal. The scheme-theoretic version of this requires a small argument.

Theorem 4.3.1: Quotient by a Normal Subgroup

Let G be an affine algebraic group and $N \trianglelefteq G$ a closed normal subgroup. Then:

1. The quotient G/N carries a unique structure of algebraic group such that the quotient map $\pi : G \rightarrow G/N$ is a homomorphism.
2. The sequence $1 \rightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} G/N \rightarrow 1$ is a short exact sequence of algebraic groups.
3. G/N is affine.

Proof:

(1): We must define morphisms $\mu_{G/N} : G/N \times G/N \rightarrow G/N$ and $\iota_{G/N} : G/N \rightarrow G/N$ (multiplication and inverse). Consider the composition

$$G \times G \xrightarrow{\mu_G} G \xrightarrow{\pi} G/N$$

This map is constant on left $N \times N$ -cosets: for $n_1, n_2 \in N(R)$,

$$\pi(\mu_G(g_1n_1, g_2n_2)) = \pi(g_1n_1g_2n_2) = \pi(g_1g_2 \cdot g_2^{-1}n_1g_2 \cdot n_2) = \pi(g_1g_2)$$

where we used the normality of N (so $g_2^{-1}n_1g_2 \in N(R)$) and the fact that π kills N . Therefore $\pi \circ \mu_G$ factors through $G/N \times G/N$ by the universal property of the quotient (theorem 4.2.2(2)), applied to the product $G \times G \rightarrow G/N \times G/N$. This gives the multiplication $\mu_{G/N}$. The inverse is constructed similarly from $\pi \circ \iota_G$.

The group axioms (associativity, identity, inverse) for G/N follow from those for G by the faithfulness of the universal property: two morphisms $G/N \rightarrow G/N$ are equal if and only if they become equal after precomposing with π (since π is an epimorphism in the category of k -schemes, being faithfully flat). For instance, associativity of $\mu_{G/N}$ follows from:

$$\mu_{G/N} \circ (\mu_{G/N} \times \text{id}) \circ (\pi \times \pi \times \pi) = \pi \circ \mu_G \circ (\mu_G \times \text{id}) = \pi \circ \mu_G \circ (\text{id} \times \mu_G) = \mu_{G/N} \circ (\text{id} \times \mu_{G/N}) \circ (\pi \times \pi \times \pi)$$

and the fact that $\pi \times \pi \times \pi$ is an epimorphism.

(2): The map $\iota : N \hookrightarrow G$ is a closed immersion identifying $N = \ker(\pi)$ (by construction of G/N

as the quotient by the equivalence relation $g_1 \sim g_2$ iff $g_1^{-1}g_2 \in N$). The map π is faithfully flat by corollary 4.2.8. So $1 \rightarrow N \rightarrow G \rightarrow G/N \rightarrow 1$ is a short exact sequence in the sense of definition 4.1.5.

(3): Affineness of G/N is a deeper result. When G is affine and N is normal, the ring of N -invariants $\mathcal{O}(G)^N$ is a finitely generated k -algebra (by a theorem of Nagata on invariant rings for reductive groups, or by direct argument using the Hopf algebra structure), and $G/N \cong \text{Spec } \mathcal{O}(G)^N$. We defer the proof to the references (see Waterhouse [12] or Milne [8]), noting that the key input is that the quotient map $\pi : G \rightarrow G/N$ is affine (its fibers are translates of the affine group N).

Statement (3) is important: it says that quotients of affine groups by *normal* subgroups remain affine. This is *not* true for non-normal subgroups: the quotient GL_n/B_n is projective, not affine (example 4.2.4). The distinction between “ G/N is affine” (normal case) and “ G/H is quasi-projective” (general case) is a fundamental structural fact.

Identification of Homogeneous Spaces

We can now fulfill the promise made in Chapter 2 (§2.4): if G acts transitively on a variety X with stabilizer H at a point x , then X is isomorphic to G/H as a G -variety.

Theorem 4.3.2: Homogeneous Spaces are Quotients

Let G be an affine algebraic group acting on a k -variety X via a transitive action, and let $x \in X(k)$ with stabilizer $H = G_x$. Then the orbit map $\varphi_x : G \rightarrow X$ induces an isomorphism of G -varieties

$$\bar{\varphi}_x : G/H \xrightarrow{\sim} X$$

Proof:

The orbit map $\varphi_x : G \rightarrow X$, $g \mapsto g \cdot x$, is constant on left H -cosets: $\varphi_x(gh) = gh \cdot x = g \cdot x = \varphi_x(g)$ for $h \in H(R)$. By the universal property of G/H (theorem 4.2.2(2)), φ_x factors through a morphism $\bar{\varphi}_x : G/H \rightarrow X$ with $\varphi_x = \bar{\varphi}_x \circ \pi$.

We claim $\bar{\varphi}_x$ is an isomorphism. First, $\bar{\varphi}_x$ is bijective on k -valued points: surjectivity follows from transitivity of the action, and injectivity from the stabilizer condition ($\varphi_x(g_1) = \varphi_x(g_2)$ iff $g_1^{-1}g_2 \cdot x = x$ iff $g_1^{-1}g_2 \in H$ iff $g_1H = g_2H$).

It remains to show $\bar{\varphi}_x$ is an isomorphism of schemes, not just a bijection. Both G/H and X are smooth varieties of the same dimension $\dim G - \dim H$ (by theorem 4.2.2(3) and the orbit-stabilizer formula proposition 2.4.4). The morphism $\bar{\varphi}_x$ is G -equivariant (since φ_x is) and bijective. A bijective morphism of smooth varieties that is G -equivariant is an isomorphism: indeed, $\bar{\varphi}_x$ is a bijective morphism of smooth varieties, hence birational. By Zariski’s Main Theorem (see [3]), a bijective birational morphism from a normal variety (smooth implies normal) to a separated variety is an open immersion. Since $\bar{\varphi}_x$ is also surjective, it is an isomorphism.

This theorem closes the loop opened in Chapter 2. We can now state the identifications that were deferred:

Corollary 4.3.3: Projective Spaces and Grassmannians as Quotients

Let k be an algebraically closed field.

1. The orbit map $\mathrm{GL}_n \rightarrow \mathbb{P}^{n-1}$, $g \mapsto g \cdot [e_1]$, induces an isomorphism of GL_n -varieties:

$$\mathrm{GL}_n/P_1 \xrightarrow{\sim} \mathbb{P}^{n-1}$$

where P_1 is the parabolic subgroup stabilizing $\langle e_1 \rangle$ (as in example 2.4.13).

2. More generally, for $1 \leq d \leq n-1$, the orbit map induces:

$$\mathrm{GL}_n/P_d \xrightarrow{\sim} \mathrm{Gr}(d, n)$$

where P_d is the parabolic stabilizing $\langle e_1, \dots, e_d \rangle$ (as in example 2.4.14).

3. The orbit map $\mathrm{SL}_2 \rightarrow \mathbb{P}^1$, $g \mapsto g \cdot [e_1]$, induces:

$$\mathrm{SL}_2/B \xrightarrow{\sim} \mathbb{P}^1$$

where B is the upper-triangular Borel (as in example 4.2.5).

Proof:

Each is a direct application of theorem 4.3.2: in each case, the group acts transitively (verified in Chapter 2) and the stabilizer was identified there. The theorem promotes the set-theoretic bijection to an isomorphism of varieties.

The Isomorphism Theorems

The classical isomorphism theorems of group theory carry over to algebraic groups, with “quotient by a normal subgroup” replaced by the construction of §4.2 and “surjectivity” replaced by “faithful flatness.”

Theorem 4.3.4: First Isomorphism Theorem

Let $\varphi : G \rightarrow H$ be a homomorphism of affine algebraic groups. Then:

1. The image $\varphi(G)$ is a closed subgroup of H .
2. The kernel $N = \ker(\varphi)$ is a closed normal subgroup of G .
3. There is a canonical isomorphism of algebraic groups:

$$G/\ker(\varphi) \xrightarrow{\sim} \varphi(G)$$

Proof:

(2) is immediate: $N(R) = \ker(\varphi_R : G(R) \rightarrow H(R))$ is a normal subgroup of $G(R)$ for all R (as noted in example 4.1.4(3)), and $N \hookrightarrow G$ is a closed immersion (the fiber of φ over e_H is a closed subscheme).

(1): The image $\varphi(G)$ is a constructible subset of H (by Chevalley's theorem on constructible images). It is also a subgroup of H at the level of R -valued points for all R . We claim it is closed. Since $\varphi(G)$ is a constructible subgroup, it contains a dense open subset of its closure $\overline{\varphi(G)}$. But $\overline{\varphi(G)}$ is also a subgroup (the closure of a constructible subgroup of an algebraic group is a subgroup — this is a standard result, see Milne [8]). Then $\varphi(G)$ is an open subgroup of $\overline{\varphi(G)}$, hence closed (the complement is a union of cosets of $\varphi(G)$, each open, hence the complement is open, so $\varphi(G)$ is closed in $\overline{\varphi(G)}$). Since $\overline{\varphi(G)}$ is closed in H , $\varphi(G)$ is closed in H .

(3): The homomorphism φ is constant on N -cosets: $\varphi(gn) = \varphi(g)\varphi(n) = \varphi(g)$ for $n \in N(R)$. By the universal property of G/N (theorem 4.2.2(2)), φ factors through $\overline{\varphi} : G/N \rightarrow H$ with $\varphi = \overline{\varphi} \circ \pi$. The image of $\overline{\varphi}$ is $\varphi(G)$, and $\overline{\varphi}$ is injective on R -valued points: if $\overline{\varphi}(g_1N) = \overline{\varphi}(g_2N)$, then $\varphi(g_1) = \varphi(g_2)$, so $g_1^{-1}g_2 \in N$, i.e. $g_1N = g_2N$. Both G/N and $\varphi(G)$ are smooth of the same dimension ($\dim G - \dim N$), and $\overline{\varphi}$ is a bijective homomorphism between them. By the same Zariski's Main Theorem argument as in theorem 4.3.2, this is an isomorphism.

Theorem 4.3.5: Second Isomorphism Theorem

Let G be an affine algebraic group, $H \subseteq G$ a closed subgroup, and $N \trianglelefteq G$ a closed normal subgroup. Then:

1. The product HN is a closed subgroup of G .
2. $H \cap N$ is a closed normal subgroup of H .
3. There is a canonical isomorphism of algebraic groups:

$$H/(H \cap N) \xrightarrow{\sim} HN/N$$

Proof:

(1): The product HN is the image of the multiplication map $H \times N \rightarrow G$, $(h, n) \mapsto hn$. Since N is normal, $HN(R) = H(R) \cdot N(R) = \{hn : h \in H(R), n \in N(R)\}$ is a subgroup of $G(R)$ for all R . It is closed by the same constructibility argument as in the first isomorphism theorem: the image of the morphism $H \times N \rightarrow G$ is a constructible subgroup, hence closed.

(2): $H \cap N$ is the fiber product $H \times_G N$, which is a closed subscheme of H . Normality in H follows from the fact that N is normal in G : for $h \in H(R)$ and $n \in (H \cap N)(R) \subseteq N(R)$, we have $hnh^{-1} \in N(R)$ (since $N \trianglelefteq G$) and $hnh^{-1} \in H(R)$ (since H is a subgroup). So $hnh^{-1} \in (H \cap N)(R)$.

(3): Consider the composition $H \hookrightarrow G \xrightarrow{\pi} G/N$ where π is the quotient map. The kernel of $\pi|_H$ is $H \cap N$. The image is $\pi(H) = HN/N$ (since $\pi(h) = hN$, and $\pi(H)$ consists of all cosets hN).

with $h \in H$, which is the same as cosets $hnN = hN$ for $hn \in HN$). By the first isomorphism theorem applied to $\pi|_H : H \rightarrow G/N$, we get $H/(H \cap N) \cong \pi(H) = HN/N$.

Theorem 4.3.6: Third Isomorphism Theorem

Let G be an affine algebraic group and $N \subseteq M \subseteq G$ closed normal subgroups (with $N \trianglelefteq G$ and $M \trianglelefteq G$). Then M/N is a closed normal subgroup of G/N , and there is a canonical isomorphism:

$$(G/N)/(M/N) \xrightarrow{\sim} G/M$$

Proof:

The quotient map $\pi_M : G \rightarrow G/M$ kills N (since $N \subseteq M = \ker(\pi_M)$), so it factors through G/N : there exists a homomorphism $\bar{\pi} : G/N \rightarrow G/M$ with $\pi_M = \bar{\pi} \circ \pi_N$. The kernel of $\bar{\pi}$ consists of cosets $gN \in G/N$ with $g \in M$, which is exactly M/N . Since $\bar{\pi}$ is faithfully flat (as $\pi_M = \bar{\pi} \circ \pi_N$ is faithfully flat and π_N is faithfully flat, $\bar{\pi}$ is faithfully flat by cancellation), the first isomorphism theorem gives $(G/N)/\ker(\bar{\pi}) = (G/N)/(M/N) \cong G/M$.

The Lie Algebra of a Quotient

Having the quotient G/N as an algebraic group, we can compute its Lie algebra. The answer is exactly what one expects:

Corollary 4.3.7: Lie Algebra of a Quotient

Let $N \trianglelefteq G$ be a closed normal subgroup of an affine algebraic group G , with both G and N smooth. Then:

$$\mathrm{Lie}(G/N) \cong \mathrm{Lie}(G)/\mathrm{Lie}(N) = \mathfrak{g}/\mathfrak{n}$$

Proof:

The short exact sequence $1 \rightarrow N \rightarrow G \rightarrow G/N \rightarrow 1$ gives, by the left-exactness of Lie (proposition 3.1.12 and ?? 4), an exact sequence $0 \rightarrow \mathfrak{n} \rightarrow \mathfrak{g} \rightarrow \mathrm{Lie}(G/N)$. By dimension counting: all three groups are smooth, so $\dim \mathrm{Lie}(G/N) = \dim(G/N) = \dim G - \dim N = \dim \mathfrak{g} - \dim \mathfrak{n}$. Since the image of $\mathfrak{g} \rightarrow \mathrm{Lie}(G/N)$ has dimension $\dim \mathfrak{g} - \dim \mathfrak{n} = \dim \mathrm{Lie}(G/N)$, the map $\mathfrak{g} \rightarrow \mathrm{Lie}(G/N)$ is surjective. Therefore $\mathrm{Lie}(G/N) \cong \mathfrak{g}/\mathfrak{n}$.

In characteristic 0, the smoothness hypothesis is automatic by Cartier's theorem (theorem 3.3.2), so the formula $\mathrm{Lie}(G/N) \cong \mathfrak{g}/\mathfrak{n}$ holds unconditionally.

Example 4.3.8: Lie Algebra of PGL_n

The projective general linear group is $\mathrm{PGL}_n = \mathrm{GL}_n/Z(\mathrm{GL}_n) = \mathrm{GL}_n/\mathbb{G}_m$, where $\mathbb{G}_m$ is embedded as scalar matrices. By the corollary:

$$\mathrm{Lie}(\mathrm{PGL}_n) \cong \mathfrak{gl}_n/k \cdot I_n = \mathfrak{gl}_n/\mathfrak{z}(\mathfrak{gl}_n) \cong \mathfrak{pgl}_n$$

When $\text{char } k \nmid n$, the trace map splits the inclusion $k \cdot I_n \hookrightarrow \mathfrak{gl}_n$, so $\mathfrak{pgl}_n \cong \mathfrak{sl}_n$ as Lie algebras. But when $\text{char } k \mid n$, the scalar matrix I_n has trace $n = 0$, so $k \cdot I_n \subseteq \mathfrak{sl}_n$, and the quotient $\mathfrak{pgl}_n$ is *not* isomorphic to $\mathfrak{sl}_n$ (it has dimension $n^2 - 1$ while $\mathfrak{sl}_n$ also has dimension $n^2 - 1$, but the quotient $\mathfrak{sl}_n/k \cdot I_n$ has dimension $n^2 - 2$; rather, $\mathfrak{pgl}_n = \mathfrak{gl}_n/k \cdot I_n$ and $\mathfrak{sl}_n \hookrightarrow \mathfrak{pgl}_n$ is an inclusion of codimension 0 that is *not* an isomorphism if we track the Lie bracket on the center). This is one of the characteristic- p subtleties encountered in example 3.2.3.

Fibers and the Dimension Principle

We close the chapter by recording the key geometric property of quotient maps that will be used throughout the structure theory: all fibers of the quotient map are “the same.”

Proposition 4.3.9: Fiber Structure of Quotient Maps

Let G be an affine algebraic group and $H \subseteq G$ a closed subgroup. Then:

1. Every fiber of $\pi : G \rightarrow G/H$ is a translate of H : the fiber over $\pi(g)$ is gH .
2. All fibers are isomorphic to H as varieties.
3. For any closed subgroup K with $H \subseteq K \subseteq G$, there is a natural G -equivariant surjection $G/H \rightarrow G/K$ with fibers isomorphic to K/H .

Proof:

(1) and (2) follow from the torsor property (proposition 4.2.7): the right H -action on G is free with orbits equal to the left cosets gH , and each coset is isomorphic to H via right translation by g^{-1} .

(3): The projection $G \rightarrow G/K$ is constant on left H -cosets (since $H \subseteq K$), so it factors through G/H : there is a unique G -equivariant morphism $\psi : G/H \rightarrow G/K$ with $\pi_K = \psi \circ \pi_H$. The fiber of ψ over the point $\pi_K(g) \in G/K$ consists of all cosets $g'H$ with $g'K = gK$, i.e. $g' \in gK$. The set of such cosets is $\{gkH : k \in K\} = gK/H$, which is isomorphic to K/H as a variety (by left translation by g^{-1}).

This “tower property” of quotients is exactly analogous to the Galois correspondence in field theory: given a chain of subgroups $H \subseteq K \subseteq G$, there is a corresponding chain of quotient maps $G/H \rightarrow G/K \rightarrow G/G = \{*\}$, with each map having fibers isomorphic to the “intermediate quotient.”

Example 4.3.10: The Tower $\text{GL}_n/B \rightarrow \text{GL}_n/P \rightarrow \{*\}$

Let $B \subseteq P \subseteq \text{GL}_n$ where B is the Borel (upper-triangular) and $P = P_d$ is the parabolic subgroup stabilizing $\langle e_1, \dots, e_d \rangle$. The tower gives:

$$\text{Fl}(1, \dots, n-1; n) = \text{GL}_n/B \xrightarrow{\psi} \text{GL}_n/P_d = \text{Gr}(d, n) \rightarrow \{*\}$$

The map ψ sends a complete flag $0 \subset V_1 \subset \dots \subset V_{n-1} \subset k^n$ to the d -dimensional subspace V_d . The fiber of ψ over a point $V_d \in \text{Gr}(d, n)$ is the variety of complete flags *refining* V_d , which is isomorphic to $P_d/B \cong \text{Fl}(1, \dots, d-1; d) \times \text{Fl}(1, \dots, n-d-1; n-d)$ (a product of smaller flag varieties). This “fibration of flag varieties over Grassmannians” is a central tool in Schubert

calculus and the geometry of flag varieties.

Exercise 4.3.1

1. Let $G = \mathbb{G}_m \times \mathbb{G}_m$ and $N = \{(t, t^{-1}) : t \in \mathbb{G}_m\}$. Show that $N \trianglelefteq G$ and compute G/N explicitly. Determine its Lie algebra.
2. Let $\varphi : \mathrm{GL}_n \rightarrow \mathrm{GL}_n$ be defined by $\varphi(g) = (\det g)^{-1} \cdot g$ (“projecting to determinant 1”). Show that φ is a homomorphism with image SL_n and kernel $\mathbb{G}_m$ (embedded as scalar matrices). Conclude $\mathrm{GL}_n/\mathbb{G}_m \cong \mathrm{SL}_n$... but wait, is this correct? (*Hint*: check carefully whether φ is actually a group homomorphism.)
3. Let $G = \mathrm{GL}_n$, $M = \mathrm{SL}_n$, and $N = \mu_n$ (embedded in SL_n as scalar matrices). Apply the third isomorphism theorem to compute $(\mathrm{GL}_n/\mu_n)/(\mathrm{SL}_n/\mu_n)$. What familiar group do you get?
4. Let $G = \mathrm{SO}_3$ act on the unit sphere $S^2 = \{v \in k^3 : q(v) = 1\}$ (where q is the standard quadratic form, $\mathrm{char} k \neq 2$). Show the action is transitive, identify the stabilizer of e_1 , and conclude that $S^2 \cong \mathrm{SO}_3/\mathrm{SO}_2$ as algebraic varieties.
5. Let $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$ be a short exact sequence of *smooth* algebraic groups over a field of characteristic $p > 0$. Show that the Lie algebra sequence $0 \rightarrow \mathfrak{n} \rightarrow \mathfrak{g} \rightarrow \mathfrak{q} \rightarrow 0$ is still exact. (*Hint*: the smoothness hypothesis ensures the dimension count works, even though Cartier’s theorem is unavailable.)
6. Consider the tower $\mathrm{SL}_3/B \rightarrow \mathrm{SL}_3/P \rightarrow \{*\}$ where B is the upper-triangular Borel and $P = P_1$ is the parabolic stabilizing $\langle e_1 \rangle$. Show that $\mathrm{SL}_3/P \cong \mathbb{P}^2$, that SL_3/B is the complete flag variety $\mathrm{Fl}(1, 2; 3)$, and that the fiber of $\mathrm{Fl}(1, 2; 3) \rightarrow \mathbb{P}^2$ over a point $[\ell] \in \mathbb{P}^2$ is isomorphic to $\mathbb{P}^1$. What is $\dim \mathrm{Fl}(1, 2; 3)$?

5

Jordan Decomposition and Diagonalizable Groups

In Chapter 1, we introduced the composition series of an affine algebraic group G :

$$e \trianglelefteq U \trianglelefteq N \trianglelefteq G^\circ \trianglelefteq G$$

where U is unipotent, N/U is a torus, and G°/N is semisimple. This decomposition breaks a general algebraic group into three atomic pieces: tori, unipotent groups, and semisimple groups. The present chapter studies the first two.

The starting point is the *Jordan decomposition*: every element g of a linear algebraic group factors uniquely as $g = g_s g_u$, where g_s is semisimple (diagonalizable over $\bar{k}$), g_u is unipotent, and the two factors commute. This is a familiar fact from linear algebra — the multiplicative Jordan normal form — but the key theorem is that the decomposition is *intrinsic*: it does not depend on the choice of embedding $G \hookrightarrow \mathrm{GL}_n$. With this decomposition in hand, we study each piece systematically. Diagonalizable groups and tori are completely controlled by their character lattices via an anti-equivalence of categories, and their representations decompose into weight spaces — a mechanism that will produce root spaces in Chapter 7. Unipotent groups are the opposite extreme: their only semisimple element is the identity, and in characteristic 0 they are entirely determined by their (nilpotent) Lie algebras via the exponential map.

5.1 Jordan Decomposition

The Linear Algebra

Let k be a perfect field and V a finite-dimensional k -vector space. Recall from linear algebra (see [5]) that an endomorphism $g \in \text{End}_k(V)$ is called:

1. *Semisimple* if g is diagonalizable over $\bar{k}$ (equivalently, the minimal polynomial of g has distinct roots in $\bar{k}$, equivalently $V \otimes_k \bar{k}$ has a basis of eigenvectors for g).
2. *Nilpotent* if $g^n = 0$ for some $n \geq 1$.
3. *Unipotent* if $g - \text{id}_V$ is nilpotent, i.e. $(g - \text{id}_V)^n = 0$ for some n .

The additive and multiplicative Jordan decompositions are:

Theorem 5.1.1: Jordan Decomposition in $\text{End}_k(V)$ and $\text{GL}(V)$

Let V be a finite-dimensional vector space over a perfect field k .

1. (Additive) For every $X \in \text{End}_k(V)$, there exist unique $X_s, X_n \in \text{End}_k(V)$ with X_s semisimple, X_n nilpotent, $X_s X_n = X_n X_s$, and $X = X_s + X_n$.
2. (Multiplicative) For every $g \in \text{GL}(V)$, there exist unique $g_s, g_u \in \text{GL}(V)$ with g_s semisimple, g_u unipotent, $g_s g_u = g_u g_s$, and $g = g_s g_u$.

Moreover, X_s and X_n (resp. g_s and g_u) are polynomials in X (resp. g and g^{-1}). In particular, they commute with every endomorphism that commutes with X (resp. g).

Proof:

We prove the additive version; the multiplicative version follows by writing $g = g_s(1 + g_s^{-1}X_n)$ where $X = g_s + X_n$ is the additive decomposition of g viewed as an endomorphism (using the fact that g_s is invertible, which follows from g being invertible and X_n being nilpotent).

Existence. We may assume $k = \bar{k}$ (since the decomposition, once shown to be unique, must be Galois-invariant, hence defined over k when k is perfect). Over $\bar{k}$, let the distinct eigenvalues of X be $\lambda_1, \dots, \lambda_r$ with generalized eigenspaces $V_i = \ker(X - \lambda_i)^{n_i}$. Then $V = V_1 \oplus \dots \oplus V_r$. Define X_s to act as $\lambda_i \cdot \text{id}$ on V_i , and $X_n = X - X_s$. Then X_s is semisimple (diagonal in any basis adapted to the decomposition), X_n is nilpotent (on each V_i , $(X_n|_{V_i})^{n_i} = (X - \lambda_i)^{n_i}|_{V_i} = 0$), and $X_s X_n = X_n X_s$ (since both preserve each V_i and X_s is scalar on V_i).

X_s is a polynomial in X . By the Chinese Remainder Theorem, the distinct polynomials $(t - \lambda_i)^{n_i}$ are pairwise coprime, so there exists a polynomial $p(t) \in \bar{k}[t]$ with $p(t) \equiv \lambda_i \pmod{(t - \lambda_i)^{n_i}}$ for each i . Then $p(X)$ acts as $\lambda_i \cdot \text{id}$ on each V_i , so $X_s = p(X)$. When k is perfect, the Galois action permutes eigenvalues in a way that preserves the decomposition, so p can be chosen in $k[t]$ (see [5]).

Uniqueness. Suppose $X = S + N$ with S semisimple, N nilpotent, and $SN = NS$. Since S

and N commute, they preserve each other's eigenspaces. On each generalized eigenspace V_i of X : S acts semisimply with a single eigenvalue (since S commutes with X and preserves V_i), which must be λ_i (since $X = S + N$ and N is nilpotent on V_i). So $S|_{V_i} = \lambda_i \cdot \text{id} = X_s|_{V_i}$, giving $S = X_s$ and $N = X_n$.

The statement that g_s and g_u are polynomials in g (and g^{-1}) is crucial: it means they commute with everything that commutes with g , and in particular they lie in any subalgebra or subgroup that g lies in (provided that subalgebra is defined by commutation conditions). This is the mechanism that will make the decomposition intrinsic.

Example 5.1.2: Jordan Decomposition in GL_2

Let $g = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \in \text{GL}_2(k)$ with $\text{char } k \neq 2$. The eigenvalue is $\lambda = 2$ with multiplicity 2. The additive decomposition of g (as an endomorphism) is:

$$g = \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}}_{g_s = 2I} + \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{X_n}$$

The multiplicative decomposition is $g = g_s \cdot g_u$ where:

$$g_s = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad g_u = g_s^{-1}g = \begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}$$

Note that $g_u - I$ is nilpotent (it squares to zero), confirming g_u is unipotent.

Example 5.1.3: Diagonal and Unipotent Parts in GL_n

More generally, the Jordan normal form of $g \in \text{GL}_n(\bar{k})$ is $g = DU$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is the diagonal of eigenvalues and $U = D^{-1}g$ is upper-triangular with 1's on the diagonal (hence unipotent). Then $g_s = D$ and $g_u = U$. The decomposition is canonical: it does not depend on the choice of Jordan basis, only on g .

Intrinsic Jordan Decomposition

The linear-algebraic decomposition depends on viewing g as an element of $\text{GL}(V)$ for some specific V . The fundamental theorem says this dependence is illusory: if G is any affine algebraic group and $g \in G(k)$, then g has a unique decomposition $g = g_s g_u$ that is compatible with *every* representation of G .

Theorem 5.1.4: Intrinsic Jordan Decomposition

Let G be an affine algebraic group over a perfect field k and let $g \in G(k)$. Then there exist unique elements $g_s, g_u \in G(k)$ such that:

1. For every representation $\rho : G \rightarrow \mathrm{GL}(V)$, $\rho(g_s)$ is semisimple and $\rho(g_u)$ is unipotent.
2. $g = g_s g_u$ and $g_s g_u = g_u g_s$.

The elements g_s and g_u are called the *semisimple part* and *unipotent part* of g , respectively.

Proof:

By the linearization theorem (lemma 1.1.8), there exists a faithful representation $\rho_0 : G \hookrightarrow \mathrm{GL}(V_0)$. Inside $\mathrm{GL}(V_0)$, the element $\rho_0(g)$ has a multiplicative Jordan decomposition $\rho_0(g) = s \cdot u$ with s semisimple and u unipotent (theorem 5.1.1). We need to show that $s, u \in \rho_0(G) \subseteq \mathrm{GL}(V_0)$, and that their preimages $g_s, g_u \in G(k)$ are independent of the choice of ρ_0 .

Step 1: s and u lie in G . We use the comodule characterization of G as a closed subgroup of $\mathrm{GL}(V_0)$. The group $G = \mathrm{Spec} A$ is cut out by the Hopf ideal $I \subseteq \mathcal{O}(\mathrm{GL}(V_0))$, meaning $g \in G(k)$ if and only if $f(g) = 0$ for all $f \in I$. Equivalently (by definition 1.2.6), $G(R) \subseteq \mathrm{GL}(V_0)(R)$ is the subgroup defined by the vanishing of I for all R .

We claim that s and u also annihilate I . This rests on the following key lemma:

Key Lemma. Let $g \in \mathrm{GL}(V)$ with Jordan decomposition $g = su$. If $W \subseteq V$ is a subspace with $g(W) \subseteq W$, then $s(W) \subseteq W$ and $u(W) \subseteq W$. Similarly, if $\varphi : V \rightarrow V'$ is any linear map with $\varphi \circ g = g' \circ \varphi$ for some $g' \in \mathrm{GL}(V')$, then $\varphi \circ s = s' \circ \varphi$ and $\varphi \circ u = u' \circ \varphi$, where $g' = s'u'$ is the Jordan decomposition of g' .

Proof of Key Lemma. Since s and u are polynomials in g (and g^{-1}) by theorem 5.1.1, any subspace preserved by g is also preserved by s and u . For the equivariance statement: $\varphi \circ g = g' \circ \varphi$ implies $\varphi \circ g^n = (g')^n \circ \varphi$ for all n , hence $\varphi \circ p(g) = p(g') \circ \varphi$ for any polynomial p . Taking p to be the polynomial giving the semisimple part: $\varphi \circ s = s' \circ \varphi$ (the semisimple part of g' is $p(g')$) for the *same* polynomial, by the uniqueness of the decomposition and the fact that φ intertwines g and g' ... actually, this requires φ to be an isomorphism, or a more careful argument). $\square$

We apply the Key Lemma to the comodule structure. The coordinate ring $A = \mathcal{O}(G)$ is a right $\mathcal{O}(\mathrm{GL}(V_0))$ -comodule via the comultiplication (the regular representation). The element $g \in G(k) \subseteq \mathrm{GL}(V_0)(k)$ acts on A via the left-regular action. By the Fundamental Finiteness Theorem (theorem 2.2.6), every element of A lies in a finite-dimensional G -stable (hence g -stable) subspace. On any such subspace W , the action of g has a Jordan decomposition, and by the Key Lemma (applied to the inclusion $W \hookrightarrow A$ and the g -equivariance), the semisimple and unipotent parts of $g|_W$ are the restrictions of $s|_A$ and $u|_A$ to W .

Now, the ideal $I \subseteq \mathcal{O}(\mathrm{GL}(V_0))$ defining G is g -stable (since $g \in G$ and G is a subgroup). By the Key Lemma, I is also s -stable and u -stable. This means s and u preserve I , and hence define elements of $G(k)$ (they annihilate I when evaluated as k -points). Set $g_s = s|_G$ and $g_u = u|_G$.

Step 2: Independence of the embedding. Let $\rho : G \rightarrow \mathrm{GL}(V)$ be any representation. The composition $\rho \circ \rho_0^{-1} : \rho_0(G) \rightarrow \mathrm{GL}(V)$ is a morphism of algebraic groups, hence a “polynomial map” on the matrix entries. Since $g_s = p(\rho_0(g))$ for some polynomial p (the polynomial from theorem 5.1.1), the element $\rho(g_s) = \rho(p(\rho_0(g))) = p'(\rho(g))$ for some polynomial p' (obtained by composing p with the polynomial map $\rho \circ \rho_0^{-1}$). But more directly: the Key Lemma applied to the G -equivariant map $\rho : V_0 \rightarrow V$ (or rather, to the induced map on the tensor product $V_0 \otimes V$ carrying the diagonal G -action) shows that $\rho(g_s)$ is the semisimple part of $\rho(g)$ and $\rho(g_u)$ is the unipotent part.

Step 3: Uniqueness. If $g = g'_s g'_u$ is another such decomposition, then for the faithful representation ρ_0 , both $\rho_0(g'_s)$ and $\rho_0(g_s)$ are the semisimple part of $\rho_0(g) \in \mathrm{GL}(V_0)$. By uniqueness of the Jordan decomposition in $\mathrm{GL}(V_0)$, $\rho_0(g'_s) = \rho_0(g_s)$, and since ρ_0 is faithful, $g'_s = g_s$ (and hence $g'_u = g_u$).

The proof is somewhat technical, but the essential point is clean: the semisimple and unipotent parts of g are polynomials in g , so they “go wherever g goes” — they are preserved by every morphism that preserves g . Since the closed subgroup $G \subseteq \mathrm{GL}(V_0)$ is defined by conditions that g satisfies, its semisimple and unipotent parts satisfy these conditions too.

5.1.5 Remark: The Role of Perfectness The assumption that k is perfect is used to ensure that the polynomial $p(t)$ giving $g_s = p(g)$ has coefficients in k (not just in $\bar{k}$). Over a non-perfect field, the Galois action on eigenvalues may not preserve the decomposition, and the Jordan decomposition can fail to be defined over k . For most applications in this book (including the structure theory of Chapters 6–7), we work over algebraically closed fields, where perfectness is automatic.

Proposition 5.1.6: Functoriality of the Jordan Decomposition

Let $\varphi : G \rightarrow H$ be a homomorphism of affine algebraic groups over a perfect field k . Then for any $g \in G(k)$:

$$\varphi(g_s) = \varphi(g)_s \quad \text{and} \quad \varphi(g_u) = \varphi(g)_u$$

That is, homomorphisms preserve the Jordan decomposition.

Proof:

The homomorphism φ is a morphism of algebraic groups, hence the composition with any faithful representation $\rho : H \hookrightarrow \mathrm{GL}(W)$ gives a representation $\rho \circ \varphi : G \rightarrow \mathrm{GL}(W)$. By the intrinsic property (theorem 5.1.4(1)), $(\rho \circ \varphi)(g_s) = \rho(\varphi(g_s))$ is the semisimple part of $(\rho \circ \varphi)(g) = \rho(\varphi(g))$. But the semisimple part of $\rho(\varphi(g))$ is $\rho(\varphi(g)_s)$ (again by theorem 5.1.4(1)). Since ρ is faithful, $\varphi(g_s) = \varphi(g)_s$.

An important special case: if $H \subseteq G$ is a closed subgroup and $g \in H(k)$, then the semisimple and unipotent parts of g computed in H agree with those computed in G . The decomposition is independent not only of the representation, but also of the ambient group.

Jordan Decomposition for Lie Algebras

In characteristic 0, there is an additive analogue for Lie algebras.

Proposition 5.1.7: Jordan Decomposition in $\mathfrak{g}$ (Characteristic 0)

Let G be an affine algebraic group over a field k of characteristic 0, and let $\mathfrak{g} = \text{Lie}(G)$. For every $X \in \mathfrak{g}$, the additive Jordan decomposition $\text{ad}(X) = \text{ad}(X)_s + \text{ad}(X)_n$ in $\text{End}_k(\mathfrak{g})$ lifts to a decomposition $X = X_s + X_n$ in $\mathfrak{g}$ such that:

1. $\text{ad}(X_s) = \text{ad}(X)_s$ is semisimple and $\text{ad}(X_n) = \text{ad}(X)_n$ is nilpotent.
2. For every representation $\rho : G \rightarrow \text{GL}(V)$, $d\rho(X_s)$ is semisimple and $d\rho(X_n)$ is nilpotent in $\text{End}_k(V)$.

Proof:

The proof follows the same strategy as the group case, using the faithful representation $d\rho_0 : \mathfrak{g} \hookrightarrow \mathfrak{gl}(V_0)$ (which is injective by proposition 3.1.12) to embed $\mathfrak{g}$ into a matrix algebra, decomposing there, and using the “polynomial” nature of the semisimple part to pull back. The characteristic 0 assumption ensures $d\rho_0$ is faithful (by Cartier’s theorem, theorem 3.3.2) and that the polynomial $p(t)$ giving $X_s = p(X)$ has coefficients in k . We omit the details, which are parallel to the group case.

In characteristic $p > 0$, the additive Jordan decomposition on $\mathfrak{g}$ still exists inside $\mathfrak{gl}(V)$ for any representation, but it need not lift to an intrinsic decomposition in $\mathfrak{g}$ (because $\mathfrak{g}$ may not be “large enough” — the polynomial $p(t)$ may not preserve $\mathfrak{g} \subseteq \mathfrak{gl}(V)$). For the purposes of this book, we will use the Lie-algebraic Jordan decomposition only in characteristic 0.

Example 5.1.8: Jordan Decomposition in $\mathfrak{sl}_2$

Let $X = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \in \mathfrak{sl}_2$ (with $\text{char } k \neq 2$). The eigenvalues are 1 and -1 , so X is already semisimple: $X_s = X$ and $X_n = 0$. The standard basis element $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is nilpotent ($e^2 = 0$), so $e_s = 0$ and $e_n = e$. These are the two extremes: $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is semisimple, and e, f are nilpotent. In Chapter 7, the semisimple elements will span the *Cartan subalgebra*, and the nilpotent elements will generate the *root spaces*.

Example 5.1.9: Semisimple and Unipotent Elements of GL_n

In $\text{GL}_n(k)$ (with $k = \bar{k}$):

1. The semisimple elements are the diagonalizable matrices. They are Zariski-dense in GL_n (since the set of matrices with n distinct eigenvalues is open and dense).
2. The unipotent elements form a closed subvariety $\mathcal{U}_n = \{g \in \text{GL}_n \mid (g - I)^n = 0\}$. This is the variety of matrices whose only eigenvalue is 1.
3. The set of semisimple elements is *not* closed in general. For instance, in GL_2 , the semisimple element $\text{diag}(\lambda, \mu)$ with $\lambda \neq \mu$ degenerates to $\text{diag}(\lambda, \lambda) + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ (a non-semisimple element) as $\mu \rightarrow \lambda$ along a curve in the orbit closure. So the set of semisimple elements is constructible but not closed.

Commuting Semisimple and Unipotent Parts

The Jordan decomposition respects the group structure in a strong sense:

Proposition 5.1.10: Jordan Decomposition and Commuting Elements

Let G be an affine algebraic group over a perfect field and let $g, h \in G(k)$. If $gh = hg$, then:

$$g_s h_s = h_s g_s, \quad g_s h_u = h_u g_s, \quad g_u h_s = h_s g_u, \quad g_u h_u = h_u g_u$$

In words: if g and h commute, then all four combinations of their semisimple and unipotent parts commute.

Proof:

Embed $G \hookrightarrow \mathrm{GL}(V)$ faithfully. Since $gh = hg$ in $\mathrm{GL}(V)$, the endomorphisms g and h commute. By theorem 5.1.1, g_s is a polynomial in g , hence commutes with everything that commutes with g — in particular with h , h_s , and h_u . The same argument with g_u , h_s , and h_u gives all four commutation relations.

This proposition is essential for the structure theory: it says that inside an abelian subgroup of G , the semisimple parts form an abelian subgroup and the unipotent parts form an abelian subgroup. When we study maximal tori in Chapter 6, this will ensure that the torus elements and unipotent elements inside a solvable group decouple cleanly.

Exercise 5.1.1

1. Compute the multiplicative Jordan decomposition of the following matrices in $\mathrm{GL}_3(\mathbb{C})$:

$$g_1 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \quad g_2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad g_3 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

2. Let $G \subseteq \mathrm{GL}_n$ be a closed subgroup and $g \in G(k)$ with k perfect. The proof of theorem 5.1.4 shows $g_s, g_u \in G(k)$. Give a direct proof of this for $G = \mathrm{SL}_n$: show that if $g \in \mathrm{SL}_n(k)$, then $g_s, g_u \in \mathrm{SL}_n(k)$. (*Hint*: $\det(g_s) \det(g_u) = \det(g) = 1$; now use the fact that $\det(g_s)$ is semisimple and $\det(g_u)$ is unipotent in $k^\times$.)
3. Let $X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \in \mathfrak{sl}_3$. Compute $\mathrm{ad}(X) : \mathfrak{sl}_3 \rightarrow \mathfrak{sl}_3$ explicitly (as an 8×8 matrix in the standard basis $\{E_{ij} - \frac{1}{3}\delta_{ij}I\}$, or simply in $\mathfrak{gl}_3$). Verify that $\mathrm{ad}(X)$ is nilpotent. What is the nilpotency index (the smallest m with $\mathrm{ad}(X)^m = 0$)?
4. Let k be a field of characteristic $p > 0$. Show that the p -th power map $F : \mathrm{GL}_n(k) \rightarrow \mathrm{GL}_n(k)$, $g \mapsto g^p$, does *not* preserve the Jordan decomposition in general: $(g^p)_s \neq (g_s)^p$ for some g . (*Hint*: take $g = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and compute g^p .)
5. Let G be a commutative affine algebraic group over an algebraically closed field k . Define $G_s = \{g \in G(k) : g = g_s\}$ (the semisimple elements) and $G_u = \{g \in G(k) : g = g_u\}$ (the unipotent elements). Show that G_s and G_u are subgroups of $G(k)$ and that the map $G_s \times G_u \rightarrow G(k)$, $(s, u) \mapsto su$, is a bijection.

5.2 Diagonalizable Groups and Tori

An element $g \in \mathrm{GL}_n(k)$ is semisimple if and only if it is diagonalizable (over $\bar{k}$), and commuting semisimple elements are simultaneously diagonalizable. The algebraic groups built entirely from semisimple elements — the *diagonalizable groups* — are therefore the groups that can be simultaneously conjugated into the diagonal torus. They are the simplest algebraic groups beyond finite groups, and yet they carry a rich structure: they are completely classified by their *character groups*, finitely generated abelian groups that encode all the representation-theoretic information. The connected ones — the *tori* — will play a central role in the structure theory, as “maximal tori” inside reductive groups provide the Cartan subalgebras that drive the root space decomposition.

Characters and Cocharacters

Definition 5.2.1: Character

Let G be an affine algebraic group over k . A *character* of G is a homomorphism of algebraic groups $\chi : G \rightarrow \mathbb{G}_m$. The set of all characters forms an abelian group under pointwise multiplication:

$$X(G) := \mathrm{Hom}_{\mathbf{Grp}}(G, \mathbb{G}_m)$$

called the *character group* (or *character lattice*) of G . The group operation is $(\chi_1 \cdot \chi_2)(g) = \chi_1(g)\chi_2(g)$ for $g \in G(R)$.

On the Hopf algebra side, a character $\chi : G \rightarrow \mathbb{G}_m$ corresponds to a Hopf algebra homomorphism $\chi^* : k[t, t^{-1}] \rightarrow \mathcal{O}(G)$, which is determined by the image $c = \chi^*(t) \in \mathcal{O}(G)$. The Hopf algebra axioms for $k[t, t^{-1}]$ (namely $\Delta(t) = t \otimes t$, $\epsilon(t) = 1$, $S(t) = t^{-1}$) force c to be a *group-like element*: $\Delta(c) = c \otimes c$, $\epsilon(c) = 1$, and $S(c) = c^{-1}$. Conversely, every group-like element of $\mathcal{O}(G)$ defines a character. We saw this correspondence in ?? 2: characters of G correspond to one-dimensional representations.

Definition 5.2.2: Cocharacter

A *cocharacter* (or *one-parameter subgroup*) of G is a homomorphism $\lambda : \mathbb{G}_m \rightarrow G$. The set of all cocharacters is denoted:

$$X_\vee(G) := \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{G}_m, G)$$

There is a natural pairing between characters and cocharacters: given $\chi \in X(G)$ and $\lambda \in X_\vee(G)$, the composition $\chi \circ \lambda : \mathbb{G}_m \rightarrow \mathbb{G}_m$ is an endomorphism of $\mathbb{G}_m$. Every such endomorphism is of the form $t \mapsto t^n$ for a unique $n \in \mathbb{Z}$, so the pairing

$$\langle \cdot, \cdot \rangle : X(G) \times X_\vee(G) \longrightarrow \mathbb{Z}, \quad \langle \chi, \lambda \rangle = n \quad \text{where } \chi \circ \lambda(t) = t^n$$

is a well-defined $\mathbb{Z}$ -bilinear map.

Diagonalizable Groups

Definition 5.2.3: Diagonalizable Group

An affine algebraic group G over k is called *diagonalizable* if G is isomorphic (over $\bar{k}$) to a closed subgroup of a diagonal torus $\mathbb{G}_m^n$ for some n . Equivalently, every representation of G decomposes as a direct sum of one-dimensional representations.

The equivalence of the two conditions is the key observation. If $G \subseteq \mathbb{G}_m^n$ is a closed subgroup of the diagonal torus, then any representation $\rho : G \rightarrow \mathrm{GL}(V)$ extends (via the inclusion) to a representation of $\mathbb{G}_m^n$ on V . Since the elements of $\mathbb{G}_m^n$ are simultaneously diagonalizable, V decomposes into simultaneous eigenspaces, each of which is a one-dimensional representation. Conversely, if every representation decomposes into one-dimensional sub-representations, then the faithful representation $G \hookrightarrow \mathrm{GL}(V)$ (from the linearization theorem) puts G inside a diagonal torus after choosing a basis of eigenvectors.

The structure of diagonalizable groups is completely controlled by their character groups:

Theorem 5.2.4: Classification of Diagonalizable Groups

The functor $G \mapsto X(G)$ is an anti-equivalence of categories:

$$\left\{ \begin{array}{c} \text{Diagonalizable groups} \\ \text{over } k \end{array} \right\}^{\mathrm{op}} \xrightarrow{\sim} \left\{ \begin{array}{c} \text{Finitely generated} \\ \text{abelian groups} \end{array} \right\}$$

The quasi-inverse sends a finitely generated abelian group M to the diagonalizable group $D(M) := \mathrm{Spec} k[M]$, where $k[M]$ is the group algebra of M with Hopf algebra structure $\Delta(e_m) = e_m \otimes e_m$, $\epsilon(e_m) = 1$, $S(e_m) = e_{-m}$ for $m \in M$.

Proof:

We first construct the functors and then verify they are quasi-inverse.

From groups to characters. Given a diagonalizable group G , its character group $X(G)$ is the group of group-like elements in $\mathcal{O}(G)$. Since every representation decomposes into one-dimensional sub-representations, $\mathcal{O}(G)$ (as a comodule over itself via the regular representation) decomposes into one-dimensional comodules. The group-like elements span $\mathcal{O}(G)$ as a k -vector space, and the Hopf algebra structure on $\mathcal{O}(G)$ is completely determined by the multiplication of these group-like elements. Specifically, $\mathcal{O}(G) \cong k[X(G)]$ as Hopf algebras.

From abelian groups to group schemes. Given a finitely generated abelian group M , the group algebra $k[M] = \bigoplus_{m \in M} k \cdot e_m$ with multiplication $e_m \cdot e_{m'} = e_{m+m'}$ is a commutative k -algebra (since M is abelian). The Hopf structure $\Delta(e_m) = e_m \otimes e_m$ makes each e_m group-like, and $D(M) = \mathrm{Spec} k[M]$ is a diagonalizable group. Functorially:

$$D(M)(R) = \mathrm{Hom}_{\mathbf{Grp}}(M, R^\times) = \mathrm{Hom}_{\mathbf{Grp}}(M, \mathbb{G}_m(R))$$

for any k -algebra R . This is the group of “ M -graded units” in R .

Quasi-inverse. Starting with G diagonalizable: $D(X(G)) = \operatorname{Spec} k[X(G)] \cong \operatorname{Spec} \mathcal{O}(G) = G$. Starting with M a finitely generated abelian group: $X(D(M)) = \operatorname{Hom}(D(M), \mathbb{G}_m) = \operatorname{Hom}_{\operatorname{Hopf}}(k[t, t^{-1}], k[M])$. A Hopf map $k[t, t^{-1}] \rightarrow k[M]$ sends $t \mapsto e_m$ for some $m \in M$ (the group-like elements of $k[M]$ are exactly $\{e_m : m \in M\}$). This gives $X(D(M)) \cong M$.

Anti-equivalence. A homomorphism $\varphi : G \rightarrow H$ of diagonalizable groups induces a group homomorphism $X(\varphi) : X(H) \rightarrow X(G)$ by $\chi \mapsto \chi \circ \varphi$ (precomposition), reversing arrows. This is functorial and compatible with the quasi-inverse by construction.

Example 5.2.5: The Fundamental Examples

1. $G = \mathbb{G}_m$: The coordinate ring is $k[t, t^{-1}]$ with group-like elements $\{t^n : n \in \mathbb{Z}\}$. So $X(\mathbb{G}_m) = \mathbb{Z}$, and the anti-equivalence sends $\mathbb{Z} \mapsto \mathbb{G}_m$. A character $\chi_n : \mathbb{G}_m \rightarrow \mathbb{G}_m$ corresponding to $n \in \mathbb{Z}$ is $\chi_n(t) = t^n$.
2. $G = \mathbb{G}_m^r$ (the split torus of rank r): $X(\mathbb{G}_m^r) = \mathbb{Z}^r$, with the character $\chi_{(n_1, \dots, n_r)}(t_1, \dots, t_r) = t_1^{n_1} \cdots t_r^{n_r}$.
3. $G = \mu_n$ (over k with $\operatorname{char} k \nmid n$): Functorially, $\mu_n(R) = \{r \in R^\times : r^n = 1\} = \operatorname{Hom}_{\mathbf{Grp}}(\mathbb{Z}/n\mathbb{Z}, R^\times)$. So $\mu_n = D(\mathbb{Z}/n\mathbb{Z})$ and $X(\mu_n) = \mathbb{Z}/n\mathbb{Z}$.
4. $G = D(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}) = \mu_2 \times \mu_3 \cong \mu_6$: The character group of a product is the direct sum, and D sends direct sums to products.

The anti-equivalence translates group-theoretic notions into abelian group theory:

Diagonalizable group G	Character group $X(G)$
Closed subgroup $H \subseteq G$	Quotient $X(G) \twoheadrightarrow X(H)$
Quotient $G \twoheadrightarrow Q$	Subgroup $X(Q) \hookrightarrow X(G)$
G is connected (a torus)	$X(G)$ is torsion-free ($\cong \mathbb{Z}^r$)
G is finite	$X(G)$ is finite
Short exact seq. $1 \rightarrow H \rightarrow G \rightarrow Q \rightarrow 1$	Short exact seq. $0 \rightarrow X(Q) \rightarrow X(G) \rightarrow X(H) \rightarrow 0$

Tori

Definition 5.2.6: Torus (Refined)

An algebraic *torus* over k is a connected diagonalizable group. Equivalently, a torus is an affine algebraic group T such that $T_{\bar{k}} \cong \mathbb{G}_m^r$ for some r . The integer r is the *rank* of T .

By the classification theorem, tori correspond to free abelian groups of finite rank: $X(T) \cong \mathbb{Z}^r$. A torus is called *split* if $T \cong \mathbb{G}_m^r$ over k itself (not just over $\bar{k}$). Over an algebraically closed field, every torus is split.

Over a non-algebraically closed field, non-split tori arise naturally. Recall from example 1.2.4 the Weil restriction $\operatorname{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m, \mathbb{C}})$: this is a 2-dimensional algebraic group over $\mathbb{R}$ with $\operatorname{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m, \mathbb{C}})(\mathbb{R}) = \mathbb{C}^\times$.

Over $\mathbb{C}$, it splits as $\mathbb{G}_m \times \mathbb{G}_m$, but over $\mathbb{R}$ it is non-split: there is no $\mathbb{R}$ -isomorphism $\text{Res}_{\mathbb{C}/\mathbb{R}}(\mathbb{G}_{m,\mathbb{C}}) \cong \mathbb{G}_{m,\mathbb{R}}^2$. The Galois group $\text{Gal}(\mathbb{C}/\mathbb{R}) = \{1, \sigma\}$ acts on $X(T_{\mathbb{C}}) = \mathbb{Z}^2$ by swapping the two factors, and this action encodes the “twisting” of the torus.

The structure of non-split tori over a general field k is controlled by the Galois action on $X(T) = X(T_{\bar{k}})$. We will not develop this theory further (it belongs to the arithmetic side of the subject), but the reader should keep in mind that over non-algebraically closed fields, tori carry a Galois-theoretic richness beyond their geometry.

Proposition 5.2.7: Basic Properties of Tori

Let T be an algebraic torus over k .

1. T is commutative.
2. Every homomorphism $T \rightarrow T'$ between tori is determined by the induced map $X(T') \rightarrow X(T)$ on character groups.
3. Every closed subgroup of T is diagonalizable (hence a product of a smaller torus and a finite diagonalizable group).
4. T has no unipotent elements other than the identity: $T_u = \{e\}$.

Proof:

(1): Over $\bar{k}$, $T \cong \mathbb{G}_m^r$ is visibly commutative. Commutativity is preserved by base change and descent.

(2): This is a special case of the anti-equivalence (theorem 5.2.4): homomorphisms $T \rightarrow T'$ correspond bijectively to group homomorphisms $X(T') \rightarrow X(T)$.

(3): A closed subgroup $H \subseteq T$ corresponds, under the anti-equivalence, to a quotient $X(T) \twoheadrightarrow X(H)$. Since $X(T) \cong \mathbb{Z}^r$ is free, $X(H)$ is a finitely generated abelian group (a quotient of $\mathbb{Z}^r$), so $H = D(X(H))$ is diagonalizable.

(4): In $\mathbb{G}_m^r(\bar{k}) = (\bar{k}^\times)^r$, a unipotent element $(t_1, \dots, t_r)$ satisfies $(t_i - 1)^N = 0$ for some N and all i . Since $\bar{k}^\times$ has no nilpotent elements (it is a group under multiplication), this forces $t_i = 1$ for all i .

Cartier Duality for Finite Diagonalizable Groups

For finite diagonalizable group schemes, the classification admits a beautiful reinterpretation as a duality.

Proposition 5.2.8: Cartier Duality

Let G be a finite commutative group scheme over k . The *Cartier dual* of G is the group scheme $G^\vee := \text{Hom}_{\mathbf{Grp}}(G, \mathbb{G}_m)$ (the functor $R \mapsto \text{Hom}_{\mathbf{Grp}_R}(G_R, \mathbb{G}_{m,R})$). If G is diagonalizable with $G = D(M)$ for a finite abelian group M , then:

$$G^\vee \cong D(M^\vee)$$

where $M^\vee = \text{Hom}_{\mathbb{Z}}(M, \mathbb{Z}/n\mathbb{Z})$ is the Pontryagin dual (for M killed by n). In particular:

1. $(\mu_n)^\vee \cong \mathbb{Z}/n\mathbb{Z}$ (the constant group scheme), when $\text{char } k \nmid n$.
2. $(\mathbb{Z}/n\mathbb{Z})^\vee \cong \mu_n$, when $\text{char } k \nmid n$.
3. $(G^\vee)^\vee \cong G$ (Cartier duality is an involution).

Proof:

By the anti-equivalence, $G = D(M) = \text{Spec } k[M]$ with $G(R) = \text{Hom}_{\mathbf{Grp}}(M, R^\times)$. The Cartier dual $G^\vee(R) = \text{Hom}_{\mathbf{Grp}_R}(G_R, \mathbb{G}_{m,R})$ consists of characters of G over R , which by the anti-equivalence correspond to group homomorphisms $\mathbb{Z} \rightarrow X(G_R) = M$ tensored with R ... more directly, for $G = \mu_n = D(\mathbb{Z}/n\mathbb{Z})$: $G^\vee(R) = \text{Hom}_{\mathbf{Grp}_R}(\mu_{n,R}, \mathbb{G}_{m,R})$. A homomorphism $\mu_n \rightarrow \mathbb{G}_m$ over R sends t (with $t^n = 1$) to some $u \in R^\times$. The homomorphism property requires $u^n = 1$... but actually we want the “characters of μ_n as a group scheme,” not just on R -points.

The cleanest approach: for $G = D(M)$ finite, the Cartier dual is $G^\vee = D(M^\vee)$ where $M^\vee = \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$ (or equivalently $\text{Hom}(M, \mu_\infty(\bar{k}))$). When $M = \mathbb{Z}/n\mathbb{Z}$ and $\text{char } k \nmid n$, the group μ_n is étale and $\mathbb{Z}/n\mathbb{Z}$ is the constant group scheme, and $(\mathbb{Z}/n\mathbb{Z})^\vee = D(\text{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z})) = D(\mathbb{Z}/n\mathbb{Z}) = \mu_n$, while $(\mu_n)^\vee = D((\mathbb{Z}/n\mathbb{Z})^\vee) = D(\mathbb{Z}/n\mathbb{Z})$... The Pontryagin dual of $\mathbb{Z}/n\mathbb{Z}$ is $\mathbb{Z}/n\mathbb{Z}$, so $(\mu_n)^\vee = D(\mathbb{Z}/n\mathbb{Z}) = \mu_n$? No: $\mu_n = D(\mathbb{Z}/n\mathbb{Z})$, and the character group of μ_n is $X(\mu_n) = \mathbb{Z}/n\mathbb{Z}$, so $(\mu_n)^\vee$ has character group $(\mathbb{Z}/n\mathbb{Z})^\vee \cong \mathbb{Z}/n\mathbb{Z}$, giving $(\mu_n)^\vee \cong D(\mathbb{Z}/n\mathbb{Z}) = \mu_n$.

Actually, the duality exchanges “multiplicative type” and “étale type.” Let us state it more carefully. The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k = \text{Spec } (k^n)$ (when $\text{char } k \nmid n$) represents the functor $R \mapsto \mathbb{Z}/n\mathbb{Z}$ (the constant functor). Its Cartier dual is μ_n because a homomorphism $(\mathbb{Z}/n\mathbb{Z})_k \rightarrow \mathbb{G}_m$ is determined by the image of $1 \in \mathbb{Z}/n\mathbb{Z}$, which must be an n -th root of unity. Conversely, $\mu_n^\vee \cong (\mathbb{Z}/n\mathbb{Z})_k$ because a homomorphism $\mu_n \rightarrow \mathbb{G}_m$ is determined by its effect on t (the tautological coordinate with $t^n = 1$), and the possible images are the n group homomorphisms $\mu_n \rightarrow \mathbb{G}_m$ given by $t \mapsto t^j$ for $j \in \mathbb{Z}/n\mathbb{Z}$. The biduality $(G^\vee)^\vee \cong G$ follows formally.

Cartier duality is the algebraic-group analogue of Pontryagin duality for finite abelian groups. In characteristic p , the duality extends to non-étale group schemes: $(\alpha_p)^\vee \cong \alpha_p$ (the group α_p is self-dual), and $(\mu_p)^\vee$ is the constant group scheme $\mathbb{Z}/p\mathbb{Z}$ in a suitable sense (using the flat topology rather than the étale topology). We will not develop these extensions, but they underlie the deeper duality theories (Cartier duality for p -divisible groups, duality of abelian varieties) that appear in the abelian varieties book.

Weight Spaces and the Decomposition of Representations

The most important property of tori for the structure theory is that their representations decompose completely into one-dimensional pieces, each labeled by a character. This is the mechanism that produces root spaces in Chapter 7.

Theorem 5.2.9: Weight Space Decomposition

Let T be a split torus over k and V a finite-dimensional representation of T . Then V decomposes as a direct sum of *weight spaces*:

$$V = \bigoplus_{\chi \in X(T)} V_{\chi}$$

where $V_{\chi} = \{v \in V \mid t \cdot v = \chi(t) \cdot v \text{ for all } t \in T(R), \text{ all } R\}$. The characters χ with $V_{\chi} \neq 0$ are called the *weights* of V , and $\dim V_{\chi}$ is the *multiplicity* of the weight χ .

Proof:

Since $T \cong \mathbb{G}_m^r$ is split, a representation $\rho : T \rightarrow \mathrm{GL}(V)$ corresponds to r commuting automorphisms $\rho(t_1, 1, \dots, 1), \dots, \rho(1, \dots, 1, t_r)$ of V , each given by the action of a single copy of $\mathbb{G}_m$.

For a single $\mathbb{G}_m$ -action on V , the coaction $\rho : V \rightarrow V \otimes_k k[t, t^{-1}]$ decomposes V into weight spaces $V_n = \{v \in V : \rho(v) = v \otimes t^n\}$ for $n \in \mathbb{Z} = X(\mathbb{G}_m)$. This was established in the exercises of Chapter 2 (?? 1): the coaction condition $\rho(v) = v \otimes t^n$ means $g \cdot v = g^n v$ for all $g \in \mathbb{G}_m(R) = R^{\times}$.

The decomposition $V = \bigoplus_{n \in \mathbb{Z}} V_n$ is a decomposition into eigenspaces for the $\mathbb{G}_m$ -action. For $T = \mathbb{G}_m^r$, we iterate: the r commuting $\mathbb{G}_m$ -actions give a simultaneous eigenspace decomposition $V = \bigoplus_{\chi \in \mathbb{Z}^r} V_{\chi}$, where $V_{\chi} = V_{n_1} \cap \dots \cap V_{n_r}$ for $\chi = (n_1, \dots, n_r)$. The character $\chi = (n_1, \dots, n_r) \in X(T) = \mathbb{Z}^r$ acts by $\chi(t_1, \dots, t_r) = t_1^{n_1} \dots t_r^{n_r}$, confirming $t \cdot v = \chi(t)v$ for $v \in V_{\chi}$.

That only finitely many V_{χ} are nonzero follows from V being finite-dimensional.

Example 5.2.10: Weights of the Adjoint Action of a Maximal Torus on $\mathfrak{gl}_n$

Let $G = \mathrm{GL}_n$ and $T = \mathbb{G}_m^n \subseteq \mathrm{GL}_n$ the diagonal torus, with elements $t = \mathrm{diag}(t_1, \dots, t_n)$. The adjoint action of T on $\mathfrak{gl}_n = M_n(k)$ is:

$$\mathrm{Ad}(t)(E_{ij}) = tE_{ij}t^{-1} = t_i t_j^{-1} E_{ij}$$

where E_{ij} is the elementary matrix with 1 in position (i, j) and 0 elsewhere. The character group is $X(T) = \mathbb{Z}^n$ with standard basis $\{\varepsilon_1, \dots, \varepsilon_n\}$, where $\varepsilon_i(t) = t_i$. The weight of E_{ij} is:

$$\chi_{ij} = \varepsilon_i - \varepsilon_j \in X(T)$$

The weight space decomposition of $\mathfrak{gl}_n$ is:

$$\mathfrak{gl}_n = \mathfrak{t} \oplus \bigoplus_{i \neq j} k \cdot E_{ij}$$

where $\mathfrak{t} = \mathrm{Lie}(T) = \{\text{diagonal matrices}\}$ is the 0-weight space (with weight $\varepsilon_i - \varepsilon_i = 0$ for each E_{ii}), and each off-diagonal E_{ij} spans a one-dimensional weight space with weight $\varepsilon_i - \varepsilon_j$.

The nonzero weights $\{\varepsilon_i - \varepsilon_j : i \neq j\}$ are the *roots* of GL_n . They form a root system of type A_{n-1} in the hyperplane $\{\sum n_i = 0\} \subseteq \mathbb{Z}^n$. The full theory will be developed in Chapter 7, but the reader should appreciate the mechanism: the adjoint action of the torus on the Lie algebra produces a decomposition into one-dimensional eigenspaces, and the “eigenvalues” (characters) organize into a combinatorial structure.

Example 5.2.11: Weights of the Standard Representation

The standard representation $V = k^n$ of GL_n has weight decomposition under $T = \mathbb{G}_m^n$:

$$V = \bigoplus_{i=1}^n k \cdot e_i, \quad \text{with weight } \varepsilon_i \text{ for } e_i$$

since $\mathrm{diag}(t_1, \dots, t_n) \cdot e_i = t_i e_i$.

The symmetric power $\mathrm{Sym}^2(V)$ has weights $\{\varepsilon_i + \varepsilon_j : 1 \leq i \leq j \leq n\}$ (each appearing with multiplicity 1), while the exterior power $\bigwedge^2(V)$ has weights $\{\varepsilon_i + \varepsilon_j : 1 \leq i < j \leq n\}$. More generally, $\mathrm{Sym}^d(V)$ and $\bigwedge^d(V)$ have weights that are sums of d elements from $\{\varepsilon_1, \dots, \varepsilon_n\}$ (with and without repetition, respectively).

Example 5.2.12: The Torus Acting on Itself by Conjugation

Since a torus T is commutative, the adjoint action $\mathrm{Ad} : T \rightarrow \mathrm{GL}(\mathfrak{t})$ is trivial (example 3.2.10): $\mathrm{Ad}(t)(X) = X$ for all $t \in T$ and $X \in \mathfrak{t}$. So $\mathfrak{t}$ is entirely the 0-weight space: all weights of the adjoint action of T on $\mathfrak{t}$ are zero. The interesting weights will come from the adjoint action of T on the *ambient* Lie algebra $\mathfrak{g}$ (as in example 5.2.10), not from T acting on its own Lie algebra.

Rigidity of Tori

Tori have a remarkable rigidity property: their automorphism groups are discrete, and in particular they have no infinitesimal automorphisms. This is in sharp contrast to unipotent groups, which are highly deformable.

Proposition 5.2.13: Rigidity of Tori

Let T be a torus and G any affine algebraic group. Then $\mathrm{Hom}(G, T)$ is a discrete (finite) set. In particular:

1. $\mathrm{Aut}(T) \cong \mathrm{GL}_r(\mathbb{Z})$ (the automorphism group of $\mathbb{Z}^r$), which is discrete.
2. If G is connected and T is a torus in G (i.e. $T \subseteq G$ is a closed subgroup that is a torus), then $T \subseteq Z_G(T)^\circ$ (the torus lies in the connected component of its own centralizer).
3. Any two conjugate tori in G are conjugate by an element of G° .

Proof:

A homomorphism $\varphi : G \rightarrow T$ corresponds, via the anti-equivalence, to a group homomorphism $X(T) \rightarrow X(G)$. But $X(T) = \mathbb{Z}^r$ is a free abelian group, and $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}^r, X(G))$ is determined by the images of the r generators, each of which must be a character of G . Since the character group $X(G)$ is a finitely generated abelian group (hence discrete), the set $\text{Hom}(G, T) = \text{Hom}(\mathbb{Z}^r, X(G))$ is discrete.

For (1): $\text{Aut}(T) = \text{Hom}(T, T)^{\times}$ corresponds to $\text{Aut}(X(T)) = \text{Aut}(\mathbb{Z}^r) = \text{GL}_r(\mathbb{Z})$.

For (2): The conjugation action of G on T (via inner automorphisms) gives a morphism $G \rightarrow \text{Aut}(T)$. Since G is connected and $\text{Aut}(T)$ is discrete, this morphism must be constant, hence trivial. So G centralizes T : $gTg^{-1} = T$ and $gtg^{-1} = t$ for all $g \in G(R)$, $t \in T(R)$. This means $G \subseteq Z_G(T)$, and in particular $T \subseteq Z_G(T)^{\circ}$.

Wait — this argument shows that if G *itself* is connected and maps to $\text{Aut}(T)$, the image is trivial. More precisely: the conjugation map $c : G \rightarrow \text{Aut}(T)$ restricted to $Z_G(T)^{\circ}$ is trivial, and the image of G° under c is a connected subgroup of the discrete group $\text{Aut}(T)$, hence trivial. So G° centralizes T , giving $T \subseteq Z_G(T)^{\circ}$ (when $T \subseteq G^{\circ}$).

Statement (3) follows: if $T_1 = gT_2g^{-1}$ for some $g \in G$, write $g = g_0 \cdot h$ with $g_0 \in G^{\circ}$ and h in a finite set of coset representatives. The conjugation $c_{g_0} : T_2 \rightarrow T_1$ is a morphism between tori, hence determined by its effect on characters. Since G° is connected and acts by automorphisms on the set of tori, the discrete nature of $\text{Aut}(T)$ constrains the possible conjugators.

The rigidity of tori is the reason that “maximal tori” are well-behaved: in Chapter 6, we will prove that all maximal tori in a connected algebraic group G are conjugate, and the rigidity ensures that the conjugacy class of a maximal torus is a robust invariant of G .

Exercise 5.2.1

1. Compute the character group $X(G)$ for: (a) $G = \mathbb{G}_m^2$; (b) $G = \mu_6$ (with $\text{char } k \nmid 6$); (c) $G = \mathbb{G}_m \times \mu_3$ (with $\text{char } k \neq 3$).
2. Let $T = \mathbb{G}_m^2$ with $X(T) = \mathbb{Z}^2$. Enumerate all cocharacters $\lambda \in X_{\vee}(T) = \mathbb{Z}^2$ with $\|\lambda\| \leq 2$. For each, compute the pairing $\langle \chi, \lambda \rangle$ where $\chi = (1, -1) \in X(T)$.
3. Classify all closed subgroups of $\mathbb{G}_m^2$ (over an algebraically closed field k) by classifying all quotients of $\mathbb{Z}^2$. Which are tori? Which are finite?
4. Let $T = \mathbb{G}_m^2$ act on $V = k^2$ via $\rho(t_1, t_2) = \text{diag}(t_1 t_2, t_1 t_2^{-1})$. Compute the weights of V , $V^{\vee}$, $\text{Sym}^2(V)$, and $\bigwedge^2(V)$.
5. Let $T = \left\{ \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} : t \in \mathbb{G}_m \right\} \subseteq \text{SL}_2$. This is a maximal torus in SL_2 . Compute $X(T)$ and the weight decomposition of $\mathfrak{sl}_2$ under the adjoint action of T . Verify that the nonzero weights are ± 2 (matching the computation of example 3.2.9).
6. Let T be a torus in a connected algebraic group G , and let $N_G(T) = \{g \in G : gTg^{-1} = T\}$ be the normalizer. Show that $N_G(T)/Z_G(T)$ is a finite group. (*Hint*: elements of $N_G(T)$ act on T by automorphisms; use the rigidity of tori and the fact that $Z_G(T)$ acts trivially.) This quotient will be the *Weyl group* in Chapter 7.

5.3 Unipotent Groups

Unipotent groups are the opposite extreme from tori: where tori consist entirely of semisimple elements, unipotent groups consist entirely of elements whose semisimple part is the identity. Where tori are rigid and completely determined by discrete data (the character lattice), unipotent groups are “soft” and, in characteristic 0, completely determined by continuous data (the Lie algebra). The simplest examples are $\mathbb{G}_a$ (the additive group) and U_n (the group of upper-triangular unipotent matrices), and in a precise sense all connected unipotent groups are built from copies of $\mathbb{G}_a$.

Definitions and First Properties

Definition 5.3.1: Unipotent Elements and Unipotent Groups (Intrinsic)

Let G be an affine algebraic group over k .

1. An element $g \in G(k)$ is called *unipotent* if $g_s = e$ in the Jordan decomposition (theorem 5.1.4), or equivalently, $\rho(g)$ is unipotent for every representation $\rho : G \rightarrow \mathrm{GL}(V)$.
2. The group G is called *unipotent* if every element of $G(\bar{k})$ is unipotent.

This refines the embedding-dependent definition from Chapter 1 (definition 1.1.11). The intrinsic definition makes no reference to a specific embedding $G \hookrightarrow \mathrm{GL}_n$: the unipotence is detected by the Jordan decomposition, which by theorem 5.1.4 is independent of the representation.

Proposition 5.3.2: Equivalent Characterizations of Unipotent Groups

Let G be an affine algebraic group over an algebraically closed field k . The following are equivalent:

1. G is unipotent.
2. For some (equivalently, every) faithful representation $\rho : G \hookrightarrow \mathrm{GL}(V)$, the image $\rho(G)$ is conjugate to a subgroup of U_n (upper-triangular unipotent matrices).
3. The only irreducible representation of G is the trivial representation.
4. For every representation V of G , the space of invariants V^G is nonzero.

Proof:

(1) $\Rightarrow$ (2): Let $\rho : G \hookrightarrow \mathrm{GL}(V)$ be a faithful representation with $\dim V = n$. Every element of $\rho(G)$ is unipotent, so $(\rho(g) - I)^n = 0$ for all $g \in G(\bar{k})$. We need to find a basis in which all elements of $\rho(G)$ are simultaneously upper-triangular. Consider the subspace $V^G = \{v \in V : g \cdot v = v \text{ for all } g\}$. If $V^G \neq 0$, pick $v_1 \in V^G$ and consider G acting on V/kv_1 , which is again a unipotent action. By induction on $\dim V$, we obtain a full flag $0 \subset kv_1 \subset kv_1 + kv_2 \subset \cdots \subset V$ stabilized by G , meaning G is upper-triangular in the basis $\{v_1, \dots, v_n\}$.

It remains to show $V^G \neq 0$. We prove this by a dimension argument. For $g \in G(\bar{k})$, the endomorphism $\rho(g) - I$ is nilpotent, so $\mathrm{tr}(\rho(g)) = n \neq 0$ (all eigenvalues are 1). Consider

the “average” of $\rho(g) - I$ over G ... actually, the cleanest argument uses the Lie algebra in characteristic 0, and a direct induction in general. Since every $g \in G$ acts unipotently on V , the subspace $\ker(\rho(g) - I)$ is nonzero for each g . For a connected unipotent group, one uses the Lie–Kolchin theorem (which we prove in Chapter 6) to conclude that there is a common eigenvector. Alternatively, we can argue by induction on $\dim G$: if G is connected and unipotent, it has a nontrivial normal connected subgroup N (e.g., the last nontrivial term of the derived series, or the center $Z(G)^\circ$), and by induction $V^N \neq 0$; then G/N acts on V^N unipotently, and by induction $V^{G/N} = (V^N)^{G/N} = V^G \neq 0$.

(2) $\Rightarrow$ (1): If $G \subseteq U_n$, every element has all eigenvalues equal to 1, hence is unipotent.

(1) $\Rightarrow$ (3): Let V be an irreducible representation of G . By (1) $\Rightarrow$ (4) (once we establish it), $V^G \neq 0$. But V^G is a subrepresentation (the trivial one), and V is irreducible, so $V = V^G$, meaning G acts trivially on V . Thus V is the trivial representation.

(3) $\Rightarrow$ (4): By the Fundamental Finiteness Theorem (theorem 2.2.6), V contains a finite-dimensional sub-comodule, which in turn contains an irreducible sub-comodule (by choosing one of minimal dimension). By (3), this irreducible sub-comodule is trivial, so it is contained in V^G .

(4) $\Rightarrow$ (1): For any $g \in G(\bar{k})$, consider the one-dimensional representation $V = \bar{k}$ on which g acts by multiplication by g_s (the semisimple part, viewed as a scalar in $\bar{k}^\times$ via some character). If $g_s \neq e$, this representation has $V^G = 0$ (since g_s acts nontrivially), contradicting (4). Hence $g_s = e$ for all g , meaning G is unipotent.

Example 5.3.3: The Basic Unipotent Groups

1. $\mathbb{G}_a$: The additive group $\mathbb{G}_a(R) = (R, +)$ is unipotent. The embedding $\mathbb{G}_a \hookrightarrow \mathrm{GL}_2$ via $t \mapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ lands in U_2 .
2. U_n : The group of $n \times n$ upper-triangular matrices with 1's on the diagonal. This is the prototypical unipotent group. It has dimension $\binom{n}{2}$ (the number of strictly upper-triangular entries).
3. The *Heisenberg group*: $H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in k \right\} \subseteq U_3$. This is a 3-dimensional non-commutative unipotent group. Its center is $Z(H) = \left\{ \begin{pmatrix} 1 & 0 & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\} \cong \mathbb{G}_a$, and $H/Z(H) \cong \mathbb{G}_a^2$.
4. In characteristic $p > 0$: the infinitesimal group $\alpha_p = \mathrm{Spec} k[t]/(t^p)$ (example 3.3.4) is unipotent. It is connected, 0-dimensional, and its only k -point is the identity. This is a unipotent group that is “invisible” to classical points.

Representations of Unipotent Groups

The characterization (3) above — unipotent groups have no nontrivial irreducible representations — means that the representation theory of unipotent groups is entirely about *extensions*: every representation is a successive extension of copies of the trivial representation.

Corollary 5.3.4: Representations of Unipotent Groups are Unipotent

Let G be a unipotent group and V a finite-dimensional representation of G . Then V admits a *composition series* of subrepresentations

$$0 = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = V$$

with each successive quotient V_i/V_{i-1} isomorphic to the trivial representation k . Equivalently, in some basis, every $g \in G$ acts by an upper-triangular unipotent matrix.

Proof:

By proposition 5.3.2(4), $V^G \neq 0$. Set $V_1 = V^G$ (or a one-dimensional subspace thereof if V^G is larger). Then G acts trivially on V_1 , and G acts unipotently on V/V_1 (since G is still unipotent). By induction on $\dim V$, V/V_1 has a composition series with trivial quotients. Pulling back gives the series for V .

This should be contrasted with the representation theory of tori (theorem 5.2.9), where every representation *splits* as a direct sum of one-dimensional pieces. For unipotent groups, the representations are filterable but not splittable: the extensions are typically nontrivial. We already saw this in example 2.2.10, where the two-dimensional representation $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ of $\mathbb{G}_a$ admits a composition series $0 \subset ke_1 \subset k^2$ with trivial quotients, but the extension does not split.

Structure of Connected Unipotent Groups

Connected unipotent groups admit composition series not just at the level of representations, but at the level of the group itself.

Theorem 5.3.5: Composition Series for Unipotent Groups

Let G be a connected unipotent group over a perfect field k . Then G admits a composition series of closed connected normal subgroups

$$\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_m = G$$

such that each successive quotient G_i/G_{i-1} is isomorphic to $\mathbb{G}_a$.

Proof Sketch:

We use the *descending central series* of G . Define $\mathcal{C}^1(G) = G$ and $\mathcal{C}^{i+1}(G) = [\mathcal{C}^i(G), G]$ (the closed subgroup generated by commutators). Since G is unipotent, this series terminates: $\mathcal{C}^{N+1}(G) = \{e\}$ for some N (this is because the adjoint action of G on $\mathfrak{g}$ is nilpotent — every $\text{ad}(X)$ is nilpotent — which implies the derived series terminates). Each $\mathcal{C}^i(G)/\mathcal{C}^{i+1}(G)$ is a connected commutative unipotent group (commutative because it is a quotient of two consecutive terms of the central series).

It remains to show that a connected commutative unipotent group U admits a composition series with quotients $\mathbb{G}_a$. When $\text{char } k = 0$, the exponential map (see below) provides an isomorphism

$U \cong \mathbb{G}_a^{\dim U}$, so U has a filtration with $\mathbb{G}_a$ -quotients by taking hyperplane sections. When $\text{char } k = p > 0$, one uses the Frobenius: the kernel $\ker(F : U \rightarrow U^{(p)})$ is a nontrivial normal subgroup, and induction on the order of U (or its dimension) completes the argument. In this case, the quotients may also include copies of α_p (an infinitesimal group scheme), but if we restrict to smooth U , the quotients are all $\mathbb{G}_a$.

The theorem says that connected unipotent groups are “built from $\mathbb{G}_a$ ” in the same way that connected tori are “built from $\mathbb{G}_m$ ”: the building blocks are iterated extensions. But where tori are always *split* (the extensions are trivial: $\mathbb{G}_m^r$ is a direct product), unipotent groups can have nontrivial extensions, producing non-commutative groups like the Heisenberg group.

The Lie Algebra of a Unipotent Group

The key algebraic property of unipotent groups is that their Lie algebras are *nilpotent*: the bracket operation, iterated enough times, always gives zero.

Definition 5.3.6: Nilpotent Lie Algebra

A Lie algebra $\mathfrak{g}$ is called *nilpotent* if the *descending central series* $\mathfrak{g}^1 = \mathfrak{g}$, $\mathfrak{g}^{i+1} = [\mathfrak{g}, \mathfrak{g}^i]$ terminates: $\mathfrak{g}^{N+1} = 0$ for some N . Equivalently, $\text{ad}(X_1) \circ \text{ad}(X_2) \circ \cdots \circ \text{ad}(X_N) = 0$ for all $X_i \in \mathfrak{g}$ and sufficiently large N .

Proposition 5.3.7: Lie Algebra of a Unipotent Group is Nilpotent

Let G be a unipotent affine algebraic group. Then $\mathfrak{g} = \text{Lie}(G)$ is a nilpotent Lie algebra. More precisely, if $G \subseteq U_n$ (upper-triangular unipotent matrices), then $\mathfrak{g} \subseteq \mathfrak{u}_n = \{\text{strictly upper-triangular matrices}\}$, and $\mathfrak{g}^n = 0$.

Proof:

By proposition 5.3.2(2), there exists a faithful representation $\rho : G \hookrightarrow \text{GL}(V)$ such that $\rho(G) \subseteq U_n$. On Lie algebras, $d\rho : \mathfrak{g} \hookrightarrow \mathfrak{gl}(V)$ maps $\mathfrak{g}$ into $\mathfrak{u}_n$ (the Lie algebra of U_n , computed in ?? 2: the strictly upper-triangular matrices). The Lie bracket on $\mathfrak{u}_n$ is $[X, Y] = XY - YX$, and for strictly upper-triangular matrices, the product XY is “more upper-triangular” than X and Y individually (the nonzero entries move further from the diagonal). After at most $n - 1$ iterated brackets, we reach the zero matrix. Hence $\mathfrak{g}^n = 0$.

The Exponential Map in Characteristic 0

In characteristic 0, the relationship between unipotent groups and nilpotent Lie algebras is as tight as possible: they are equivalent, via the exponential map.

For a nilpotent matrix $X \in \mathfrak{u}_n(k)$ with $X^n = 0$, the exponential series

$$\exp(X) = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots + \frac{X^{n-1}}{(n-1)!}$$

is a *finite* sum (all terms from X^n onward vanish), so it makes algebraic sense over any field where the denominators $m!$ are invertible — that is, in characteristic 0. The inverse is the logarithm:

$$\log(I + X) = X - \frac{X^2}{2} + \frac{X^3}{3} - \cdots + (-1)^{n-2} \frac{X^{n-1}}{n-1}$$

again a finite sum for nilpotent X .

Theorem 5.3.8: Exponential Isomorphism (Characteristic 0)

Let G be a connected unipotent group over a field k of characteristic 0, and let $\mathfrak{g} = \text{Lie}(G)$. Then the exponential map

$$\exp : \mathfrak{g} \xrightarrow{\sim} G$$

is an isomorphism of k -varieties (though not of groups, unless G is commutative). Its inverse is the logarithm $\log : G \xrightarrow{\sim} \mathfrak{g}$.

Proof Sketch:

Embed $G \hookrightarrow U_n \subseteq \text{GL}_n$ via a faithful representation. On the ambient group U_n , the exponential $\exp : \mathfrak{u}_n \rightarrow U_n$ is a polynomial isomorphism of varieties (a finite sum, with polynomial inverse $\log$). It restricts to a morphism $\exp : \mathfrak{g} \rightarrow G$ because $\mathfrak{g}$ is a Lie subalgebra of $\mathfrak{u}_n$ and G is a closed subgroup of U_n . The key point is that the Baker–Campbell–Hausdorff (BCH) formula

$$\log(\exp(X)\exp(Y)) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}([X, [X, Y]] - [Y, [X, Y]]) + \cdots$$

is again a finite sum (since $\mathfrak{g}$ is nilpotent), and it shows that the group law on G is expressed entirely in terms of the Lie bracket on $\mathfrak{g}$. In particular, $\exp$ maps $\mathfrak{g}$ bijectively onto G (bijectivity follows from $\log$ being an inverse), and it is a morphism of varieties (being a polynomial map).

That $\exp$ is not a group homomorphism (unless G is commutative) is visible from the BCH formula: $\exp(X)\exp(Y) = \exp(X + Y + \frac{1}{2}[X, Y] + \cdots) \neq \exp(X + Y)$ in general. But it is an isomorphism of the underlying *varieties*, which suffices to transfer information between G and $\mathfrak{g}$.

Corollary 5.3.9: Unipotent Groups in Characteristic 0 are Determined by $\mathfrak{g}$

Over a field of characteristic 0, the functor $G \mapsto \text{Lie}(G)$ is an equivalence of categories:

$$\left\{ \begin{array}{c} \text{Connected unipotent} \\ \text{algebraic groups} \end{array} \right\} \xleftrightarrow{\sim} \left\{ \begin{array}{c} \text{Finite-dimensional} \\ \text{nilpotent Lie algebras} \end{array} \right\}$$

The quasi-inverse sends a nilpotent Lie algebra $\mathfrak{g}$ to the group with underlying variety $\mathfrak{g}$ and multiplication given by the BCH formula.

This is a remarkable contrast with tori: tori are classified by discrete data ($X(T) \cong \mathbb{Z}^r$, a lattice), while unipotent groups in characteristic 0 are classified by continuous data ($\mathfrak{g}$, a nilpotent Lie algebra). Together, they illustrate the two extremes of the Jordan decomposition: the “semisimple side” is rigid and combinatorial, the “unipotent side” is soft and Lie-theoretic.

Failure of the Exponential in Characteristic p

In characteristic $p > 0$, the exponential map breaks down because the denominators $m!$ are not invertible for $m \geq p$. This has dramatic consequences for the structure theory.

Example 5.3.10: The Exponential Fails for U_2 in Characteristic p

Let k have characteristic p and consider $\mathfrak{u}_2 = \{X = \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix} : a \in k\}$. The formal exponential gives:

$$\exp(X) = I + X = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

which is well-defined (only the first two terms survive since $X^2 = 0$). So for U_2 , the exponential works fine even in characteristic p .

Now consider $\mathfrak{u}_3$. For $X = \begin{pmatrix} 0 & a & c \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}$:

$$\exp(X) = I + X + \frac{X^2}{2} = \begin{pmatrix} 1 & a & c + ab/2 \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

When $\text{char } k = 2$, the term $ab/2$ is undefined (division by $2 = 0$). The exponential map $\mathfrak{u}_3 \rightarrow U_3$ does not exist as a morphism of varieties in characteristic 2.

Example 5.3.11: Non-Isomorphic Unipotent Groups with the Same Lie Algebra

In characteristic p , the functor Lie is not faithful on unipotent groups. The simplest example: the group schemes α_p and $\mathbb{G}_a[F] := \ker(F : \mathbb{G}_a \rightarrow \mathbb{G}_a)$ (the Frobenius kernel of $\mathbb{G}_a$) are both connected unipotent group schemes of dimension 0 with $\text{Lie} = k$. But $\alpha_p = \mathbb{G}_a[F]$ (they are the same group scheme by definition), so this is not yet a counterexample.

A genuine example: consider the two-dimensional unipotent groups over a field of characteristic $p \geq 3$:

$$G_1 = \mathbb{G}_a \times \mathbb{G}_a, \quad G_2 = \{(a, b) \in \mathbb{G}_a^2 : \text{mult. } (a_1, b_1)(a_2, b_2) = (a_1 + a_2, b_1 + b_2 + a_1^p a_2)\}$$

Both have Lie algebra $\mathfrak{g} = k^2$ with trivial bracket (since both are extensions of $\mathbb{G}_a$ by $\mathbb{G}_a$, the Lie algebra is abelian). But G_1 is commutative while G_2 is not (G_2 is a nontrivial extension using the p -th power, which is invisible to Lie). The groups G_1 and G_2 are not isomorphic, yet $\text{Lie}(G_1) \cong \text{Lie}(G_2)$.

These examples reinforce the moral from Chapter 3 (§3.3): in characteristic p , the Lie algebra is an imperfect tool. For unipotent groups, the correct replacement is the p -Lie algebra (restricted Lie algebra), which encodes the p -th power operation and determines the infinitesimal structure more faithfully. A full treatment belongs to Part II.

Interaction Between Tori and Unipotent Groups

The Jordan decomposition says that inside a commutative group, the semisimple and unipotent elements form complementary subgroups (?? 5). For general (non-commutative) groups, the interaction is more subtle, but one clean statement holds:

Proposition 5.3.12: Tori and Unipotent Groups Have Trivial Intersection

Let G be an affine algebraic group over an algebraically closed field k , $T \subseteq G$ a torus, and $U \subseteq G$ a unipotent subgroup. If $T \cap U$ is a subgroup scheme, then $T \cap U = \{e\}$.

Proof:

Every element of $(T \cap U)(\bar{k})$ is simultaneously semisimple (since T is a torus and all its elements are semisimple, proposition 5.2.7(4)) and unipotent (since U is unipotent). An element that is both semisimple and unipotent must be the identity: in any faithful representation, its only eigenvalue is 1 (unipotent) and it is diagonalizable (semisimple), so it is the identity matrix. Hence $(T \cap U)(\bar{k}) = \{e\}$.

At the scheme level: $T \cap U$ is a closed subgroup of T (hence diagonalizable by proposition 5.2.7(3)) and also a closed subgroup of U (hence unipotent). A diagonalizable unipotent group scheme has $X(T \cap U) = 0$ (every character of a unipotent group is trivial, since a unipotent element cannot map to a nontrivial element of $\mathbb{G}_m$), so $T \cap U = D(0) = \{e\}$.

This proposition is the group-theoretic reason that the Jordan decomposition works: the “torus part” and the “unipotent part” of an element cannot interfere with each other. In Chapter 6, this will ensure that the decomposition of a solvable group B into its torus T and unipotent radical U is well-behaved: $T \cap U = \{e\}$ and (in many cases) $B = T \ltimes U$.

Example 5.3.13: The Borel as a Semidirect Product

Let $B_n \subseteq \mathrm{GL}_n$ be the upper-triangular Borel subgroup. Define $T = \mathbb{G}_m^n$ (the diagonal torus) and $U = U_n$ (the upper-triangular unipotent matrices). Then $T \cap U = \{I\}$ and $B_n = T \ltimes U$ (semidirect product): every upper-triangular invertible matrix factors uniquely as a diagonal matrix times a unipotent upper-triangular matrix. Functorially, for any k -algebra R :

$$B_n(R) = T(R) \ltimes U_n(R), \quad \begin{pmatrix} a_1 & * & \cdots \\ 0 & a_2 & \cdots \\ \vdots & & \ddots \end{pmatrix} = \begin{pmatrix} a_1 & & \\ & a_2 & \\ & & \ddots \end{pmatrix} \begin{pmatrix} 1 & a_1^{-1}* & \cdots \\ 0 & 1 & \cdots \\ \vdots & & \ddots \end{pmatrix}$$

The torus T acts on U by conjugation: $\mathrm{diag}(t_1, \dots, t_n) \cdot (I + X) = I + tXt^{-1}$, where the (i, j) -entry of tXt^{-1} is $t_i t_j^{-1} X_{ij}$. The weights of this action are exactly the *positive roots* $\varepsilon_i - \varepsilon_j$ for $i < j$ (cf. example 5.2.10).

This example is the concrete prototype for the general theory of Chapter 6: every Borel subgroup B of a reductive group decomposes as $B = T \ltimes U$ (its maximal torus times its unipotent radical), and the weights of T acting on $\mathrm{Lie}(U)$ are the positive roots.

Exercise 5.3.1

1. Let $G = U_3$ (the 3×3 upper-triangular unipotent group) act on $V = k^3$ by left multiplication. Write down the composition series $0 \subset V_1 \subset V_2 \subset V_3 = V$ with trivial quotients. Is this representation decomposable (i.e. does $V \cong W \oplus W'$ for nonzero subrepresentations)?
2. Let H be the Heisenberg group (the 3×3 upper-triangular unipotent group with 1's on the diagonal and the $(1, 3)$ -entry free). Compute $\text{Lie}(H)$ and verify it is nilpotent. What is the nilpotency class (the smallest N with $\mathfrak{g}^{N+1} = 0$)? Compute the descending central series of H (as an algebraic group).
3. (Requires $\text{char } k = 0$.) Use the BCH formula to compute the group law on the Heisenberg group H from ?? 2 via the exponential map. That is, express $\exp(X) \cdot \exp(Y) = \exp(Z)$ where $X, Y \in \text{Lie}(H)$ and Z is given by the BCH formula.
4. Show that a unipotent group G has no nontrivial characters: $X(G) = 0$. (*Hint*: a character $\chi : G \rightarrow \mathbb{G}_m$ would make $\mathbb{G}_m$ a quotient of G ; use the fact that a quotient of a unipotent group is unipotent, and $\mathbb{G}_m$ is not unipotent.)
5. Let G be a connected commutative unipotent group over an algebraically closed field k of characteristic 0. Show that $G \cong \mathbb{G}_a^n$ for some n . (*Hint*: use the exponential isomorphism $G \cong \mathfrak{g}$; since G is commutative, $\mathfrak{g}$ is abelian, so $\mathfrak{g} \cong k^n$ and the BCH formula reduces to $\exp(X)\exp(Y) = \exp(X + Y)$.)
6. Let $B \subseteq \text{SL}_3$ be the upper-triangular Borel subgroup. Write $B = T \ltimes U$ where T is the diagonal torus and U is the unipotent radical. Compute $\dim T$, $\dim U$, and verify $\dim B = \dim T + \dim U$. Determine the weights of the adjoint action of T on $\text{Lie}(U)$.

6

Solvable Groups and Borel Subgroups

We have now assembled the two atomic building blocks — tori (Chapter 5, §5.2) and unipotent groups (Chapter 5, §5.3) — and the next step is to understand how they fit together. The answer is the *solvable groups*: groups that admit a composition series with commutative quotients, or equivalently, groups whose derived series terminates. The prototype is the upper-triangular Borel subgroup $B_n \subseteq \mathrm{GL}_n$, which decomposes as a semidirect product $B_n = T \ltimes U_n$ of a torus by a unipotent group (example 5.3.13).

This chapter has two goals. First, we prove the *Lie–Kolchin theorem*, which says that every connected solvable group is conjugate into B_n (the “upper-triangular” form), and derive the structure theorem for solvable groups and Borel’s fixed-point theorem. Second, we use these tools to study the *Borel subgroups* of a general algebraic group G : maximal connected solvable subgroups. The main results are that G/B is a projective variety (the *flag variety*), that all Borel subgroups are conjugate, and that all maximal tori are conjugate. These conjugacy theorems are the gateway to the classification theory of Chapter 7: they ensure that the root system, defined via a choice of maximal torus and Borel, is independent of that choice.

6.1 Solvable Groups and the Lie–Kolchin Theorem

Solvable Algebraic Groups

Definition 6.1.1: Solvable Algebraic Group

Let G be an affine algebraic group. The *derived series* of G is defined by $\mathcal{D}^0(G) = G$ and $\mathcal{D}^{i+1}(G) = [\mathcal{D}^i(G), \mathcal{D}^i(G)]$ (the closed subgroup generated by commutators). The group G is called *solvable* if $\mathcal{D}^n(G) = \{e\}$ for some $n \geq 0$.

Functorially, $\mathcal{D}^{i+1}(G)(R)$ is the subgroup of $G(R)$ generated by the commutators $[g, h] = ghg^{-1}h^{-1}$ with $g, h \in \mathcal{D}^i(G)(R)$. The derived series is a descending chain of closed connected normal subgroups of G (connectedness is preserved at each step because the commutator map $G \times G \rightarrow G$ is a morphism, and the image of a connected scheme is connected).

Example 6.1.2: Solvable Groups

1. *Commutative groups* are solvable: $\mathcal{D}^1(G) = [G, G] = \{e\}$, so the derived series terminates in one step. In particular, tori and $\mathbb{G}_a$ are solvable.
2. *Unipotent groups* are solvable: the descending central series $\mathcal{C}^i(G)$ refines the derived series (since $\mathcal{D}^i(G) \subseteq \mathcal{C}^i(G)$), and we proved in theorem 5.3.5 that the central series terminates.
3. The *Borel subgroup* $B_n \subseteq \mathrm{GL}_n$ of upper-triangular matrices is solvable. We have $\mathcal{D}^1(B_n) = [B_n, B_n] = U_n$ (the commutator of two upper-triangular matrices is upper-triangular unipotent), and U_n is solvable by (2).
4. GL_n is *not* solvable for $n \geq 2$: $\mathcal{D}^1(\mathrm{GL}_n) = \mathrm{SL}_n$ (the commutator subgroup consists of determinant-1 matrices), and $\mathcal{D}^1(\mathrm{SL}_n) = \mathrm{SL}_n$ for $n \geq 2$ (since SL_n is its own commutator subgroup, being “almost simple”). So the derived series stabilizes at SL_n and never reaches $\{e\}$.

Proposition 6.1.3: Basic Properties of Solvable Groups

1. Subgroups and quotients of solvable groups are solvable.
2. Extensions of solvable groups are solvable: if $1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1$ is a short exact sequence with N and Q solvable, then G is solvable.
3. In any affine algebraic group G , there exists a unique maximal connected solvable normal subgroup, called the *radical* of G , denoted $R(G)$.

Proof:

(1): If $H \subseteq G$, then $\mathcal{D}^i(H) \subseteq \mathcal{D}^i(G)$, so $\mathcal{D}^n(G) = \{e\}$ implies $\mathcal{D}^n(H) = \{e\}$. If $\pi : G \rightarrow Q$ is surjective, then $\pi(\mathcal{D}^i(G)) = \mathcal{D}^i(Q)$ (surjective maps preserve commutators), so $\mathcal{D}^n(G) = \{e\}$ implies $\mathcal{D}^n(Q) = \{e\}$.

(2): If $\mathcal{D}^m(N) = \{e\}$ and $\mathcal{D}^r(Q) = \{e\}$, then $\mathcal{D}^r(G)$ maps to $\mathcal{D}^r(Q) = \{e\}$ under π , so $\mathcal{D}^r(G) \subseteq N$. Then $\mathcal{D}^{r+m}(G) = \mathcal{D}^m(\mathcal{D}^r(G)) \subseteq \mathcal{D}^m(N) = \{e\}$.

(3): Let R_1, R_2 be connected solvable normal subgroups of G . The product R_1R_2 is a connected normal subgroup (it is the image of $R_1 \times R_2 \rightarrow G$, which is connected), and it is solvable by (2): the sequence $1 \rightarrow R_1 \rightarrow R_1R_2 \rightarrow R_1R_2/R_1 \rightarrow 1$ has R_1 solvable and $R_1R_2/R_1 \cong R_2/(R_1 \cap R_2)$ solvable (a quotient of R_2). Since G is Noetherian (its subgroups satisfy the ascending chain condition on closed subgroups), the collection of connected solvable normal subgroups has a maximal element $R(G) = R_1R_2 \cdots$.

The Lie–Kolchin Theorem

The fundamental theorem about solvable groups says that in any representation, they can be simultaneously put in upper-triangular form. This is the group-theoretic analogue of the linear-algebraic fact that commuting matrices are simultaneously triangularizable.

Theorem 6.1.4: Lie–Kolchin Theorem

Let G be a connected solvable affine algebraic group over an algebraically closed field k , and let $\rho : G \rightarrow \mathrm{GL}(V)$ be a representation on a finite-dimensional vector space $V \neq 0$. Then there exists a complete flag $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ (with $\dim V_i = i$) that is stabilized by G . Equivalently, in some basis of V , every element of $\rho(G)$ is upper-triangular.

Proof:

We argue by induction on $\dim V$. The base case $\dim V = 1$ is trivial (every endomorphism of a 1-dimensional space is upper-triangular).

Step 1: Find a common eigenvector. We need to show that V contains a one-dimensional subrepresentation: a line $L = kv$ such that $G \cdot L = L$, i.e. $g \cdot v = \chi(g)v$ for some character $\chi : G \rightarrow \mathbb{G}_m$.

We induct on $\dim G$. If $\dim G = 0$, then G is finite and connected over $k = \bar{k}$, so $G = \{e\}$ and every vector is an eigenvector.

If $\dim G > 0$, consider the commutator $N = [G, G]^\circ$ (the identity component of the derived subgroup). Since G is solvable, $N \subsetneq G$: the derived series is strictly decreasing until it reaches $\{e\}$, so $[G, G] \neq G$ (otherwise the series would stabilize). Moreover, N is connected, solvable, normal, and $\dim N < \dim G$.

By the induction hypothesis applied to N acting on V , there exists a character $\chi : N \rightarrow \mathbb{G}_m$ and a nonzero eigenvector $v \in V$ with $n \cdot v = \chi(n)v$ for all $n \in N$. Define the eigenspace:

$$W_\chi = \{w \in V \mid n \cdot w = \chi(n)w \text{ for all } n \in N\}$$

This is nonzero (it contains v).

Step 2: G preserves W_χ . For $g \in G(k)$ and $w \in W_\chi$, we show $g \cdot w \in W_\chi$. For any $n \in N(k)$:

$$n \cdot (g \cdot w) = g \cdot (g^{-1}ng \cdot w) = g \cdot \chi(g^{-1}ng)w$$

Since $N \trianglelefteq G$, we have $g^{-1}ng \in N$. We claim $\chi(g^{-1}ng) = \chi(n)$. The map $g \mapsto (n \mapsto \chi(g^{-1}ng))$ defines, for each $g \in G$, a character of N . This gives a morphism $G \rightarrow X(N)$ (the character group of N). By the rigidity of tori (proposition 5.2.13), $X(N)$ is discrete. Since G is connected, a morphism from G to a discrete set must be constant. At $g = e$, the character is χ itself. So $\chi(g^{-1}ng) = \chi(n)$ for all $g \in G$, and:

$$n \cdot (g \cdot w) = \chi(n)(g \cdot w)$$

Hence $g \cdot w \in W_\chi$, and W_χ is G -stable.

Step 3: Reduce to a smaller representation. If $W_\chi = V$, then N acts by scalars on V (every element acts as $\chi(n) \cdot \text{id}_V$), and the action of G on V factors through G/N . Since G/N is a connected solvable group of smaller dimension, the induction hypothesis on $\dim G$ gives a common eigenvector for G/N , hence for G .

If $W_\chi \subsetneq V$, then W_χ is a G -stable subspace of dimension $< \dim V$. By induction on $\dim V$, G has a common eigenvector in W_χ .

In either case, we obtain a line $V_1 = kv_1$ that is G -stable.

Step 4: Build the flag. The quotient V/V_1 is a representation of G of dimension $n - 1$. By induction on $\dim V$, G preserves a complete flag in V/V_1 . Pulling back gives a complete flag $V_1 \subset V_2 \subset \cdots \subset V_n = V$ stabilized by G .

The proof combines two inductions: one on $\dim V$ (to build the flag step by step) and one on $\dim G$ (to find the initial common eigenvector). The key mechanism is the rigidity of characters: conjugation by the connected group G cannot change the (discrete) character χ of the normal subgroup N .

Corollary 6.1.5: Connected Solvable Groups are Triangularizable

Let G be a connected solvable affine algebraic group over $k = \bar{k}$. Then for any faithful representation $\rho : G \hookrightarrow \text{GL}(V)$, the image $\rho(G)$ is conjugate to a subgroup of the upper-triangular Borel $B_n \subseteq \text{GL}_n$ (where $n = \dim V$).

Proof:

The Lie–Kolchin theorem gives a G -stable flag. Choosing a basis adapted to this flag puts G in upper-triangular form.

Structure of Connected Solvable Groups

The Lie–Kolchin theorem gives us a concrete picture of connected solvable groups: they are “built from a torus and a unipotent part.” We now make this precise.

Definition 6.1.6: Unipotent Radical of a Solvable Group

Let G be a connected solvable affine algebraic group. The *unipotent radical* $R_u(G)$ is the maximal connected normal unipotent subgroup of G .

The existence of $R_u(G)$ follows from the same ascending chain argument as for the radical (proposition 6.1.3(3)): the product of two connected normal unipotent subgroups is again connected normal unipotent (normality is clear, and the product of unipotent groups in a solvable group is unipotent, which follows from the Lie–Kolchin theorem: in upper-triangular form, the product of unipotent upper-triangular matrices is unipotent upper-triangular).

Theorem 6.1.7: Structure Theorem for Connected Solvable Groups

Let G be a connected solvable affine algebraic group over an algebraically closed field k . Then:

1. $R_u(G) = G_u := \{g \in G(k) \mid g \text{ is unipotent}\}$, the set of all unipotent elements, which is a closed connected normal subgroup.
2. $G/R_u(G)$ is a torus.
3. There exists a maximal torus $T \subseteq G$ such that $G = T \ltimes R_u(G)$ (semidirect product). Any two such tori are conjugate by an element of $R_u(G)$.

Proof:

By the Lie–Kolchin theorem, embed $G \hookrightarrow B_n$ for some n . Inside $B_n = T_0 \ltimes U_n$, the unipotent elements form U_n (a matrix is unipotent iff its diagonal entries are all 1). So $G_u = G \cap U_n$ is a closed subgroup.

(1): We show G_u is connected, normal, and equals $R_u(G)$. Normality: if $g \in G$ and $u \in G_u$, then gug^{-1} is unipotent (by proposition 5.1.6, conjugation preserves the Jordan decomposition), so $gug^{-1} \in G_u$. The quotient G/G_u is a connected solvable group all of whose elements are semisimple: in upper-triangular form, quotienting by $G_u = G \cap U_n$ removes the unipotent part, leaving only diagonal elements. A connected group consisting entirely of semisimple elements is a torus (this uses the Lie–Kolchin theorem: the elements are simultaneously diagonalizable, and a connected subgroup of a diagonal torus is a torus). So G/G_u is a torus, hence G_u is connected (the fibers of $G \rightarrow G/G_u$ are cosets of G_u , all isomorphic to G_u ; since G and G/G_u are connected, G_u is connected). Being connected normal unipotent, $G_u \subseteq R_u(G)$. Conversely, $R_u(G) \subseteq G_u$ (every element of $R_u(G)$ is unipotent). So $R_u(G) = G_u$.

(2) follows: $G/R_u(G) = G/G_u$ is a torus.

(3): Inside $B_n = T_0 \ltimes U_n$, the projection $\pi : B_n \rightarrow T_0$ restricts to a homomorphism $\pi|_G : G \rightarrow T_0$ with kernel $G \cap U_n = R_u(G)$. The image $\pi(G)$ is a connected closed subgroup of T_0 , hence a torus T' . The inclusion $T' \hookrightarrow T_0 \hookrightarrow B_n$ composed with the embedding $G \hookrightarrow B_n$ provides a torus $T \subseteq G$ mapping isomorphically onto $G/R_u(G)$ via $\pi|_G$. Since $T \cap R_u(G) = \{e\}$ (proposition 5.3.12) and $\dim T + \dim R_u(G) = \dim G$, we get $G = T \ltimes R_u(G)$.

Conjugacy of maximal tori: if T_1, T_2 are two maximal tori in G , both map isomorphically onto $G/R_u(G)$. The two sections $T_1, T_2 \hookrightarrow G$ of the projection $G \rightarrow G/R_u(G)$ differ by conjugation by an element of $R_u(G)$. This can be proved by induction on $\dim R_u(G)$, using the fact that $H^1(T, \mathbb{G}_a) = 0$ (the first cohomology of a torus acting rationally on an affine space vanishes, since the torus representation on $\mathbb{G}_a$ decomposes into weight spaces and each contributes trivially

to cohomology).

Example 6.1.8: The Structure of B_2

Let $B_2 = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : ad \neq 0 \right\} \subseteq \mathrm{GL}_2$. The derived subgroup is $[B_2, B_2] = \left\{ \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \right\} = U_2 \cong \mathbb{G}_a$ (the commutator of two upper-triangular matrices is upper-triangular unipotent). The unipotent radical is $R_u(B_2) = U_2$. The maximal torus is $T = \left\{ \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \right\} \cong \mathbb{G}_m^2$. We have $B_2 = T \ltimes U_2$, confirming the structure theorem. The torus acts on $U_2 \cong \mathbb{G}_a$ by $\mathrm{diag}(a, d) \cdot \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & ad^{-1}t \\ 0 & 1 \end{pmatrix}$, so the weight of T on $\mathrm{Lie}(U_2) = k$ is $\varepsilon_1 - \varepsilon_2$ — the single positive root of GL_2 .

Borel’s Fixed-Point Theorem

The Lie–Kolchin theorem says that a connected solvable group has a common eigenvector in any *linear* representation. Borel’s fixed-point theorem is the geometric generalization: a connected solvable group has a *fixed point* on any complete variety.

Theorem 6.1.9: Borel’s Fixed-Point Theorem

Let G be a connected solvable affine algebraic group over an algebraically closed field k , and let X be a non-empty complete G -variety. Then G has a fixed point: $X^G \neq \emptyset$.

Proof:

By the Closed Orbit Lemma (theorem 2.4.15), X contains a closed orbit $O = G \cdot x$. Since X is complete and O is closed in X , O is complete. Since O is a single G -orbit, $O \cong G/H$ for some closed subgroup H (theorem 4.3.2), so O is a complete homogeneous space for G .

We claim O is a point. Suppose for contradiction that $\dim O > 0$. Since O is projective (a complete variety), there exists a very ample line bundle $\mathcal{L}$ on O , giving a closed immersion $O \hookrightarrow \mathbb{P}(H^0(O, \mathcal{L})^\vee)$. The G -action on O lifts to a linear action on the finite-dimensional vector space $V = H^0(O, \mathcal{L})^\vee$: this is because $O = G/H$ is a homogeneous space and $\mathcal{L}$ admits a G -linearization (see [3] for the general theory of G -linearized line bundles on homogeneous spaces).

By the Lie–Kolchin theorem (theorem 6.1.4), G has a common eigenvector $v \in V$, hence a fixed point $[v] \in \mathbb{P}(V)$. Since O is a G -stable closed subvariety of $\mathbb{P}(V)$ and $[v]$ is a G -fixed point of $\mathbb{P}(V)$, we need $[v] \in O$ (a fixed point of $\mathbb{P}(V)$ that lies in O is a fixed point of O). But does $[v]$ necessarily lie in O ?

We argue more carefully. The embedding $O \hookrightarrow \mathbb{P}(V)$ is equivariant, and G acts on $\mathbb{P}(V)$ linearly. By Lie–Kolchin, G preserves a complete flag $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$. In particular, G fixes the point $[V_1] \in \mathbb{P}(V)$. The closed subvariety $O \subseteq \mathbb{P}(V)$ is G -stable, and the intersection $O \cap \{[V_1]\}$ is either empty or $\{[V_1]\}$. If it is non-empty, we have a G -fixed point on O , contradicting $\dim O > 0$.

If $[V_1] \notin O$, consider the projection from $[V_1]$: this is a G -equivariant rational map $\mathbb{P}(V) \dashrightarrow \mathbb{P}(V/V_1)$ defined away from $[V_1]$. Since $[V_1] \notin O$, this restricts to a G -equivariant morphism $O \rightarrow \mathbb{P}(V/V_1)$. The image is a closed G -stable subvariety of $\mathbb{P}(V/V_1)$ (since O is complete), and

G acts on V/V_1 with a flag of length $n - 1$. By induction on $n = \dim V$, G has a fixed point in the image, and pulling back (using completeness of O) gives a fixed point on O .

In either case, O contains a G -fixed point. But $O \cong G/H$ is a homogeneous space, and a fixed point means $H = G$, so $\dim O = 0$. Contradiction. Therefore $\dim O = 0$ from the start, and $O = \{x\}$ is a G -fixed point.

The proof uses the projective geometry of X (via ample line bundles) to reduce the geometric fixed-point problem to the linear Lie–Kolchin theorem. This is a fundamental instance of the principle “geometry over algebra”: the complete variety’s projective embedding provides the bridge between group actions on varieties and group actions on vector spaces.

Corollary 6.1.10: Solvable Groups Preserve Flags

Let G be a connected solvable group acting on a flag variety $\mathrm{Fl}(d_1, \dots, d_r; n)$ or a Grassmannian $\mathrm{Gr}(d, n)$. Then G has a fixed point.

Proof:

These varieties are projective (example 4.2.4, corollary 4.3.3), so theorem 6.1.9 applies directly.

Exercise 6.1.1

1. Compute the derived series of $B_3 \subseteq \mathrm{GL}_3$ (the 3×3 upper-triangular Borel). At which step does it reach $\{e\}$? Compare with the descending central series of U_3 from ?? 2.
2. Give a direct proof of the Lie–Kolchin theorem for $G = B_2$ (the 2×2 upper-triangular Borel) acting on $V = k^2$. That is, without appealing to the general theorem, find a common eigenvector for all elements of B_2 .
3. Show that the Lie–Kolchin theorem fails without the connectedness hypothesis. (*Hint*: the symmetric group S_3 is solvable; consider its two-dimensional irreducible representation.)
4. Let $G = \mathrm{GL}_n$. Show that $R(G) = Z(G) = \mathbb{G}_m$ (scalar matrices). Deduce that GL_n is not solvable for $n \geq 2$, but its radical is a torus.
5. Let $G = B_2$ act on $\mathbb{P}^1$ via $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \cdot [x : y] = [ax + by : dy]$. Find the fixed points of this action directly, verifying Borel’s fixed-point theorem.
6. Let $G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a \in \mathbb{G}_m, b \in \mathbb{G}_a \right\}$ (the “ $ax + b$ ” group). Show G is connected solvable. Compute $R_u(G)$, find a maximal torus T , and verify $G = T \ltimes R_u(G)$. Determine the weight of T on $\mathrm{Lie}(R_u(G))$.

6.2 Borel Subgroups and Flag Varieties

We now turn from the internal structure of solvable groups to the question of how solvable subgroups sit inside a general affine algebraic group G . The key objects are the *Borel subgroups* — the maximal connected solvable subgroups — and the quotients G/B , which are the *flag varieties*. The main

results of this section are: G/B is projective, all Borel subgroups are conjugate, and all maximal tori are conjugate. These are proved using Borel's fixed-point theorem as the principal tool, and they are the structural foundations on which the root system classification of Chapter 7 rests.

Throughout this section, k denotes an algebraically closed field and G a connected affine algebraic group over k .

Borel Subgroups

Definition 6.2.1: Borel Subgroup

A *Borel subgroup* of G is a maximal connected solvable closed subgroup $B \subseteq G$. Equivalently, B is a closed connected solvable subgroup that is not properly contained in any other closed connected solvable subgroup.

Borel subgroups exist by a dimension argument: the collection of connected solvable closed subgroups is non-empty (it contains $\{e\}$, or any torus, or any copy of $\mathbb{G}_a$) and partially ordered by inclusion. Since $\dim H \leq \dim G$ for any closed subgroup H , a chain of connected solvable subgroups has bounded dimension, and a maximal element exists.

Example 6.2.2: Borel Subgroups of Classical Groups

1. $G = \mathrm{GL}_n$: The standard Borel is $B_n = \{\text{upper-triangular invertible matrices}\}$. It is connected solvable (example 6.1.2(3)), and it is maximal because any connected solvable subgroup of GL_n is conjugate into B_n by the Lie–Kolchin theorem (corollary 6.1.5), hence has dimension $\leq \dim B_n = \binom{n+1}{2}$.
2. $G = \mathrm{SL}_n$: The standard Borel is $B = B_n \cap \mathrm{SL}_n$ (upper-triangular matrices of determinant 1). It has dimension $\binom{n+1}{2} - 1 = \binom{n}{2} + (n-1)$.
3. $G = \mathrm{Sp}_{2n}$: The standard Borel consists of those elements of Sp_{2n} that preserve the standard isotropic flag. In a suitable basis, these are the “upper-triangular” elements of Sp_{2n} , forming a group of dimension n^2 .
4. G commutative: If G is commutative (e.g. a torus), then G itself is solvable, so $B = G$. A commutative group has a unique Borel subgroup: itself.

Projectivity of G/B

The first main theorem says that the quotient of G by a Borel subgroup is always projective. This is the result that makes the “geometry over algebra” program work: it gives us a projective variety G/B on which we can use the tools of projective geometry (ample line bundles, intersection theory, Schubert calculus) to study G .

Theorem 6.2.3: G/B is Projective

Let G be a connected affine algebraic group and $B \subseteq G$ a Borel subgroup. Then G/B is a projective variety.

Proof:

By theorem 4.2.2, the quotient G/B exists as a quasi-projective variety: there is a representation V of G and a line $L \subseteq V$ with $B = \text{Stab}_G(L)$ (by Chevalley's theorem, theorem 4.2.1), and G/B is the orbit $G \cdot [L] \subseteq \mathbb{P}(V)$, which is locally closed. We need to show this orbit is in fact *closed* in $\mathbb{P}(V)$.

By the Closed Orbit Lemma (theorem 2.4.15), the closure $\overline{G \cdot [L]}$ is a G -stable closed subvariety of $\mathbb{P}(V)$, and the boundary $\overline{G \cdot [L]} \setminus G \cdot [L]$ is a union of orbits of strictly lower dimension. It suffices to show that there are *no* orbits of lower dimension, i.e. $G \cdot [L]$ is already closed.

Consider any closed orbit $O \subseteq \overline{G \cdot [L]}$ (which exists by the Closed Orbit Lemma). The Borel subgroup B acts on $\overline{G \cdot [L]}$ by restriction of the G -action, and O is a closed B -stable subvariety of the projective variety $\overline{G \cdot [L]}$. Since O is projective (closed in a projective variety) and B is connected solvable, Borel's fixed-point theorem (theorem 6.1.9) gives a B -fixed point $[v] \in O$.

A B -fixed point in $\mathbb{P}(V)$ is a line $kv \subseteq V$ stabilized by B : for every $b \in B(R)$, $b \cdot v = \chi(b)v$ for some character $\chi : B \rightarrow \mathbb{G}_m$. In particular, $B \subseteq \text{Stab}_G(kv)$. But B is a *maximal* connected solvable subgroup, and $\text{Stab}_G(kv)$ contains B and is a closed subgroup of G . If $\text{Stab}_G(kv)^\circ$ were solvable, maximality of B would force $\text{Stab}_G(kv)^\circ = B$, meaning $[v] \in G \cdot [L]$ (since both $[L]$ and $[v]$ have the same stabilizer up to connected components).

We need to verify that $\text{Stab}_G(kv)^\circ$ is indeed solvable, or at least that $[v]$ lies in $G \cdot [L]$. Here we use the specific choice of L from Chevalley's theorem more carefully. In the proof of theorem 4.2.1, the line $L = \bigwedge^d V_0$ was constructed so that $B = \text{Stab}_G(L)$ *exactly* (not just the connected component). The orbit $G \cdot [L]$ parametrizes all lines of the form $g \cdot L$ for $g \in G$, and two points $g_1 \cdot [L] = g_2 \cdot [L]$ iff $g_1^{-1}g_2 \in B$.

Now, the B -fixed point $[v] \in O \subseteq \overline{G \cdot [L]}$ satisfies $B \subseteq \text{Stab}_G(kv)$. The orbit $G \cdot [v] = O$ has stabilizer $\text{Stab}_G(kv)$, and $\dim O = \dim G - \dim \text{Stab}_G(kv) \leq \dim G - \dim B = \dim(G/B)$. But $G \cdot [L]$ has $\dim(G \cdot [L]) = \dim G - \dim B$ as well (since $\text{Stab}_G(L) = B$). If $O \neq G \cdot [L]$, then O would be a *different* orbit contained in $\overline{G \cdot [L]}$ with $\dim O \leq \dim(G \cdot [L])$. By the Closed Orbit Lemma, boundary orbits have strictly smaller dimension, so we need $\dim O < \dim(G \cdot [L])$, meaning $\dim \text{Stab}_G(kv) > \dim B$.

Suppose $\dim \text{Stab}_G(kv) > \dim B$. Since $B \subseteq \text{Stab}_G(kv)^\circ$ and $\dim \text{Stab}_G(kv)^\circ > \dim B$, the connected group $\text{Stab}_G(kv)^\circ$ properly contains B . But B is a Borel subgroup (maximal connected solvable), so $\text{Stab}_G(kv)^\circ$ is *not* solvable. This means $\text{Stab}_G(kv)^\circ$ is a connected non-solvable group properly containing B .

We now use the following lemma:

Lemma. If B is a Borel subgroup of G and P is a closed subgroup with $B \subseteq P \subseteq G$, then B is

also a Borel subgroup of P° .

Proof of Lemma. B is connected, solvable, and contained in P° . If $B' \subseteq P^\circ$ were a larger connected solvable subgroup containing B , then $B' \subseteq G$ would also be connected solvable, contradicting the maximality of B in G . $\square$

By the Lemma, B is a Borel subgroup of $H := \text{Stab}_G(kv)^\circ$. The quotient H/B is a quasi-projective variety of positive dimension (since $\dim H > \dim B$). Repeating the argument: H/B is quasi-projective and B acts on it by left translation. But H/B embeds into G/B , and the orbit $G \cdot [L]$ restricted to H gives $H \cdot [L] = H/B$ (since $\text{Stab}_H(L) = H \cap B = B$). We have an inclusion $H/B \subseteq \overline{G \cdot [L]}$ and H/B is a B -orbit in $\overline{G \cdot [L]}$. Its closure is H -stable and closed in $\overline{G \cdot [L]}$, hence projective. By Borel's fixed-point theorem, B has a fixed point on $\overline{H/B}$, which must be a point with stabilizer $\supsetneq B$, and we can repeat.

This argument produces an ascending chain $B \subseteq H_1 \subseteq H_2 \subseteq \dots$ of stabilizers, each properly containing the previous. Since $\dim G$ is finite, this chain terminates, which is a contradiction. Therefore no orbit in $\overline{G \cdot [L]}$ other than $G \cdot [L]$ can exist, and $G \cdot [L]$ is closed.

Since $G \cdot [L]$ is closed in $\mathbb{P}(V)$ and $\mathbb{P}(V)$ is projective, $G/B \cong G \cdot [L]$ is projective.

The projectivity of G/B is one of the most consequential results in the structure theory. It means that the flag variety G/B is a “compact” geometric object (in the algebraic sense), on which intersection theory, Schubert calculus, and the Borel–Weil theorem can be applied. The proof demonstrates the power of Borel's fixed-point theorem: by showing that B always finds a fixed point on any orbit closure, it forces the orbit to be closed.

Conjugacy of Borel Subgroups

The second main theorem says that all Borel subgroups of G are conjugate. This is an essential structural result: it means that the “shape” of a Borel subgroup is an invariant of G , not of an arbitrary choice.

Theorem 6.2.4: Conjugacy of Borel Subgroups

Let G be a connected affine algebraic group over an algebraically closed field k . Then any two Borel subgroups of G are conjugate: if B, B' are Borel subgroups, there exists $g \in G(k)$ with $B' = gBg^{-1}$.

Proof:

Consider the projective variety G/B (theorem 6.2.3). The Borel subgroup B' acts on G/B by left translation: $b' \cdot gB = (b'g)B$. Since G/B is projective and B' is connected solvable, Borel's fixed-point theorem (theorem 6.1.9) gives a fixed point $gB \in G/B$, meaning $b'gB = gB$ for all $b' \in B'$. Equivalently, $g^{-1}b'g \in B$ for all $b' \in B'$, i.e. $g^{-1}B'g \subseteq B$.

Since $g^{-1}B'g$ is connected solvable (conjugation is an isomorphism) and B is a Borel, maximality of B and $g^{-1}B'g$ (both are Borel subgroups, and one is contained in the other) forces $g^{-1}B'g = B$, i.e. $B' = gBg^{-1}$.

The proof is a beautiful application of the “geometry over algebra” principle: the conjugacy of Borel subgroups, a purely group-theoretic statement, is proved by finding a fixed point on the *projective variety* G/B . Without the projectivity of G/B , Borel’s fixed-point theorem could not be applied.

Conjugacy of Maximal Tori

Every Borel subgroup contains a maximal torus (theorem 6.1.7(3)), and all maximal tori inside a fixed Borel are conjugate (by elements of the unipotent radical). Combined with the conjugacy of Borel subgroups, this gives:

Theorem 6.2.5: Conjugacy of Maximal Tori

Let G be a connected affine algebraic group over an algebraically closed field k . Then:

1. Every maximal torus of G is contained in a Borel subgroup.
2. Any two maximal tori of G are conjugate.

Proof:

(1): Let T be a maximal torus of G . Since T is connected solvable (it is commutative), it is contained in some connected solvable subgroup. By Zorn’s lemma (or the finite-dimensionality argument), T is contained in a maximal connected solvable subgroup, i.e. a Borel subgroup B .

(2): Let T_1, T_2 be maximal tori. By (1), $T_i \subseteq B_i$ for Borel subgroups B_1, B_2 . By theorem 6.2.4, there exists $g \in G(k)$ with $B_2 = gB_1g^{-1}$. Then gT_1g^{-1} is a maximal torus in B_2 (conjugation preserves tori and maximality). But T_2 is also a maximal torus in B_2 . By the conjugacy of maximal tori inside a connected solvable group (theorem 6.1.7(3)), there exists $u \in R_u(B_2)$ with $T_2 = u(gT_1g^{-1})u^{-1} = (ug)T_1(ug)^{-1}$.

The conjugacy of maximal tori has a profound consequence: the *rank* of G (the dimension of a maximal torus) is well-defined, as is the isomorphism type of the character lattice $X(T)$. When we define root systems in Chapter 7, the conjugacy theorem will guarantee that the root system is independent of the choice of maximal torus and Borel subgroup.

Example 6.2.6: Maximal Tori in GL_n and SL_n

In GL_n , the standard maximal torus is the diagonal torus $T = \mathbb{G}_m^n$, of rank n . Any other maximal torus is of the form gTg^{-1} for some $g \in GL_n(k)$: it consists of the matrices simultaneously diagonalizable by the basis $\{ge_1, \dots, ge_n\}$.

In SL_n , the standard maximal torus is $T = \{\text{diag}(t_1, \dots, t_n) : t_1 \cdots t_n = 1\} \cong \mathbb{G}_m^{n-1}$, of rank $n - 1$. Its character lattice is $X(T) = \mathbb{Z}^n / \mathbb{Z}(1, 1, \dots, 1)$.

Parabolic Subgroups

Between the Borel B and the full group G , there are intermediate subgroups that also give rise to projective quotients.

Definition 6.2.7: Parabolic Subgroup

A closed subgroup $P \subseteq G$ is called *parabolic* if G/P is a projective variety. Equivalently, P is parabolic if and only if P contains a Borel subgroup.

The equivalence of the two definitions is a nontrivial fact. One direction is clear: if $P \supseteq B$, then G/P is a quotient of G/B (via the natural map $G/B \rightarrow G/P$, which is a fiber bundle with fiber P/B), and a quotient of a projective variety by a group action is projective (the map $G/B \rightarrow G/P$ is proper with connected fibers, so the image is projective). For the other direction: if G/P is projective and B is a Borel, then B acts on G/P and Borel's fixed-point theorem gives a B -fixed point gP , meaning $g^{-1}Bg \subseteq P$, so P contains a conjugate of B (hence P contains a Borel).

Example 6.2.8: Parabolic Subgroups of GL_n

The parabolic subgroups of GL_n containing the standard Borel B_n are in bijection with subsets $I \subseteq \{1, \dots, n-1\}$. The subset $I = \{d_1, \dots, d_r\}$ (with $d_1 < \dots < d_r$) corresponds to the parabolic P_I that is the stabilizer of the partial flag

$$\langle e_1, \dots, e_{d_1} \rangle \subset \langle e_1, \dots, e_{d_2} \rangle \subset \dots \subset \langle e_1, \dots, e_{d_r} \rangle \subset k^n$$

The quotient GL_n/P_I is the partial flag variety $\mathrm{Fl}(d_1, \dots, d_r; n)$. Special cases:

1. $I = \{1, 2, \dots, n-1\}$ gives $P_I = B_n$ and $\mathrm{GL}_n/B_n = \mathrm{Fl}(1, 2, \dots, n-1; n)$, the complete flag variety.
2. $I = \{d\}$ gives $P_I = P_d$ (a maximal parabolic) and $\mathrm{GL}_n/P_d = \mathrm{Gr}(d, n)$, the Grassmannian.
3. $I = \emptyset$ gives $P_I = \mathrm{GL}_n$ and $\mathrm{GL}_n/\mathrm{GL}_n = \{*\}$.

The number of parabolic subgroups containing B_n is 2^{n-1} (the number of subsets of $\{1, \dots, n-1\}$). Each gives a different partial flag variety, forming a lattice under the natural projections $\mathrm{Fl}(d_1, \dots, d_r; n) \rightarrow \mathrm{Fl}(d_{i_1}, \dots, d_{i_s}; n)$ for $\{i_1, \dots, i_s\} \subseteq \{1, \dots, r\}$.

Reductive and Semisimple Groups

The radical $R(G)$ and the unipotent radical $R_u(G)$ measure the “solvable” and “unipotent” content of G . The groups where these are trivial are the ones that carry the most interesting structure theory.

Definition 6.2.9: Unipotent Radical (General)

Let G be a connected affine algebraic group. The *unipotent radical* $R_u(G)$ is the maximal connected normal unipotent subgroup of G . Equivalently, $R_u(G) = R_u(R(G))$: the unipotent radical of the radical.

The unipotent radical is always contained in the radical ($R_u(G) \subseteq R(G)$, since a connected normal unipotent subgroup is in particular solvable), and the quotient $R(G)/R_u(G)$ is a torus (by the structure theorem for solvable groups, theorem 6.1.7(2)).

Definition 6.2.10: Reductive and Semisimple Groups

Let G be a connected affine algebraic group.

1. G is called *reductive* if $R_u(G) = \{e\}$ (the unipotent radical is trivial).
2. G is called *semisimple* if $R(G) = \{e\}$ (the radical is trivial).

Semisimple implies reductive (since $R_u(G) \subseteq R(G)$), but not conversely: a torus is reductive (it has no unipotent elements, so $R_u = \{e\}$) but not semisimple (it is its own radical: $R(T) = T$). The relationship between the three classes is:

$$\{\text{semisimple}\} \subset \{\text{reductive}\} \subset \{\text{affine algebraic groups}\}$$

Example 6.2.11: Classification of Low-Dimensional Cases

1. GL_n : Reductive but not semisimple. The radical is $R(\text{GL}_n) = Z(\text{GL}_n) = \mathbb{G}_m$ (?? 4), a torus, so $R_u(\text{GL}_n) = \{e\}$. The center is nontrivial, so $R(\text{GL}_n) \neq \{e\}$.
2. SL_n ($n \geq 2$): Semisimple. The center $Z(\text{SL}_n) = \mu_n$ is finite (not connected), so it is not contained in $R(\text{SL}_n)$, which must be connected. One shows $R(\text{SL}_n) = \{e\}$ by verifying that SL_n has no nontrivial connected solvable normal subgroups.
3. Sp_{2n} ($n \geq 1$): Semisimple. Similar reasoning.
4. SO_n ($n \geq 3$, char $\neq 2$): Semisimple.
5. B_n : Not reductive. The unipotent radical $R_u(B_n) = U_n \neq \{e\}$.
6. $\mathbb{G}_m^r$: Reductive (a torus), but not semisimple for $r \geq 1$.

A reductive group G satisfies a clean structural decomposition: the radical $R(G)$ is a central torus (since $R_u(G) = \{e\}$, the radical is a torus by theorem 6.1.7(2), and a normal torus in a connected group is central by proposition 5.2.13), and the quotient $G/R(G)$ is semisimple. More precisely:

Proposition 6.2.12: Structure of Reductive Groups

Let G be a connected reductive group. Then:

1. $R(G) = Z(G)^\circ$ (the radical is the identity component of the center).
2. The derived subgroup $[G, G]$ is semisimple.
3. The multiplication map $R(G) \times [G, G] \rightarrow G$ is surjective with finite kernel. Equivalently, $G = R(G) \cdot [G, G]$ and $R(G) \cap [G, G]$ is finite.

Proof Sketch:

(1): The radical $R(G)$ is a connected solvable normal subgroup. Since G is reductive, $R_u(R(G)) = R_u(G) = \{e\}$, so $R(G)$ is a torus. A normal torus in a connected group is central (proposition 5.2.13), so $R(G) \subseteq Z(G)^\circ$. Conversely, $Z(G)^\circ$ is connected, solvable (commutative), and

normal, so $Z(G)^\circ \subseteq R(G)$.

(2): The quotient $G/R(G)$ is semisimple (its radical is trivial by maximality of $R(G)$). The derived subgroup $[G, G]$ surjects onto $[G/R(G), G/R(G)] = G/R(G)$ (since $G/R(G)$ is semisimple, it equals its own derived subgroup). The kernel of $[G, G] \rightarrow G/R(G)$ is $[G, G] \cap R(G)$, which is finite (a closed subgroup of the torus $R(G)$ that lies in $[G, G]$; since $[G, G]$ has no nontrivial torus quotient — its abelianization is finite — this intersection is finite). Hence $[G, G]$ is an extension of the semisimple group $G/R(G)$ by a finite group, and one checks that $[G, G]$ is itself semisimple.

(3): Every $g \in G$ can be written as $g = z \cdot [g, \dots]$ modulo the derived subgroup; more precisely, $G = R(G) \cdot [G, G]$ follows from the surjectivity of $[G, G] \rightarrow G/R(G)$ and $R(G) \rightarrow G/[G, G]$. The kernel of $R(G) \times [G, G] \rightarrow G$ is $\{(z, z^{-1}) : z \in R(G) \cap [G, G]\}$, which is finite.

This decomposition says that a reductive group is “almost” a product of a torus and a semisimple group, up to a finite central overlap. For example, $\mathrm{GL}_n = \mathbb{G}_m \cdot \mathrm{SL}_n$ with $\mathbb{G}_m \cap \mathrm{SL}_n = \mu_n$.

The terms “reductive” and “semisimple” now justify the book’s focus: Part I culminates in Chapter 7 with the classification of semisimple (equivalently, reductive modulo center) groups via root systems. The representation theory of Part II will study how these groups act, with the Borel subgroup B and the flag variety G/B as the primary geometric tools.

Example 6.2.13: GL_n as a Reductive Group

The group GL_n has $R(\mathrm{GL}_n) = Z(\mathrm{GL}_n)^\circ = \mathbb{G}_m$ (scalar matrices), $[G, G] = \mathrm{SL}_n$, and $\mathrm{GL}_n = \mathbb{G}_m \cdot \mathrm{SL}_n$ with $\mathbb{G}_m \cap \mathrm{SL}_n = \mu_n$. The Borel subgroup B_n decomposes as $T \ltimes U_n$ where $T = \mathbb{G}_m^n$ is the maximal torus and U_n is the unipotent radical. The flag variety is $\mathrm{GL}_n/B_n = \mathrm{SL}_n/(B_n \cap \mathrm{SL}_n) = \mathrm{Fl}(1, \dots, n-1; n)$, a smooth projective variety of dimension $\binom{n}{2}$.

The root system will be defined in Chapter 7 by decomposing $\mathfrak{gl}_n$ under the adjoint action of T : the nonzero weights $\varepsilon_i - \varepsilon_j$ ($i \neq j$) are the roots (example 5.2.10), the weights with $i < j$ are the positive roots (corresponding to $\mathrm{Lie}(U_n)$), and the Weyl group $W = N_G(T)/Z_G(T) \cong S_n$ (?? 6) acts by permuting the ε_i . The conjugacy theorems of this section ensure that this data is independent of the choice of T and B .

Exercise 6.2.1

1. Let $G = \mathrm{SL}_3$ with standard Borel B (upper-triangular, determinant 1). Compute $\dim B$, $\dim G/B$, the rank of G (dimension of a maximal torus), and verify $\dim G = \dim B + \dim G/B$.
2. Let $G = \mathrm{SL}_2$ with standard Borel $B = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \right\}$ and non-standard Borel $B' = \left\{ \begin{pmatrix} a & 0 \\ c & a^{-1} \end{pmatrix} \right\}$ (lower-triangular). Find an explicit $g \in \mathrm{SL}_2(k)$ with $gBg^{-1} = B'$, verifying the conjugacy theorem.
3. How many parabolic subgroups of SL_4 contain the standard Borel? List them and describe the corresponding partial flag varieties with their dimensions.
4. Determine which of the following groups are reductive, semisimple, both, or neither: (a) $\mathrm{GL}_1 = \mathbb{G}_m$; (b) PGL_n ; (c) $\mathbb{G}_a$; (d) $\mathbb{G}_m \times \mathrm{SL}_2$; (e) $B_2 \subseteq \mathrm{GL}_2$.

5. Let $G = \mathrm{SL}_3$ with maximal torus $T = \{\mathrm{diag}(t_1, t_2, (t_1 t_2)^{-1})\}$. Compute the normalizer $N_G(T)$ and the Weyl group $W = N_G(T)/Z_G(T)$. Show that $W \cong S_3$ and describe its action on $X(T)$.
6. Let $G = \mathrm{Sp}_4$ (the 4×4 symplectic group). What is the rank of G ? What is $\dim G/B$? The flag variety G/B parametrizes complete isotropic flags $0 \subset L_1 \subset L_2 \subset k^4$ where L_1 is a line and L_2 is a Lagrangian plane (with $L_1 \subset L_2$ and L_2 isotropic). How many maximal parabolic subgroups does G have?

Root Systems and Classification

This is the culminating chapter of Part I. Everything we have built — the functor of points and Hopf algebras (Chapter 1), representations and comodules (Chapter 2), the adjoint representation and Lie algebras (Chapter 3), quotient constructions (Chapter 4), the Jordan decomposition and torus theory (Chapter 5), and the Borel subgroup machinery (Chapter 6) — converges here into a single combinatorial invariant: the *root system*. A root system is a finite configuration of vectors in a Euclidean space, subject to crystallographic symmetry conditions. The classification of root systems is a finite computation, and it yields the classification of semisimple algebraic groups — one of the great achievements of twentieth-century mathematics.

The mechanism is the one we have been previewing since Chapter 3: the adjoint action of a maximal torus T on the Lie algebra $\mathfrak{g}$ decomposes $\mathfrak{g}$ into weight spaces, and the nonzero weights — the *roots* — organize into a combinatorial structure governed by the *Weyl group* $W = N_G(T)/Z_G(T)$. The conjugacy theorems of Chapter 6 ensure that the root system is independent of the choice of T and B , and the classification theorem asserts that it is a *complete* invariant: two semisimple groups with isomorphic root systems are isomorphic (up to isogeny).

We develop the theory in three stages. First, we extract the root system from a reductive group and compute it for all the classical families. Second, we develop the abstract combinatorial theory of root systems and classify them via Dynkin diagrams. Third, we introduce root data and state the isomorphism and existence theorems, completing the classification.

7.1 Roots and the Weyl Group

Throughout this section, k denotes an algebraically closed field, G a connected reductive group over k , $T \subseteq G$ a maximal torus of rank $r = \dim T$, and $B \supseteq T$ a Borel subgroup containing T . The existence of T and B was established in Chapter 6 (theorem 6.2.5, definition 6.2.1), and the conjugacy theorems guarantee that the structures we define will be independent of these choices (up

to canonical isomorphism).

The Root Space Decomposition

The maximal torus T acts on the Lie algebra $\mathfrak{g} = \text{Lie}(G)$ via the adjoint representation $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ (definition 3.2.1). Since T is a torus, the weight space decomposition (theorem 5.2.9) gives:

$$\mathfrak{g} = \bigoplus_{\alpha \in X(T)} \mathfrak{g}_{\alpha}$$

where $\mathfrak{g}_{\alpha} = \{X \in \mathfrak{g} \mid \text{Ad}(t)(X) = \alpha(t)X \text{ for all } t \in T\}$ is the weight space for the character $\alpha \in X(T)$. The 0-weight space has a natural identification:

Proposition 7.1.1: The Zero Weight Space

$\mathfrak{g}_0 = \{X \in \mathfrak{g} \mid \text{Ad}(t)(X) = X \text{ for all } t \in T\} = \text{Lie}(Z_G(T)) = \mathfrak{t}$, the Lie algebra of T .

Proof:

The set $\mathfrak{g}_0$ consists of elements fixed by the adjoint action of T , which is exactly the Lie algebra of the centralizer $Z_G(T)$ (since $\text{Ad}(t)(X) = X$ for all t means X commutes with T at the infinitesimal level). For a reductive group, $Z_G(T) = T$ (the centralizer of a maximal torus is the torus itself — this is because $Z_G(T)$ is a connected reductive subgroup containing T as a maximal torus, and by theorem 6.1.7, $Z_G(T)^\circ$ is commutative, hence a torus, hence equal to T by maximality). Therefore $\mathfrak{g}_0 = \text{Lie}(T) = \mathfrak{t}$.

The nonzero weights are the central definition of this chapter:

Definition 7.1.2: Roots

The *roots* of G with respect to T are the nonzero characters $\alpha \in X(T)$ such that $\mathfrak{g}_{\alpha} \neq 0$. The set of all roots is denoted

$$\Phi = \Phi(G, T) = \{\alpha \in X(T) \setminus \{0\} \mid \mathfrak{g}_{\alpha} \neq 0\}$$

For each $\alpha \in \Phi$, the subspace $\mathfrak{g}_{\alpha}$ is called the *root space* of α .

The root space decomposition of $\mathfrak{g}$ is thus:

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha}$$

with $\dim \mathfrak{g} = \text{rank}(G) + |\Phi|$. This is the decomposition we computed for GL_n in example 5.2.10 and previewed for SL_2 in example 3.2.9. We now develop the general theory.

Fundamental Properties of Roots

The roots of a reductive group satisfy a collection of structural properties that will eventually be axiomatized into the notion of an “abstract root system” in §7.2. We state the properties here as theorems about G and prove them using the tools of previous chapters.

Theorem 7.1.3: Structural Properties of Roots

Let G be a connected reductive group with maximal torus T and root system $\Phi = \Phi(G, T)$.

1. Each root space is one-dimensional: $\dim_k \mathfrak{g}_\alpha = 1$ for every $\alpha \in \Phi$.
2. If $\alpha \in \Phi$, then $-\alpha \in \Phi$ (roots come in opposite pairs).
3. If $\alpha \in \Phi$ and $c\alpha \in \Phi$ for some $c \in \mathbb{Z}$, then $c = \pm 1$. (No “multiples” of a root, other than its negative, are roots.)
4. Φ is finite, and Φ spans $X(T) \otimes_{\mathbb{Z}} \mathbb{Q}$ if G is semisimple.
5. For each $\alpha \in \Phi$, there exists a *coroot* $\alpha^\vee \in X_\vee(T)$ such that $\langle \alpha, \alpha^\vee \rangle = 2$ and the reflection $s_\alpha : X(T) \otimes \mathbb{Q} \rightarrow X(T) \otimes \mathbb{Q}$ defined by $s_\alpha(\lambda) = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha$ satisfies $s_\alpha(\Phi) = \Phi$.

Proof Sketch:

The proofs of these properties use the SL_2 -subgroups of G . For each root $\alpha \in \Phi$, there exists a homomorphism $\varphi_\alpha : \mathrm{SL}_2 \rightarrow G$ such that φ_α maps the standard maximal torus of SL_2 into T and the root spaces of SL_2 onto $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$. The existence of φ_α is a deep result (it follows from the structure theory of rank-1 semisimple groups, which shows that the subgroup generated by the root groups U_α and $U_{-\alpha}$ is isomorphic to SL_2 or PGL_2).

Given φ_α , the properties follow from the representation theory of SL_2 :

1. The root space $\mathfrak{g}_\alpha$ is the image of the 1-dimensional root space of SL_2 (which is ke in the notation of example 3.2.9), hence one-dimensional.
2. The image of $f \in \mathfrak{sl}_2$ lands in $\mathfrak{g}_{-\alpha}$, so $-\alpha \in \Phi$.
3. If $c\alpha$ were a root with $c \neq \pm 1$, then $\mathfrak{g}$ would contain a representation of SL_2 (via φ_α) with a weight equal to $2c$ (since α corresponds to weight 2 for SL_2). But the weights of any finite-dimensional SL_2 -representation are integers in the range $\{-d, -d+2, \dots, d-2, d\}$ for some d , and $2c$ with $c \neq \pm 1$ and $|c| > 1$ would force roots $2\alpha, 3\alpha, \dots$, contradicting finiteness.
4. Finiteness: $\Phi \subseteq X(T)$ and $\mathfrak{g}$ is finite-dimensional, so $|\Phi| \leq \dim \mathfrak{g} - \mathrm{rank} G < \infty$. For the spanning property when G is semisimple: if Φ did not span $X(T) \otimes \mathbb{Q}$, then T would have a nontrivial subtorus acting trivially on $\mathfrak{g}$ (the annihilator of Φ in $X_\vee(T)$), hence lying in the center $Z(G)$. But $Z(G)$ is finite for a semisimple group (proposition 6.2.12), contradicting the subtorus being nontrivial.
5. The coroot $\alpha^\vee$ is defined as the cocharacter $\varphi_\alpha \circ \lambda_0$, where $\lambda_0 : \mathbb{G}_m \rightarrow \mathrm{SL}_2$ is the standard cocharacter $t \mapsto \mathrm{diag}(t, t^{-1})$. The pairing $\langle \alpha, \alpha^\vee \rangle = 2$ follows from the SL_2 -computation (?? 5). The reflection s_α is the image in W of the Weyl group element $\varphi_\alpha \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix} \right) \in N_G(T)$, and it permutes Φ because W acts on $\mathfrak{g}$ by automorphisms that preserve weight spaces.

The appearance of SL_2 -subgroups is the structural heart of the theory: each root α generates a copy of SL_2 (or PGL_2) inside G , and the entire root system can be understood as the pattern of how these SL_2 -copies overlap and interact. This is sometimes called the “ SL_2 -philosophy” of reductive group

theory.

Positive Roots and Simple Roots

The choice of Borel subgroup $B \supseteq T$ determines a partition of the roots into “positive” and “negative” halves.

Definition 7.1.4: Positive and Negative Roots

Let $B \supseteq T$ be a Borel subgroup. The *positive roots* (with respect to B) are:

$$\Phi^+ = \Phi^+(G, T, B) = \{\alpha \in \Phi \mid \mathfrak{g}_\alpha \subseteq \text{Lie}(R_u(B))\}$$

The *negative roots* are $\Phi^- = -\Phi^+ = \{-\alpha \mid \alpha \in \Phi^+\}$. We have $\Phi = \Phi^+ \sqcup \Phi^-$ (disjoint union).

The definition is geometric: the positive roots are the weights of T on the Lie algebra of the unipotent radical of B . Since $B = T \ltimes R_u(B)$ (theorem 6.1.7), we have $\text{Lie}(B) = \mathfrak{t} \oplus \text{Lie}(R_u(B))$, and hence:

$$\text{Lie}(B) = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha, \quad \mathfrak{g} = \bigoplus_{\alpha \in \Phi^-} \mathfrak{g}_\alpha \oplus \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha$$

The dimension of the flag variety is $\dim G/B = \dim G - \dim B = |\Phi^-| = |\Phi^+| = |\Phi|/2$.

A different choice of Borel $B' \supseteq T$ gives a different set of positive roots $(\Phi^+)' = \Phi^+(G, T, B')$, but the conjugacy of Borel subgroups (theorem 6.2.4) ensures that the root system Φ itself is unchanged. The positive roots determined by B and B' are related by a Weyl group element.

Definition 7.1.5: Simple Roots

A root $\alpha \in \Phi^+$ is called *simple* if it cannot be expressed as $\alpha = \beta + \gamma$ with $\beta, \gamma \in \Phi^+$. The set of simple roots is denoted $\Delta = \{\alpha_1, \dots, \alpha_r\}$.

Proposition 7.1.6: Properties of Simple Roots

1. $|\Delta| = r = \text{rank}(G)$ when G is semisimple ($|\Delta| = \dim T$ in general).
2. Δ is a basis for $X(T) \otimes_{\mathbb{Z}} \mathbb{Q}$ (when G is semisimple).
3. Every positive root is a non-negative integer linear combination of simple roots: if $\alpha \in \Phi^+$, then $\alpha = \sum_{i=1}^r n_i \alpha_i$ with $n_i \in \mathbb{Z}_{\geq 0}$.
4. The simple roots are linearly independent over $\mathbb{Q}$.

Proof Sketch:

If $\alpha \in \Phi^+$ is not simple, then $\alpha = \beta + \gamma$ with $\beta, \gamma \in \Phi^+$. Since β, γ are positive roots with strictly smaller “height” (the sum of their coordinates in any basis) than α , induction on the height shows every positive root is a non-negative integer combination of simple roots. Linear independence follows from the fact that simple roots cannot be further decomposed, and the count $|\Delta| = r$ follows from the spanning property of Φ and the non-negative-combination property.

The Weyl Group

Definition 7.1.7: Weyl Group

The *Weyl group* of G with respect to T is the finite group

$$W = W(G, T) = N_G(T)/Z_G(T)$$

where $N_G(T) = \{g \in G \mid gTg^{-1} = T\}$ is the normalizer and $Z_G(T) = T$ is the centralizer.

That W is finite was established in ?? 6: $N_G(T)$ acts on T by automorphisms, $\text{Aut}(T) \cong \text{GL}_r(\mathbb{Z})$ is discrete, and the kernel of $N_G(T) \rightarrow \text{Aut}(T)$ is $Z_G(T) = T$, so W injects into $\text{GL}_r(\mathbb{Z})$ as a finite subgroup. The Weyl group acts on $X(T)$ by $w \cdot \chi = \chi \circ \text{Ad}(n_w)^{-1}$ where $n_w \in N_G(T)$ is any lift of w , and this action permutes the roots: $w(\Phi) = \Phi$ for all $w \in W$.

The Weyl group is generated by the *simple reflections*:

Theorem 7.1.8: Weyl Group is Generated by Simple Reflections

The Weyl group W is generated by the reflections $\{s_\alpha : \alpha \in \Delta\}$ corresponding to the simple roots. Moreover, W acts simply transitively on the set of Weyl chambers (connected components of $X(T) \otimes \mathbb{R} \setminus \bigcup_{\alpha \in \Phi} \ker(\alpha)$), and the order of W equals the number of Weyl chambers.

Proof Sketch:

Each simple reflection s_{α_i} permutes $\Phi^+ \setminus \{\alpha_i\}$ (it sends α_i to $-\alpha_i$ and permutes the other positive roots). Given any $w \in W$, the element w permutes Φ and sends Φ^+ to some other set of positive roots $w(\Phi^+)$. The “length” of w is the number of positive roots sent to negative roots: $\ell(w) = |\Phi^+ \cap w^{-1}(\Phi^-)|$. One shows by induction on $\ell(w)$ that w is a product of simple reflections: if $\ell(w) > 0$, there exists a simple root α_i with $w(\alpha_i) \in \Phi^-$, and then $\ell(s_{\alpha_i}w) < \ell(w)$.

The Root System of SL_n (Type A_{n-1})

We now compute the root system for each classical family, beginning with the most fundamental case.

Example 7.1.9: Roots of SL_n

Let $G = \text{SL}_n$ with maximal torus $T = \{\text{diag}(t_1, \dots, t_n) : t_1 \cdots t_n = 1\} \cong \mathbb{G}_m^{n-1}$. The character lattice is $X(T) = \mathbb{Z}^n / \mathbb{Z}(1, \dots, 1)$, with the images of the standard characters ε_i satisfying $\varepsilon_1 + \cdots + \varepsilon_n = 0$.

The adjoint action of T on $\mathfrak{sl}_n = \{X \in M_n(k) : \text{tr}(X) = 0\}$ was computed in example 5.2.10: the weight of E_{ij} (with $i \neq j$) is $\varepsilon_i - \varepsilon_j$. Therefore:

$$\Phi = \{\varepsilon_i - \varepsilon_j \mid 1 \leq i, j \leq n, i \neq j\}$$

This is a root system of $n(n-1)$ roots in the $(n-1)$ -dimensional space $V = \{\sum x_i = 0\} \subseteq \mathbb{R}^n$. Its rank is $n-1$.

The standard Borel $B = B_n \cap \mathrm{SL}_n$ (upper-triangular) determines the positive roots $\Phi^+ = \{\varepsilon_i - \varepsilon_j : i < j\}$ (these are the weights of T on $\mathrm{Lie}(U_n) = \{\text{strictly upper-triangular matrices}\}$). The simple roots are:

$$\Delta = \{\alpha_1, \dots, \alpha_{n-1}\}, \quad \alpha_i = \varepsilon_i - \varepsilon_{i+1}$$

(the “adjacent” differences). Every positive root $\varepsilon_i - \varepsilon_j$ with $i < j$ is a sum $\alpha_i + \alpha_{i+1} + \dots + \alpha_{j-1}$ of consecutive simple roots.

The Weyl group is $W \cong S_n$ (the symmetric group), acting on $X(T)$ by permuting the ε_i . The simple reflection s_{α_i} corresponds to the transposition $(i \ i+1) \in S_n$, and the transpositions of adjacent elements generate S_n .

The coroots are $\alpha_i^\vee = \varepsilon_i^\vee - \varepsilon_{i+1}^\vee \in X_\vee(T)$, where $\varepsilon_i^\vee : \mathbb{G}_m \rightarrow T$ is the cocharacter $t \mapsto \mathrm{diag}(1, \dots, t, \dots, 1)$ (with t in the i -th position and t^{-1} in the n -th position to preserve $\det = 1$). The pairing is $\langle \alpha_i, \alpha_j^\vee \rangle = \delta_{ij} \cdot 2 - \delta_{|i-j|,1}$, giving the Cartan matrix:

$$A = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & \cdots & -1 & 2 \end{pmatrix}$$

This is the Cartan matrix of type A_{n-1} .

The Root System of Sp_{2n} (Type C_n)

Example 7.1.10: Roots of Sp_{2n}

Let $G = \mathrm{Sp}_{2n}$ with the symplectic form $\Omega = \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix}$ as in §1.1.1. The maximal torus is

$$T = \{\mathrm{diag}(t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})\} \cong \mathbb{G}_m^n$$

with character lattice $X(T) = \mathbb{Z}^n$ and standard basis $\varepsilon_i(t) = t_i$.

The Lie algebra $\mathfrak{sp}_{2n}$ consists of matrices X with $X^T \Omega + \Omega X = 0$ (example 3.1.8). Writing $X = \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix}$ with $B = B^T$ and $C = C^T$, the adjoint action of T on these blocks gives:

1. The entries of A : A_{ij} has weight $\varepsilon_i - \varepsilon_j$ (for $i \neq j$) and weight 0 (for $i = j$).
2. The entries of B : B_{ij} has weight $\varepsilon_i + \varepsilon_j$ (for $i \leq j$).
3. The entries of C : C_{ij} has weight $-\varepsilon_i - \varepsilon_j$ (for $i \leq j$).

Therefore the roots are:

$$\Phi = \{\pm \varepsilon_i \pm \varepsilon_j : i \neq j\} \cup \{\pm 2\varepsilon_i\}$$

This is a root system of $2n^2$ roots in $\mathbb{R}^n$, of rank n .

The standard Borel (upper-triangular in the symplectic form) gives positive roots $\Phi^+ = \{\varepsilon_i - \varepsilon_j :$

$i < j\} \cup \{\varepsilon_i + \varepsilon_j : i \leq j\}$ and simple roots:

$$\Delta = \{\alpha_1, \dots, \alpha_n\}, \quad \alpha_i = \begin{cases} \varepsilon_i - \varepsilon_{i+1} & 1 \leq i \leq n-1 \\ 2\varepsilon_n & i = n \end{cases}$$

The Weyl group is $W \cong S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$, the group of *signed permutations*: it permutes the ε_i and independently changes their signs. The order is $|W| = 2^n \cdot n!$.

The Cartan matrix has entries $\langle \alpha_i, \alpha_j^\vee \rangle$ forming the type- C_n matrix, characterized by the last simple root being *long* ($\langle \alpha_n, \alpha_n \rangle = 2$ times the length of the short roots).

The Root System of SO_{2n+1} (Type B_n)

Example 7.1.11: Roots of SO_{2n+1}

Let $G = \mathrm{SO}_{2n+1}$ (with $\mathrm{char} k \neq 2$) preserving the form $s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & J \\ 0 & J & 0 \end{pmatrix}$ from §1.1.1. The maximal torus is

$$T = \{\mathrm{diag}(1, t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})\} \cong \mathbb{G}_m^n$$

(the first coordinate is fixed at 1). The roots are:

$$\Phi = \{\pm \varepsilon_i \pm \varepsilon_j : i \neq j\} \cup \{\pm \varepsilon_i\}$$

This differs from type C_n in having *short* roots $\pm \varepsilon_i$ instead of long roots $\pm 2\varepsilon_i$. The total number is $2n^2$, and the rank is n .

The simple roots are:

$$\Delta = \{\alpha_1, \dots, \alpha_n\}, \quad \alpha_i = \begin{cases} \varepsilon_i - \varepsilon_{i+1} & 1 \leq i \leq n-1 \\ \varepsilon_n & i = n \end{cases}$$

The Weyl group is again $W \cong S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$ (signed permutations), of order $2^n \cdot n!$, identical to type C_n as an abstract group.

The root systems B_n and C_n are *dual*: the coroots of B_n form a root system of type C_n and vice versa. As abstract root systems they are non-isomorphic (for $n \geq 3$; for $n = 2$, $B_2 \cong C_2$), since B_n has short roots $\pm \varepsilon_i$ while C_n has long roots $\pm 2\varepsilon_i$. However, their Weyl groups are isomorphic.

The Root System of SO_{2n} (Type D_n)

Example 7.1.12: Roots of SO_{2n}

Let $G = \mathrm{SO}_{2n}$ preserving the form $s = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}$ (with $\mathrm{char} k \neq 2$). The maximal torus is

$$T = \{\mathrm{diag}(t_1, \dots, t_n, t_1^{-1}, \dots, t_n^{-1})\} \cong \mathbb{G}_m^n$$

The roots are:

$$\Phi = \{\pm\varepsilon_i \pm \varepsilon_j : 1 \leq i < j \leq n\}$$

There are no roots of the form $\pm\varepsilon_i$ or $\pm 2\varepsilon_i$ — all roots have the same length. The total number is $2n(n-1)$, and the rank is n .

The simple roots are:

$$\Delta = \{\alpha_1, \dots, \alpha_n\}, \quad \alpha_i = \begin{cases} \varepsilon_i - \varepsilon_{i+1} & 1 \leq i \leq n-1 \\ \varepsilon_{n-1} + \varepsilon_n & i = n \end{cases}$$

Note that $\alpha_n = \varepsilon_{n-1} + \varepsilon_n$ is different from the last simple root of types B_n and C_n : it is a *sum* rather than a single ε_i or $2\varepsilon_i$. This reflects the “branching” of the Dynkin diagram at the last node.

The Weyl group is $W \cong S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^{n-1}$: it consists of signed permutations with an *even* number of sign changes. The order is $|W| = 2^{n-1} \cdot n!$, which is half the size of the Weyl groups of B_n and C_n .

Summary of the Classical Root Systems

Type	Group	Rank	$ \Phi $	$ W $	Roots
A_{n-1}	SL_n	$n-1$	$n(n-1)$	$n!$	$\varepsilon_i - \varepsilon_j$
B_n	SO_{2n+1}	n	$2n^2$	$2^n n!$	$\pm\varepsilon_i \pm \varepsilon_j, \pm\varepsilon_i$
C_n	Sp_{2n}	n	$2n^2$	$2^n n!$	$\pm\varepsilon_i \pm \varepsilon_j, \pm 2\varepsilon_i$
D_n	SO_{2n}	n	$2n(n-1)$	$2^{n-1} n!$	$\pm\varepsilon_i \pm \varepsilon_j$

These are the four “classical” infinite families of root systems. The classification in §7.2 will show that, beyond these, there are exactly five “exceptional” root systems (G_2, F_4, E_6, E_7, E_8) that do not belong to any infinite family.

Exercise 7.1.1

1. Let $G = \mathrm{SL}_3$ with maximal torus $T = \{\mathrm{diag}(t_1, t_2, (t_1 t_2)^{-1})\}$ and standard Borel B (upper-triangular). List all roots, positive roots, and simple roots explicitly. Draw the root system in the plane $\{\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0\} \subseteq \mathbb{R}^3$. Verify that $|\Phi^+| = 3 = \dim \mathrm{SL}_3/B$.
2. For $G = \mathrm{SL}_3$ with the roots from ?? 1, compute all the coroots $\alpha^\vee \in X_\vee(T)$. Verify $\langle \alpha_i, \alpha_j^\vee \rangle$ gives the Cartan matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ of type A_2 .
3. Let $W = S_3$ be the Weyl group of SL_3 . The simple reflections are $s_1 = s_{\alpha_1}$ and $s_2 = s_{\alpha_2}$. Compute $s_1(\alpha_2)$ and $s_2(\alpha_1)$. Write every element of W as a product of s_1 and s_2 , and compute $\ell(w)$ for each $w \in W$.
4. Verify the root system computation for Sp_4 (type C_2) by direct computation. Write down the maximal torus, compute the adjoint action on $\mathfrak{sp}_4$, and identify all 8 roots. Draw the root

system in $\mathbb{R}^2$.

5. The root systems B_2 and C_2 have the same underlying set of 8 roots up to scaling: $\{\pm\varepsilon_1 \pm \varepsilon_2, \pm\varepsilon_1, \pm\varepsilon_2\}$ for B_2 and $\{\pm\varepsilon_1 \pm \varepsilon_2, \pm 2\varepsilon_1, \pm 2\varepsilon_2\}$ for C_2 . Show that $B_2 \cong C_2$ as abstract root systems (there is a linear isomorphism of the ambient space sending one to the other). What happens for B_3 and C_3 ?
6. Compute the root system of GL_n (which is reductive but not semisimple). Show that $\Phi(\mathrm{GL}_n, T) = \Phi(\mathrm{SL}_n, T')$ where $T' = T \cap \mathrm{SL}_n$. Why doesn't Φ span $X(T) \otimes \mathbb{Q}$ for GL_n ? (*Hint*: the center $Z(\mathrm{GL}_n) = \mathbb{G}_m$ provides a “missing direction”.)

7.2 Abstract Root Systems and Dynkin Diagrams

In the previous section, we extracted a finite set of vectors $\Phi \subseteq X(T) \otimes \mathbb{R}$ from the adjoint action of a maximal torus on the Lie algebra of a reductive group. We now step away from algebraic groups entirely and study the combinatorial structures — *root systems* — that arise from this construction. The axioms are motivated by the properties proved in theorem 7.1.3: the roots come in pairs $\pm\alpha$, each root determines a reflection that preserves the whole system, and the “projection” of any root onto any other is an integer. The remarkable fact is that these axioms are so restrictive that only finitely many root systems exist in each rank, and they can be completely classified by a simple combinatorial device: the Dynkin diagram.

Axioms for Root Systems

Let V be a finite-dimensional real vector space equipped with a positive-definite inner product $(\cdot, \cdot)$. For a nonzero vector $\alpha \in V$, define the *coroot* (or *coweight*)

$$\alpha^\vee = \frac{2\alpha}{(\alpha, \alpha)}$$

and the *reflection* in the hyperplane orthogonal to α :

$$s_\alpha : V \rightarrow V, \quad s_\alpha(\lambda) = \lambda - (\lambda, \alpha^\vee) \alpha = \lambda - \frac{2(\lambda, \alpha)}{(\alpha, \alpha)} \alpha$$

The reflection s_α fixes $\ker(\alpha^\vee)$ pointwise and sends $\alpha \mapsto -\alpha$. It is an orthogonal transformation of order 2.

Definition 7.2.1: Root System

A (*reduced, crystallographic*) root system in V is a finite subset $\Phi \subseteq V$ satisfying:

1. (Spanning) Φ spans V and $0 \notin \Phi$.
2. (Reflection invariance) For every $\alpha \in \Phi$, the reflection s_α preserves Φ : $s_\alpha(\Phi) = \Phi$.
3. (Integrality) For every $\alpha, \beta \in \Phi$, the number $\langle \beta, \alpha^\vee \rangle := (\beta, \alpha^\vee) = \frac{2(\beta, \alpha)}{(\alpha, \alpha)}$ is an integer.
4. (Reduced) If $\alpha \in \Phi$ and $c\alpha \in \Phi$ for $c \in \mathbb{R}$, then $c = \pm 1$.

The *rank* of Φ is $\dim V$.

Axiom (3) is the *crystallographic condition*: it says that the angle between any two roots is constrained to a discrete set of values. Specifically, if $\alpha, \beta \in \Phi$ with $\alpha \neq \pm\beta$, then:

$$\langle \beta, \alpha^\vee \rangle \cdot \langle \alpha, \beta^\vee \rangle = \frac{4(\alpha, \beta)^2}{(\alpha, \alpha)(\beta, \beta)} = 4 \cos^2 \theta$$

where θ is the angle between α and β . Since both factors on the left are integers, the product $4 \cos^2 \theta$ is a non-negative integer. Since $\cos^2 \theta \leq 1$, we have:

$$\langle \beta, \alpha^\vee \rangle \cdot \langle \alpha, \beta^\vee \rangle \in \{0, 1, 2, 3\}$$

(It cannot be 4 because that would require $\theta = 0$ or π , meaning $\alpha = \pm\beta$, which we excluded.) This gives exactly four possible angle/length-ratio combinations for a pair of non-proportional roots, and this finiteness is what makes the classification possible.

7.2.2 Remark: Reduced vs. Non-Reduced Axiom (4) (the “reduced” condition) excludes root systems containing both α and 2α . If we drop this axiom, we get *non-reduced* root systems, the only new case being BC_n (a superposition of B_n and C_n). Non-reduced root systems arise in the theory of algebraic groups over non-algebraically closed fields (specifically, in the theory of *relative* root systems), but they do not arise from the absolute root system of a split reductive group. We restrict to reduced root systems throughout.

The Weyl Group and Chambers

Definition 7.2.3: Weyl Group of a Root System

The *Weyl group* of a root system Φ is the subgroup $W = W(\Phi) \subseteq \text{GL}(V)$ generated by all reflections $\{s_\alpha : \alpha \in \Phi\}$.

The Weyl group is a finite group (it permutes the finite set Φ , and is therefore a subgroup of $S_{|\Phi|}$) of orthogonal transformations of V . It acts faithfully on Φ (since Φ spans V , a linear transformation fixing Φ pointwise is the identity).

The *hyperplane arrangement* $\mathcal{H} = \{\ker(\alpha^\vee) : \alpha \in \Phi\}$ cuts V into finitely many open connected regions called *Weyl chambers*. Each chamber is a connected component of $V \setminus \bigcup_{H \in \mathcal{H}} H$. The Weyl group permutes the chambers, and:

Proposition 7.2.4: Weyl Chambers and the Weyl Group

The Weyl group W acts simply transitively on the set of Weyl chambers. In particular, the number of Weyl chambers equals $|W|$.

Proof Sketch:

Transitivity: given two chambers C, C' , there exists a sequence of reflections carrying C to C' (“walk” from a point in C to a point in C' , crossing one hyperplane at a time; each crossing corresponds to a reflection). Simple transitivity: if $w \in W$ fixes a chamber C , then w fixes all points in C , hence fixes V (since C is open and w is linear), so $w = \text{id}$.

Bases and the Cartan Matrix

A choice of Weyl chamber determines a set of simple roots, analogous to how a choice of Borel subgroup determined simple roots in §7.1.

Definition 7.2.5: Base of a Root System

A *base* (or *system of simple roots*) for Φ is a subset $\Delta = \{\alpha_1, \dots, \alpha_r\} \subseteq \Phi$ such that:

1. Δ is a basis of V .
2. Every root $\alpha \in \Phi$ can be written as $\alpha = \sum_{i=1}^r n_i \alpha_i$ with all $n_i \in \mathbb{Z}_{\geq 0}$ or all $n_i \in \mathbb{Z}_{\leq 0}$.

The roots with all $n_i \geq 0$ are the *positive roots* $\Phi^+(\Delta)$, and those with all $n_i \leq 0$ are the *negative roots* $\Phi^-(\Delta) = -\Phi^+(\Delta)$.

Proposition 7.2.6: Existence and Uniqueness of Bases

1. Every root system has a base.
2. Bases correspond bijectively to Weyl chambers: the base Δ corresponding to a chamber C consists of the roots $\alpha \in \Phi$ such that $\ker(\alpha^\vee)$ is a wall of C and α is positive on C .
3. Any two bases are related by a unique element of W : if Δ, Δ' are bases, there exists a unique $w \in W$ with $w(\Delta) = \Delta'$.

Proof Sketch:

For (1), choose any vector $\xi \in V$ not lying on any root hyperplane (this is possible since Φ is finite and the hyperplanes have codimension 1). This ξ lies in some Weyl chamber C . Define $\Phi^+ = \{\alpha \in \Phi : (\alpha, \xi) > 0\}$ and let Δ be the set of positive roots that cannot be written as sums of other positive roots. One verifies that Δ is a base. Statement (2) follows from this construction, and (3) follows from the simple transitivity of W on chambers.

The base Δ encodes the root system through the *Cartan matrix*:

Definition 7.2.7: Cartan Matrix

Let $\Delta = \{\alpha_1, \dots, \alpha_r\}$ be a base for Φ . The *Cartan matrix* of Φ (with respect to Δ) is the $r \times r$ integer matrix

$$A = (a_{ij})_{1 \leq i, j \leq r}, \quad a_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle = \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)}$$

The Cartan matrix has the following properties: $a_{ii} = 2$ for all i ; $a_{ij} \leq 0$ for $i \neq j$ (because simple roots make obtuse angles: $(\alpha_i, \alpha_j) \leq 0$ for $i \neq j$, which follows from the fact that $\alpha_i - \alpha_j$ is not a root); and $a_{ij} = 0 \iff a_{ji} = 0$.

The Cartan matrix determines the root system up to isomorphism. In fact, even less information suffices:

Proposition 7.2.8: The Cartan Matrix Determines the Root System

Two root systems with the same Cartan matrix (up to simultaneous re-ordering of the indices) are isomorphic.

Proof Sketch:

The Weyl group is generated by the simple reflections $s_1, \dots, s_r$ (this is theorem 7.1.8 in the abstract setting), and the action of each s_i on the simple roots is determined by the Cartan matrix: $s_i(\alpha_j) = \alpha_j - a_{ji}\alpha_i$. Since $\Phi = W \cdot \Delta$ (every root is in the W -orbit of some simple root), the Cartan matrix determines Φ via the action of W .

The Dynkin Diagram

The Cartan matrix is an $r \times r$ array of integers, but most of its entries are 0 (most pairs of simple roots are orthogonal). A more efficient encoding is the *Dynkin diagram*, which records only the nonzero off-diagonal information.

Definition 7.2.9: Dynkin Diagram

The *Dynkin diagram* of a root system Φ (with base $\Delta = \{\alpha_1, \dots, \alpha_r\}$) is a graph with r nodes (one per simple root) and edges determined by the Cartan matrix:

1. If $a_{ij} = a_{ji} = 0$ (i.e. $\alpha_i \perp \alpha_j$): no edge between nodes i and j .
2. If $a_{ij} = a_{ji} = -1$ (i.e. $(\alpha_i, \alpha_i) = (\alpha_j, \alpha_j)$ and the angle is 120°): a single edge between i and j .
3. If $a_{ij} = -1$ and $a_{ji} = -2$ (i.e. $|\alpha_j| = \sqrt{2}|\alpha_i|$ and the angle is 135°): a double edge with an arrow pointing from j to i (from the long root to the short root).
4. If $a_{ij} = -1$ and $a_{ji} = -3$ (i.e. $|\alpha_j| = \sqrt{3}|\alpha_i|$ and the angle is 150°): a triple edge with an arrow pointing from j to i .

The arrow convention is: the arrow points toward the *shorter* root. When all roots have the same length (as in types A , D , E), the diagram has no arrows and all edges are single. When there are two root lengths (types B , C , F_4 , G_2), the arrow distinguishes the long and short roots.

A root system is *irreducible* if it cannot be decomposed as an orthogonal direct sum $\Phi = \Phi_1 \sqcup \Phi_2$ with $(\Phi_1, \Phi_2) = 0$. Equivalently, the Dynkin diagram is connected. Every root system is a direct sum of irreducible ones (corresponding to the connected components of the Dynkin diagram), so it suffices to classify the irreducible (connected) cases.

Classification in Rank 2

Before stating the general classification, we carry out the rank-2 case completely. This is a clean exercise in plane geometry that illustrates all the constraints.

Let Φ be an irreducible root system of rank 2 in $V = \mathbb{R}^2$, with base $\Delta = \{\alpha_1, \alpha_2\}$. Since Φ is irreducible, α_1 and α_2 are not orthogonal: $a_{12} \neq 0$. The constraint $a_{12}a_{21} \in \{1, 2, 3\}$ (since both $a_{12}, a_{21} \leq -1$ and $a_{12}a_{21} \leq 3$) gives three possibilities:

Case 1: $a_{12}a_{21} = 1$, so $a_{12} = a_{21} = -1$. The angle between α_1 and α_2 is 120° and $|\alpha_1| = |\alpha_2|$. The Weyl group is generated by two reflections through lines meeting at 60° , giving $|W| = 6$ and $W \cong S_3$. The root system has 6 roots forming a regular hexagon. This is **type** A_2 , the root system of SL_3 .

Case 2: $a_{12}a_{21} = 2$, so (up to relabeling) $a_{12} = -1$ and $a_{21} = -2$. The angle is 135° and $|\alpha_2| = \sqrt{2}|\alpha_1|$. The Weyl group has $|W| = 8$ and is the dihedral group D_4 (symmetries of a square). The root system has 8 roots. This is **type** $B_2 = C_2$, the root system of $SO_5 \cong Sp_4$.

Case 3: $a_{12}a_{21} = 3$, so (up to relabeling) $a_{12} = -1$ and $a_{21} = -3$. The angle is 150° and $|\alpha_2| = \sqrt{3}|\alpha_1|$. The Weyl group has $|W| = 12$ and is the dihedral group D_6 (symmetries of a regular hexagon). The root system has 12 roots. This is **type** G_2 , the smallest exceptional root system.

These are the *only* irreducible rank-2 root systems. We also have the reducible case $A_1 \times A_1$ ($a_{12} = a_{21} = 0$, two orthogonal roots), which has 4 roots and $W \cong (\mathbb{Z}/2\mathbb{Z})^2$.

Example 7.2.10: The Root System G_2

Let α_1 be a short root and α_2 a long root with $|\alpha_2| = \sqrt{3}|\alpha_1|$ and angle 150° . The 12 roots are:

$$\pm\alpha_1, \quad \pm\alpha_2, \quad \pm(\alpha_1 + \alpha_2), \quad \pm(2\alpha_1 + \alpha_2), \quad \pm(3\alpha_1 + \alpha_2), \quad \pm(3\alpha_1 + 2\alpha_2)$$

The positive roots (with respect to the base $\{\alpha_1, \alpha_2\}$) are $\alpha_1, \alpha_2, \alpha_1 + \alpha_2, 2\alpha_1 + \alpha_2, 3\alpha_1 + \alpha_2, 3\alpha_1 + 2\alpha_2$. The highest root (the root with the largest coefficients) is $3\alpha_1 + 2\alpha_2$.

The Weyl group $W(G_2)$ is the dihedral group of order 12, isomorphic to $S_3 \times \mathbb{Z}/2\mathbb{Z}$. It is generated by the two simple reflections $s_1 = s_{\alpha_1}$ and $s_2 = s_{\alpha_2}$ with the relation $(s_1s_2)^6 = \text{id}$ (the product s_1s_2 is a rotation by 30° , and 12 such rotations give the identity... actually (s_1s_2) is a rotation by $\pi/6 = 30^\circ$, so $(s_1s_2)^6$ rotates by 180° ; the correct relation is $(s_1s_2)^6 = \text{id}$, giving the dihedral group D_6 of order 12).

The root system G_2 arises from the exceptional algebraic group of the same name, the automorphism group of the octonion algebra. We shall not construct G_2 as an algebraic group in this book, but its existence is guaranteed by the existence theorem stated in §7.3.

The Classification Theorem

The rank-2 analysis extends to all ranks. The argument proceeds by showing that the Cartan matrix of an irreducible root system is severely constrained: since $a_{ij}a_{ji} \in \{0, 1, 2, 3\}$ for all $i \neq j$, the Dynkin diagram is a connected graph with single, double, or triple edges and at most one arrow per edge. The positive-definiteness of the inner product imposes further constraints (the Cartan matrix must have positive determinant), and a case analysis reduces the classification to a finite check.

Theorem 7.2.11: Classification of Irreducible Root Systems

The irreducible (reduced, crystallographic) root systems are classified by their Dynkin diagrams. The complete list of connected Dynkin diagrams is:

Classical types (four infinite families):

A_n ($n \geq 1$): $\circ - \circ - \cdots - \circ$ (n nodes in a line)

B_n ($n \geq 2$): $\circ - \circ - \cdots - \circ \Rightarrow \circ$ (n nodes, double edge with arrow at the end)

C_n ($n \geq 3$): $\circ - \circ - \cdots - \circ \Leftarrow \circ$ (n nodes, double edge with arrow at the end, reversed)

D_n ($n \geq 4$): $\circ - \circ - \cdots - \frac{\circ}{\circ}$ (n nodes, branching at the end)

Exceptional types (five isolated cases):

G_2 $\circ \Rightarrow \circ$ (triple edge, 2 nodes)

F_4 $\circ - \circ \Rightarrow \circ - \circ$ (4 nodes, double edge in the middle)

E_6 $\circ - \circ - \overset{\circ}{\circ} - \circ - \circ$ (6 nodes, one branch of length 1)

E_7 $\circ - \circ - \overset{\circ}{\circ} - \circ - \circ - \circ$ (7 nodes)

E_8 $\circ - \circ - \overset{\circ}{\circ} - \circ - \circ - \circ - \circ$ (8 nodes)

There are no others. In particular, the only diagrams with multiple edges are B_n , C_n , F_4 (double edges), and G_2 (triple edge). The only diagrams with branching are D_n and E_6, E_7, E_8 .

Proof Strategy:

The proof proceeds in stages, each eliminating large families of potential diagrams.

Step 1: Constraints on edges. The condition $a_{ij}a_{ji} \in \{0, 1, 2, 3\}$ means each pair of connected nodes has a single edge ($a_{ij}a_{ji} = 1$), a double edge ($a_{ij}a_{ji} = 2$), or a triple edge ($a_{ij}a_{ji} = 3$). No quadruple or higher edges exist.

Step 2: No cycles. The positive-definiteness of the bilinear form implies that the underlying graph of the Dynkin diagram is a *tree* (a connected graph with no cycles). Indeed, if the diagram contained a cycle $\alpha_{i_1} - \alpha_{i_2} - \cdots - \alpha_{i_m} - \alpha_{i_1}$, the vector $v = \alpha_{i_1} + \cdots + \alpha_{i_m}$ would satisfy $(v, v) \leq 0$ (using the obtuse-angle property of simple roots and the cycle condition), contradicting positive-definiteness.

Step 3: At most one multiple edge. If the diagram has a double or triple edge, it can have at most one such edge, and the rest of the edges are single. This is proved by bounding the “total energy” $\sum_{i \neq j} a_{ij}a_{ji}$ using the positive-definiteness constraint.

Step 4: Branching constraints. The diagram has at most one branching point (a node of degree ≥ 3), and the branch lengths satisfy $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$ where p, q, r are the lengths of the three branches. The solutions are $(1, q, r)$ for any q, r (type D), $(2, 2, r)$ for $r \geq 2$... actually, the precise condition is: if the three branch lengths from the central node are $p-1, q-1, r-1$ (so the total nodes are $p+q+r-2$), then $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$. The integer solutions (with $p \leq q \leq r$) are:

1. $(1, 1, r)$ for any $r \geq 1$: this gives A_n (a line, i.e. one branch has length 0).
2. $(1, 2, r)$ for $r \geq 2$: this gives D_n (with $n = r + 2$).
3. $(1, 2, 3)$: gives E_6 .
4. $(1, 2, 4)$: gives E_7 .
5. $(1, 2, 5)$: gives E_8 .

No further solutions exist ($(1, 2, 6)$ gives $\frac{1}{2} + \frac{1}{3} + \frac{1}{7} < 1$... actually $(1, 2, 6)$ gives $1 + \frac{1}{2} + \frac{1}{6} = \frac{5}{3} > 1$; the correct constraint is that the associated quadratic form is positive-definite, which excludes $(1, 3, 3)$, $(2, 2, 2)$, $(1, 2, 6)$ by explicit determinant computation).

Step 5: Multiple edges with chains. Diagrams with a double edge and additional nodes are constrained: the double edge can have chains of arbitrary length on one side only. This gives B_n and C_n (double edge at the end) and F_4 (double edge in the middle with exactly one node on each side). Triple edges allow no additional nodes: G_2 is the only possibility.

The full proof is a finite (but combinatorially detailed) verification that these are the only diagrams satisfying all constraints simultaneously. We refer to Humphreys [6] or Bourbaki [1] for the complete argument.

The classification is one of the most beautiful results in mathematics: a simple set of axioms about reflections and integrality leads, through elementary but careful combinatorics, to a complete and finite list. The four classical families (A_n, B_n, C_n, D_n) correspond to the matrix groups $(\text{SL}, \text{SO}_{\text{odd}}, \text{Sp}, \text{SO}_{\text{even}})$, and the five exceptional types $(G_2, F_4, E_6, E_7, E_8)$ correspond to algebraic groups that cannot be described as matrix groups preserving a classical bilinear form. Their existence is guaranteed by the existence theorem in §7.3.

Numerical Data of the Root Systems

We collect the key numerical invariants for reference:

Type	Rank	$ \Phi $	$ W $	Root lengths
A_n	n	$n(n+1)$	$(n+1)!$	all equal
B_n	n	$2n^2$	$2^n n!$	two (short : long = $1 : \sqrt{2}$)
C_n	n	$2n^2$	$2^n n!$	two (short : long = $1 : \sqrt{2}$)
D_n	n	$2n(n-1)$	$2^{n-1} n!$	all equal
G_2	2	12	12	two (short : long = $1 : \sqrt{3}$)
F_4	4	48	1152	two (short : long = $1 : \sqrt{2}$)
E_6	6	72	51840	all equal
E_7	7	126	2903040	all equal
E_8	8	240	696729600	all equal

The simply-laced types (A , D , E) have all roots of equal length; the non-simply-laced types (B , C , F_4 , G_2) have two root lengths. The Weyl group orders grow rapidly: $|W(E_8)| = 696,729,600 = 2^{14} \cdot 3^5 \cdot 5^2 \cdot 7$.

Exercise 7.2.1

1. Verify that the root system $\Phi = \{\pm\varepsilon_i \pm \varepsilon_j : i \neq j\} \cup \{\pm\varepsilon_i\}$ of type B_n (example 7.1.11) satisfies all four axioms of definition 7.2.1. In particular, verify the integrality condition $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$ for a pair of roots of different lengths.
2. Draw the root system G_2 in the plane $\mathbb{R}^2$ (choose coordinates so that the short roots have length 1 and the long roots have length $\sqrt{3}$). Identify all 12 roots, the 6 positive roots, and the 2 simple roots. Compute the Cartan matrix and verify it is $\begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$.
3. Verify that the Weyl group of type D_n has order $2^{n-1} \cdot n!$ by showing it is the group of $n \times n$ signed permutation matrices with an *even* number of sign changes. (*Hint*: the reflection $s_{\varepsilon_i - \varepsilon_j}$ transposes the i -th and j -th coordinates, while $s_{\varepsilon_i + \varepsilon_j}$ transposes them and changes both signs.)
4. The *highest root* $\tilde{\alpha}$ of a root system is the unique root $\tilde{\alpha} = \sum n_i \alpha_i$ with $\sum n_i$ maximal. Compute the highest root for types A_3 , B_3 , C_3 , and G_2 .
5. The *dual root system* $\Phi^\vee$ is the set of coroots $\{\alpha^\vee : \alpha \in \Phi\}$. Show that $\Phi^\vee$ is itself a root system. Determine the type of $\Phi^\vee$ when Φ has type A_n , B_n , C_n , D_n , G_2 . (The answer reveals the duality $B_n \leftrightarrow C_n$ and the self-duality of A_n , D_n , G_2 .)
6. Prove that the Dynkin diagram of an irreducible root system contains no cycles. (*Hint*: if $\alpha_1, \dots, \alpha_m$ form a cycle, consider $v = \alpha_1 + \dots + \alpha_m$ and compute (v, v) , using $(\alpha_i, \alpha_{i+1}) \leq -\frac{1}{2}|\alpha_i||\alpha_{i+1}|$ for adjacent simple roots and $(\alpha_i, \alpha_j) \leq 0$ for all pairs.)

7.3 Root Data and the Classification of Reductive Groups

The root system Φ captures the “shape” of a reductive group G , but it does not capture G completely. Two non-isomorphic groups can share the same root system: for instance, SL_n and PGL_n both have root system A_{n-1} , yet they are not isomorphic (the former has center μ_n , the latter is centerless). The missing information is the *lattice*: the root system Φ sits inside the character lattice $X(T)$, and the position of Φ relative to $X(T)$ determines the group up to isomorphism. This extra data is encoded in the *root datum*, and the main theorems of this section assert that root data classify reductive groups completely.

Root Data

Definition 7.3.1: Root Datum

A *root datum* is a quadruple $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ where:

1. X and $X^\vee$ are free abelian groups of finite rank, equipped with a perfect pairing $\langle \cdot, \cdot \rangle : X \times X^\vee \rightarrow \mathbb{Z}$ (i.e. $X^\vee \cong \mathrm{Hom}_{\mathbb{Z}}(X, \mathbb{Z})$).
2. $\Phi \subseteq X$ and $\Phi^\vee \subseteq X^\vee$ are finite subsets (the *roots* and *coroots*), equipped with a bijection $\Phi \xrightarrow{\sim} \Phi^\vee, \alpha \mapsto \alpha^\vee$.
3. For every $\alpha \in \Phi$: $\langle \alpha, \alpha^\vee \rangle = 2$.
4. For every $\alpha \in \Phi$, the reflection $s_\alpha : X \rightarrow X$ defined by $s_\alpha(\lambda) = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha$ preserves Φ , and the dual reflection $s_\alpha^\vee : X^\vee \rightarrow X^\vee$ defined by $s_\alpha^\vee(\mu) = \mu - \langle \alpha, \mu \rangle \alpha^\vee$ preserves $\Phi^\vee$.

The root datum is called *reduced* if $\alpha \in \Phi$ and $2\alpha \in \Phi$ imply $\alpha = 0$.

The key difference from an abstract root system is that the roots live in a *specific lattice* X , not merely in a real vector space. The lattice X determines the “size” of the center: a larger lattice (more characters) corresponds to a smaller center (fewer relations among characters).

Given a connected reductive group G with maximal torus T , the associated root datum is:

$$\Psi(G, T) = (X(T), \Phi(G, T), X_\vee(T), \Phi^\vee(G, T))$$

where $X(T)$ is the character lattice (definition 5.2.1), $X_\vee(T)$ is the cocharacter lattice (definition 5.2.2), Φ are the roots (definition 7.1.2), $\Phi^\vee$ are the coroots (theorem 7.1.3(5)), and the pairing is $\langle \chi, \lambda \rangle$ from §5.2. The axioms of a root datum are satisfied by the properties established in theorem 7.1.3.

Root Data vs. Root Systems

A root datum determines a root system (embed Φ into $V = X \otimes_{\mathbb{Z}} \mathbb{R}$ and forget the lattice structure), but different root data can give the same root system. The root *system* classifies groups up to *isogeny* (a surjective homomorphism with finite kernel), while the root *datum* classifies groups up to *isomorphism*.

Example 7.3.2: SL_n vs. PGL_n vs. GL_n

All three groups have root system of type A_{n-1} (the roots $\varepsilon_i - \varepsilon_j$ with $i \neq j$). But their root data differ:

SL_n : The maximal torus $T = \{\mathrm{diag}(t_1, \dots, t_n) : \prod t_i = 1\}$ has $X(T) = \mathbb{Z}^n / \mathbb{Z}(1, \dots, 1)$. The roots $\varepsilon_i - \varepsilon_j$ span a sublattice of index 1 in $X(T)$ (they generate all of $X(T)$). This is the *simply connected* root datum of type A_{n-1} : the character lattice $X(T)$ is the *weight lattice* P (the largest lattice compatible with the integrality condition).

PGL_n : The maximal torus is $T' = T/Z(\mathrm{SL}_n) = T/\mu_n$, and $X(T') \subseteq X(T)$ is the sublattice generated by the roots $\varepsilon_i - \varepsilon_j$ themselves. This is the *adjoint* root datum: the character lattice $X(T')$ equals the *root lattice* Q (the smallest lattice containing the roots). The quotient $P/Q \cong \mathbb{Z}/n\mathbb{Z}$ measures the difference between the two lattices and is isomorphic to $Z(\mathrm{SL}_n) = \mu_n$.

GL_n : The maximal torus $T = \mathbb{G}_m^n$ has $X(T) = \mathbb{Z}^n$, which strictly contains the root lattice Q (the roots $\varepsilon_i - \varepsilon_j$ span only the hyperplane $\sum x_i = 0$). The extra direction is the determinant character $\det = \varepsilon_1 + \dots + \varepsilon_n$, reflecting the nontrivial center $Z(\mathrm{GL}_n) = \mathbb{G}_m$. This is neither simply connected nor adjoint: GL_n is reductive but not semisimple, and its root datum records the extra torus direction.

The isogeny $\mathrm{SL}_n \rightarrow \mathrm{PGL}_n$ (with kernel μ_n) corresponds to the inclusion of root data $Q \hookrightarrow P$ (the adjoint lattice sits inside the weight lattice). Every lattice L with $Q \subseteq L \subseteq P$ gives a reductive group “between” PGL_n and SL_n (an *intermediate isogeny form*). For n prime, there are only two choices (Q and P), but for composite n , the subgroups of $P/Q \cong \mathbb{Z}/n\mathbb{Z}$ give additional forms.

More generally, for a semisimple root system Φ , the lattices between the root lattice Q and the weight lattice P are in bijection with the subgroups of the finite abelian group P/Q (the *fundamental group* of the root system). Each such lattice gives a different root datum, hence a different semisimple group. The two extreme cases have names:

Definition 7.3.3: Simply Connected and Adjoint

A semisimple group G is called:

1. *Simply connected* if $X(T) = P$ (the weight lattice), equivalently $Z(G)$ is as large as possible (equal to P/Q).
2. *Adjoint* if $X(T) = Q$ (the root lattice), equivalently $Z(G) = \{e\}$ (the center is trivial).

For each root system Φ , the simply connected group is denoted G_{sc} and the adjoint group G_{ad} . Every semisimple group with root system Φ is a quotient of G_{sc} by a subgroup of $Z(G_{sc})$.

Type	G_{sc} (simply connected)	G_{ad} (adjoint)	P/Q
A_n	SL_{n+1}	PGL_{n+1}	$\mathbb{Z}/(n+1)\mathbb{Z}$
B_n	$Spin_{2n+1}$	SO_{2n+1}	$\mathbb{Z}/2\mathbb{Z}$
C_n	Sp_{2n}	PGL_{2n}	$\mathbb{Z}/2\mathbb{Z}$
D_n (n odd)	$Spin_{2n}$	PGL_{2n}	$\mathbb{Z}/4\mathbb{Z}$
D_n (n even)	$Spin_{2n}$	PGL_{2n}	$(\mathbb{Z}/2\mathbb{Z})^2$
G_2	G_2	G_2	1
F_4	F_4	F_4	1
E_6	E_6^{sc}	E_6^{ad}	$\mathbb{Z}/3\mathbb{Z}$
E_7	E_7^{sc}	E_7^{ad}	$\mathbb{Z}/2\mathbb{Z}$
E_8	E_8	E_8	1

When $P/Q = 1$ (types G_2, F_4, E_8), the simply connected and adjoint forms coincide: there is a unique semisimple group with the given root system.

The Isomorphism Theorem

The first main theorem says that the root datum is a *complete* invariant: it determines the reductive group up to isomorphism.

Theorem 7.3.4: Isomorphism Theorem

Let G and G' be connected reductive groups over an algebraically closed field k , with maximal tori $T \subseteq G$ and $T' \subseteq G'$. Let $\Psi = \Psi(G, T)$ and $\Psi' = \Psi(G', T')$ be the associated root data. Then every isomorphism of root data $\varphi : \Psi \xrightarrow{\sim} \Psi'$ (i.e. an isomorphism $\varphi : X(T') \xrightarrow{\sim} X(T)$ carrying Φ' to Φ and $(\Phi')^\vee$ to $\Phi^\vee$, and preserving the pairing) is induced by an isomorphism of algebraic groups $f : G \xrightarrow{\sim} G'$ with $f(T) = T'$.

More precisely: the natural map

$$\left\{ f : G \xrightarrow{\sim} G' \mid f(T) = T' \right\} / \text{Inn}(T') \longrightarrow \text{Isom}(\Psi', \Psi)$$

is a bijection, where $\text{Inn}(T')$ denotes inner automorphisms by elements of T' (which act trivially on Ψ').

Proof Key ideas:

The proof proceeds in two stages.

Stage 1: The case $G = G', T = T'$ (automorphisms). An isomorphism of root data $\varphi : \Psi \rightarrow \Psi$ is an automorphism of the root datum. This is an element of $\text{Aut}(\Psi) = W \rtimes \text{Aut}(\text{Dynkin diagram})$ (the Weyl group acts by inner automorphisms of Ψ , and the diagram automorphisms are the “outer” part). One shows that every such automorphism lifts to an automorphism of

G : the Weyl group elements lift to inner automorphisms $\text{Inn}(n_w)$ for $n_w \in N_G(T)$, and the diagram automorphisms lift to “graph automorphisms” of G (for instance, the nontrivial diagram automorphism of A_n corresponds to the outer automorphism $g \mapsto (g^T)^{-1}$ of SL_{n+1}).

Stage 2: The general case. Given an isomorphism $\varphi : \Psi' \rightarrow \Psi$, one constructs an isomorphism $f : G \rightarrow G'$ by first building isomorphisms of the SL_2 -subgroups (one for each simple root), then gluing them using the *Bruhat decomposition*: $G = \bigsqcup_{w \in W} BwB$ (the decomposition of G into double cosets of B). The Bruhat decomposition gives a combinatorial “skeleton” of G in terms of B , T , and the root groups U_α (one for each root), and the root datum determines the gluing data completely. The isomorphism φ provides compatible identifications of the root groups, which extend to a global isomorphism $f : G \rightarrow G'$.

The full proof is substantial and requires the detailed theory of root groups and the Bruhat decomposition. We refer to Springer [9] or Milne [8] for the complete argument.

The Existence Theorem

The second main theorem is the converse: every root datum actually arises from a reductive group.

Theorem 7.3.5: Existence Theorem

Let $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ be a reduced root datum. Then there exists a connected reductive group G over any algebraically closed field k , together with a maximal torus $T \subseteq G$, such that $\Psi(G, T) \cong \Psi$.

Proof Key ideas:

For the classical root data (A_n, B_n, C_n, D_n) , the groups were constructed explicitly in Chapter 1 as matrix groups (SL , SO , Sp , and their variants). The content of the existence theorem is that the five exceptional root data $(G_2, F_4, E_6, E_7, E_8)$ also correspond to algebraic groups.

The general construction is due to Chevalley (1955). The idea is to build G directly from the root datum by generators and relations, working inside the “Chevalley algebra” — a Lie algebra over $\mathbb{Z}$ associated to the root system. Specifically:

1. Start with the complex semisimple Lie algebra $\mathfrak{g}_{\mathbb{C}}$ of the given type (constructed, for instance, via the Serre relations from the Cartan matrix).
2. Choose a *Chevalley basis*: a basis $\{h_i, e_\alpha\}$ of $\mathfrak{g}_{\mathbb{C}}$ (indexed by simple coroots h_i and roots $\alpha \in \Phi$) with the property that all structure constants are *integers*.
3. Define $\mathfrak{g}_{\mathbb{Z}} = \bigoplus \mathbb{Z}h_i \oplus \bigoplus_{\alpha} \mathbb{Z}e_\alpha$ (the “Chevalley $\mathbb{Z}$ -form”). For any field k , set $\mathfrak{g}_k = \mathfrak{g}_{\mathbb{Z}} \otimes_{\mathbb{Z}} k$.
4. The group $G(k)$ is generated by the “root group elements” $x_\alpha(t) = \exp(t \cdot e_\alpha)$ for $\alpha \in \Phi$ and $t \in k$. Here $\exp$ is the formal exponential, which terminates because e_α acts nilpotently on every finite-dimensional representation. The group G is defined as the group scheme representing the functor $R \mapsto \langle x_\alpha(t) : \alpha \in \Phi, t \in R \rangle$.

5. One verifies that G is a connected reductive group with root datum Ψ , by checking the commutation relations of the root group elements (the “Steinberg relations”) and constructing the maximal torus and Borel subgroup.

The beauty of Chevalley’s construction is that it works *uniformly over any field* (even over $\mathbb{Z}$!), producing “split” reductive groups. This is why the classification of split reductive groups is the same in every characteristic: the Chevalley groups $G(k)$ depend only on the root datum Ψ and the field k , not on the characteristic of k .

For the complete construction, see Steinberg [10] or Milne [8]. The original paper is Chevalley [2].

The Classification

Combining the isomorphism and existence theorems, we obtain the classification of split reductive groups:

Theorem 7.3.6: Classification of Split Reductive Groups

The map $G \mapsto \Psi(G, T)$ is a bijection:

$$\left\{ \begin{array}{l} \text{Split connected reductive} \\ \text{groups over } k \text{ (up to isom.)} \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{l} \text{Reduced root data} \\ \text{(up to isom.)} \end{array} \right\}$$

In particular, for *semisimple* groups, the classification reduces to: a choice of root system (one of the Dynkin diagrams from theorem 7.2.11), together with a choice of lattice X between the root lattice Q and the weight lattice P .

For semisimple groups, the lattice choice corresponds to a subgroup of the fundamental group P/Q , which is a finite abelian group computed from the Dynkin diagram. The extreme choices give the simply connected and adjoint forms.

Example 7.3.7: The Complete List for Type A_2

The root system A_2 has $P/Q \cong \mathbb{Z}/3\mathbb{Z}$. The subgroups of $\mathbb{Z}/3\mathbb{Z}$ are $\{0\}$ and $\mathbb{Z}/3\mathbb{Z}$ itself (since 3 is prime, there are no intermediate subgroups). The two semisimple groups with root system A_2 are:

1. SL_3 (simply connected, $X(T) = P$, center μ_3).
2. PGL_3 (adjoint, $X(T) = Q$, center trivial).

These are connected by the isogeny $\mathrm{SL}_3 \rightarrow \mathrm{PGL}_3$ with kernel μ_3 .

Example 7.3.8: The Complete List for Type D_4

The root system D_4 has $P/Q \cong (\mathbb{Z}/2\mathbb{Z})^2$. The subgroups of $(\mathbb{Z}/2\mathbb{Z})^2$ are: $\{0\}$ (one), three subgroups isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and $(\mathbb{Z}/2\mathbb{Z})^2$ itself. This gives 5 lattices between Q and P , hence *five* semisimple groups with root system D_4 :

1. Spin_8 (simply connected).
2. Three “half-spin” groups HSpin_8 (intermediate, each with center $\mathbb{Z}/2\mathbb{Z}$).
3. PGL_8 (adjoint, centerless).

The three intermediate forms are permuted by the *triality automorphism* of D_4 (the S_3 -symmetry of the D_4 Dynkin diagram, which has three “legs” meeting at a central node). This phenomenon — multiple intermediate isogeny forms permuted by diagram automorphisms — is unique to D_4 among the classical types.

Beyond Split Groups

The classification theorem, as stated, applies to *split* reductive groups: those possessing a maximal torus T that is split over the ground field k (i.e. $T \cong \mathbb{G}_m^r$ over k itself). Over a non-algebraically closed field, there are additional “non-split” or “twisted” forms of each group, classified by Galois cohomology.

For a split group G over k , the set of k -forms (reductive groups G' over k with $G'_k \cong G_k$) is classified by the pointed set $H^1(\mathrm{Gal}(\bar{k}/k), \mathrm{Aut}(G_{\bar{k}}))$ — the first Galois cohomology of the automorphism group. The automorphism group $\mathrm{Aut}(G)$ fits into an exact sequence

$$1 \longrightarrow G_{ad}(k) \longrightarrow \mathrm{Aut}(G)(k) \longrightarrow \mathrm{Aut}(\Psi) \longrightarrow 1$$

where G_{ad} is the adjoint form (inner automorphisms) and $\mathrm{Aut}(\Psi)$ is the group of diagram automorphisms (outer automorphisms). The inner forms (those where the Galois action on Ψ is trivial) are classified by $H^1(k, G_{ad})$, while the outer forms involve nontrivial Galois action on the Dynkin diagram.

This is the connection to the twisted finite groups and Galois cohomology discussed in Chapter 1 (§1.1.1): the “twisting” of algebraic groups by Galois cocycles is the higher-dimensional analogue of the twisting of finite group schemes that we studied there. A full development of the theory of non-split groups over arbitrary fields belongs to the next stage of the subject (the theory of *reductive groups over general fields*, as developed by Borel–Tits), and is beyond the scope of this book.

Closing Remarks

We close Part I by summarizing the path we have traveled. Starting from the definition of an affine algebraic group (Chapter 1), we developed its representation theory (Chapter 2), extracted its infinitesimal invariant the Lie algebra (Chapter 3), constructed quotients (Chapter 4), decomposed its elements via Jordan decomposition and studied the atomic pieces — tori and unipotent groups (Chapter 5) — proved the Lie–Kolchin theorem, Borel’s fixed-point theorem, and the conjugacy theorems (Chapter 6), and finally arrived at the root system, which classifies semisimple groups via Dynkin diagrams (Chapter 7).

The two main theorems — isomorphism and existence — assert that the passage from geometry (algebraic groups) to combinatorics (root data) loses no information and omits nothing: every root datum arises, and groups with the same root datum are isomorphic.

Part II of this book will reverse the arrow: starting from the root datum and the combinatorics of the Weyl group, we will develop the *representation theory* of reductive groups. The central objects will

be the *highest weight modules* (finite-dimensional representations parametrized by dominant weights in $X(T)$), the *Weyl character formula* (an explicit formula for the character of a highest weight module in terms of the Weyl group), and the geometry of the flag variety G/B (Schubert calculus, the Borel–Weil–Bott theorem, and geometric representation theory). The root datum, constructed here as a classification tool, will become the organizing principle for all of representation theory.

Exercise 7.3.1

1. Write down the root datum $\Psi(\mathrm{SL}_2, T)$ explicitly: $X(T)$, $X_\vee(T)$, Φ , $\Phi^\vee$, and the pairing. Then write down $\Psi(\mathrm{PGL}_2, T')$. Show that the two root data are not isomorphic (even though the root systems are both A_1). What is the relationship between them?
2. For the root system A_2 , compute the weight lattice P and root lattice Q explicitly (as sublattices of $\mathbb{R}^3$ in the hyperplane $\sum x_i = 0$). Verify that $P/Q \cong \mathbb{Z}/3\mathbb{Z}$. Identify the *fundamental weights* $\omega_1, \omega_2 \in P$ (defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$).
3. The isogeny $\pi : \mathrm{SL}_n \rightarrow \mathrm{PGL}_n$ (with kernel μ_n) induces a map on root data. Describe this map explicitly: what does $\pi^* : X(T') \rightarrow X(T)$ look like on the character lattices? Verify that π^* sends the root lattice Q into the weight lattice P , and that the cokernel $P/\pi^*(Q)$ is isomorphic to μ_n .
4. For the root system G_2 (rank 2, 12 roots), the fundamental group P/Q is trivial. Conclude that there is a *unique* semisimple group with root system G_2 (up to isomorphism) over any algebraically closed field. This group is denoted G_2 .
5. Let $G = \mathrm{GL}_n$ with maximal torus $T = \mathbb{G}_m^n$. Write down the root datum $\Psi(\mathrm{GL}_n, T)$. Show that $X(T) = \mathbb{Z}^n$, $\Phi = \{\varepsilon_i - \varepsilon_j\}$, and $\Phi^\vee = \{\varepsilon_i^\vee - \varepsilon_j^\vee\}$, and verify the pairing $\langle \varepsilon_i, \varepsilon_j^\vee \rangle = \delta_{ij}$. Explain why this root datum is not semisimple (i.e. Φ does not span $X(T) \otimes \mathbb{Q}$) and identify the “extra” direction.
6. The Dynkin diagram of D_4 has a symmetry group isomorphic to S_3 (permuting the three outer nodes). This gives a group of outer automorphisms of Spin_8 called *triality*. Show that triality permutes the three 8-dimensional representations of Spin_8 : the vector representation, and the two half-spin representations. (*Hint*: the three outer nodes of D_4 correspond to the three fundamental weights $\omega_1, \omega_3, \omega_4$, each of which defines an 8-dimensional representation; triality permutes these weights.)

Part II

Representation Theory

The goal of Part II is to classify, compute with, and geometrically construct the representations of a reductive algebraic group. Where Part I was the “anatomy” — dismantling the group into its root system — Part II is the “action”: we now ask how the group acts on vector spaces, and what the combinatorics of the root system can tell us about these actions.

The starting point is the weight space decomposition from Chapter 5: any representation V of a reductive group G , when restricted to a maximal torus T , breaks into eigenspaces $V = \bigoplus V_\lambda$ indexed by characters $\lambda \in X(T)$. The root system Φ — the set of weights of the adjoint representation — is the “skeleton” around which all other representations are organized. The irreducible representations are classified by their *highest weights*: dominant characters in $X(T)$, one for each cone in the weight lattice determined by the Weyl chamber. This is the content of Chapter 8, which establishes the highest weight classification and proves the complete reducibility theorem in characteristic 0.

Chapter 9 turns from classification to computation. The *Weyl character formula* expresses the character of an irreducible representation as an alternating sum over the Weyl group, and it implies explicit formulas for dimensions, weight multiplicities, and tensor product decompositions. This is the most powerful computational tool in the subject, and we develop it with extensive examples for the classical groups.

Chapter 10 brings geometry back to the center. The flag variety G/B , constructed in Chapter 6 as a projective variety, carries a family of line bundles $\mathcal{L}(\lambda)$ parametrized by the weight lattice $X(T)$. The *Borel–Weil theorem* asserts that the global sections of $\mathcal{L}(\lambda)$, for dominant λ , are the irreducible representations — they are constructed geometrically, as spaces of sections on G/B , rather than algebraically via generators and relations. The *Borel–Weil–Bott theorem* extends this to higher cohomology, showing that the Weyl character formula is a shadow of the cohomological structure of line bundles on the flag variety. This closes the circle between the geometry of G/B (Part I) and the representation theory of G (Part II): the flag variety is the machine that *produces* representations.

Chapter 11 addresses the elephant in the room: what happens in positive characteristic? The Borel–Weil construction and the Weyl modules $V(\lambda)$ still exist, but they are no longer irreducible. Complete reducibility fails, the simple modules $L(\lambda)$ are mysterious proper quotients of the Weyl modules, and the problem of computing their characters remains open in general. We develop the Frobenius twist and Steinberg’s tensor product theorem — which reduce the problem to “restricted” weights — and state Lusztig’s conjecture, which gives a conjectural answer in terms of Kazhdan–Lusztig polynomials for sufficiently large primes. The chapter is deliberately open-ended: it brings the reader to the frontier of an active area of research, equipped with the tools and language to engage with the current literature.

A word on prerequisites. The reader should have Part I firmly in hand, especially the weight space decomposition (Chapter 5), the Borel subgroup theory and flag varieties (Chapter 6), and the root system combinatorics (Chapter 7). Some familiarity with sheaf cohomology on projective varieties (see [3]) is needed for Chapter 10. No prior knowledge of Lie algebra representations is assumed: we work entirely with algebraic group representations, though we occasionally note the connection to the Lie-algebraic theory for the reader’s benefit.

Highest Weight Theory

In Part I, we classified the irreducible “building blocks” of a reductive group G : the roots, which are the weights of the adjoint representation. In this chapter, we classify the irreducible “building blocks” of $\text{Rep}(G)$: the irreducible finite-dimensional representations. The answer is strikingly parallel — they are indexed by a discrete combinatorial datum, the *dominant weights* in the character lattice $X(T)$ — and the proof uses the same mechanism: the decomposition of representations into weight spaces under a maximal torus.

The chapter has three parts. First, we study the weight structure of an arbitrary representation: the weights form a finite, Weyl-group-invariant subset of $X(T)$ with a natural partial order. Second, we prove the classification theorem: irreducible representations are in bijection with dominant weights, via the *highest weight*. Third, we prove Weyl’s complete reducibility theorem in characteristic 0: every finite-dimensional representation is a direct sum of irreducibles. Together, these results give a complete and computable description of $\text{Rep}(G)$ in characteristic 0.

Throughout this chapter, k denotes an algebraically closed field, G a connected reductive group over k , $T \subseteq B \subseteq G$ a maximal torus inside a Borel subgroup, $\Phi = \Phi(G, T)$ the root system, $\Phi^+ = \Phi^+(G, T, B)$ the positive roots, and $\Delta = \{\alpha_1, \dots, \alpha_r\}$ the simple roots.

8.1 Weights of a Representation

Weight Space Decomposition for G -Modules

Let V be a finite-dimensional representation of G . Since $T \subseteq G$ is a torus, restricting the G -action to T and applying the weight space decomposition (theorem 5.2.9) gives:

$$V = \bigoplus_{\lambda \in X(T)} V_\lambda$$

where $V_\lambda = \{v \in V \mid t \cdot v = \lambda(t)v \text{ for all } t \in T(R), \text{ all } R\}$. Most of these weight spaces are zero; the finitely many λ with $V_\lambda \neq 0$ carry all the information.

Definition 8.1.1: Weights and Weight Spaces

Let V be a finite-dimensional representation of G . The *weights* of V are

$$\text{wt}(V) = \{\lambda \in X(T) \mid V_\lambda \neq 0\}$$

For each $\lambda \in \text{wt}(V)$, the subspace V_λ is the *weight space* and $m_\lambda(V) = \dim_k V_\lambda$ is the *multiplicity* of λ in V .

The weight spaces interact with the root spaces of $\mathfrak{g}$ in a predictable way:

Proposition 8.1.2: Root Operators Shift Weights

Let V be a representation of G , and let $\alpha \in \Phi$ be a root. If $X \in \mathfrak{g}_\alpha$ (a root vector) and $v \in V_\lambda$ (a weight vector), then $X \cdot v \in V_{\lambda+\alpha}$. That is, the action of a root vector of weight α shifts the weight by α :

$$\mathfrak{g}_\alpha \cdot V_\lambda \subseteq V_{\lambda+\alpha}$$

Proof:

For $t \in T(k)$ and $X \in \mathfrak{g}_\alpha$, the adjoint action gives $\text{Ad}(t)(X) = \alpha(t)X$. The infinitesimal action of X on $v \in V_\lambda$ satisfies:

$$t \cdot (X \cdot v) = (\text{Ad}(t)(X)) \cdot (t \cdot v) = (\alpha(t)X) \cdot (\lambda(t)v) = \alpha(t)\lambda(t)(X \cdot v) = (\lambda + \alpha)(t)(X \cdot v)$$

where the first equality uses the compatibility of the group action and the Lie algebra action: $t \cdot (X \cdot v) = \text{Ad}(t)(X) \cdot (t \cdot v)$, which follows from differentiating the conjugation identity $\rho(t) \circ d\rho(X) \circ \rho(t)^{-1} = d\rho(\text{Ad}(t)(X))$. Hence $X \cdot v \in V_{\lambda+\alpha}$.

This proposition is the engine of highest weight theory: the root vectors of $\mathfrak{g}$ act as “raising” and “lowering” operators, moving weight vectors from one weight space to an adjacent one (shifted by a root). The positive root vectors X_α with $\alpha \in \Phi^+$ are the raising operators, and the negative root vectors $X_{-\alpha}$ are the lowering operators.

Weyl Group Invariance

The weight set $\text{wt}(V)$ is not just any subset of $X(T)$; it is invariant under the Weyl group.

Proposition 8.1.3: W -Invariance of Weights

Let V be a finite-dimensional representation of G . Then:

1. For every $w \in W$ and $\lambda \in \text{wt}(V)$, we have $w(\lambda) \in \text{wt}(V)$ and $m_{w(\lambda)}(V) = m_\lambda(V)$. That is, the Weyl group permutes the weights and preserves multiplicities.
2. $\text{wt}(V)$ is a finite, W -stable subset of $X(T)$.

Proof:

Let $w \in W$ and choose a representative $n_w \in N_G(T)(k)$. Define $\varphi_w : V_\lambda \rightarrow V_{w(\lambda)}$ by $\varphi_w(v) = n_w \cdot v$. This is well-defined: for $v \in V_\lambda$ and $t \in T(k)$,

$$t \cdot (n_w \cdot v) = n_w \cdot (n_w^{-1} t n_w \cdot v) = n_w \cdot (w^{-1}(t) \cdot v) = n_w \cdot (\lambda(w^{-1}(t))v) = (w(\lambda))(t) \cdot n_w \cdot v$$

where we used $w^{-1}(t) = n_w^{-1} t n_w$ (the action of W on T) and $(w(\lambda))(t) = \lambda(w^{-1}(t))$ (the contragredient action on characters). So $n_w \cdot v \in V_{w(\lambda)}$.

The map φ_w is a linear isomorphism $V_\lambda \xrightarrow{\sim} V_{w(\lambda)}$ (with inverse $\varphi_{w^{-1}}$), so $\dim V_{w(\lambda)} = \dim V_\lambda$.

The Partial Order on Weights

The root lattice $Q = \mathbb{Z}\Phi \subseteq X(T)$ defines a natural partial order on $X(T)$:

Definition 8.1.4: The Dominance Partial Order

For $\lambda, \mu \in X(T)$, we write $\mu \preceq \lambda$ (and say “ μ is *dominated by* λ ”) if $\lambda - \mu$ is a non-negative integer linear combination of positive roots:

$$\mu \preceq \lambda \iff \lambda - \mu \in \mathbb{Z}_{\geq 0}\Phi^+ := \left\{ \sum_{\alpha \in \Phi^+} n_\alpha \alpha \mid n_\alpha \in \mathbb{Z}_{\geq 0} \right\}$$

Equivalently, $\lambda - \mu = \sum_{i=1}^r n_i \alpha_i$ with $n_i \in \mathbb{Z}_{\geq 0}$ (a non-negative combination of *simple* roots, since every positive root is a non-negative combination of simple roots by proposition 7.1.6).

This is a partial order on $X(T)$: reflexive ($\lambda \preceq \lambda$ since $\lambda - \lambda = 0$), antisymmetric (if $\mu \preceq \lambda$ and $\lambda \preceq \mu$, then $\lambda - \mu$ and $\mu - \lambda$ are both non-negative combinations of positive roots, which forces $\lambda = \mu$), and transitive.

The partial order is compatible with the Lie algebra action in the following sense: if $v \in V_\lambda$ and $X \in \mathfrak{g}_{-\alpha}$ (a negative root vector), then $X \cdot v \in V_{\lambda-\alpha}$, and $\lambda - \alpha \prec \lambda$ (strictly dominated). So the lowering operators move weights *down* in the partial order, and the raising operators move them *up*.

Proposition 8.1.5: Weights are Bounded Above

Let V be a finite-dimensional representation of G . The set $\text{wt}(V)$ has finitely many maximal elements with respect to $\preceq$. If V is irreducible, there is a *unique* maximal weight.

Proof:

Since $\text{wt}(V)$ is finite, it has at least one maximal element. Now suppose V is irreducible and $\lambda \in \text{wt}(V)$ is maximal. Let $v \in V_\lambda$ be nonzero. For any positive root $\alpha \in \Phi^+$ and any root vector $X_\alpha \in \mathfrak{g}_\alpha$, we have $X_\alpha \cdot v \in V_{\lambda+\alpha}$. Since λ is maximal and $\lambda + \alpha \succ \lambda$, we must have $V_{\lambda+\alpha} = 0$ (otherwise λ would not be maximal). So $X_\alpha \cdot v = 0$ for all $\alpha \in \Phi^+$.

Now consider the subspace $U \subseteq V$ generated by v under the action of $\mathfrak{g}$ (equivalently, under

the action of G , by the Fundamental Finiteness Theorem and the density of G in the formal neighborhood of e). Since V is irreducible, $U = V$. But U is spanned by elements of the form $X_{\beta_1} \cdots X_{\beta_m} \cdot v$ where $\beta_i \in \Phi^- \cup \{0\}$ (applying raising operators gives 0 since λ is maximal, so only lowering operators and Cartan elements contribute). Each such element has weight $\lambda + \beta_1 + \cdots + \beta_m \preceq \lambda$. Therefore every weight $\mu \in \text{wt}(V)$ satisfies $\mu \preceq \lambda$.

Uniqueness: if λ' were another maximal weight, then $\lambda' \preceq \lambda$ (by the above) and $\lambda \preceq \lambda'$ (by the same argument applied to λ'), so $\lambda = \lambda'$.

The unique maximal weight of an irreducible representation is the central definition of this chapter:

Definition 8.1.6: Highest Weight

Let V be a finite-dimensional irreducible representation of G . The *highest weight* of V is the unique maximal element $\lambda \in \text{wt}(V)$ with respect to the dominance order $\preceq$. A nonzero vector $v^+ \in V_\lambda$ is called a *highest weight vector*.

The highest weight vector v^+ is characterized by two properties: it has weight λ (i.e. $t \cdot v^+ = \lambda(t)v^+$ for all $t \in T$), and it is killed by all positive root vectors ($X_\alpha \cdot v^+ = 0$ for all $\alpha \in \Phi^+$). Equivalently, v^+ is a simultaneous eigenvector for $B = T \ltimes R_u(B)$: the torus T acts by λ and the unipotent radical $R_u(B)$ acts trivially (since $\text{Lie}(R_u(B)) = \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha$ and all positive root vectors kill v^+).

Proposition 8.1.7: Highest Weight Space is One-Dimensional

Let V be a finite-dimensional irreducible representation of G with highest weight λ . Then $\dim V_\lambda = 1$.

Proof:

Suppose $\dim V_\lambda \geq 2$ and let $v_1, v_2 \in V_\lambda$ be linearly independent highest weight vectors. Both are killed by $\mathfrak{g}_\alpha$ for all $\alpha \in \Phi^+$. The subspace $U_1 \subseteq V$ generated by v_1 under the action of $\mathfrak{g}$ is a nonzero subrepresentation. Since V is irreducible, $U_1 = V$.

Now consider v_2 . Since $v_2 \in V_\lambda$ and all weights of V are $\preceq \lambda$, the element v_2 cannot be in the $\mathfrak{g}$ -span of $\{X_\beta \cdot v_1 : \beta \in \Phi^-\}$ unless $v_2 \in kv_1$ (because the only way to return to weight λ from a lower weight is via raising operators, which kill v_1). More precisely: the $\mathfrak{g}$ -submodule generated by v_1 has a unique vector of weight λ up to scalar (namely v_1 itself), because any element of weight λ in U_1 must be in the kernel of all raising operators and in the span of vectors obtained by lowering and then raising back — but raising from a lower weight cannot exceed λ in an irreducible module where λ is maximal. This forces $v_2 \in kv_1$, a contradiction.

A cleaner argument: the space $V^{R_u(B)} = \{v \in V : X \cdot v = 0 \text{ for all } X \in \text{Lie}(R_u(B))\}$ of $R_u(B)$ -invariants is a representation of T (since $B = T \ltimes R_u(B)$ and T normalizes $R_u(B)$). In an irreducible G -module, $V^{R_u(B)}$ is one-dimensional: if it were higher-dimensional, the B -action on $V^{R_u(B)}$ would decompose into multiple one-dimensional T -eigenspaces, and the G -submodule generated by any single eigenvector would be a proper submodule (it could not reach the other eigenvectors by lowering operators, since those lower the weight and cannot raise back to a different maximal-weight vector). By irreducibility, $V^{R_u(B)}$ must be one-dimensional, and it equals V_λ .

Example 8.1.8: Weights of the Irreducible Representations of SL_2

For $G = \mathrm{SL}_2$ with $T = \{\mathrm{diag}(t, t^{-1})\}$, we have $X(T) = \mathbb{Z}$ (generated by ε where $\varepsilon(\mathrm{diag}(t, t^{-1})) = t$). The positive root is $\alpha = 2\varepsilon$ (so $\Phi^+ = \{2\varepsilon\}$), and the dominance order is $m \preceq n$ iff $n - m \in 2\mathbb{Z}_{\geq 0}$.

The irreducible representations are the symmetric powers $L(d) = \mathrm{Sym}^d(k^2)$ for $d \geq 0$ (?? 4). The highest weight of $L(d)$ is $d\varepsilon$ (the weight of the monomial x^d), and the weights are:

$$\mathrm{wt}(L(d)) = \{d\varepsilon, (d-2)\varepsilon, (d-4)\varepsilon, \dots, -(d-2)\varepsilon, -d\varepsilon\}$$

each with multiplicity 1. The highest weight vector is $v^+ = x^d$, and it is killed by $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ (the positive root vector). The lowering operator $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ sends $x^{d-k}y^k \mapsto (d-k)x^{d-k-1}y^{k+1}$ (up to scalar), moving down by 2ε at each step.

The dominance partial order on $\mathbb{Z}$ is $m \preceq n$ iff $n - m \in 2\mathbb{Z}_{\geq 0}$. The highest weight $d\varepsilon$ dominates all other weights $\{(d-2j)\varepsilon : 0 \leq j \leq d\}$, confirming proposition 8.1.5. Note that d must be a non-negative integer: a “weight -1 ” or a “weight $1/2$ ” cannot be the highest weight of a finite-dimensional representation. The constraint is that $\langle d\varepsilon, \alpha^\vee \rangle = d \geq 0$ (dominance with respect to Φ^+).

Example 8.1.9: Weights of the Standard Representation of SL_3

For $G = \mathrm{SL}_3$ with the standard representation $V = k^3$, the maximal torus $T = \{\mathrm{diag}(t_1, t_2, (t_1 t_2)^{-1})\}$ acts with weights $\varepsilon_1, \varepsilon_2, \varepsilon_3 = -\varepsilon_1 - \varepsilon_2$ (one for each basis vector e_i). The positive roots are $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$, and $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$.

The highest weight is ε_1 (it dominates $\varepsilon_2 = \varepsilon_1 - \alpha_1$ and $\varepsilon_3 = \varepsilon_1 - \alpha_1 - \alpha_2$). The highest weight vector is e_1 :

$$E_{12} \cdot e_1 = 0, \quad E_{13} \cdot e_1 = 0$$

(the positive root vectors E_{12} and E_{13} kill e_1). Lowering: $E_{21} \cdot e_1 = e_2$ (weight ε_2) and $E_{32} \cdot e_2 = e_3$ (weight ε_3).

In the language of dominant weights: $\varepsilon_1 = \omega_1$ is the first fundamental weight (defined by $\langle \omega_1, \alpha_1^\vee \rangle = 1$, $\langle \omega_1, \alpha_2^\vee \rangle = 0$). The standard representation is $L(\omega_1)$.

Example 8.1.10: Weights of the Adjoint Representation of SL_3

The adjoint representation $\mathfrak{g} = \mathfrak{sl}_3$ has dimension 8. The weights are the roots $\Phi = \{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2)\}$ (each with multiplicity 1) together with 0 (with multiplicity 2 = $\mathrm{rank}(\mathrm{SL}_3)$). The highest weight is $\alpha_1 + \alpha_2 = \varepsilon_1 - \varepsilon_3$ (the highest root), and the adjoint representation is $L(\alpha_1 + \alpha_2)$.

In terms of fundamental weights: $\alpha_1 + \alpha_2 = \omega_1 + \omega_2$ (the sum of the two fundamental weights). The fact that the adjoint representation has highest weight $\omega_1 + \omega_2$ means it appears in the tensor product $L(\omega_1) \otimes L(\omega_2)$ as a direct summand (in characteristic 0). Indeed, $L(\omega_1) = k^3$ (the standard representation), $L(\omega_2) = (k^3)^\vee \cong \bigwedge^2(k^3)$ (the dual, which is the second exterior power for SL_3), and $L(\omega_1) \otimes L(\omega_2) \cong \mathfrak{sl}_3 \oplus k$ (the adjoint plus the trivial representation), recovering the decomposition $\mathfrak{gl}_3 = \mathfrak{sl}_3 \oplus kI$ from example 3.2.3.

Dominant Weights

The SL_2 example showed that the highest weight of an irreducible representation must satisfy $\langle \lambda, \alpha^\vee \rangle \geq 0$ for each positive root α . This condition has a name:

Definition 8.1.11: Dominant Weights

A weight $\lambda \in X(T)$ is called *dominant* (with respect to B) if

$$\langle \lambda, \alpha_i^\vee \rangle \geq 0 \quad \text{for all simple roots } \alpha_i \in \Delta$$

The set of dominant weights is denoted $X(T)^+ = X(T)_B^+$. In terms of fundamental weights $\omega_1, \dots, \omega_r$ (defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$):

$$X(T)^+ = \left\{ \lambda = \sum_{i=1}^r n_i \omega_i \mid n_i \in \mathbb{Z}_{\geq 0} \right\} = \mathbb{Z}_{\geq 0} \omega_1 + \cdots + \mathbb{Z}_{\geq 0} \omega_r$$

when G is semisimple (so that $X(T)$ is spanned by the fundamental weights).

The dominant weights form a “cone” in $X(T)$: the set of non-negative integer combinations of the fundamental weights. Geometrically, they are the lattice points in the closure of the dominant Weyl chamber $\overline{C} = \{ \lambda \in X(T) \otimes \mathbb{R} \mid \langle \lambda, \alpha_i^\vee \rangle \geq 0 \text{ for all } i \}$.

Proposition 8.1.12: Highest Weights are Dominant

If V is a finite-dimensional irreducible representation of G with highest weight λ , then λ is dominant: $\lambda \in X(T)^+$.

Proof:

For each simple root α_i , the SL_2 -subgroup $\varphi_{\alpha_i} : \mathrm{SL}_2 \rightarrow G$ (theorem 7.1.3) restricts the representation V to an SL_2 -representation. The highest weight vector $v^+ \in V_\lambda$ generates an irreducible SL_2 -subrepresentation (it is killed by the raising operator $e_{\alpha_i} \in \mathfrak{g}_{\alpha_i}$, so it is a highest weight vector for SL_2 as well). The highest weight of this SL_2 -representation is $\langle \lambda, \alpha_i^\vee \rangle$ (by the identification of $X(T|_{\mathrm{SL}_2}) = \mathbb{Z}$ with $\mathbb{Z}$ via the coroot $\alpha_i^\vee$).

From the SL_2 classification (example 8.1.8), the highest weight of a finite-dimensional irreducible SL_2 -representation is a non-negative integer $d \geq 0$. Therefore $\langle \lambda, \alpha_i^\vee \rangle \geq 0$ for every simple root α_i , which means λ is dominant.

This brings us to the threshold of the classification theorem: we have shown that every irreducible representation has a dominant highest weight. In the next section, we will prove the converse — every dominant weight arises as the highest weight of a unique irreducible representation — completing the classification of $\mathrm{Rep}(G)$.

Example 8.1.13: Dominant Weights for SL_3

The fundamental weights are $\omega_1 = \varepsilon_1$ and $\omega_2 = \varepsilon_1 + \varepsilon_2 = -\varepsilon_3$. The dominant weights are:

$$X(T)^+ = \{ a\omega_1 + b\omega_2 \mid a, b \in \mathbb{Z}_{\geq 0} \} = \{ a\varepsilon_1 + b(\varepsilon_1 + \varepsilon_2) \mid a, b \geq 0 \}$$

The first few dominant weights and their corresponding irreducible representations are:

λ	(a, b)	$L(\lambda)$	$\dim L(\lambda)$
0	(0, 0)	trivial	1
ω_1	(1, 0)	standard k^3	3
ω_2	(0, 1)	dual $(k^3)^\vee \cong \bigwedge^2(k^3)$	3
$2\omega_1$	(2, 0)	$\text{Sym}^2(k^3)$	6
$\omega_1 + \omega_2$	(1, 1)	adjoint $\mathfrak{sl}_3$	8
$2\omega_2$	(0, 2)	$\text{Sym}^2((k^3)^\vee)$	6
$3\omega_1$	(3, 0)	$\text{Sym}^3(k^3)$	10

The dimensions can be computed from the Weyl dimension formula (Chapter 9). Note that the standard representation $L(\omega_1) = k^3$ and its dual $L(\omega_2) = (k^3)^\vee$ are the two “fundamental” representations (one for each node of the Dynkin diagram A_2), and every other irreducible representation is built from these via tensor products and direct summands.

Exercise 8.1.1

1. Let $G = \text{SL}_4$ with standard representation $V = k^4$. Compute the weights of V , $\bigwedge^2(V)$, and $\bigwedge^3(V)$. Identify the highest weight of each and verify they are the fundamental weights $\omega_1, \omega_2, \omega_3$ respectively. What is the relationship between $\bigwedge^3(V)$ and $V^\vee$?
2. For $G = \text{SL}_3$ with standard representation $V = k^3$, compute $\text{wt}(\text{Sym}^2(V))$ and verify that the highest weight is $2\omega_1$. What are the multiplicities of each weight?
3. Let V be a finite-dimensional representation of G (not necessarily irreducible) with $\text{wt}(V) = \{\lambda_1, \dots, \lambda_m\}$. Show that for each weight $\mu \in \text{wt}(V)$, the entire “ α -string through μ ” is contained in $\text{wt}(V)$: that is, for each simple root α_i , the weights $\mu, \mu - \alpha_i, \mu - 2\alpha_i, \dots, \mu - p\alpha_i$ are in $\text{wt}(V)$, where $p = \langle \mu, \alpha_i^\vee \rangle$. (*Hint*: restrict to the SL_2 -subgroup for α_i .)
4. Show that if V is a finite-dimensional (not necessarily irreducible) representation with a unique highest weight λ (i.e. $V_\lambda \neq 0$ and $V_\mu = 0$ for all $\mu \succ \lambda$), and $\dim V_\lambda = 1$, then V has a unique irreducible quotient with highest weight λ .
5. Draw the weight lattice $X(T) = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ for SL_3 in the plane, mark the dominant cone $X(T)^+$, and plot the weights of $L(\omega_1)$, $L(\omega_2)$, and $L(\omega_1 + \omega_2)$. Verify the W -invariance: for each representation, check that $W = S_3$ permutes the weight set.
6. For each of the classical groups SL_n , SO_{2n+1} , Sp_{2n} , SO_{2n} , identify the highest weight of the adjoint representation as a dominant weight in terms of the fundamental weights. (*Hint*: the highest root is the highest weight of the adjoint representation.)

8.2 Dominant Weights and the Classification

We proved in Section 8.1 that every irreducible representation has a dominant highest weight, and that this weight determines the representation uniquely (since the highest weight space is one-

dimensional and generates the representation). This gives an injection from isomorphism classes of irreducible representations to dominant weights. The content of this section is the *surjectivity*: every dominant weight $\lambda \in X(T)^+$ arises as the highest weight of some irreducible representation. The proof constructs the representation explicitly, by building a “universal” module with highest weight λ and taking its unique irreducible quotient.

Induced Modules

The construction uses the *induction* functor from B -representations to G -representations. Since the highest weight vector v^+ is a simultaneous eigenvector for B (the torus T acts by λ and $R_u(B)$ acts trivially), the one-dimensional space k_λ on which B acts by the character λ (extended trivially to $R_u(B)$) is the natural starting point.

Definition 8.2.1: Induced Module

Let $\lambda \in X(T)$ be a character. Define the one-dimensional B -representation k_λ : the vector space k on which $B = T \ltimes R_u(B)$ acts via $b \cdot z = \lambda(t)z$ for $b = tu$ with $t \in T$, $u \in R_u(B)$. The *induced module* (or *costandard module*) is:

$$H^0(\lambda) := \text{Ind}_B^G(k_\lambda) = \{f \in \mathcal{O}(G) \mid f(gb) = \lambda(b)^{-1}f(g) \text{ for all } g \in G, b \in B\}$$

with G acting by left translation: $(g' \cdot f)(g) = f(g'^{-1}g)$.

The notation $H^0(\lambda)$ anticipates the geometric interpretation: this is the space of global sections $H^0(G/B, \mathcal{L}(\lambda))$ of the line bundle $\mathcal{L}(\lambda)$ on the flag variety G/B , which we will develop in Chapter 10. For now, we work purely algebraically.

Proposition 8.2.2: Properties of the Induced Module

Let $\lambda \in X(T)$.

1. $H^0(\lambda)$ is a finite-dimensional representation of G (possibly zero).
2. $H^0(\lambda) \neq 0$ if and only if λ is dominant: $\lambda \in X(T)^+$.
3. If $\lambda \in X(T)^+$, then $H^0(\lambda)$ has a unique highest weight λ , with $\dim H^0(\lambda)_\lambda = 1$.
4. $H^0(\lambda)$ has a unique irreducible quotient, which we denote $L(\lambda)$.

Proof Sketch:

(1): The space $H^0(\lambda)$ is a subspace of $\mathcal{O}(G)$, and it is G -stable (left translation preserves the equivariance condition). Finite-dimensionality follows from the fact that $H^0(\lambda)$ is the space of global sections of a line bundle on the projective variety G/B , which is finite-dimensional by the projectivity of G/B (theorem 6.2.3) and Serre’s theorem on coherent sheaves on projective varieties (see [3]).

(2): The nonvanishing of $H^0(\lambda)$ is equivalent to the existence of a nonzero section of $\mathcal{L}(\lambda)$ on G/B . This is a consequence of the Borel–Weil theorem (Chapter 10), but can also be proved

directly: one shows that $H^0(\lambda)$ contains a unique B -eigenvector of weight λ (the “canonical section”), which exists if and only if λ is dominant. The “only if” direction follows from the SL_2 -restriction argument of proposition 8.1.12. For the “if” direction when λ is dominant, one constructs a nonzero section explicitly using the Bruhat decomposition $G = \bigsqcup_{w \in W} BwB$: the restriction of $f \in H^0(\lambda)$ to the open Bruhat cell Bw_0B (where w_0 is the longest Weyl group element) is determined by its value at w_0 , and one verifies that a nonzero choice extends to all of G .

(3): The weight λ appears in $H^0(\lambda)$ with multiplicity 1 (from the construction of the canonical section), and no weight $\mu \succ \lambda$ appears (by the equivariance condition: elements of $H^0(\lambda)$ transform by λ under B , so no higher weight can occur).

(4): Since $H^0(\lambda)_\lambda$ is one-dimensional and generates a submodule with no higher weight, the argument of ?? 4 applies: the sum of all proper submodules is again proper (it cannot contain $H^0(\lambda)_\lambda$), and the quotient by this maximal proper submodule is the unique irreducible quotient $L(\lambda)$.

The Classification Theorem

Theorem 8.2.3: Classification of Irreducible Representations

Let G be a connected reductive group over an algebraically closed field k , with maximal torus $T \subseteq B \subseteq G$ and dominant weights $X(T)^+$. The map

$$\lambda \mapsto L(\lambda)$$

is a bijection:

$$X(T)^+ \xrightarrow{1:1} \{\text{Irreducible representations of } G \mid \text{up to isomorphism}\}$$

Explicitly: every irreducible finite-dimensional representation of G is isomorphic to $L(\lambda)$ for a unique $\lambda \in X(T)^+$, where $L(\lambda)$ is the unique irreducible quotient of the induced module $H^0(\lambda)$.

Proof:

Surjectivity (every dominant weight arises): For $\lambda \in X(T)^+$, the module $H^0(\lambda) \neq 0$ (proposition 8.2.2(2)), and its unique irreducible quotient $L(\lambda)$ has highest weight λ (proposition 8.2.2(3)–(4)).

Injectivity (different dominant weights give non-isomorphic representations): If $L(\lambda) \cong L(\mu)$, then both have highest weight λ and highest weight μ . By the uniqueness of the highest weight in an irreducible representation (proposition 8.1.5), $\lambda = \mu$.

Exhaustiveness (every irreducible is some $L(\lambda)$): Let V be an irreducible representation of G . By proposition 8.1.5, V has a unique highest weight λ , and by proposition 8.1.12, $\lambda \in X(T)^+$. The highest weight vector $v^+ \in V_\lambda$ generates V (by irreducibility). We need $V \cong L(\lambda)$. Consider

the map $H^0(\lambda) \rightarrow V$ that sends the canonical section to (a nonzero multiple of) v^+ . This is... actually, the cleanest argument avoids an explicit map from $H^0(\lambda)$.

Instead: since V is irreducible with highest weight λ , any other irreducible representation V' with the same highest weight must satisfy $V' \cong V$. Indeed, V and V' both have one-dimensional λ -weight spaces, and the G -submodule of $V \oplus V'$ generated by $(v^+, (v')^+)$ (a vector of weight λ in the direct sum) maps surjectively onto both V and V' (by irreducibility of each), and its kernel intersects each summand trivially (by irreducibility again), forcing $V \cong V'$ by Schur's lemma (?? 5).

Since $L(\lambda)$ is an irreducible representation with highest weight λ (constructed above), we have $V \cong L(\lambda)$.

This theorem is the representation-theoretic analogue of the classification of reductive groups by root data (theorem 7.3.6): just as root data classify groups, dominant weights classify their irreducible representations. The two classifications are intimately related: the dominant weights lie in the character lattice $X(T)$, which is part of the root datum, and the root system Φ determines the dominance condition $\langle \lambda, \alpha_i^\vee \rangle \geq 0$.

Example 8.2.4: Irreducible Representations of SL_2

For $G = \mathrm{SL}_2$, the dominant weights are $X(T)^+ = \mathbb{Z}_{\geq 0}$ (the non-negative integers, since there is one simple root α with $\langle d, \alpha^\vee \rangle = d$). The classification gives:

$$L(d) = \mathrm{Sym}^d(k^2) \quad \text{for } d = 0, 1, 2, \dots$$

with $\dim L(d) = d + 1$. The weight set is $\{d, d - 2, \dots, -d\}$ (as in example 8.1.8). In characteristic 0, these are the *only* irreducible representations of SL_2 , and every finite-dimensional representation decomposes as a direct sum of these (by complete reducibility, theorem 8.3.1 below).

Example 8.2.5: Fundamental Representations

For each simple root $\alpha_i \in \Delta$, the fundamental weight ω_i (defined by $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$) is the “smallest” non-trivial dominant weight in the α_i -direction. The corresponding irreducible representation $L(\omega_i)$ is called the i -th *fundamental representation*. For the classical groups:

1. SL_n (type A_{n-1}): $L(\omega_i) = \bigwedge^i(k^n)$ for $1 \leq i \leq n - 1$. These are the exterior powers of the standard representation.
2. SO_{2n+1} (type B_n): $L(\omega_i) = \bigwedge^i(k^{2n+1})$ for $1 \leq i \leq n - 1$, and $L(\omega_n)$ is the *spin representation* of dimension 2^n .
3. Sp_{2n} (type C_n): $L(\omega_1) = k^{2n}$ (the standard representation). The higher fundamental representations are *not* exterior powers but rather their irreducible quotients (since $\bigwedge^i(k^{2n})$ is reducible for $i \geq 2$: the symplectic form gives a contraction map $\bigwedge^i \rightarrow \bigwedge^{i-2}$, and $L(\omega_i)$ is the kernel).
4. SO_{2n} (type D_n): $L(\omega_i) = \bigwedge^i(k^{2n})$ for $1 \leq i \leq n - 2$, and $L(\omega_{n-1}), L(\omega_n)$ are the two *half-spin representations*, each of dimension 2^{n-1} .

Every irreducible representation is a quotient of a tensor product of fundamental representations (in characteristic 0, a direct summand).

Example 8.2.6: The Representation Ring of SL_3

The irreducible representations of SL_3 are $\{L(a\omega_1 + b\omega_2) : a, b \geq 0\}$. In characteristic 0, the representation ring $R(\mathrm{SL}_3) = K_0(\mathrm{Rep}(\mathrm{SL}_3))$ is a polynomial ring:

$$R(\mathrm{SL}_3) \cong \mathbb{Z}[L(\omega_1), L(\omega_2)]$$

That is, every representation is a $\mathbb{Z}$ -linear combination of products (= tensor products) of the two fundamental representations $L(\omega_1) = k^3$ and $L(\omega_2) = (k^3)^\vee$. The tensor product decomposition $L(\omega_1) \otimes L(\omega_2) \cong L(\omega_1 + \omega_2) \oplus L(0)$ (the adjoint plus the trivial) is the simplest example of a “Clebsch–Gordan” decomposition for SL_3 .

Weyl Modules

There is a second natural module associated to a dominant weight λ , which will play a central role in the positive-characteristic theory of Chapter 11.

Definition 8.2.7: Weyl Module

Let $\lambda \in X(T)^+$ be a dominant weight. The *Weyl module* (or *standard module*) is

$$V(\lambda) := H^0(-w_0\lambda)^\vee$$

where $w_0 \in W$ is the longest element of the Weyl group and $(\cdot)^\vee$ denotes the contragredient (definition 2.3.4).

The Weyl module $V(\lambda)$ is the “universal” module with highest weight λ : it has a unique irreducible *quotient* $L(\lambda)$ (the unique simple head), while the induced module $H^0(\lambda)$ has $L(\lambda)$ as its unique irreducible quotient from a different perspective (the unique simple *socle* of $H^0(\lambda)^\vee$). In characteristic 0, $V(\lambda) = L(\lambda) = H^0(\lambda)^\vee$ (everything is irreducible). In characteristic p , $V(\lambda)$ can be strictly larger than $L(\lambda)$, and the “extra” composition factors are the subject of modular representation theory.

Proposition 8.2.8: Properties of Weyl Modules

Let $\lambda \in X(T)^+$.

1. $V(\lambda)$ has highest weight λ with $\dim V(\lambda)_\lambda = 1$.
2. $V(\lambda)$ has a unique maximal submodule $\mathrm{rad}V(\lambda)$, with $V(\lambda)/\mathrm{rad}V(\lambda) \cong L(\lambda)$.
3. The character of $V(\lambda)$ is given by the Weyl character formula (Chapter 9) and is *independent of the characteristic of k* .
4. In characteristic 0: $V(\lambda) = L(\lambda)$ (the Weyl module is irreducible).

Property (3) is crucial: the Weyl module always has the “expected” character (the one given by the Weyl formula), even in characteristic p where the module may be reducible. This means the Weyl character formula computes the character of $V(\lambda)$, not of $L(\lambda)$, in positive characteristic. The problem of computing $\mathrm{ch}(L(\lambda))$ in characteristic p — equivalently, computing the composition factors of $V(\lambda)$ — is the fundamental open problem of modular representation theory, which we discuss in

Chapter 11.

Exercise 8.2.1

1. For $G = \mathrm{SL}_2$, compute $H^0(d)$ explicitly for $d = 0, 1, 2$ by finding all $f \in \mathcal{O}(\mathrm{SL}_2)$ satisfying $f(gb) = \lambda(b)^{-1}f(g)$ for $b \in B$. Verify that $H^0(d) \cong \mathrm{Sym}^d(k^2)^\vee$.
2. Verify the fundamental representations of SL_4 : show that $L(\omega_1) = k^4$, $L(\omega_2) = \bigwedge^2(k^4)$ (dimension 6), and $L(\omega_3) = \bigwedge^3(k^4) \cong (k^4)^\vee$ (dimension 4). Compute the weights of $L(\omega_2)$ and verify the highest weight is $\omega_2 = \varepsilon_1 + \varepsilon_2$.
3. Decompose $L(\omega_1) \otimes L(\omega_1) = \mathrm{Sym}^2(k^3) \oplus \bigwedge^2(k^3)$ for SL_3 (in characteristic $\neq 2$). Identify the highest weights: $\mathrm{Sym}^2(k^3) = L(2\omega_1)$ and $\bigwedge^2(k^3) = L(\omega_2)$. Verify $\dim = 6 + 3 = 9 = 3^2$.
4. For $G = \mathrm{SL}_n$ ($n \geq 3$), show that the adjoint representation $\mathfrak{g} = \mathfrak{sl}_n$ is the irreducible representation $L(\omega_1 + \omega_{n-1})$. What happens for $n = 2$? (*Hint*: for $n = 2$, $\omega_1 + \omega_1 = 2\omega_1 = \alpha$, and $L(\alpha) = \mathrm{Sym}^2(k^2)$ has dimension $3 = \dim \mathfrak{sl}_2$.)
5. In characteristic $p = 2$, consider $G = \mathrm{SL}_2$ and $\lambda = 2$ (so $L(\lambda)$ has highest weight 2). Show that $V(2) = \mathrm{Sym}^2(k^2)$ is 3-dimensional, but $L(2) \subsetneq V(2)$: the vector xy (weight 0) generates a 1-dimensional submodule (the trivial representation), and $L(2) = V(2)/k\{xy\}$ is 2-dimensional. Verify this by computing the action of $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ on the basis $\{x^2, xy, y^2\}$ in characteristic 2.

8.3 Complete Reducibility

In characteristic 0, the representation theory of a reductive group is as clean as possible: every finite-dimensional representation is completely reducible (a direct sum of irreducibles). This is the analogue of Maschke's theorem for finite groups and Weyl's theorem for compact Lie groups. The proof uses the existence of a non-degenerate invariant bilinear form — the *Killing form* — which provides a “complement” for any subrepresentation.

Weyl's Complete Reducibility Theorem**Theorem 8.3.1: Complete Reducibility (Weyl's Theorem)**

Let G be a connected reductive group over an algebraically closed field k of characteristic 0. Then every finite-dimensional representation of G is completely reducible: for every subrepresentation $U \subseteq V$, there exists a complementary subrepresentation W with $V = U \oplus W$.

Proof:

We first prove the result for semisimple G , then extend to reductive G .

Step 1: The Casimir element. Let $\mathfrak{g} = \mathrm{Lie}(G)$ and let $\kappa : \mathfrak{g} \times \mathfrak{g} \rightarrow k$ be the Killing form, $\kappa(X, Y) = \mathrm{tr}(\mathrm{ad}(X) \circ \mathrm{ad}(Y))$. Since G is semisimple (and $\mathrm{char} k = 0$), the Killing form is non-degenerate (this is Cartan's criterion for semisimplicity; see [5]).

Choose a basis $\{X_1, \dots, X_n\}$ for $\mathfrak{g}$ and let $\{X^1, \dots, X^n\}$ be the dual basis with respect to κ : $\kappa(X_i, X^j) = \delta_{ij}$. The *Casimir element* is

$$C = \sum_{i=1}^n X_i \cdot X^i \in U(\mathfrak{g})$$

where $U(\mathfrak{g})$ is the universal enveloping algebra. The key properties of C are:

1. C is independent of the choice of basis.
2. C commutes with the $\mathfrak{g}$ -action: $[X, C] = 0$ for all $X \in \mathfrak{g}$. (This is because the Killing form is ad-invariant, proposition 3.2.11.)
3. On an irreducible representation $L(\lambda)$, C acts by a scalar $c(\lambda) \in k$ (by Schur's lemma, ?? 5). This scalar is $c(\lambda) = (\lambda, \lambda + 2\rho)$ where $\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$ and $(\cdot, \cdot)$ is the inner product on $X(T) \otimes \mathbb{R}$ induced by the Killing form.
4. $c(0) = 0$ (the Casimir acts by 0 on the trivial representation) and $c(\lambda) \neq 0$ for $\lambda \neq 0$ dominant (since $(\lambda, \lambda + 2\rho) > 0$ when λ is a nonzero dominant weight and ρ is in the interior of the dominant chamber).

Step 2: Splitting extensions. It suffices to show that every short exact sequence $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$ of G -representations splits. We proceed by induction on $\dim V$.

If W is irreducible with $W = L(\lambda)$ and $\lambda \neq 0$: the Casimir C acts on W by the nonzero scalar $c(\lambda)$ and on U by some (possibly reducible) action. Consider C acting on V : the subspace $\ker(C - c(\lambda)) \subseteq V$ maps isomorphically onto W under the projection $V \rightarrow W$ (because $C - c(\lambda)$ is zero on W and, by induction, the restriction of $C - c(\lambda)$ to U is injective on any U -quotient where C acts differently from $c(\lambda)$). This gives a section $W \rightarrow V$, splitting the sequence.

If $W = L(0) = k$ (the trivial representation): we need to split $0 \rightarrow U \rightarrow V \rightarrow k \rightarrow 0$, i.e. find a G -invariant complement to U in V , which is a G -fixed vector in V mapping to $1 \in k$. The Casimir acts by 0 on k and by 0 on U^G (the trivial sub-representations of U), so this case requires a different argument. One uses the fact that $\text{Ext}_G^1(k, U) = H^1(G, U) = 0$ for semisimple G in characteristic 0 (the vanishing of the first cohomology), which can be proved by the Casimir argument applied to the non-trivial composition factors of U .

Step 3: Extension to reductive G . If G is reductive but not semisimple, write $G = Z(G)^\circ \cdot [G, G]$ where $Z(G)^\circ$ is a torus and $[G, G]$ is semisimple (proposition 6.2.12). Any representation of G restricts to a representation of $[G, G]$, which is completely reducible by Step 2. The $Z(G)^\circ$ -action (being a torus action) is also completely reducible (theorem 5.2.9). Since the two actions commute and both are completely reducible, the combined G -action is completely reducible.

8.3.2 Remark: Failure in Characteristic p Complete reducibility fails dramatically in characteristic $p > 0$. Even for $G = \mathrm{SL}_2$, the Weyl module $V(p-1) = \mathrm{Sym}^{p-1}(k^2)$ is irreducible ($L(p-1) = V(p-1)$), but $V(p) = \mathrm{Sym}^p(k^2)$ is *not*: it contains the trivial submodule $k\{x^p + y^p\}$ (using $\mathrm{char} k = p$, so $(x+y)^p = x^p + y^p \dots$ actually x^p and y^p are SL_2 -invariant modulo lower terms). More precisely, the element x^p has weight p , but in $\mathrm{Sym}^p(k^2)$ the action of e on $x^{p-k}y^k$ involves coefficients $\binom{p-k+1}{1}$ which vanish modulo p at certain points, creating “gaps” in the weight string that allow submodules to appear. The full analysis is carried out in Chapter 11.

The failure is not merely technical: it reflects the fact that in characteristic p , the Killing form can be degenerate (for instance, $\kappa = 0$ on $\mathfrak{sl}_p$), and the Casimir element no longer provides complementary projections. This is why modular representation theory is fundamentally harder than the characteristic-0 theory.

Consequences of Complete Reducibility

Corollary 8.3.3: Structure of $\mathrm{Rep}(G)$ in Characteristic 0

Let G be a connected reductive group over an algebraically closed field of characteristic 0. Then:

1. Every finite-dimensional representation decomposes as $V \cong \bigoplus_{\lambda \in X(T)^+} L(\lambda)^{\oplus m_\lambda}$ for unique non-negative integers m_λ (the *multiplicities*).
2. The representation ring $R(G) = K_0(\mathrm{Rep}(G))$ is a free $\mathbb{Z}$ -module with basis $\{[L(\lambda)] : \lambda \in X(T)^+\}$.
3. The character map $\mathrm{ch} : R(G) \rightarrow \mathbb{Z}[X(T)]^W$ is an isomorphism: the representation ring is isomorphic to the ring of W -invariant elements in the group ring of $X(T)$.

Proof:

(1) is an immediate consequence of complete reducibility and the classification (theorem 8.2.3): decompose V into irreducibles, and each irreducible is some $L(\lambda)$.

(2) follows from (1): every element of $K_0(\mathrm{Rep}(G))$ is a $\mathbb{Z}$ -linear combination of the classes $[L(\lambda)]$, and these are linearly independent (their characters are linearly independent in $\mathbb{Z}[X(T)]$, by the fact that the highest weight λ appears with multiplicity 1 in $\mathrm{ch}(L(\lambda))$ and does not appear in $\mathrm{ch}(L(\mu))$ for $\mu \neq \lambda$).

(3): The character map sends $[V] \mapsto \mathrm{ch}(V) = \sum_{\lambda} (\dim V_{\lambda}) e^{\lambda} \in \mathbb{Z}[X(T)]$. It is a ring homomorphism (characters are additive on direct sums and multiplicative on tensor products), injective (by (2) and the linear independence of characters), and surjective onto $\mathbb{Z}[X(T)]^W$ (every W -invariant element is a character of some virtual representation, which can be verified using the Weyl character formula from Chapter 9).

Example 8.3.4: Tensor Product Decomposition for SL_2

In $\mathrm{Rep}(\mathrm{SL}_2)$, the tensor product $L(a) \otimes L(b)$ (with $a \geq b$) decomposes as:

$$L(a) \otimes L(b) \cong L(a+b) \oplus L(a+b-2) \oplus \cdots \oplus L(a-b)$$

This is the *Clebsch–Gordan formula*: $b+1$ summands, with highest weights $a+b, a+b-2, \dots, a-b$ (decreasing by 2). The formula follows from the Weyl character formula (Chapter 9) or by direct computation: the weights of $L(a) \otimes L(b)$ are $\{i+j : i \in \mathrm{wt}(L(a)), j \in \mathrm{wt}(L(b))\}$, and counting multiplicities gives the decomposition.

As a check: $\dim(L(a) \otimes L(b)) = (a+1)(b+1) = \sum_{k=0}^b (a+b-2k+1) = \sum_{k=0}^b \dim L(a+b-2k)$.

Exercise 8.3.1

1. For $G = \mathrm{SL}_2$, compute the Casimir element $C \in U(\mathfrak{sl}_2)$ using the basis $\{e, f, h\}$ and the Killing form κ from ?? 3. Show that C acts on $L(d) = \mathrm{Sym}^d(k^2)$ by the scalar $\frac{d(d+2)}{2}$ (in the normalization where $\kappa(h, h) = 8$).
2. Let $G = \mathrm{SL}_2$ in characteristic 0 and $V = k^2 \otimes k^2$ (the tensor square of the standard representation). Decompose V into irreducibles. Then verify complete reducibility by finding a G -stable complement to the subrepresentation $\mathrm{Sym}^2(k^2) \subseteq V$.
3. Show that for a semisimple group G in characteristic 0, $\mathrm{Hom}_G(L(\lambda), L(\mu)) = 0$ for $\lambda \neq \mu$ and $\mathrm{End}_G(L(\lambda)) = k$ (i.e. Schur's lemma). Deduce that $\mathrm{Ext}_G^1(L(\lambda), L(\mu)) = 0$ for all λ, μ (using the Casimir argument from the proof of theorem 8.3.1).
4. Show that the representation ring $R(\mathrm{SL}_2) \cong \mathbb{Z}[x]$ where $x = [L(1)]$ (the class of the standard representation). Express $[L(d)]$ as a polynomial in x . (*Hint*: use the Clebsch–Gordan formula from example 8.3.4 to compute $x^d = [L(1)]^d$ and solve for $[L(d)]$.)
5. Let $G = \mathrm{SL}_2$ over a field of characteristic $p = 3$, and consider $V = L(1) \otimes L(1) = k^2 \otimes k^2$ (4-dimensional). Show that V contains a subrepresentation isomorphic to $L(0)$ (the trivial representation), but the Clebsch–Gordan decomposition $L(2) \oplus L(0)$ *fails*: V is indecomposable (it does not split as a direct sum). (*Hint*: the antisymmetric tensor $e_1 \wedge e_2$ generates a trivial submodule, but there is no G -stable complement because $\mathrm{Sym}^2(k^2)$ is not irreducible in characteristic 3 — check whether $L(2)$ has dimension 3 or 2 in characteristic 3.)

9

Characters and the Weyl Character Formula

Chapter 8 classified the irreducible representations of a connected reductive group G in characteristic 0: they are indexed by dominant weights $\lambda \in X(T)^+$, and every representation decomposes as a direct sum of irreducibles. But the classification does not tell us *what* these representations look like: how many weights does $L(\lambda)$ have? What are their multiplicities? What is $\dim L(\lambda)$? How does $L(\lambda) \otimes L(\mu)$ decompose?

All of these questions are answered by the *character*, a generating function that encodes the complete weight-multiplicity data of a representation. The central result of this chapter is the *Weyl character formula*, which gives an explicit closed-form expression for the character of $L(\lambda)$ as an alternating sum over the Weyl group. From this single formula, we will extract: the *Weyl dimension formula* (the dimension of $L(\lambda)$ as a product over positive roots), *Freudenthal's formula* (a recursive computation of individual weight multiplicities), and the *tensor product decomposition rules* (how to decompose $L(\lambda) \otimes L(\mu)$ into irreducibles).

Throughout this chapter, k is an algebraically closed field of characteristic 0 (unless otherwise stated), G is a connected reductive group over k , $T \subseteq B \subseteq G$ is a maximal torus inside a Borel subgroup, and we use the notation of Chapter 7: Φ , Φ^+ , $\Delta = \{\alpha_1, \dots, \alpha_r\}$, W , and $\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$ (the *Weyl vector*, or half-sum of positive roots).

9.1 Formal Characters

The Group Ring of the Weight Lattice

To keep track of weights and their multiplicities algebraically, we work in the *group ring* of the character lattice.

Definition 9.1.1: The Group Ring

The *group ring* of $X(T)$ is the free abelian group

$$\mathbb{Z}[X(T)] = \bigoplus_{\lambda \in X(T)} \mathbb{Z} \cdot e^\lambda$$

with multiplication defined by $e^\lambda \cdot e^\mu = e^{\lambda+\mu}$ and extended $\mathbb{Z}$ -linearly. The identity element is $e^0 = 1$.

Elements of $\mathbb{Z}[X(T)]$ are formal $\mathbb{Z}$ -linear combinations of the symbols e^λ , and the multiplication is determined by the addition in $X(T)$. When $T = \mathbb{G}_m$ and $X(T) = \mathbb{Z}$, the group ring is $\mathbb{Z}[X(T)] = \mathbb{Z}[e^\varepsilon, e^{-\varepsilon}] \cong \mathbb{Z}[t, t^{-1}]$ (Laurent polynomials in one variable). For $T = \mathbb{G}_m^r$, it is $\mathbb{Z}[t_1^{\pm 1}, \dots, t_r^{\pm 1}]$ (Laurent polynomials in r variables).

The Weyl group W acts on $\mathbb{Z}[X(T)]$ by $w \cdot e^\lambda = e^{w(\lambda)}$, extended linearly. A central role is played by the subring of W -invariants:

$$\mathbb{Z}[X(T)]^W = \{f \in \mathbb{Z}[X(T)] \mid w \cdot f = f \text{ for all } w \in W\}$$

We proved in corollary 8.3.3 that the character map gives an isomorphism $R(G) \cong \mathbb{Z}[X(T)]^W$ in characteristic 0.

Characters of Representations

Definition 9.1.2: Formal Character

Let V be a finite-dimensional representation of G . The *(formal) character* of V is the element

$$\text{ch}(V) = \sum_{\lambda \in X(T)} (\dim V_\lambda) e^\lambda \in \mathbb{Z}[X(T)]$$

This is a finite sum (only finitely many weight spaces are nonzero).

The character is a “generating function” that packages all weight multiplicities into a single algebraic object. It determines the representation up to semisimplification (i.e., up to isomorphism in characteristic 0 where everything is semisimple, and up to the same composition factors with the same multiplicities in general).

Proposition 9.1.3: Basic Properties of Characters

Let V, W be finite-dimensional representations of G . Then:

1. (Additivity) $\text{ch}(V \oplus W) = \text{ch}(V) + \text{ch}(W)$. More generally, if $0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$ is a short exact sequence, then $\text{ch}(V) = \text{ch}(V') + \text{ch}(V'')$.
2. (Multiplicativity) $\text{ch}(V \otimes W) = \text{ch}(V) \cdot \text{ch}(W)$ (product in $\mathbb{Z}[X(T)]$).
3. (Duality) $\text{ch}(V^\vee) = \overline{\text{ch}(V)}$, where $-$ is the involution $e^\lambda \mapsto e^{-\lambda}$.
4. (W -invariance) $\text{ch}(V) \in \mathbb{Z}[X(T)]^W$.
5. (Dimension) Evaluating at $e^\lambda = 1$ for all λ gives $\text{ch}(V)|_{e^\lambda=1} = \dim V$.

Proof:

(1): $(V \oplus W)_\lambda = V_\lambda \oplus W_\lambda$, so $\dim(V \oplus W)_\lambda = \dim V_\lambda + \dim W_\lambda$. For a short exact sequence: $\dim V_\lambda = \dim V'_\lambda + \dim V''_\lambda$ (exactness of weight space functors, since the torus T is linearly reductive in any characteristic — its representations are completely reducible by theorem 5.2.9).

(2): $(V \otimes W)_\lambda = \bigoplus_{\mu+\nu=\lambda} V_\mu \otimes W_\nu$, so $\dim(V \otimes W)_\lambda = \sum_{\mu+\nu=\lambda} \dim V_\mu \cdot \dim W_\nu$. This is exactly the coefficient of e^λ in $\text{ch}(V) \cdot \text{ch}(W)$.

(3): $(V^\vee)_\lambda = (V_{-\lambda})^*$ (the dual of a weight- $(-\lambda)$ vector is a weight- λ functional under the contragredient action, definition 2.3.4), so $\dim(V^\vee)_\lambda = \dim V_{-\lambda}$.

(4): This is proposition 8.1.3: $\dim V_{w(\lambda)} = \dim V_\lambda$ for all $w \in W$.

(5): Setting $e^\lambda = 1$ for all λ gives $\text{ch}(V)|_1 = \sum_\lambda \dim V_\lambda = \dim V$.

Example 9.1.4: Characters of SL_2 -Representations

For $G = \text{SL}_2$ with $X(T) = \mathbb{Z}$ and $e^\varepsilon = t$, the character of $L(d) = \text{Sym}^d(k^2)$ is:

$$\text{ch}(L(d)) = t^d + t^{d-2} + \cdots + t^{-d+2} + t^{-d} = \frac{t^{d+1} - t^{-(d+1)}}{t - t^{-1}}$$

(a geometric series with $d+1$ terms). Setting $t = 1$: $\text{ch}(L(d))|_{t=1} = d+1 = \dim L(d)$. The W -invariance ($W = \{1, s\}$ with $s \cdot t = t^{-1}$) is visible: the character is a palindromic Laurent polynomial ($t^d + t^{-d} + \cdots$).

The multiplicativity: $\text{ch}(L(a)) \cdot \text{ch}(L(b)) = \text{ch}(L(a) \otimes L(b))$. The Clebsch–Gordan decomposition (example 8.3.4) reads, at the character level:

$$\frac{t^{a+1} - t^{-(a+1)}}{t - t^{-1}} \cdot \frac{t^{b+1} - t^{-(b+1)}}{t - t^{-1}} = \sum_{k=0}^b \frac{t^{a+b-2k+1} - t^{-(a+b-2k+1)}}{t - t^{-1}}$$

The reader should verify this identity of Laurent polynomials directly (it is a telescoping sum).

Example 9.1.5: Characters of SL_3 -Representations

For $G = \mathrm{SL}_3$ with $X(T) = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and $e^{\omega_1} = x$, $e^{\omega_2} = y$:

1. $\mathrm{ch}(L(\omega_1)) = e^{\varepsilon_1} + e^{\varepsilon_2} + e^{\varepsilon_3} = x + xy^{-1} + x^{-1}$ (using $\varepsilon_1 = \omega_1$, $\varepsilon_2 = \omega_1 - \alpha_1 = -\omega_1 + \omega_2$, $\varepsilon_3 = -\omega_2$). Actually, let us use the variables e^{ε_i} : $\mathrm{ch}(L(\omega_1)) = e^{\varepsilon_1} + e^{\varepsilon_2} + e^{\varepsilon_3}$, the sum of the three standard weights.
2. $\mathrm{ch}(L(\omega_2)) = e^{-\varepsilon_3} + e^{-\varepsilon_2} + e^{-\varepsilon_1} = e^{\varepsilon_1+\varepsilon_2} + e^{\varepsilon_1+\varepsilon_3} + e^{\varepsilon_2+\varepsilon_3}$ (the dual representation).
3. $\mathrm{ch}(L(\omega_1 + \omega_2))$: the adjoint representation $\mathfrak{sl}_3$ has weights $\{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2), 0, 0\}$ (the six roots plus the two-dimensional zero weight space $\mathfrak{t}$). So:

$$\mathrm{ch}(L(\omega_1 + \omega_2)) = e^{\alpha_1} + e^{\alpha_2} + e^{\alpha_1+\alpha_2} + e^{-\alpha_1} + e^{-\alpha_2} + e^{-\alpha_1-\alpha_2} + 2e^0$$

The $W = S_3$ -invariance of each character can be verified by checking that permuting the ε_i leaves the expression unchanged (which it does, since the weight sets are S_3 -orbits in each case).

The Weyl Denominator

Before stating the Weyl character formula, we introduce a key auxiliary object: the *Weyl denominator*, which appears in the denominator of the formula and has a beautiful closed-form expression.

Definition 9.1.6: The Weyl Denominator

The *Weyl denominator* is the element $D \in \mathbb{Z}[X(T)]$ defined by:

$$D = \sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)}$$

where $\rho = \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$ is the Weyl vector and $\ell(w)$ is the length of w (the number of simple reflections in a reduced expression for w , equivalently $\ell(w) = |\Phi^+ \cap w^{-1}(\Phi^-)|$).

Theorem 9.1.7: Weyl Denominator Formula

The Weyl denominator has the product expansion:

$$D = \sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)} = e^\rho \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})$$

Equivalently, after multiplying both sides by $e^{-\rho}$:

$$\sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho) - \rho} = \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})$$

Proof:

We give the proof for SL_2 and indicate the strategy for the general case.

SL_2 : Here $W = \{1, s\}$, $\Phi^+ = \{\alpha\}$, $\rho = \alpha/2$, $\ell(1) = 0$, $\ell(s) = 1$, and $s(\rho) = -\rho$. The left side is $e^\rho - e^{-\rho}$. The right side is $e^\rho(1 - e^{-\alpha}) = e^\rho - e^{\rho-\alpha} = e^{\alpha/2} - e^{-\alpha/2}$. These are equal.

General case (sketch): The product $e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha})$ expands as e^ρ times a sum over subsets $S \subseteq \Phi^+$, each contributing $(-1)^{|S|} e^{-\sum_{\alpha \in S} \alpha}$. One shows that most terms cancel, and the surviving terms are exactly $\{(-1)^{\ell(w)} e^{w(\rho)}\}_{w \in W}$. The cancellation is proved by a combinatorial argument using the fact that W acts simply transitively on the set of Weyl chambers (see Humphreys [6] for the full details).

Example 9.1.8: The Weyl Denominator for SL_3

For SL_3 : $W = S_3$, $\Phi^+ = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2\}$, $\rho = \alpha_1 + \alpha_2 = \omega_1 + \omega_2$. The six Weyl group elements and their contributions are:

w	$\ell(w)$	$w(\rho)$	$(-1)^{\ell(w)} e^{w(\rho)}$
e	0	$\alpha_1 + \alpha_2$	$+e^{\alpha_1 + \alpha_2}$
s_1	1	$-\alpha_1 + \alpha_1 + \alpha_2 = \alpha_2$	$-e^{\alpha_2}$
s_2	1	α_1	$-e^{\alpha_1}$
$s_1 s_2$	2	$-\alpha_2$	$+e^{-\alpha_2}$
$s_2 s_1$	2	$-\alpha_1$	$+e^{-\alpha_1}$
$w_0 = s_1 s_2 s_1$	3	$-\alpha_1 - \alpha_2$	$-e^{-\alpha_1 - \alpha_2}$

The denominator is:

$$D = e^{\alpha_1 + \alpha_2} - e^{\alpha_2} - e^{\alpha_1} + e^{-\alpha_2} + e^{-\alpha_1} - e^{-\alpha_1 - \alpha_2}$$

The product formula gives the same result:

$$e^\rho (1 - e^{-\alpha_1})(1 - e^{-\alpha_2})(1 - e^{-\alpha_1 - \alpha_2})$$

which the reader should expand and verify equals D .

Characters Determine Representations

The character map is injective on the representation ring:

Proposition 9.1.9: Characters Separate Representations

In characteristic 0, the character map $\text{ch} : R(G) \rightarrow \mathbb{Z}[X(T)]^W$ is injective. That is, two representations with the same character are isomorphic.

Proof:

By complete reducibility (theorem 8.3.1), every representation V decomposes as $V \cong \bigoplus_{\lambda} L(\lambda)^{\oplus m_{\lambda}}$. The character is $\text{ch}(V) = \sum_{\lambda} m_{\lambda} \text{ch}(L(\lambda))$. The characters $\{\text{ch}(L(\lambda))\}_{\lambda \in X(T)^+}$ are linearly independent in $\mathbb{Z}[X(T)]$: the highest weight λ appears in $\text{ch}(L(\lambda))$ with coefficient 1, and does not appear in $\text{ch}(L(\mu))$ for any $\mu \neq \lambda$ with $\mu \succeq \lambda$ (since λ is the *unique* highest weight of $L(\lambda)$). So the multiplicities m_{λ} are uniquely determined by $\text{ch}(V)$.

This means that computing the character of a representation is *equivalent* to decomposing it into irreducibles. The Weyl character formula, which we state in the next section, reduces this computation to a finite, explicit algebraic calculation.

9.1.10 Remark: Characters in Positive Characteristic In characteristic $p > 0$, the character still makes sense and still satisfies additivity on short exact sequences (proposition 9.1.3(1)). However, it does *not* determine the representation up to isomorphism: non-isomorphic representations can have the same character (since complete reducibility fails, a representation and its semisimplification have the same character but may not be isomorphic). The character determines only the *composition factors* and their multiplicities.

The Weyl module $V(\lambda)$ has the same character as $L(\lambda)$ would have in characteristic 0 (this is property (3) of proposition 8.2.8). But in characteristic p , $V(\lambda)$ is reducible, and $\text{ch}(V(\lambda)) = \text{ch}(L(\lambda)) + \sum_{\mu \prec \lambda} [V(\lambda) : L(\mu)] \text{ch}(L(\mu))$, where $[V(\lambda) : L(\mu)]$ are the composition multiplicities. Computing these multiplicities is the central problem of Chapter 11.

Exercise 9.1.1

1. For $G = \text{SL}_2$, verify the identity $\text{ch}(L(a)) \cdot \text{ch}(L(b)) = \sum_{k=0}^b \text{ch}(L(a+b-2k))$ (the Clebsch–Gordan formula at the character level) by direct computation with Laurent polynomials for $a = 3, b = 2$.
2. Compute $\text{ch}(L(\omega_1)) \cdot \text{ch}(L(\omega_1))$ for SL_3 (the character of $V = k^3 \otimes k^3$). Decompose the result into $\text{ch}(L(2\omega_1)) + \text{ch}(L(\omega_2))$ and verify this matches $\text{Sym}^2(k^3) \oplus \bigwedge^2(k^3)$.
3. For $G = \text{SL}_n$, show that $\text{ch}(L(\lambda)^{\vee}) = \text{ch}(L(-w_0\lambda))$, where $w_0 \in W$ is the longest element. Deduce that $L(\lambda)^{\vee} \cong L(-w_0\lambda)$. For SL_3 , what is $L(\omega_1)^{\vee}$?
4. Expand the product $e^{\rho} \prod_{\alpha \in \Phi^+} (1 - e^{-\alpha})$ for $G = \text{SL}_3$ (with $\Phi^+ = \{\alpha_1, \alpha_2, \alpha_1 + \alpha_2\}$) and verify it equals the alternating sum $\sum_{w \in W} (-1)^{\ell(w)} e^{w(\rho)}$ computed in example 9.1.8.
5. Show that for $G = \text{SL}_3$, the ring $\mathbb{Z}[X(T)]^W$ is a polynomial ring in two variables: $\mathbb{Z}[X(T)]^W = \mathbb{Z}[\sigma_1, \sigma_2]$ where $\sigma_1 = e^{\varepsilon_1} + e^{\varepsilon_2} + e^{\varepsilon_3}$ and $\sigma_2 = e^{\varepsilon_1 + \varepsilon_2} + e^{\varepsilon_1 + \varepsilon_3} + e^{\varepsilon_2 + \varepsilon_3}$ are the elementary symmetric polynomials. Identify $\sigma_1 = \text{ch}(L(\omega_1))$ and $\sigma_2 = \text{ch}(L(\omega_2))$.
6. Let $0 \rightarrow L(\mu) \rightarrow V \rightarrow L(\lambda) \rightarrow 0$ be a non-split short exact sequence of G -representations (in characteristic p). Show that $\text{ch}(V) = \text{ch}(L(\lambda)) + \text{ch}(L(\mu))$ even though $V \not\cong L(\lambda) \oplus L(\mu)$. Conclude that the character does not determine the isomorphism class of V in characteristic p .

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